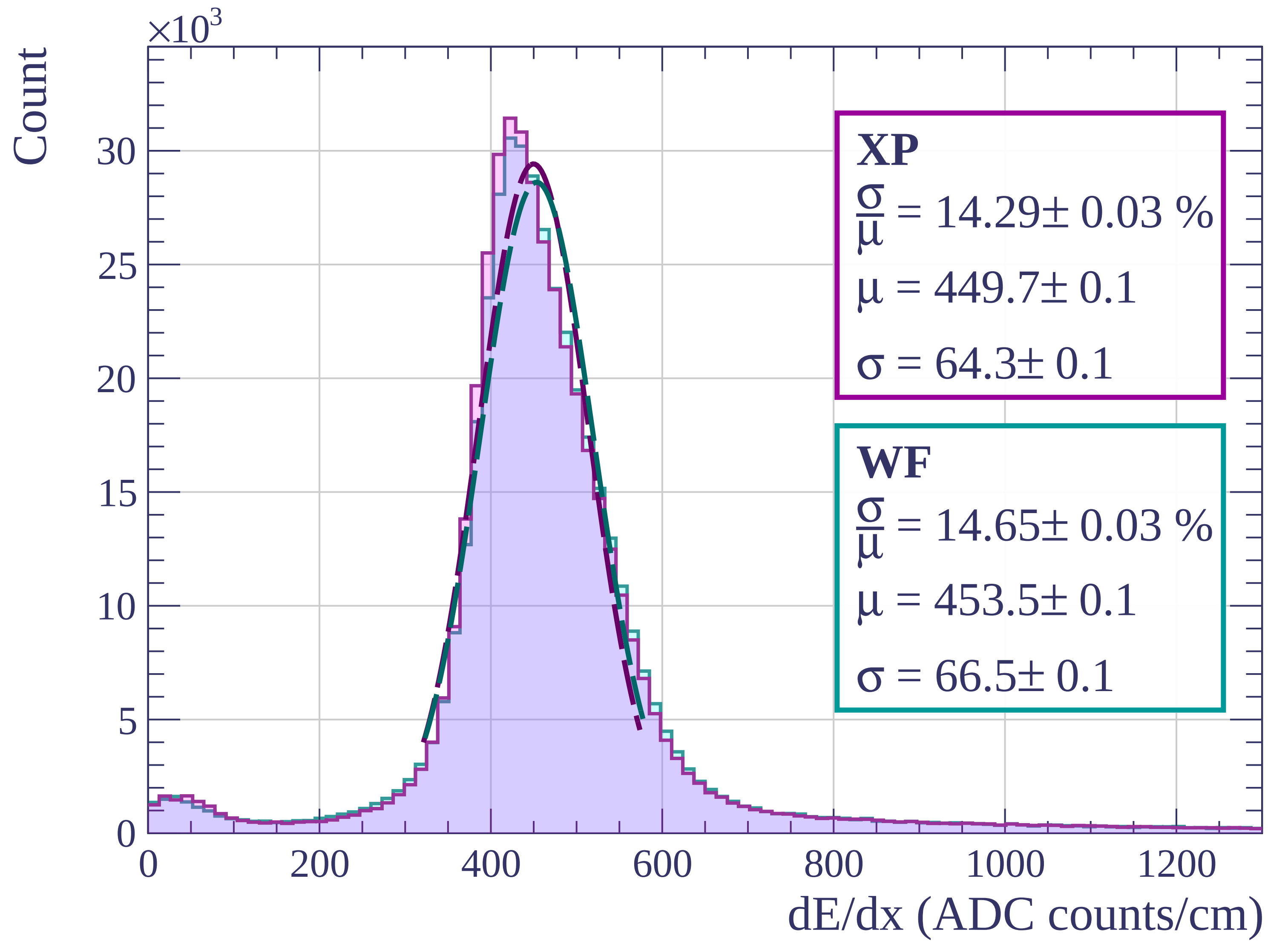
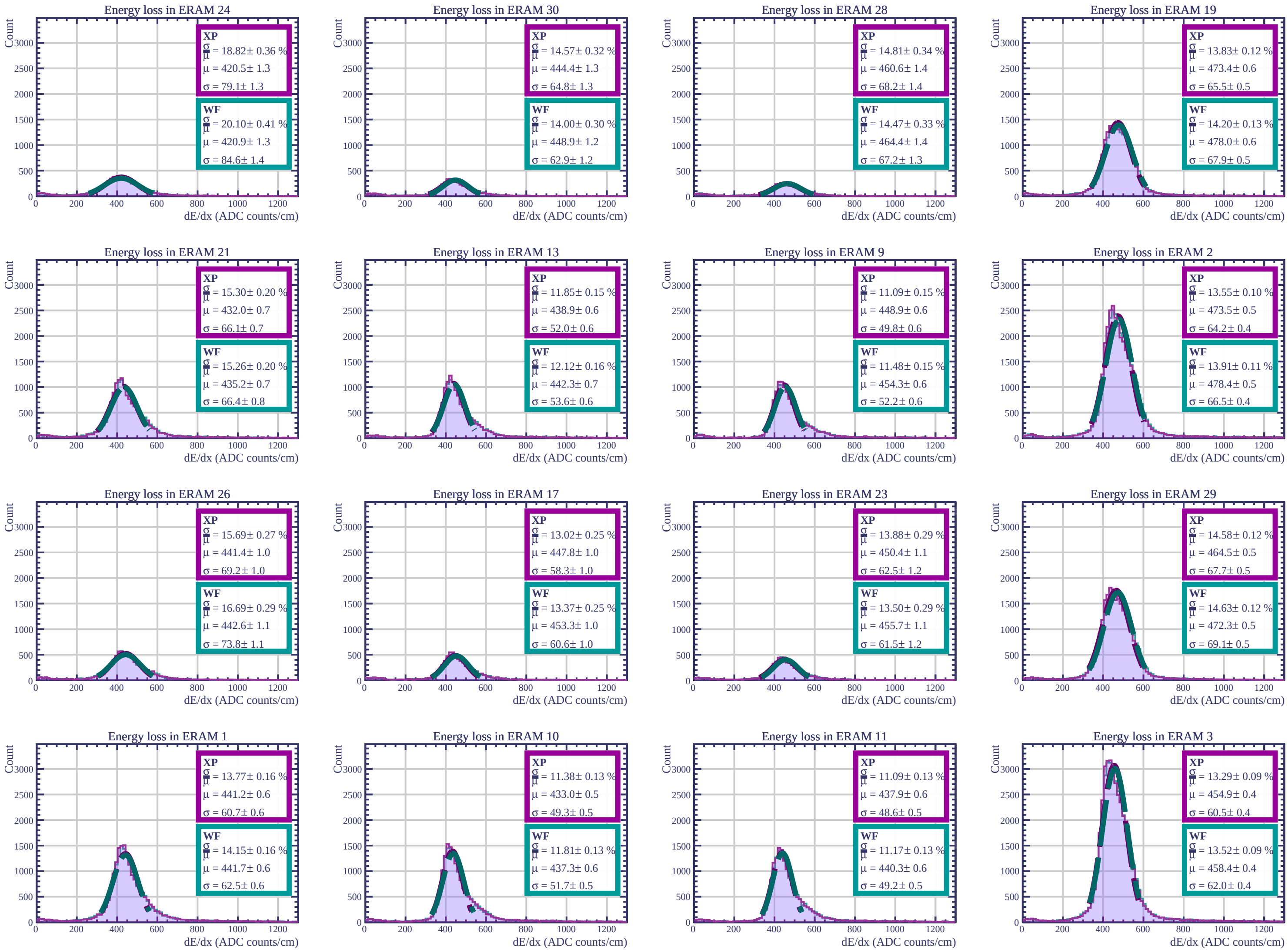
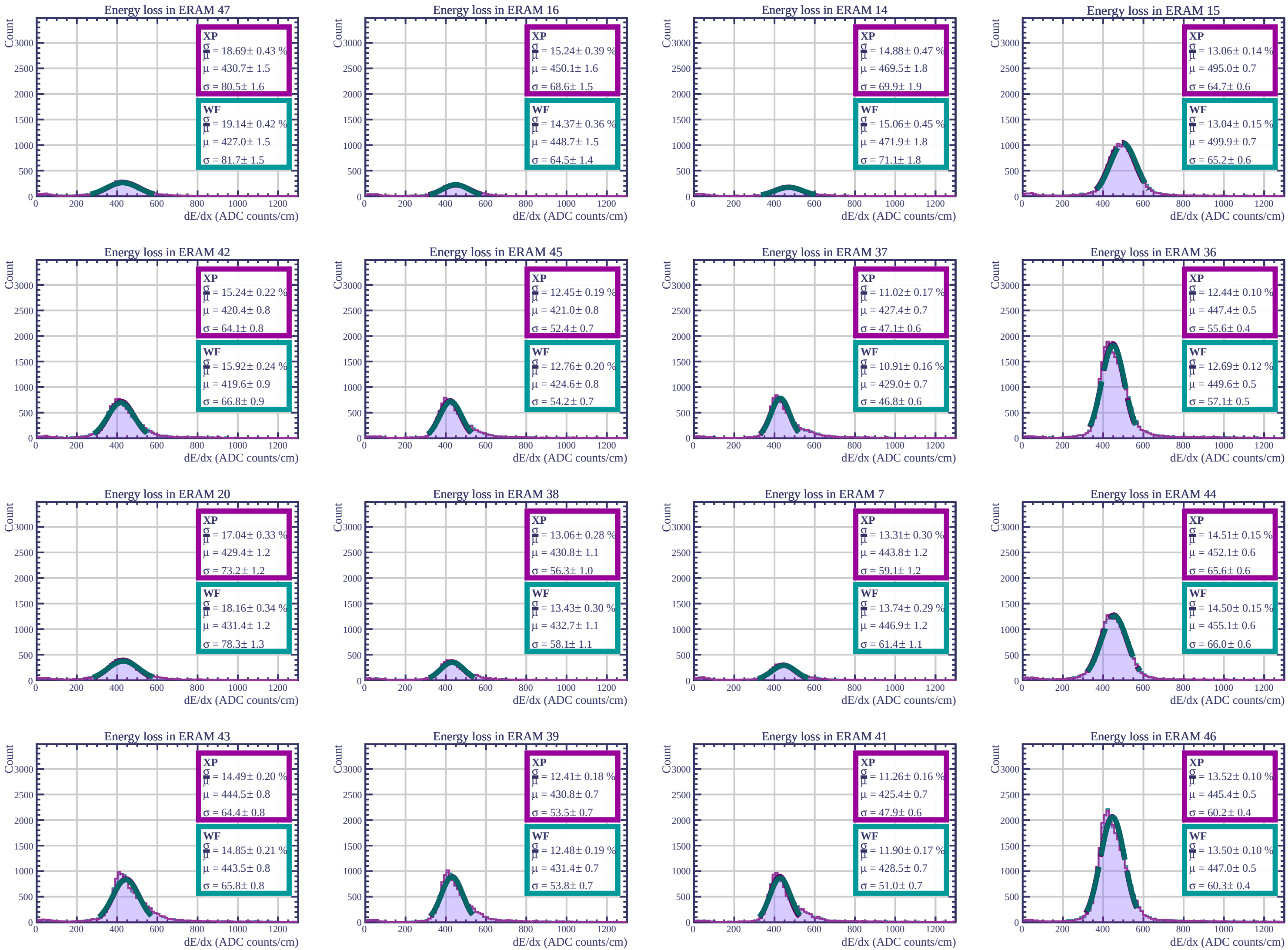
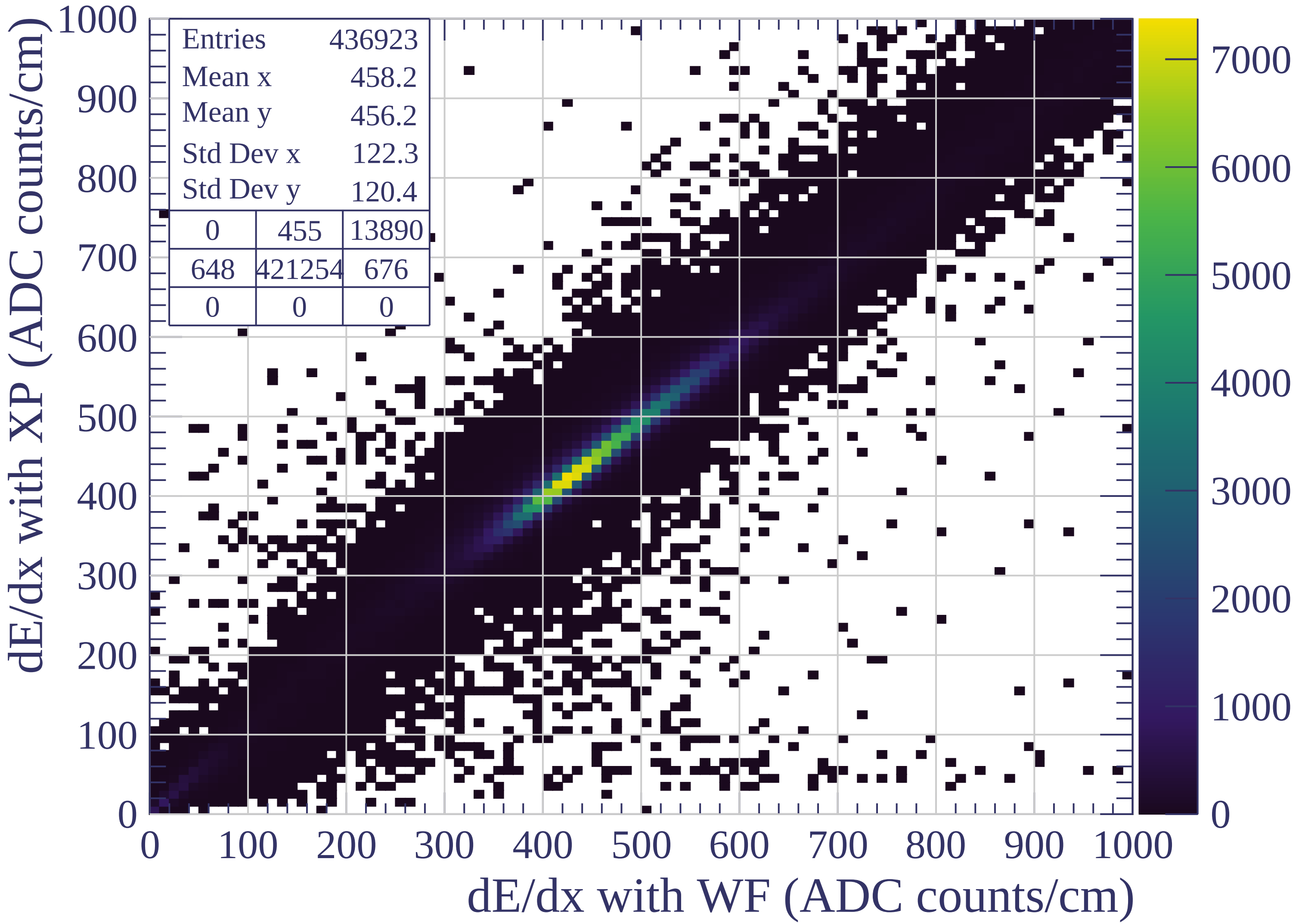


Count
×10^3
30
25
20
15
10
5
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 14.29± 0.03 %
μ = 449.7± 0.1
σ = 64.3± 0.1
WF
σ/μ = 14.65± 0.03 %
μ = 453.5± 0.1
σ = 66.5± 0.1

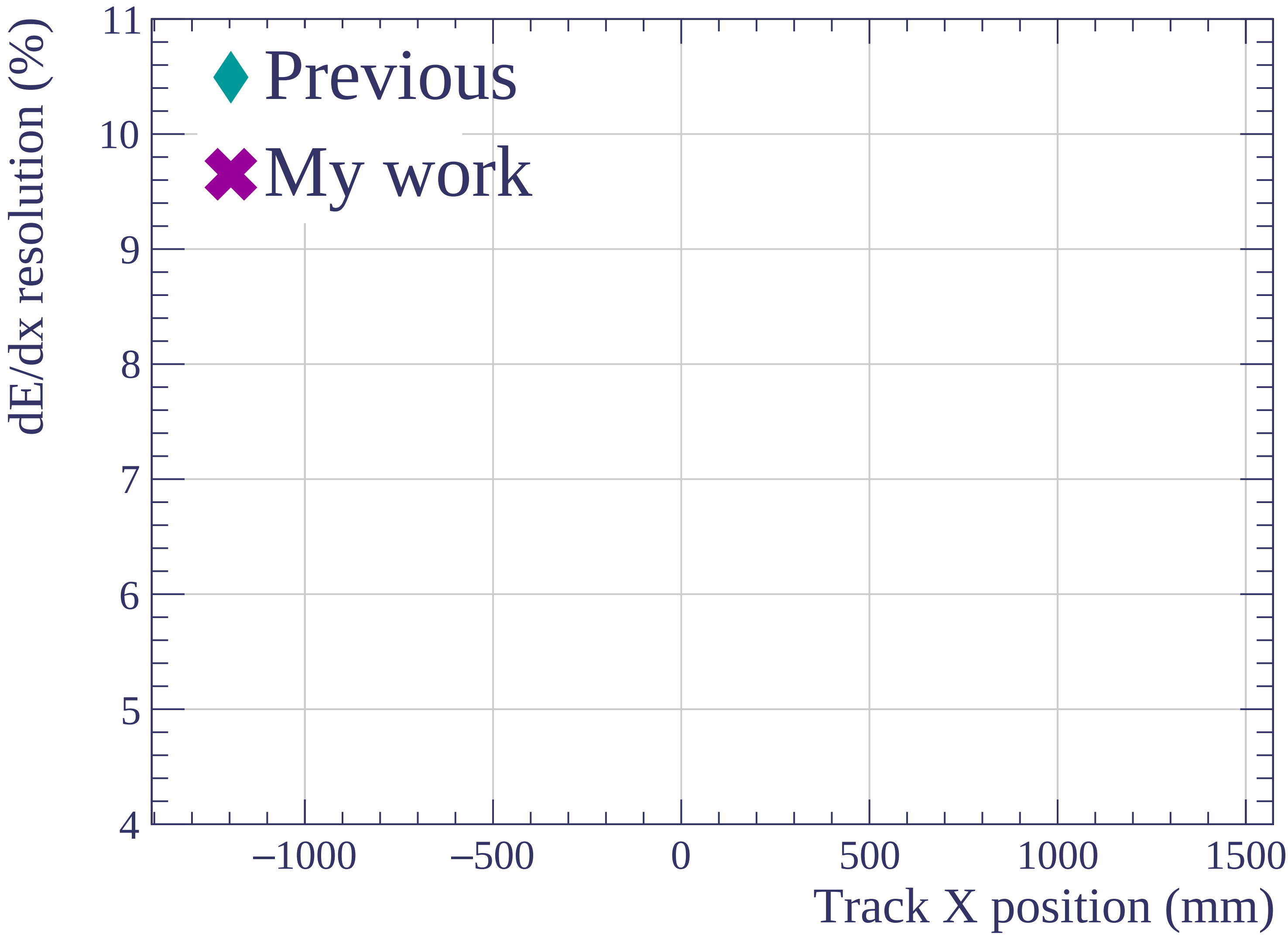


Energy loss in ERAM 24
XP
σ/μ = 18.82± 0.36 %
μ = 420.5± 1.3
σ = 79.1± 1.3
WF
σ/μ = 20.10± 0.41 %
μ = 420.9± 1.3
σ = 84.6± 1.4
Energy loss in ERAM 30
XP
σ/μ = 14.57± 0.32 %
μ = 444.4± 1.3
σ = 64.8± 1.3
WF
σ/μ = 14.00± 0.30 %
μ = 448.9± 1.2
σ = 62.9± 1.2
Energy loss in ERAM 28
XP
σ/μ = 14.81± 0.34 %
μ = 460.6± 1.4
σ = 68.2± 1.4
WF
σ/μ = 14.47± 0.33 %
μ = 464.4± 1.4
σ = 67.2± 1.3
Energy loss in ERAM 19
XP
σ/μ = 13.83± 0.12 %
μ = 473.4± 0.6
σ = 65.5± 0.5
WF
σ/μ = 14.20± 0.13 %
μ = 478.0± 0.6
σ = 67.9± 0.5
Energy loss in ERAM 21
XP
σ/μ = 15.30± 0.20 %
μ = 432.0± 0.7
σ = 66.1± 0.7
WF
σ/μ = 15.26± 0.20 %
μ = 435.2± 0.7
σ = 66.4± 0.8
Energy loss in ERAM 13
XP
σ/μ = 11.85± 0.15 %
μ = 438.9± 0.6
σ = 52.0± 0.6
WF
σ/μ = 12.12± 0.16 %
μ = 442.3± 0.7
σ = 53.6± 0.6
Energy loss in ERAM 9
XP
σ/μ = 11.09± 0.15 %
μ = 448.9± 0.6
σ = 49.8± 0.6
WF
σ/μ = 11.48± 0.15 %
μ = 454.3± 0.6
σ = 52.2± 0.6
Energy loss in ERAM 2
XP
σ/μ = 13.55± 0.10 %
μ = 473.5± 0.5
σ = 64.2± 0.4
WF
σ/μ = 13.91± 0.11 %
μ = 478.4± 0.5
σ = 66.5± 0.4
Energy loss in ERAM 26
XP
σ/μ = 15.69± 0.27 %
μ = 441.4± 1.0
σ = 69.2± 1.0
WF
σ/μ = 16.69± 0.29 %
μ = 442.6± 1.1
σ = 73.8± 1.1
Energy loss in ERAM 17
XP
σ/μ = 13.02± 0.25 %
μ = 447.8± 1.0
σ = 58.3± 1.0
WF
σ/μ = 13.37± 0.25 %
μ = 453.3± 1.0
σ = 60.6± 1.0
Energy loss in ERAM 23
XP
σ/μ = 13.88± 0.29 %
μ = 450.4± 1.1
σ = 62.5± 1.2
WF
σ/μ = 13.50± 0.29 %
μ = 455.7± 1.1
σ = 61.5± 1.2
Energy loss in ERAM 29
XP
σ/μ = 14.58± 0.12 %
μ = 464.5± 0.5
σ = 67.7± 0.5
WF
σ/μ = 14.63± 0.12 %
μ = 472.3± 0.5
σ = 69.1± 0.5
Energy loss in ERAM 1
XP
σ/μ = 13.77± 0.16 %
μ = 441.2± 0.6
σ = 60.7± 0.6
WF
σ/μ = 14.15± 0.16 %
μ = 441.7± 0.6
σ = 62.5± 0.6
Energy loss in ERAM 10
XP
σ/μ = 11.38± 0.13 %
μ = 433.0± 0.5
σ = 49.3± 0.5
WF
σ/μ = 11.81± 0.13 %
μ = 437.3± 0.6
σ = 51.7± 0.5
Energy loss in ERAM 11
XP
σ/μ = 11.09± 0.13 %
μ = 437.9± 0.6
σ = 48.6± 0.5
WF
σ/μ = 11.17± 0.13 %
μ = 440.3± 0.6
σ = 49.2± 0.5
Energy loss in ERAM 3
XP
σ/μ = 13.29± 0.09 %
μ = 454.9± 0.4
σ = 60.5± 0.4
WF
σ/μ = 13.52± 0.09 %
μ = 458.4± 0.4
σ = 62.0± 0.4
Count
dE/dx (ADC counts/cm)

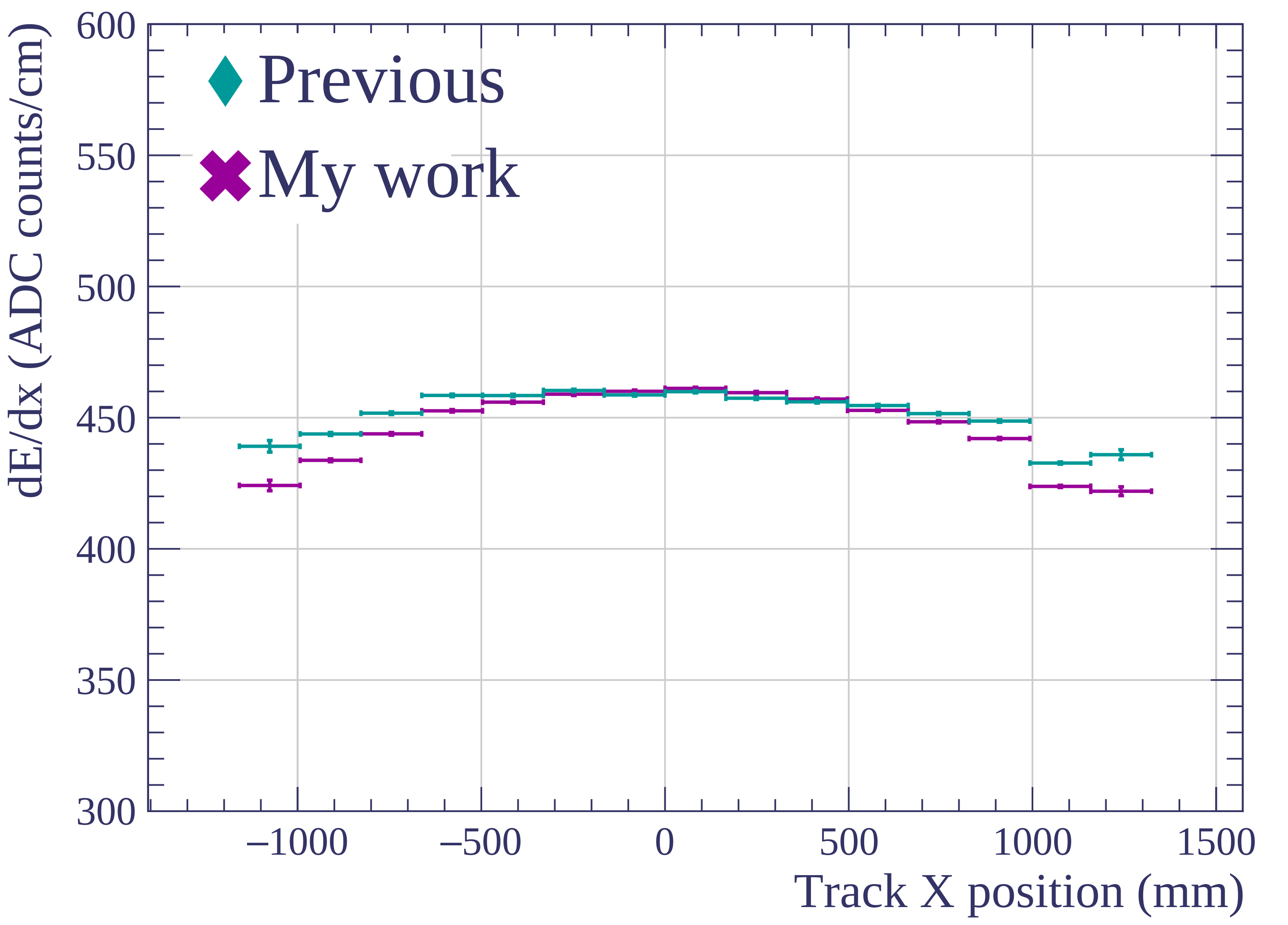

Energy loss in ERAM 47
XP σ/μ = 18.69± 0.43 % μ = 430.7± 1.5 σ = 80.5± 1.6
WF σ/μ = 19.14± 0.42 % μ = 427.0± 1.5 σ = 81.7± 1.5
Energy loss in ERAM 16
XP σ/μ = 15.24± 0.39 % μ = 450.1± 1.6 σ = 68.6± 1.5
WF σ/μ = 14.37± 0.36 % μ = 448.7± 1.5 σ = 64.5± 1.4
Energy loss in ERAM 14
XP σ/μ = 14.88± 0.47 % μ = 469.5± 1.8 σ = 69.9± 1.9
WF σ/μ = 15.06± 0.45 % μ = 471.9± 1.8 σ = 71.1± 1.8
Energy loss in ERAM 15
XP σ/μ = 13.06± 0.14 % μ = 495.0± 0.7 σ = 64.7± 0.6
WF σ/μ = 13.04± 0.15 % μ = 499.9± 0.7 σ = 65.2± 0.6
Energy loss in ERAM 42
XP σ/μ = 15.24± 0.22 % μ = 420.4± 0.8 σ = 64.1± 0.8
WF σ/μ = 15.92± 0.24 % μ = 419.6± 0.9 σ = 66.8± 0.9
Energy loss in ERAM 45
XP σ/μ = 12.45± 0.19 % μ = 421.0± 0.8 σ = 52.4± 0.7
WF σ/μ = 12.76± 0.20 % μ = 424.6± 0.8 σ = 54.2± 0.7
Energy loss in ERAM 37
XP σ/μ = 11.02± 0.17 % μ = 427.4± 0.7 σ = 47.1± 0.6
WF σ/μ = 10.91± 0.16 % μ = 429.0± 0.7 σ = 46.8± 0.6
Energy loss in ERAM 36
XP σ/μ = 12.44± 0.10 % μ = 447.4± 0.5 σ = 55.6± 0.4
WF σ/μ = 12.69± 0.12 % μ = 449.6± 0.5 σ = 57.1± 0.5
Energy loss in ERAM 20
XP σ/μ = 17.04± 0.33 % μ = 429.4± 1.2 σ = 73.2± 1.2
WF σ/μ = 18.16± 0.34 % μ = 431.4± 1.2 σ = 78.3± 1.3
Energy loss in ERAM 38
XP σ/μ = 13.06± 0.28 % μ = 430.8± 1.1 σ = 56.3± 1.0
WF σ/μ = 13.43± 0.30 % μ = 432.7± 1.1 σ = 58.1± 1.1
Energy loss in ERAM 7
XP σ/μ = 13.31± 0.30 % μ = 443.8± 1.2 σ = 59.1± 1.2
WF σ/μ = 13.74± 0.29 % μ = 446.9± 1.2 σ = 61.4± 1.1
Energy loss in ERAM 44
XP σ/μ = 14.51± 0.15 % μ = 452.1± 0.6 σ = 65.6± 0.6
WF σ/μ = 14.50± 0.15 % μ = 455.1± 0.6 σ = 66.0± 0.6
Energy loss in ERAM 43
XP σ/μ = 14.49± 0.20 % μ = 444.5± 0.8 σ = 64.4± 0.8
WF σ/μ = 14.85± 0.21 % μ = 443.5± 0.8 σ = 65.8± 0.8
Energy loss in ERAM 39
XP σ/μ = 12.41± 0.18 % μ = 430.8± 0.7 σ = 53.5± 0.7
WF σ/μ = 12.48± 0.19 % μ = 431.4± 0.7 σ = 53.8± 0.7
Energy loss in ERAM 41
XP σ/μ = 11.26± 0.16 % μ = 425.4± 0.7 σ = 47.9± 0.6
WF σ/μ = 11.90± 0.17 % μ = 428.5± 0.7 σ = 51.0± 0.7
Energy loss in ERAM 46
XP σ/μ = 13.52± 0.10 % μ = 445.4± 0.5 σ = 60.2± 0.4
WF σ/μ = 13.50± 0.10 % μ = 447.0± 0.5 σ = 60.3± 0.4
Count
dE/dx (ADC counts/cm)



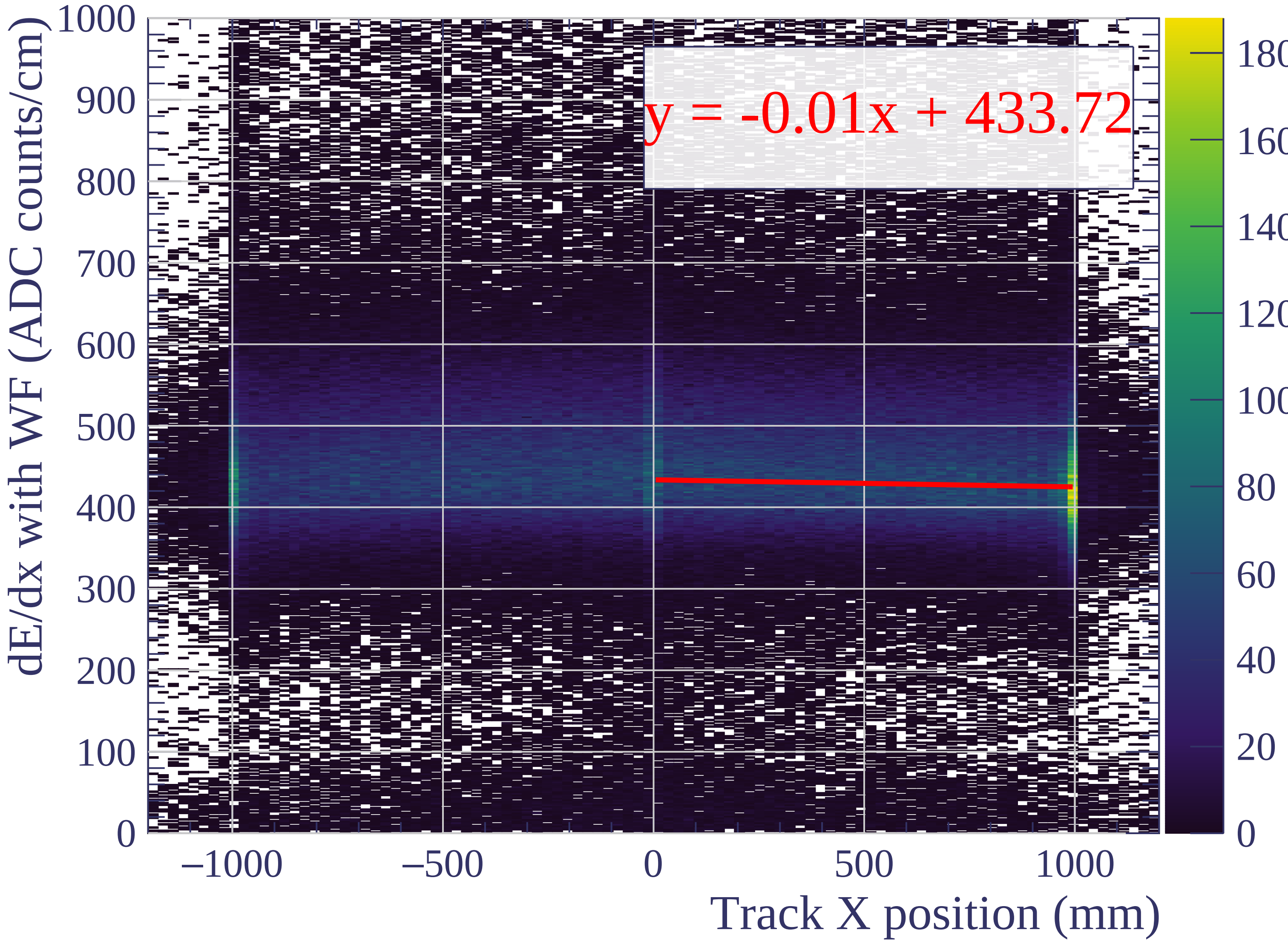
Entries 436923
Mean x 458.2
Mean y 456.2
Std Dev x 122.3
Std Dev y 120.4
0 455 13890
648 421254 676
0 0 0
dE/dx with XP (ADC counts/cm)
dE/dx with WF (ADC counts/cm)



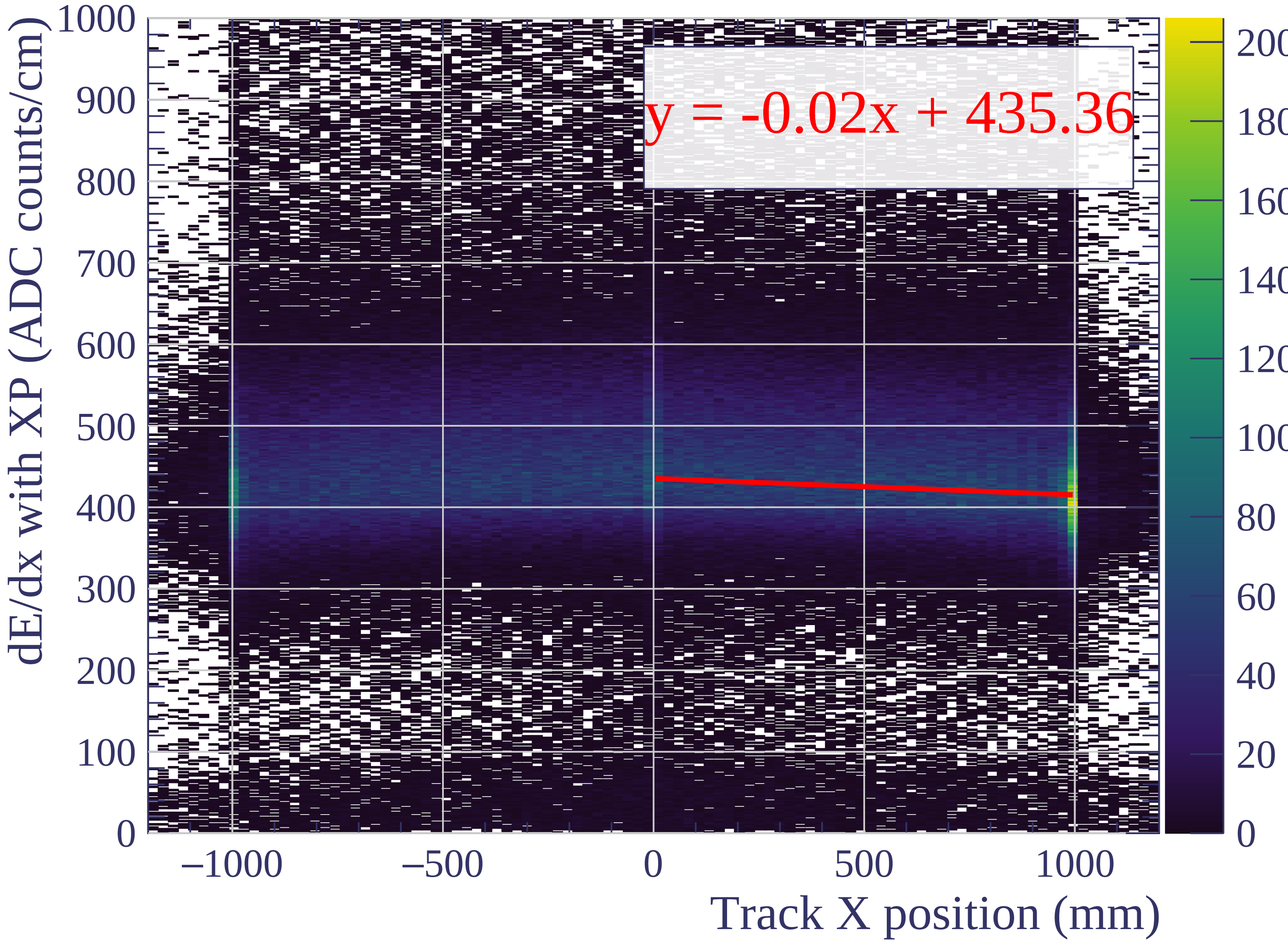
Previous
My work
11
10
9
8
7
6
5
4
−1000
−500
0
500
1000
1500
Track X position (mm)
dE/dx resolution (%)

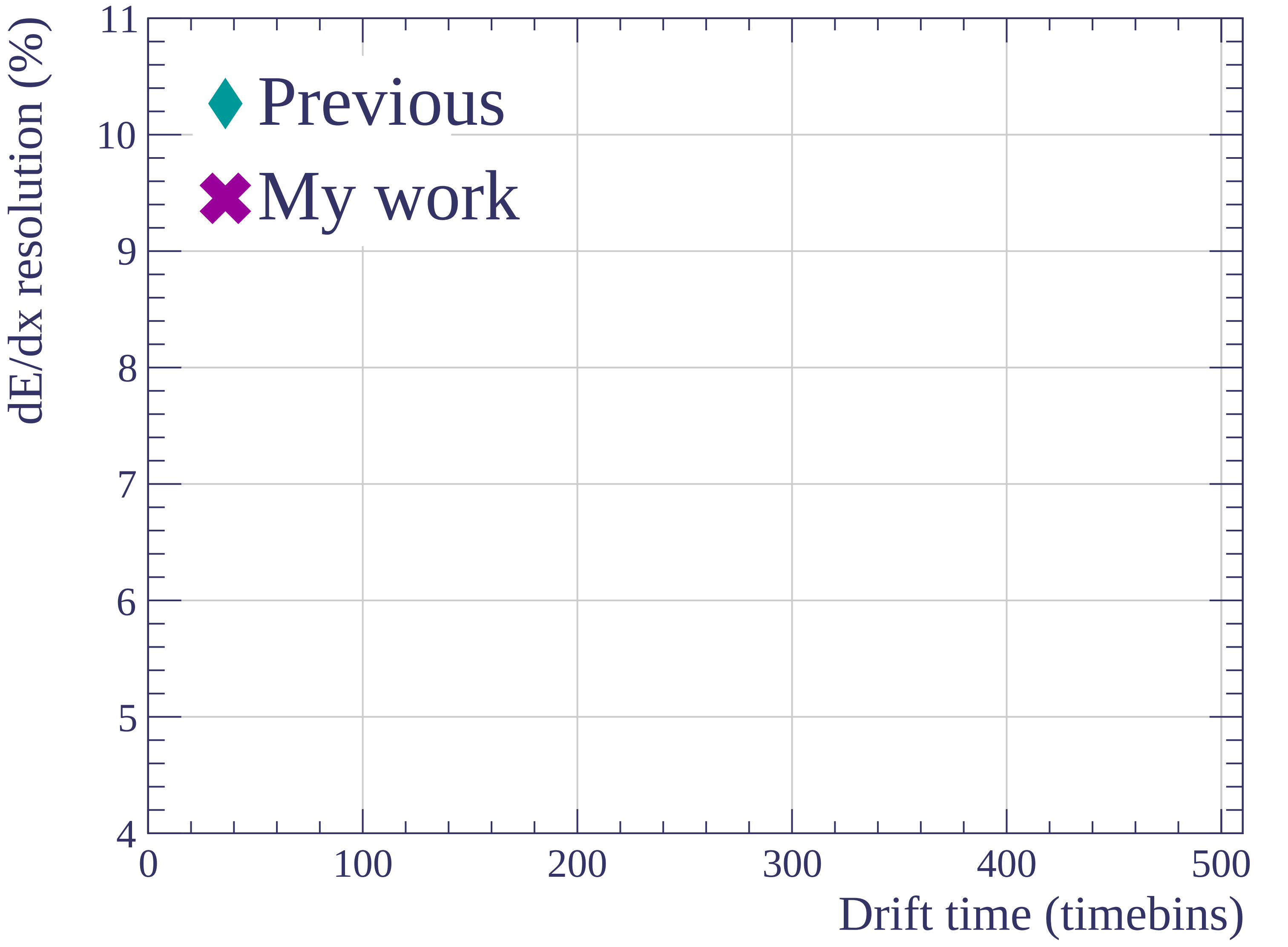


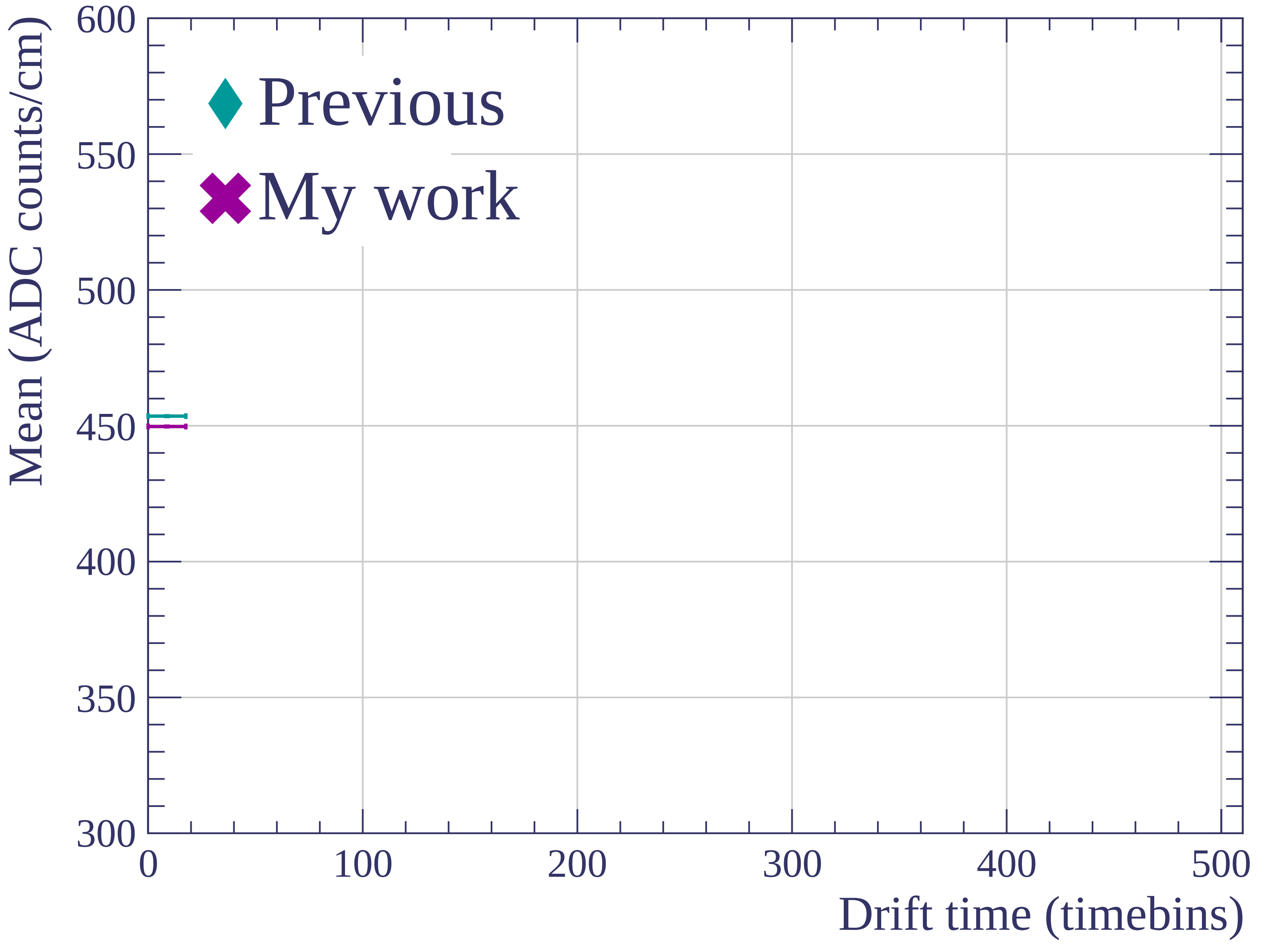
Previous
My work
dE/dx (ADC counts/cm)
600
550
500
450
400
350
300
−1000
−500
0
500
1000
1500
Track X position (mm)

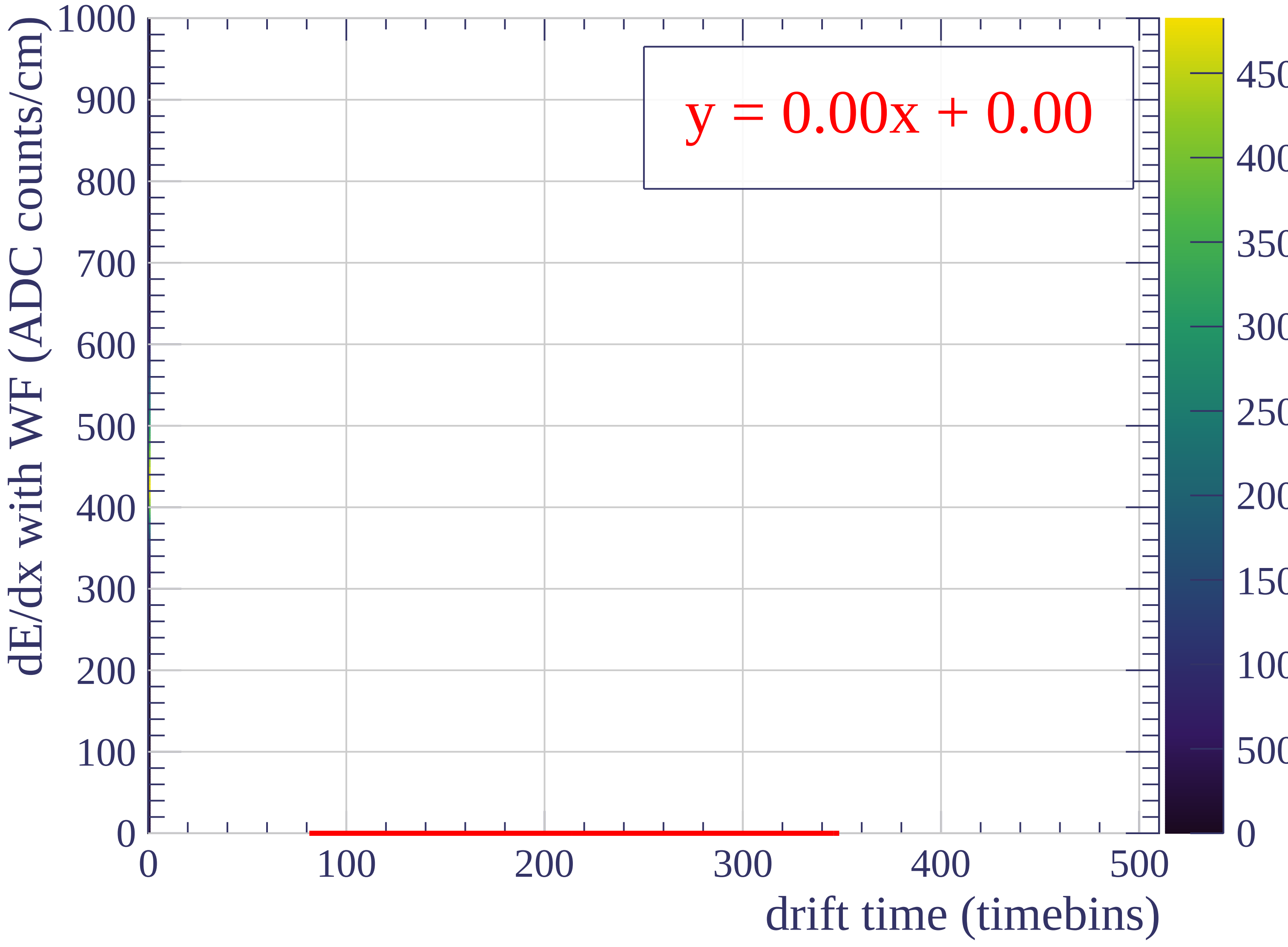


y = -0.01x + 433.72
dE/dx with WF (ADC counts/cm)
Track X position (mm)
0
100
200
300
400
500
600
700
800
900
1000
−1000
−500
0
500
1000
20
40
60
80
100
120
140
160
180

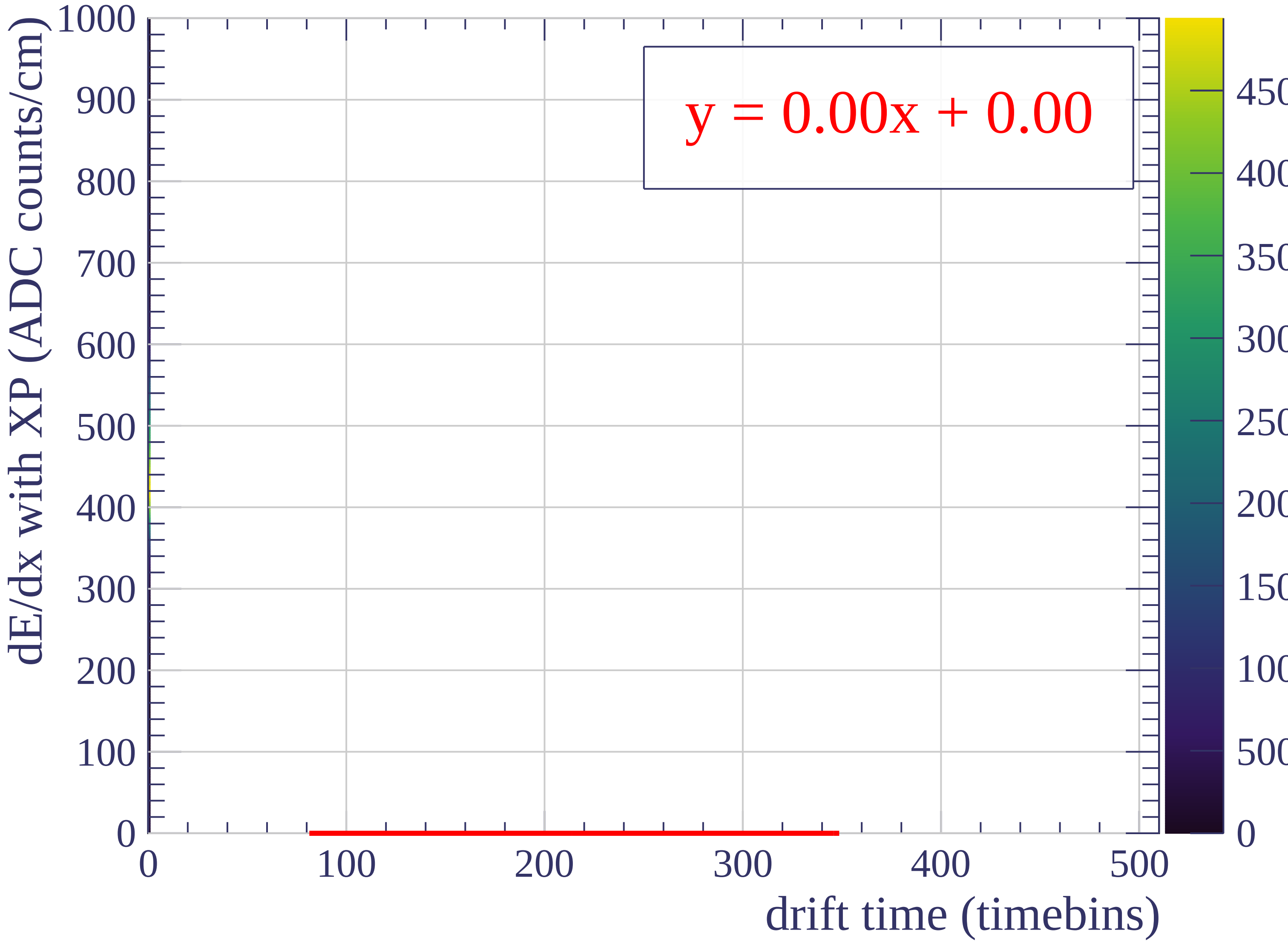

y = -0.02x + 435.36
dE/dx with XP (ADC counts/cm)
Track X position (mm)
1000
900
800
700
600
500
400
300
200
100
0
−1000
−500
0
500
1000
20
40
60
80
100
120
140
160
180
200


dE/dx resolution (%)
11
10
9
8
7
6
5
4
0
100
200
300
400
500
Drift time (timebins)
Previous
My work

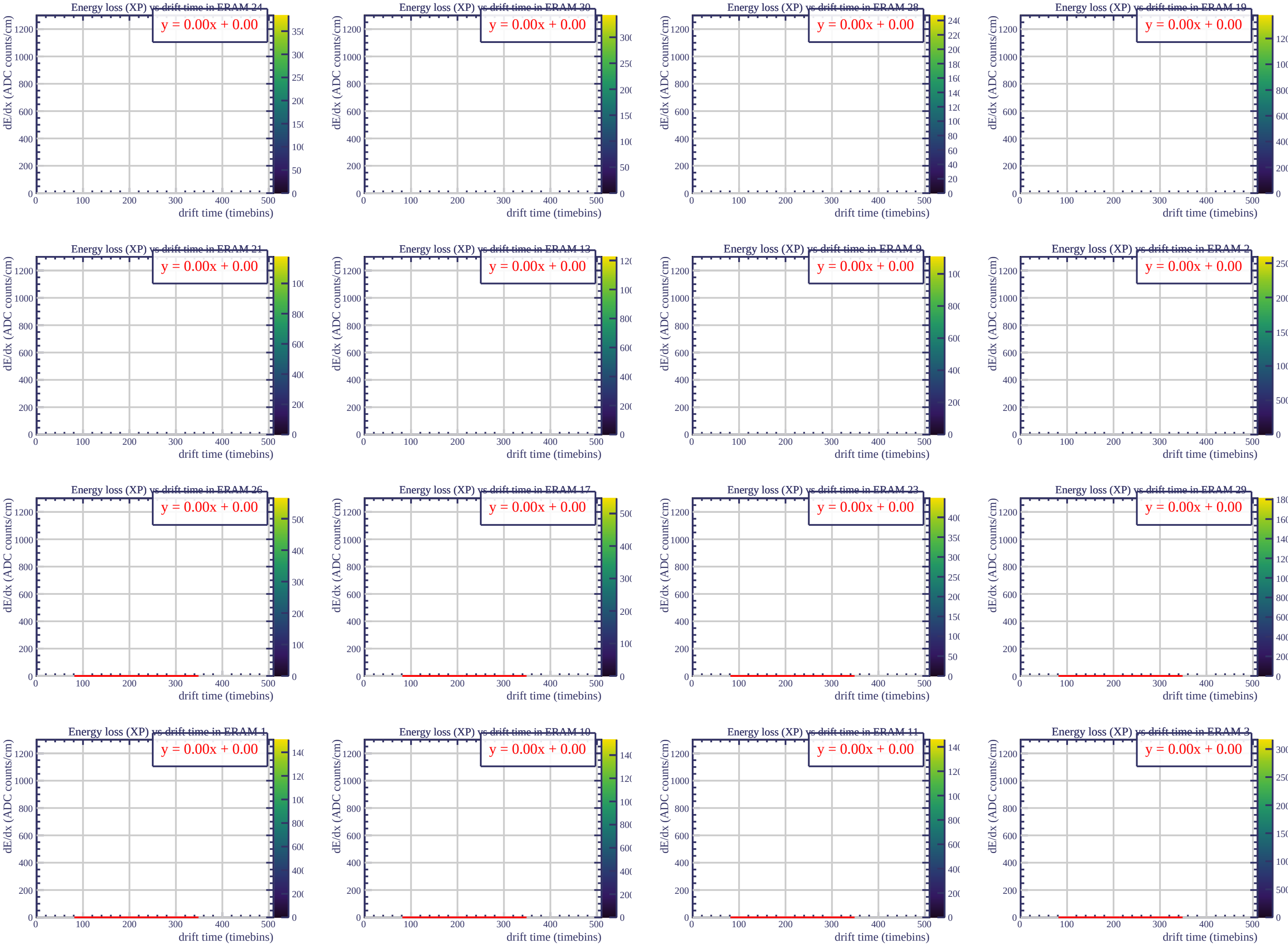

Previous
My work
Mean (ADC counts/cm)
Drift time (timebins)
600
550
500
450
400
350
300
0
100
200
300
400
500

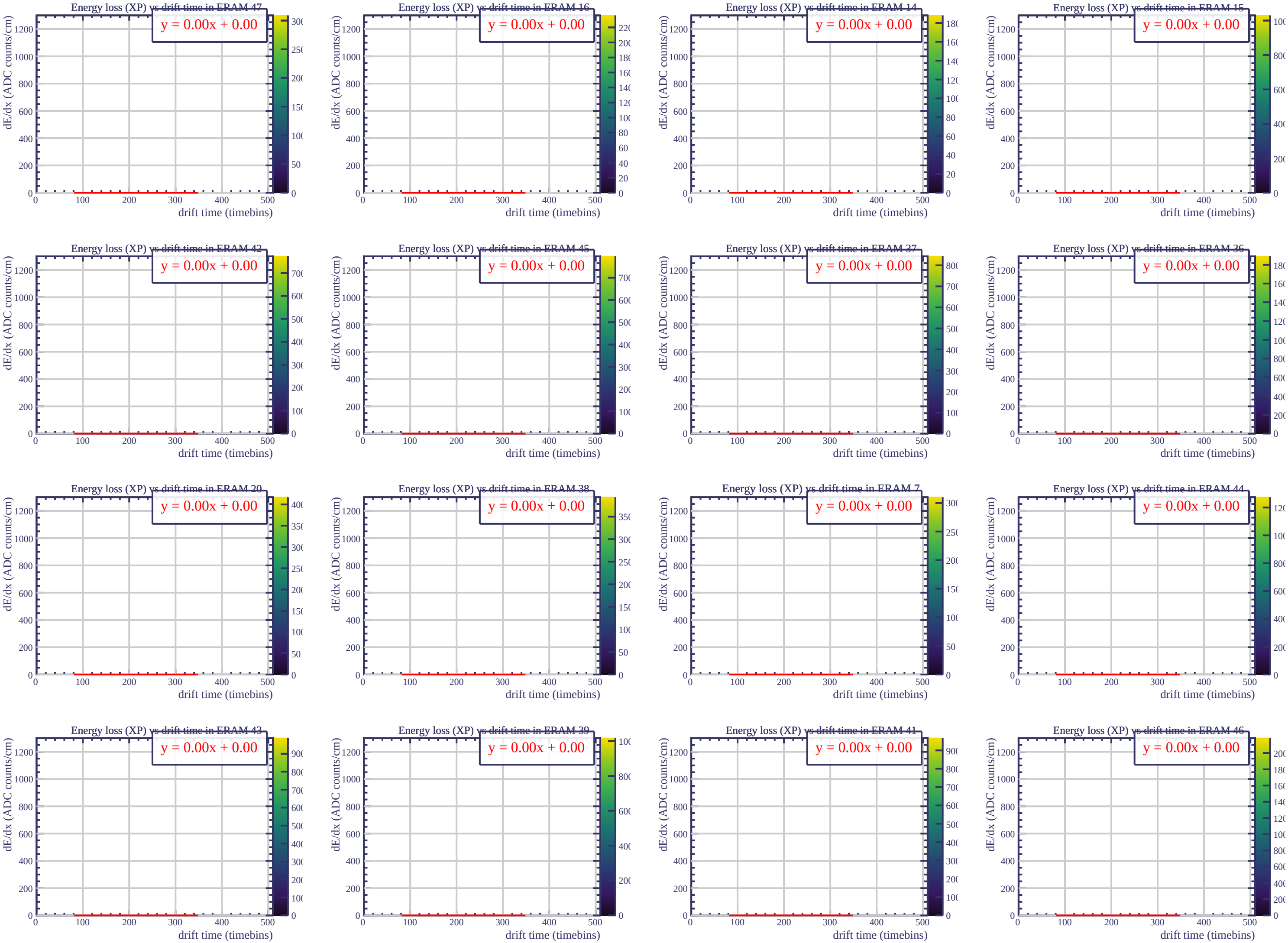


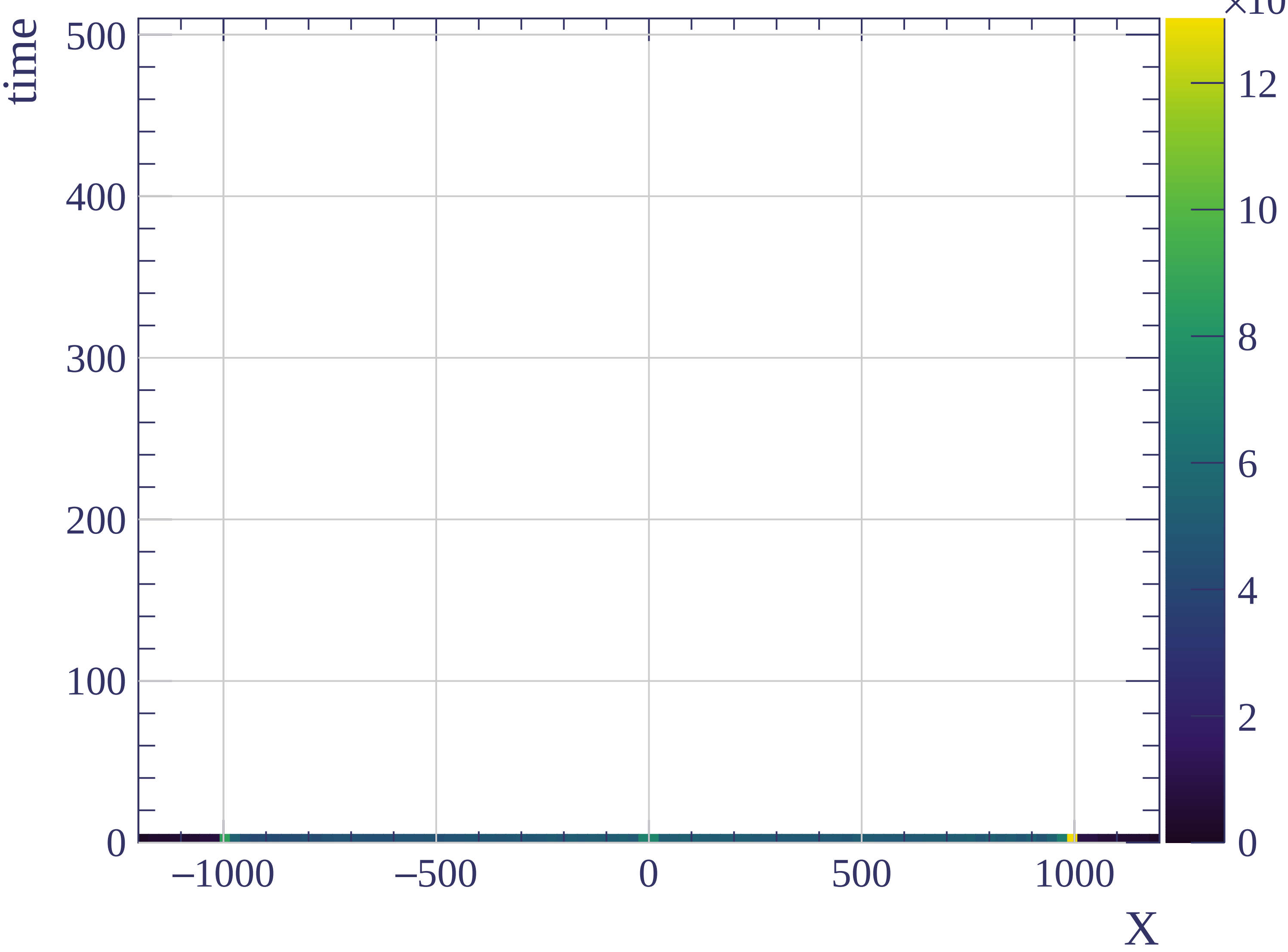
y = 0.00x + 0.00
dE/dx with WF (ADC counts/cm)
drift time (timebins)
0
100
200
300
400
500
600
700
800
900
1000



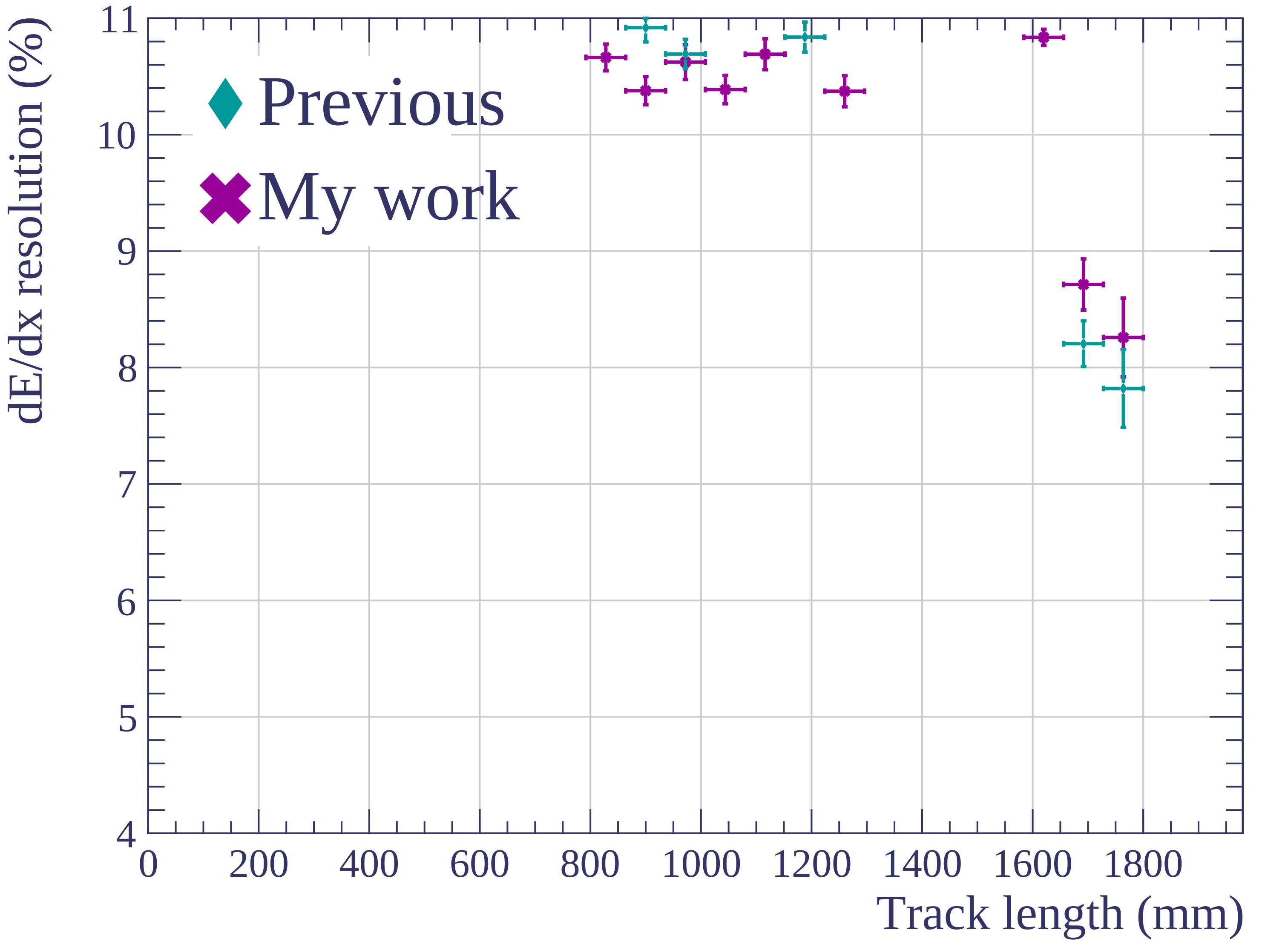
y = 0.00x + 0.00
dE/dx with XP (ADC counts/cm)
drift time (timebins)
0
100
200
300
400
500
600
700
800
900
1000
0
100
200
300
400
500
0
500
100
150
200
250
300
350
400
450


Energy loss (XP) vs drift time in ERAM 24
Energy loss (XP) vs drift time in ERAM 30
Energy loss (XP) vs drift time in ERAM 28
Energy loss (XP) vs drift time in ERAM 19
Energy loss (XP) vs drift time in ERAM 21
Energy loss (XP) vs drift time in ERAM 13
Energy loss (XP) vs drift time in ERAM 9
Energy loss (XP) vs drift time in ERAM 2
Energy loss (XP) vs drift time in ERAM 26
Energy loss (XP) vs drift time in ERAM 17
Energy loss (XP) vs drift time in ERAM 23
Energy loss (XP) vs drift time in ERAM 29
Energy loss (XP) vs drift time in ERAM 1
Energy loss (XP) vs drift time in ERAM 10
Energy loss (XP) vs drift time in ERAM 11
Energy loss (XP) vs drift time in ERAM 3
y = 0.00x + 0.00
dE/dx (ADC counts/cm)
drift time (timebins)

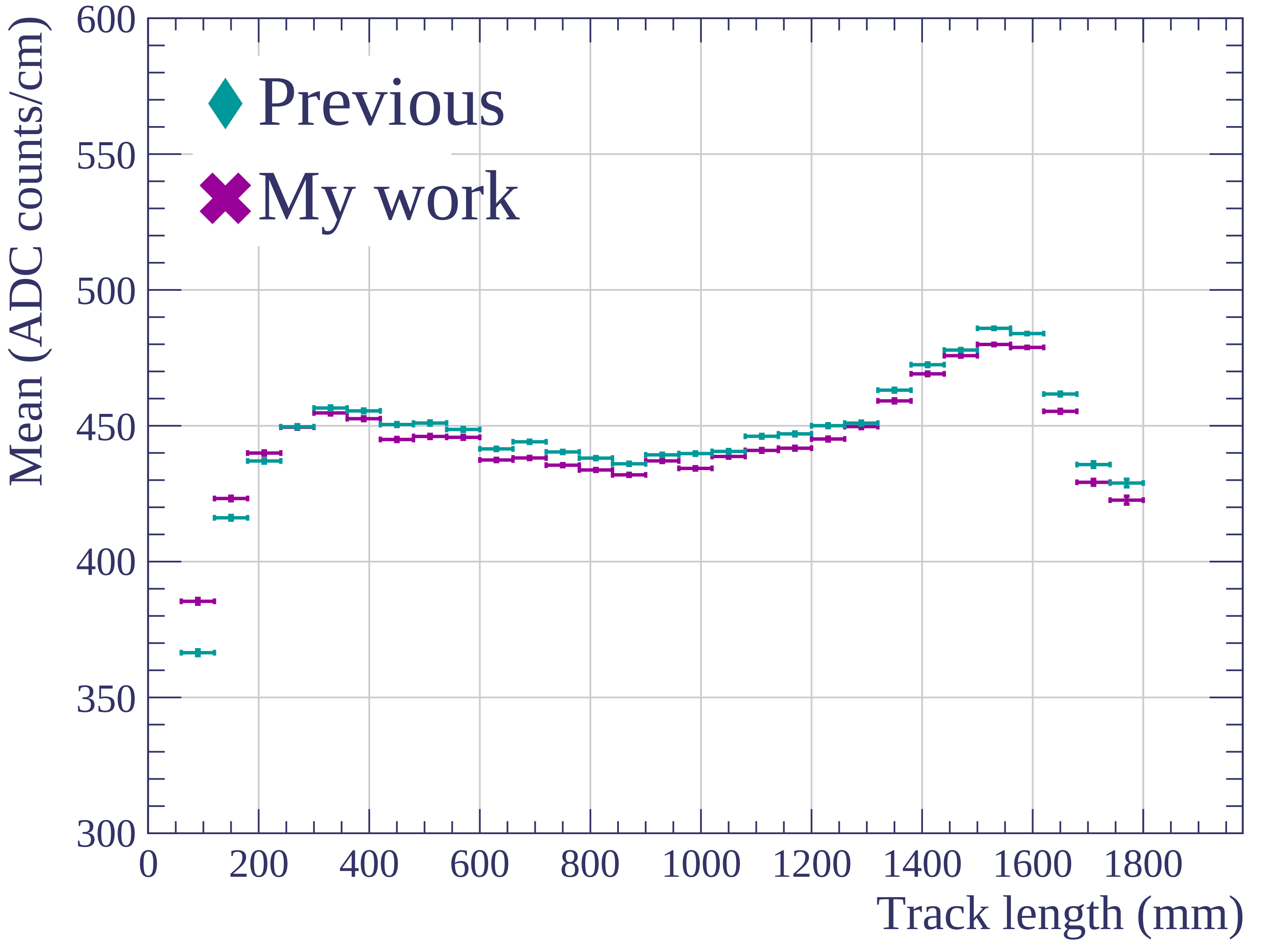

Energy loss (XP) vs drift time in ERAM 47
y = 0.00x + 0.00
Energy loss (XP) vs drift time in ERAM 16
y = 0.00x + 0.00
Energy loss (XP) vs drift time in ERAM 14
y = 0.00x + 0.00
Energy loss (XP) vs drift time in ERAM 15
y = 0.00x + 0.00
Energy loss (XP) vs drift time in ERAM 42
y = 0.00x + 0.00
Energy loss (XP) vs drift time in ERAM 45
y = 0.00x + 0.00
Energy loss (XP) vs drift time in ERAM 37
y = 0.00x + 0.00
Energy loss (XP) vs drift time in ERAM 36
y = 0.00x + 0.00
Energy loss (XP) vs drift time in ERAM 20
y = 0.00x + 0.00
Energy loss (XP) vs drift time in ERAM 38
y = 0.00x + 0.00
Energy loss (XP) vs drift time in ERAM 7
y = 0.00x + 0.00
Energy loss (XP) vs drift time in ERAM 44
y = 0.00x + 0.00
Energy loss (XP) vs drift time in ERAM 43
y = 0.00x + 0.00
Energy loss (XP) vs drift time in ERAM 39
y = 0.00x + 0.00
Energy loss (XP) vs drift time in ERAM 41
y = 0.00x + 0.00
Energy loss (XP) vs drift time in ERAM 46
y = 0.00x + 0.00
dE/dx (ADC counts/cm)
drift time (timebins)

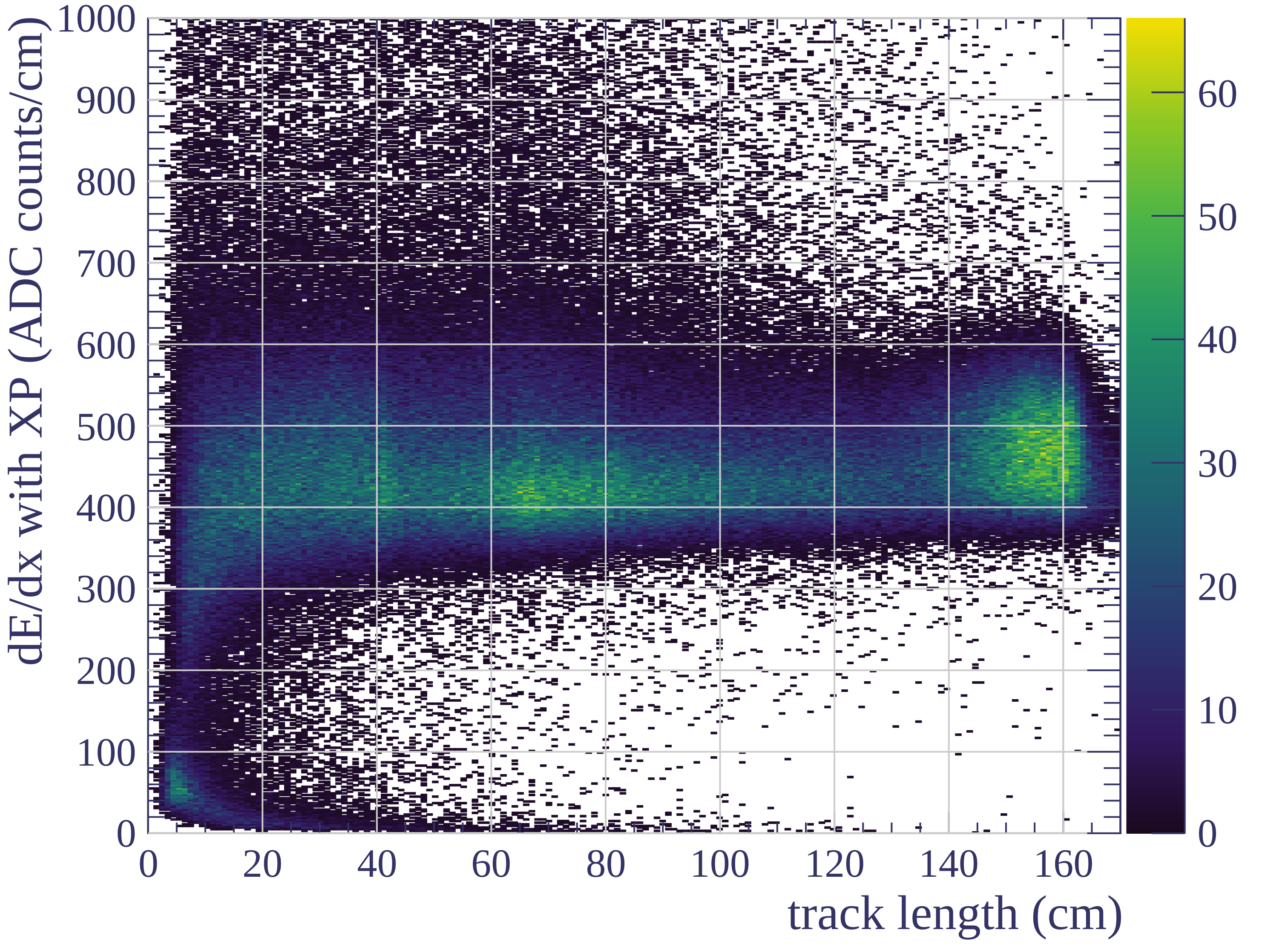


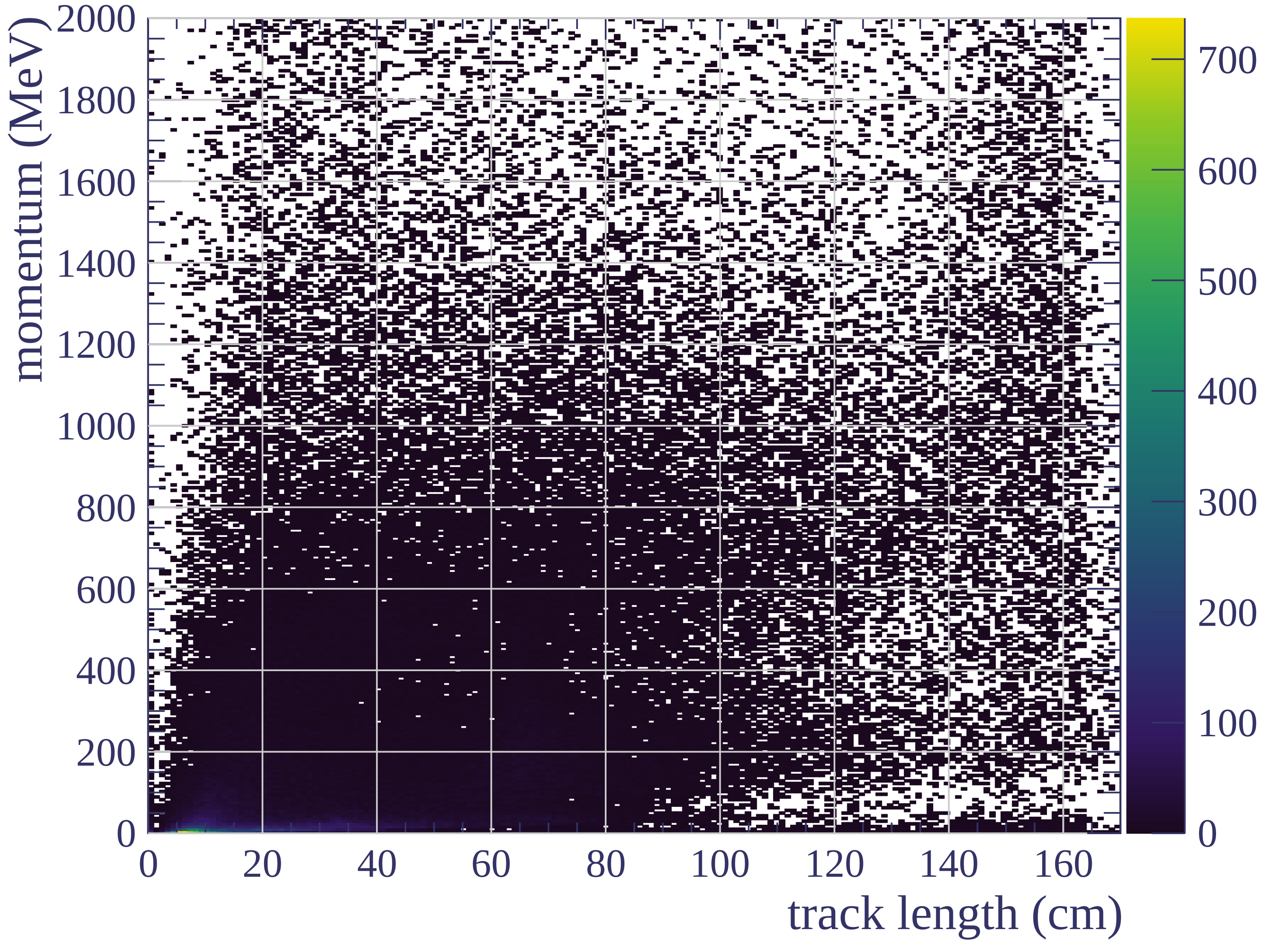
time
500
400
300
200
100
0
−1000
−500
0
500
1000
X
×10
12
10
8
6
4
2
0

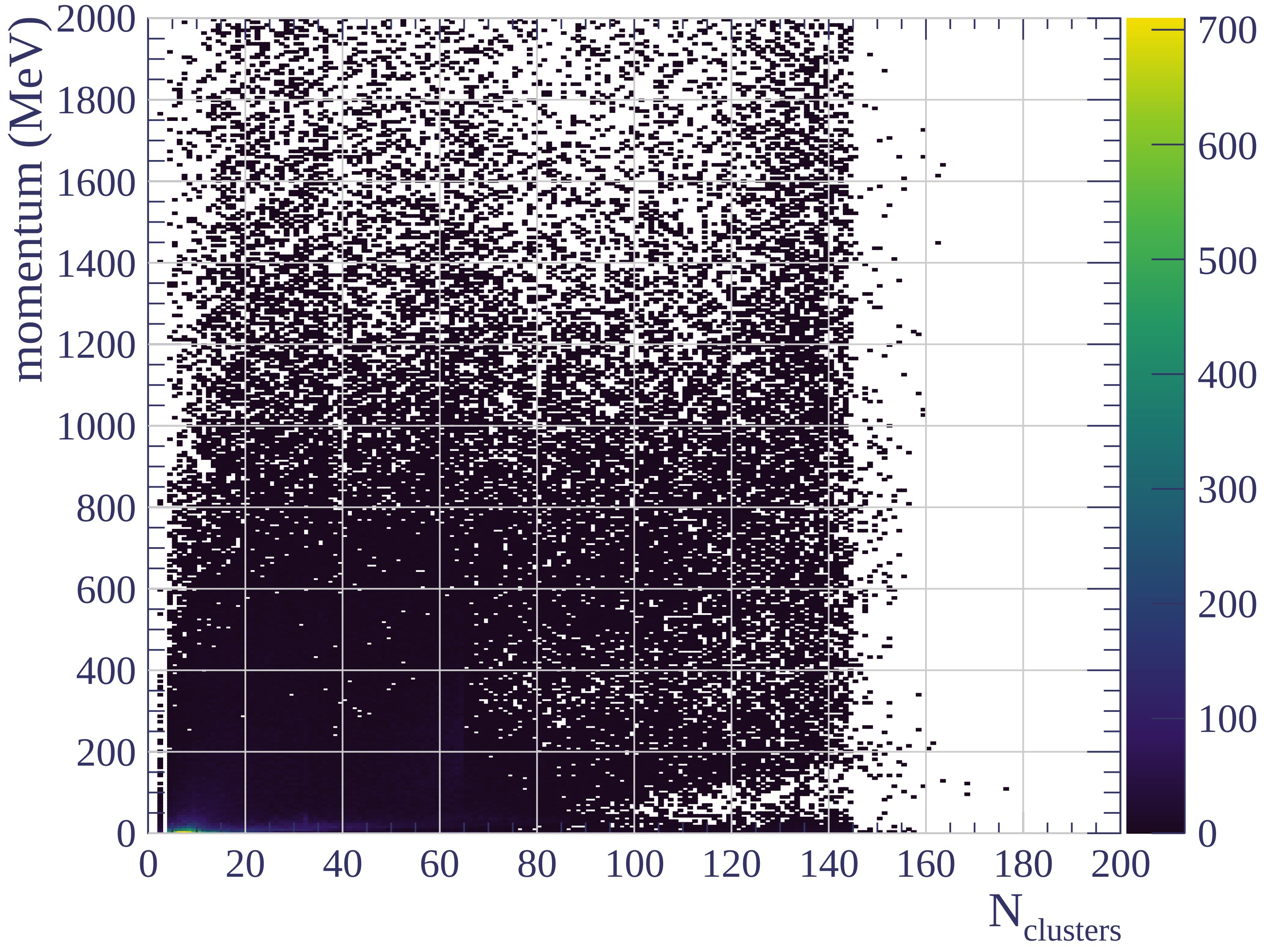


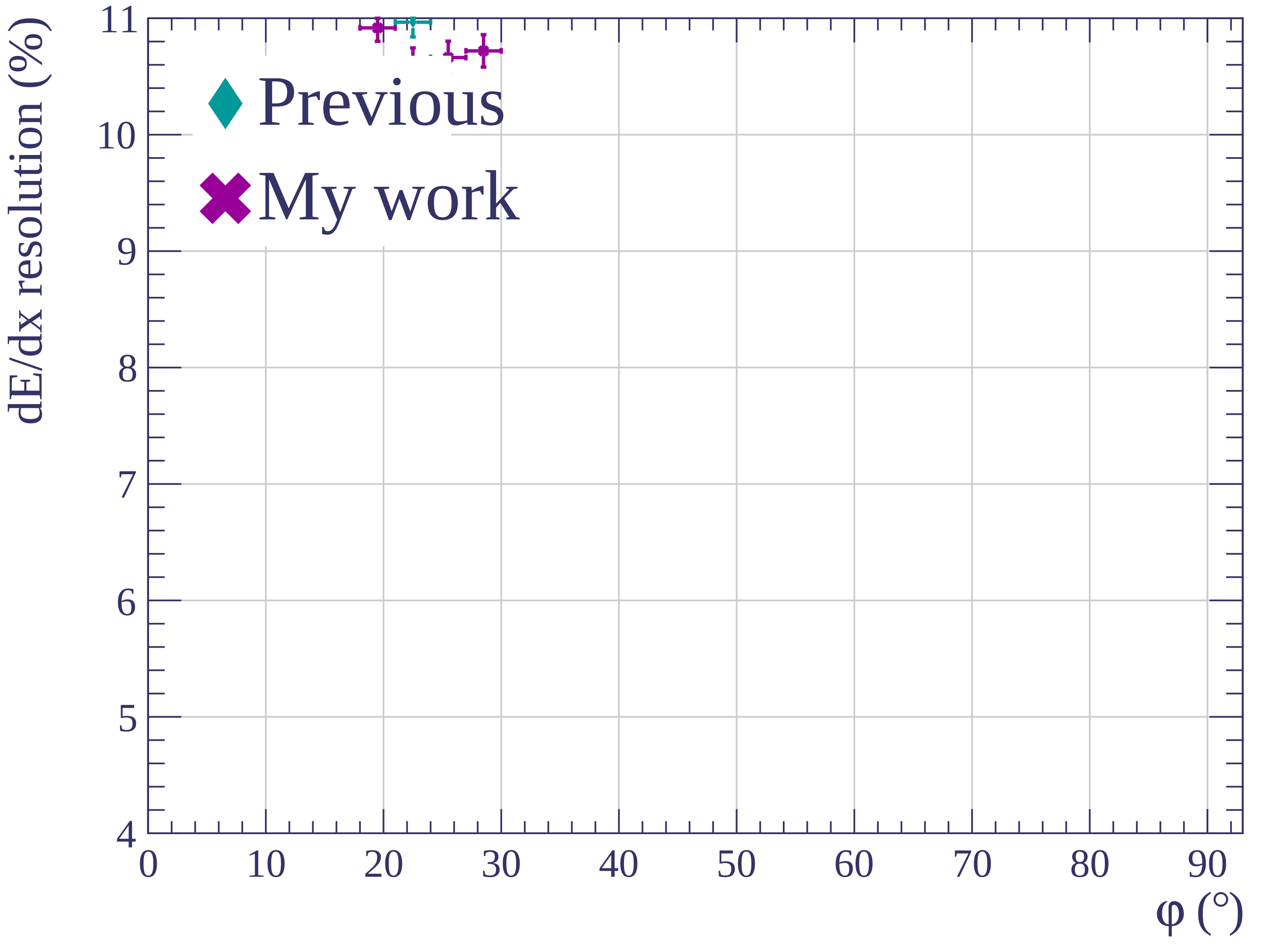
dE/dx resolution (%)
Track length (mm)
Previous
My work
4
5
6
7
8
9
10
11
0
200
400
600
800
1000
1200
1400
1600
1800


Previous
My work
Mean (ADC counts/cm)
Track length (mm)
600
550
500
450
400
350
300
0
200
400
600
800
1000
1200
1400
1600
1800

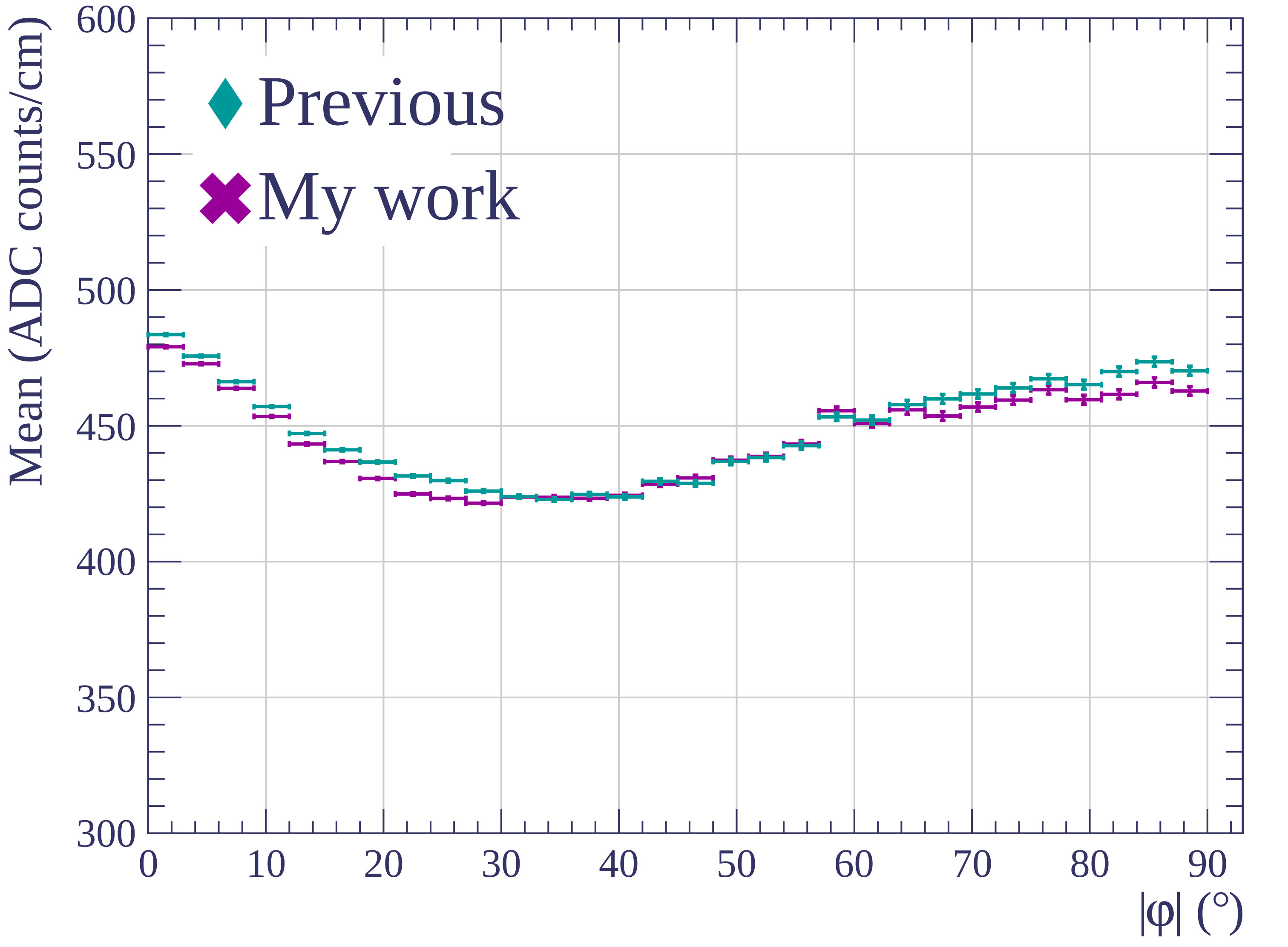

dE/dx with XP (ADC counts/cm)
track length (cm)
0
100
200
300
400
500
600
700
800
900
1000
0
20
40
60
80
100
120
140
160
0
10
20
30
40
50
60

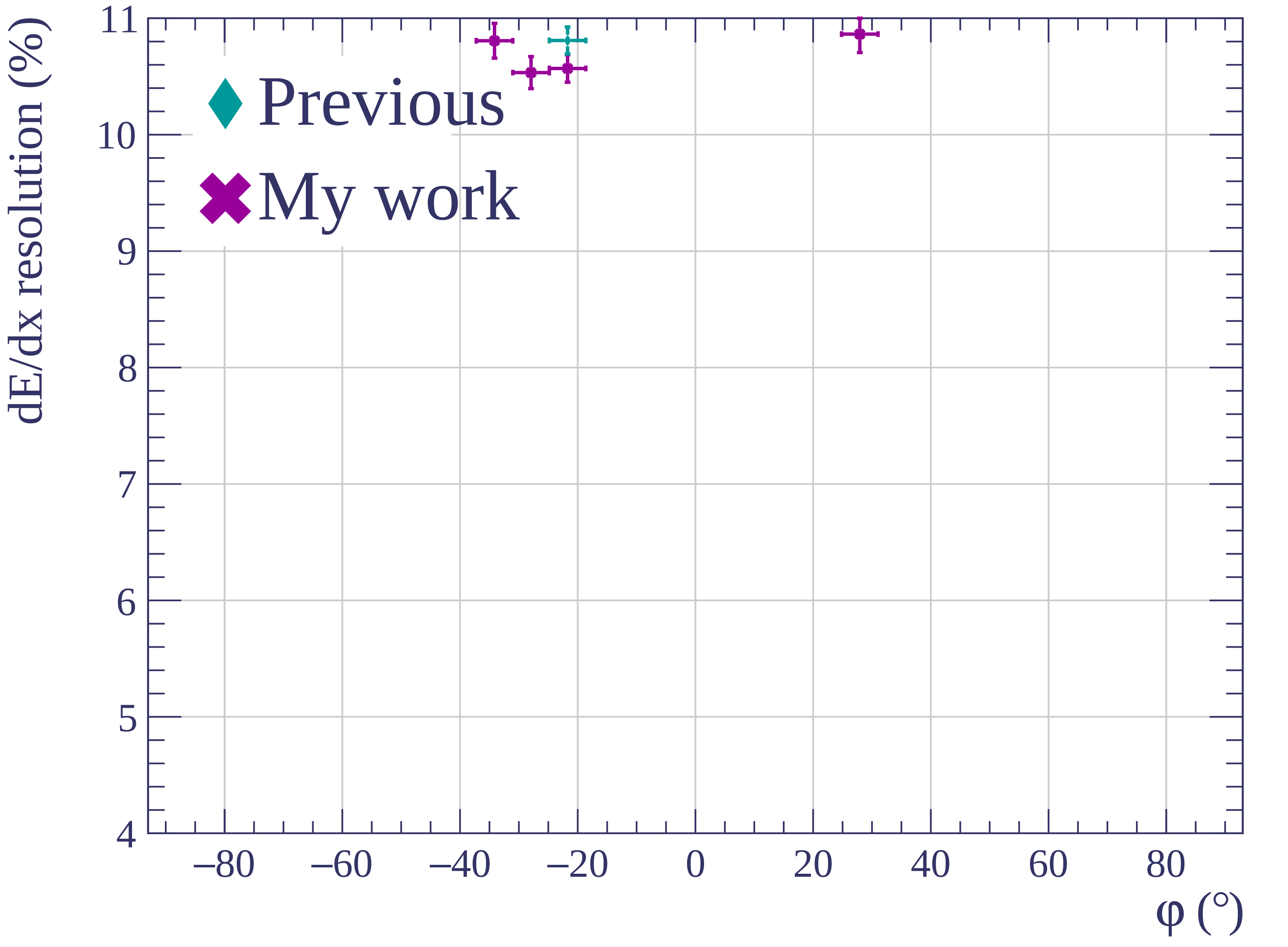

momentum (MeV)
2000
1800
1600
1400
1200
1000
800
600
400
200
0
0
20
40
60
80
100
120
140
160
track length (cm)
700
600
500
400
300
200
100
0

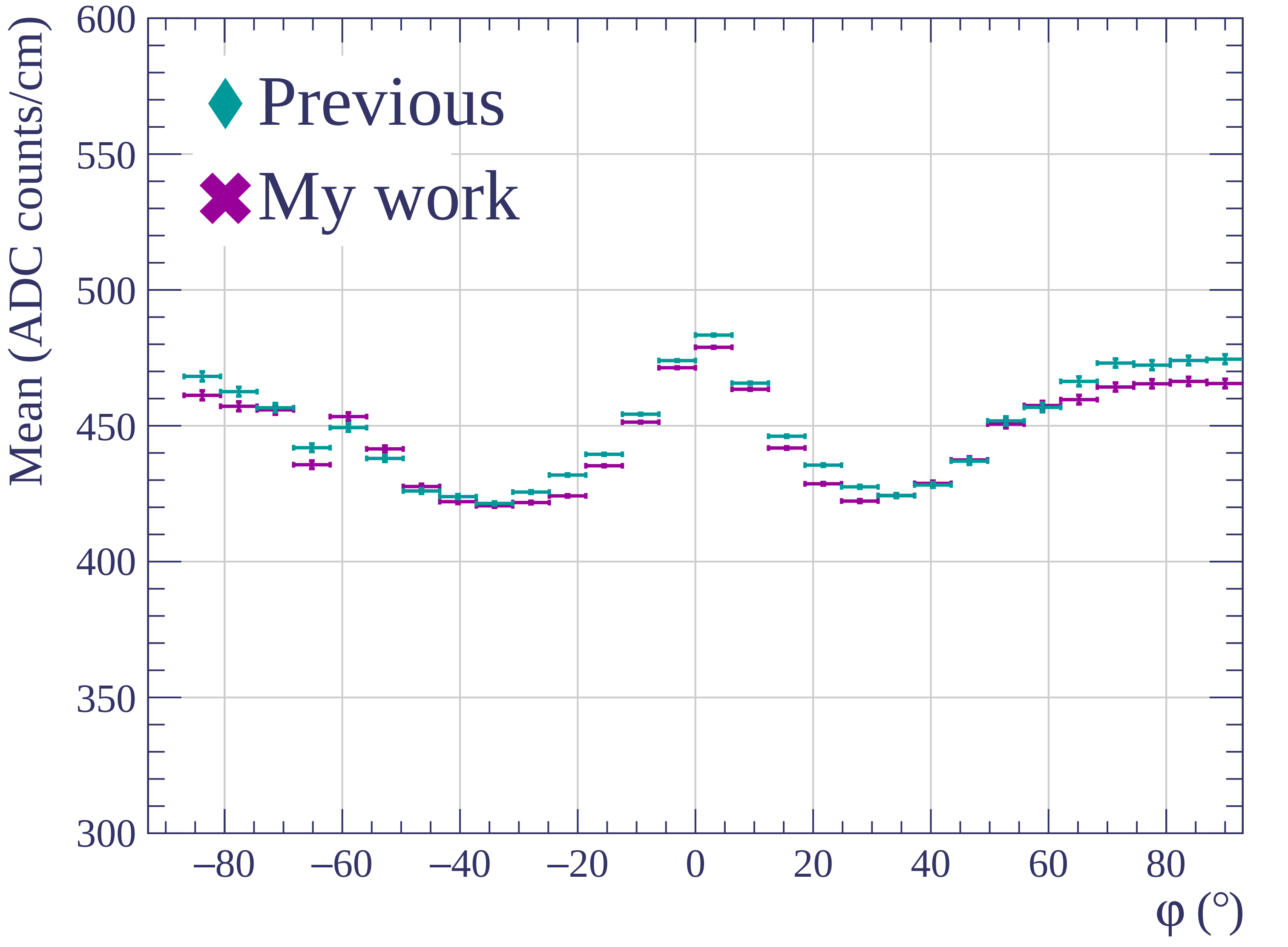

momentum (MeV)
2000
1800
1600
1400
1200
1000
800
600
400
200
0
0
20
40
60
80
100
120
140
160
180
200
$N_{clusters}$
700
600
500
400
300
200
100
0



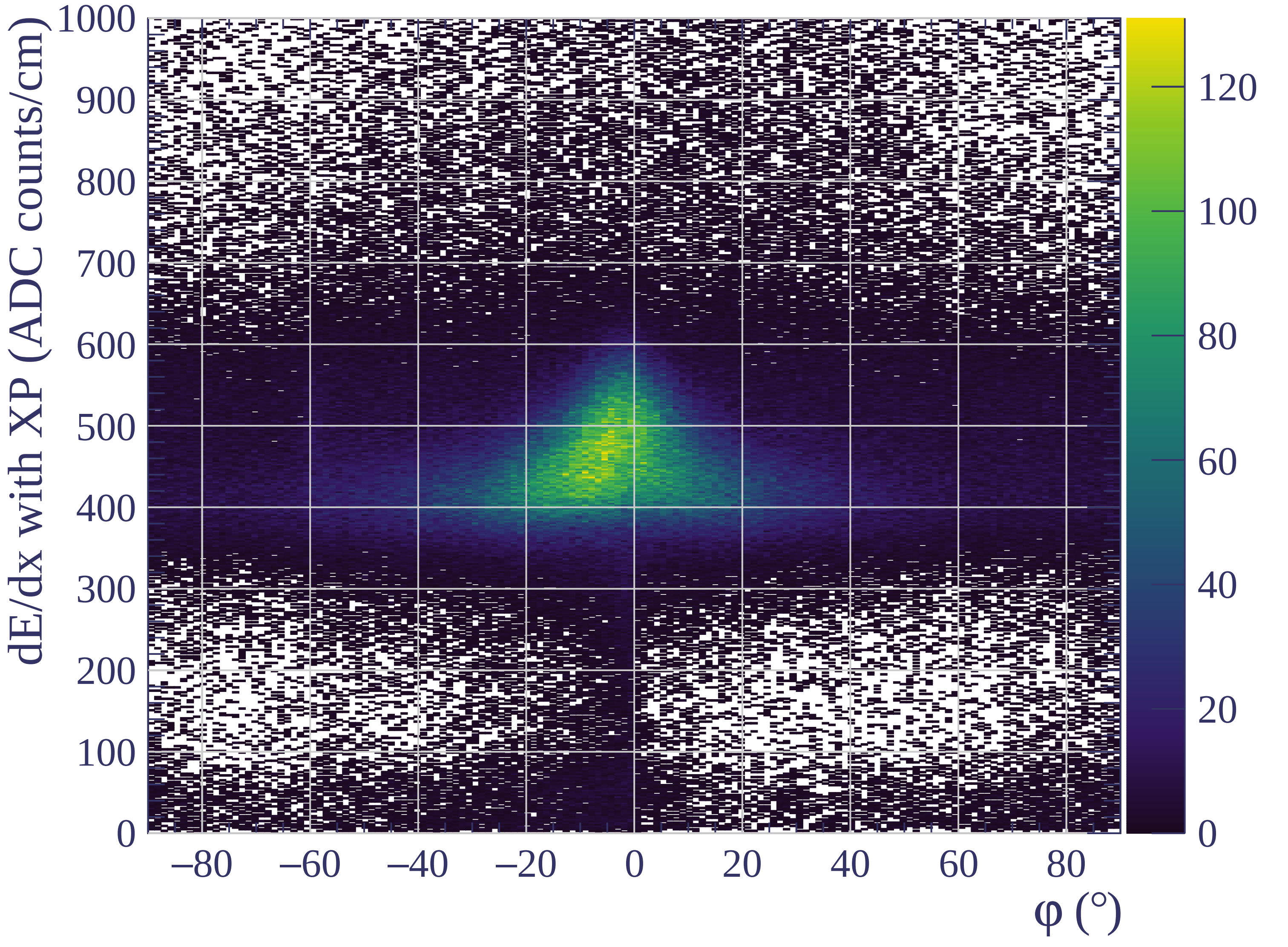
dE/dx resolution (%)
11
10
9
8
7
6
5
4
0
10
20
30
40
50
60
70
80
90
φ (°)
Previous
My work

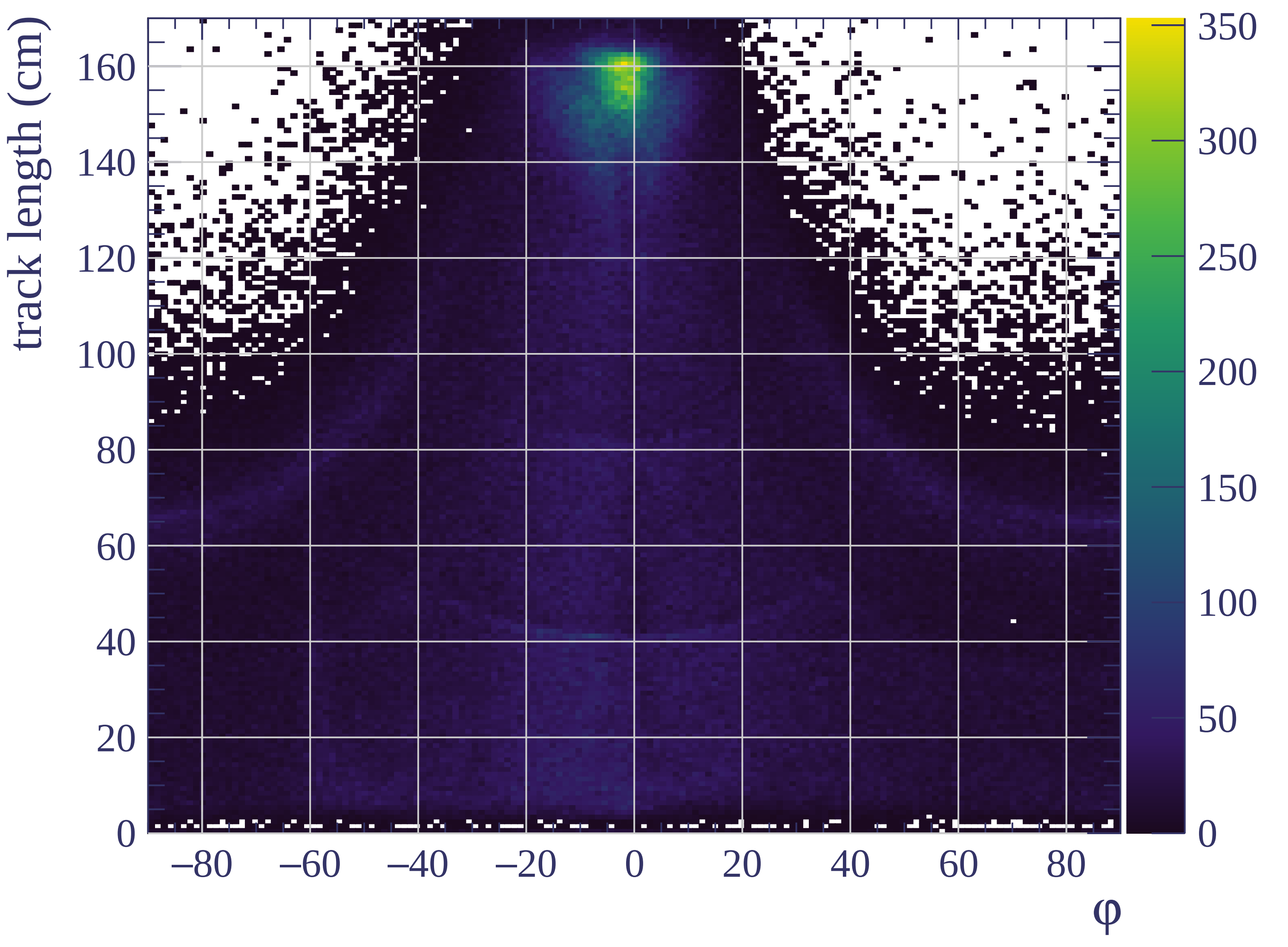

Previous
My work
Mean (ADC counts/cm)
|φ| (°)
600
550
500
450
400
350
300
0
10
20
30
40
50
60
70
80
90


Previous
My work
dE/dx resolution (%)
φ (°)
11
10
9
8
7
6
5
4
−80
−60
−40
−20
0
20
40
60
80

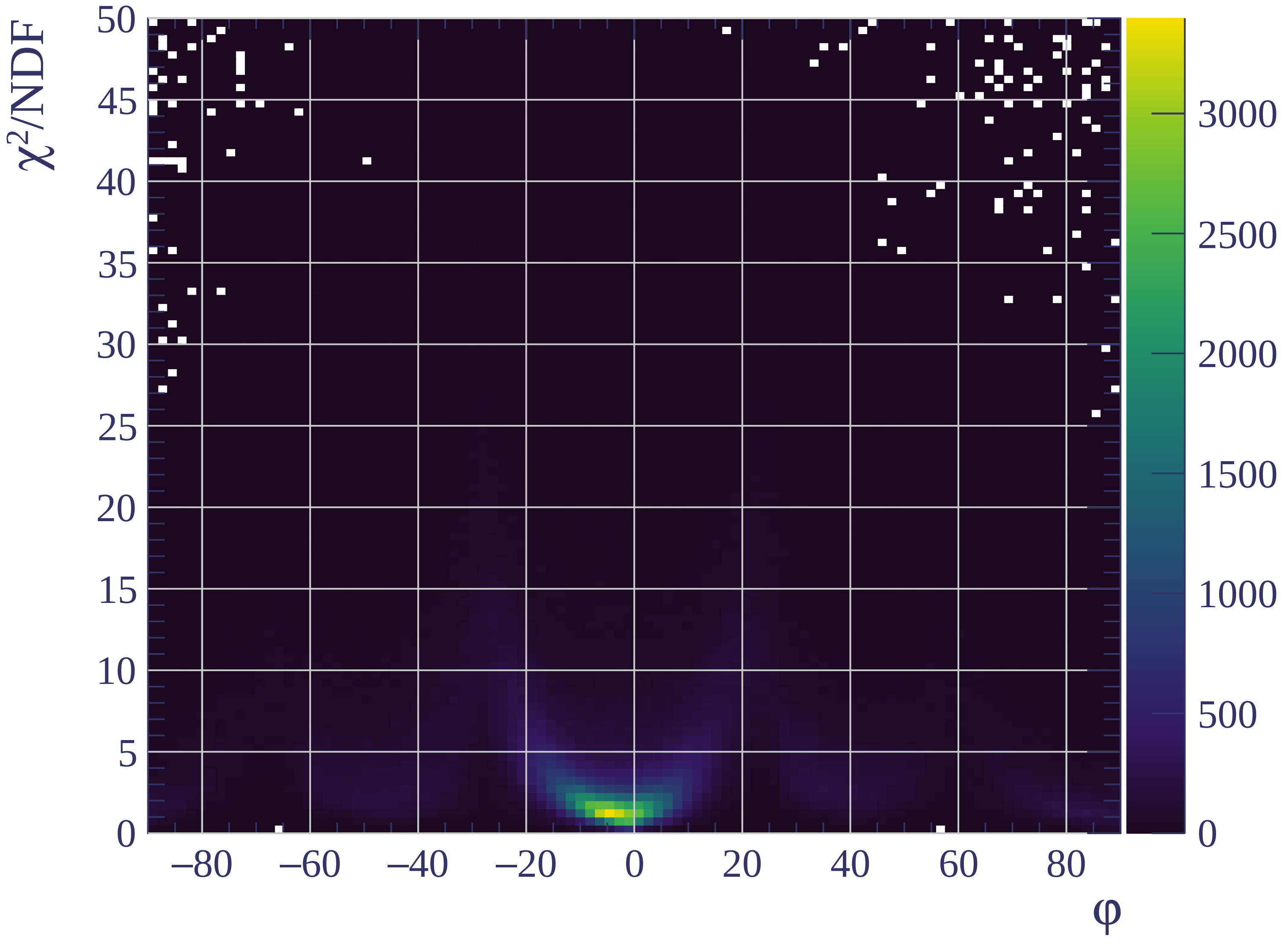


Previous
My work
Mean (ADC counts/cm)
600
550
500
450
400
350
300
−80
−60
−40
−20
0
20
40
60
80
φ (°)

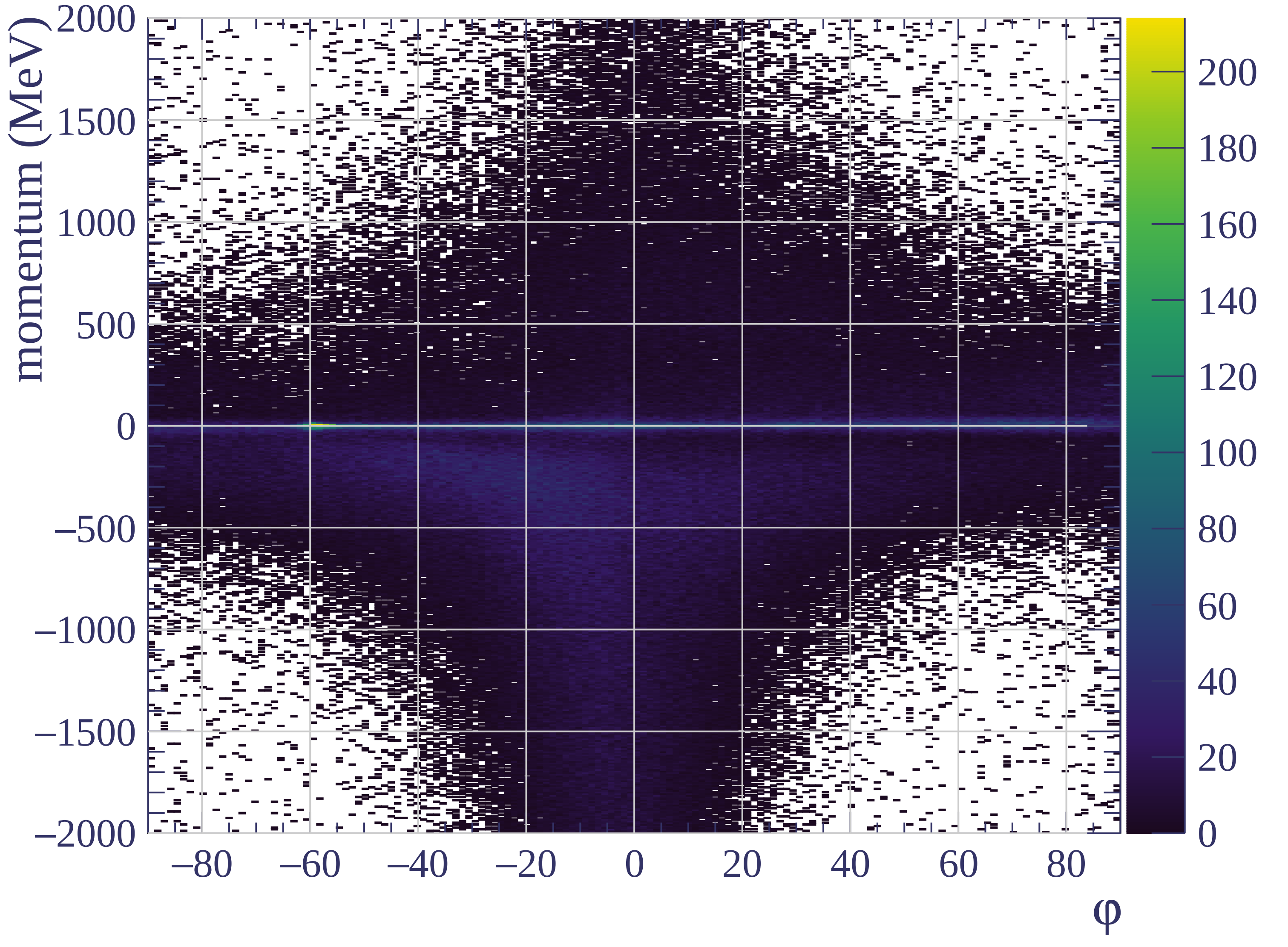

dE/dx with XP (ADC counts/cm)
1000
900
800
700
600
500
400
300
200
100
0
−80
−60
−40
−20
0
20
40
60
80
φ (°)
120
100
80
60
40
20
0



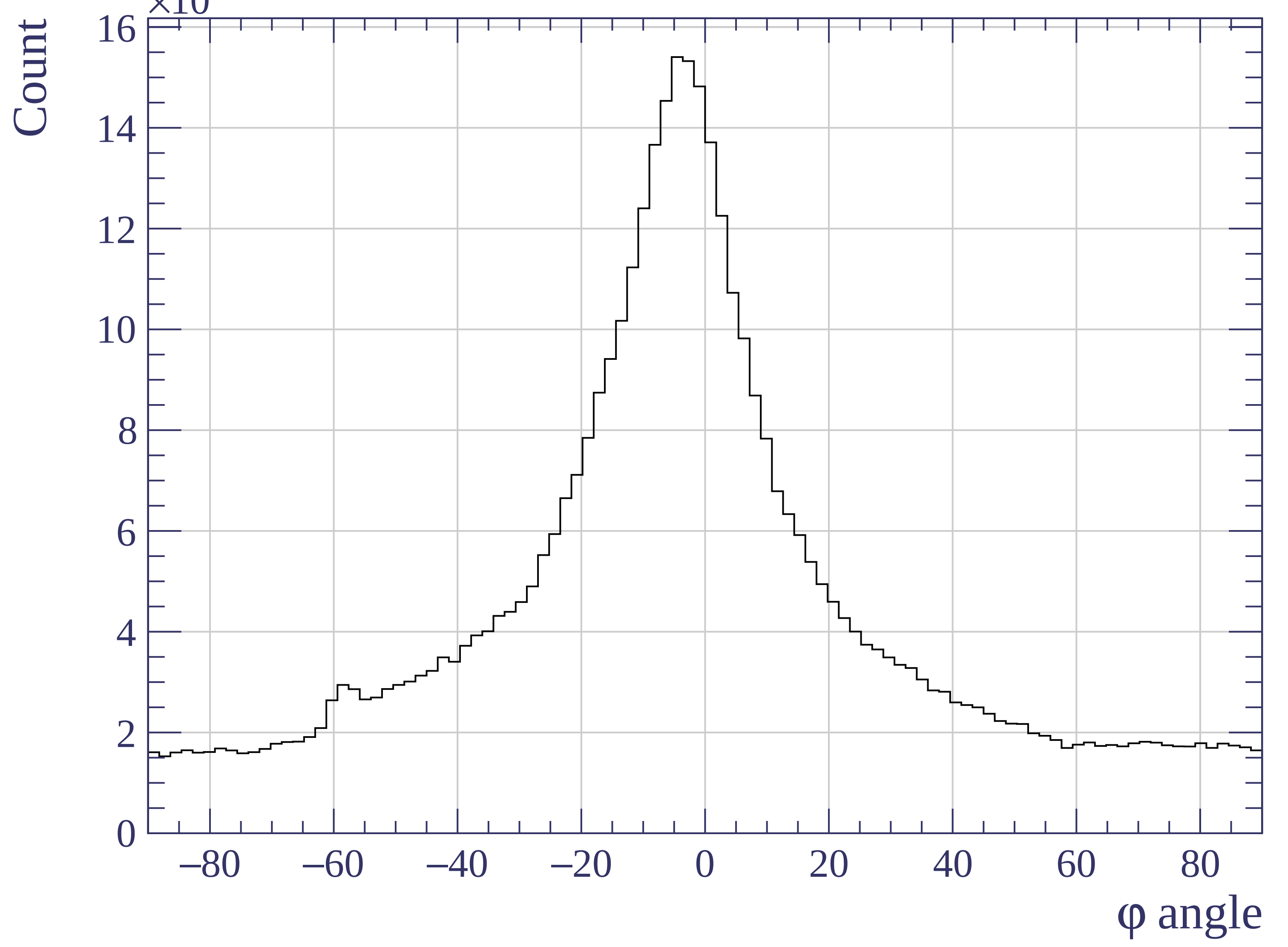
track length (cm)
0
20
40
60
80
100
120
140
160
−80
−60
−40
−20
0
20
40
60
80
φ
0
50
100
150
200
250
300
350



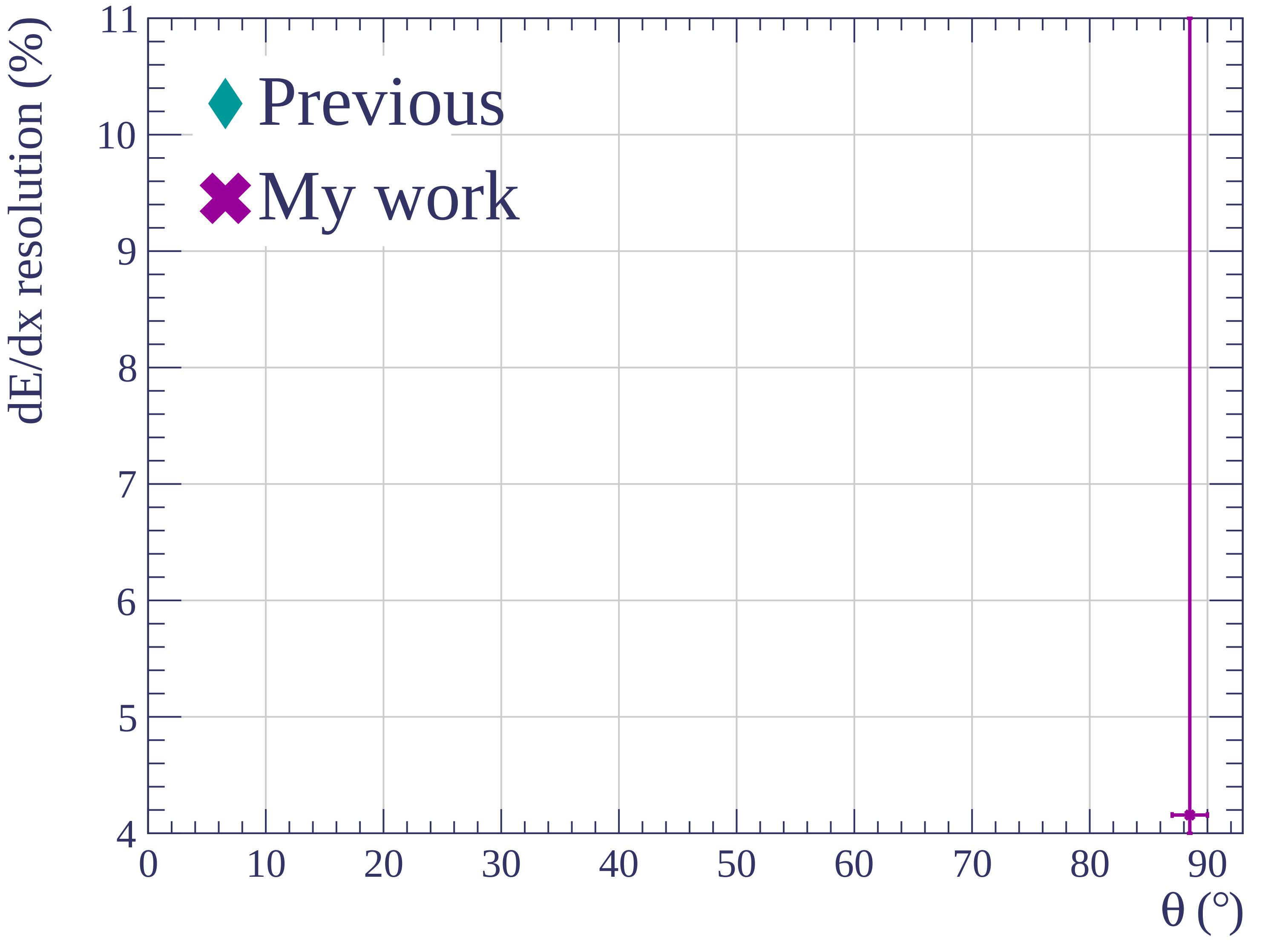
χ^2/NDF
50
45
40
35
30
25
20
15
10
5
0
−80
−60
−40
−20
0
20
40
60
80
φ
3000
2500
2000
1500
1000
500
0



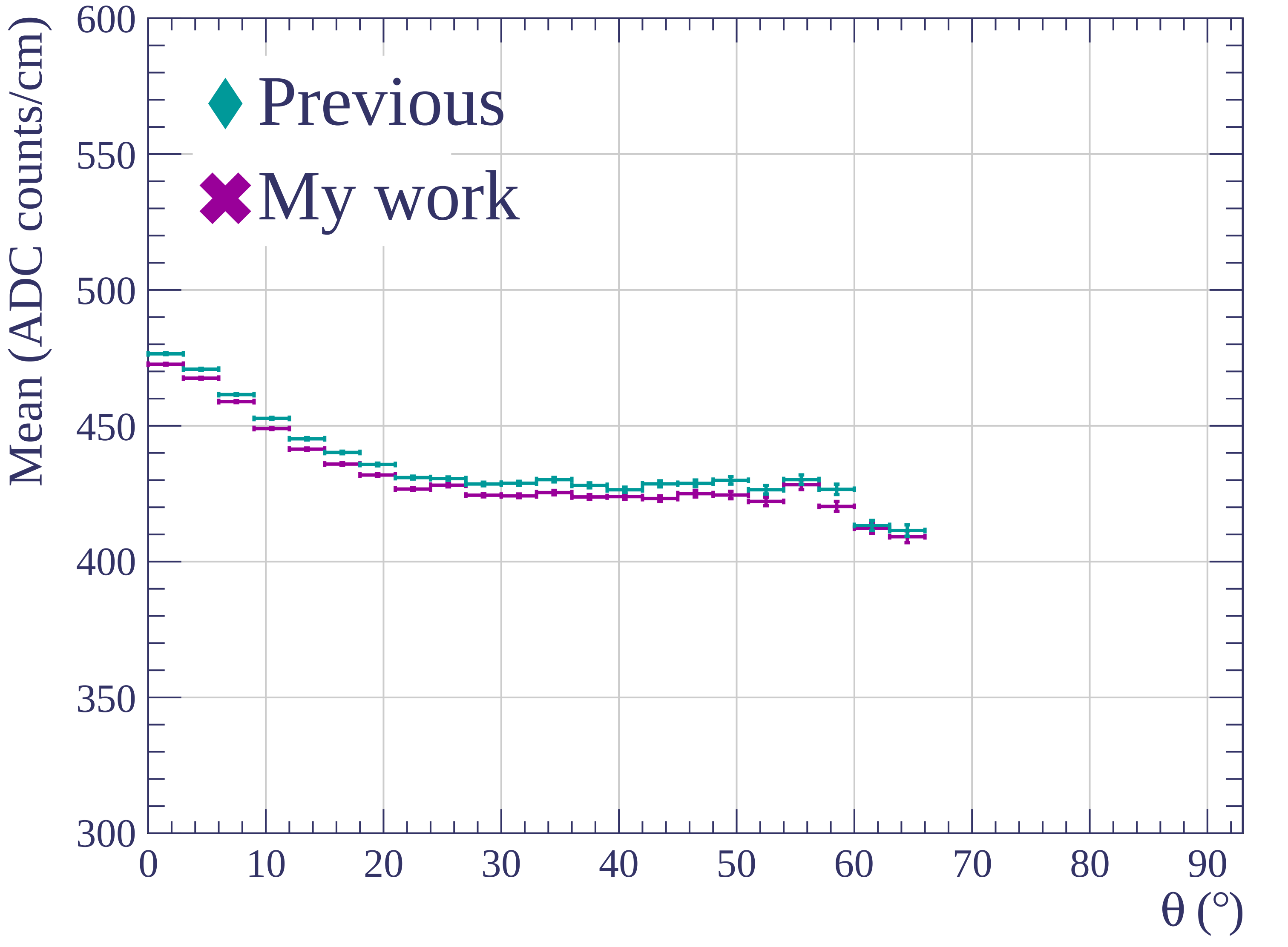
momentum (MeV)
2000
1500
1000
500
0
−500
−1000
−1500
−2000
−80
−60
−40
−20
0
20
40
60
80
φ
200
180
160
140
120
100
80
60
40
20
0

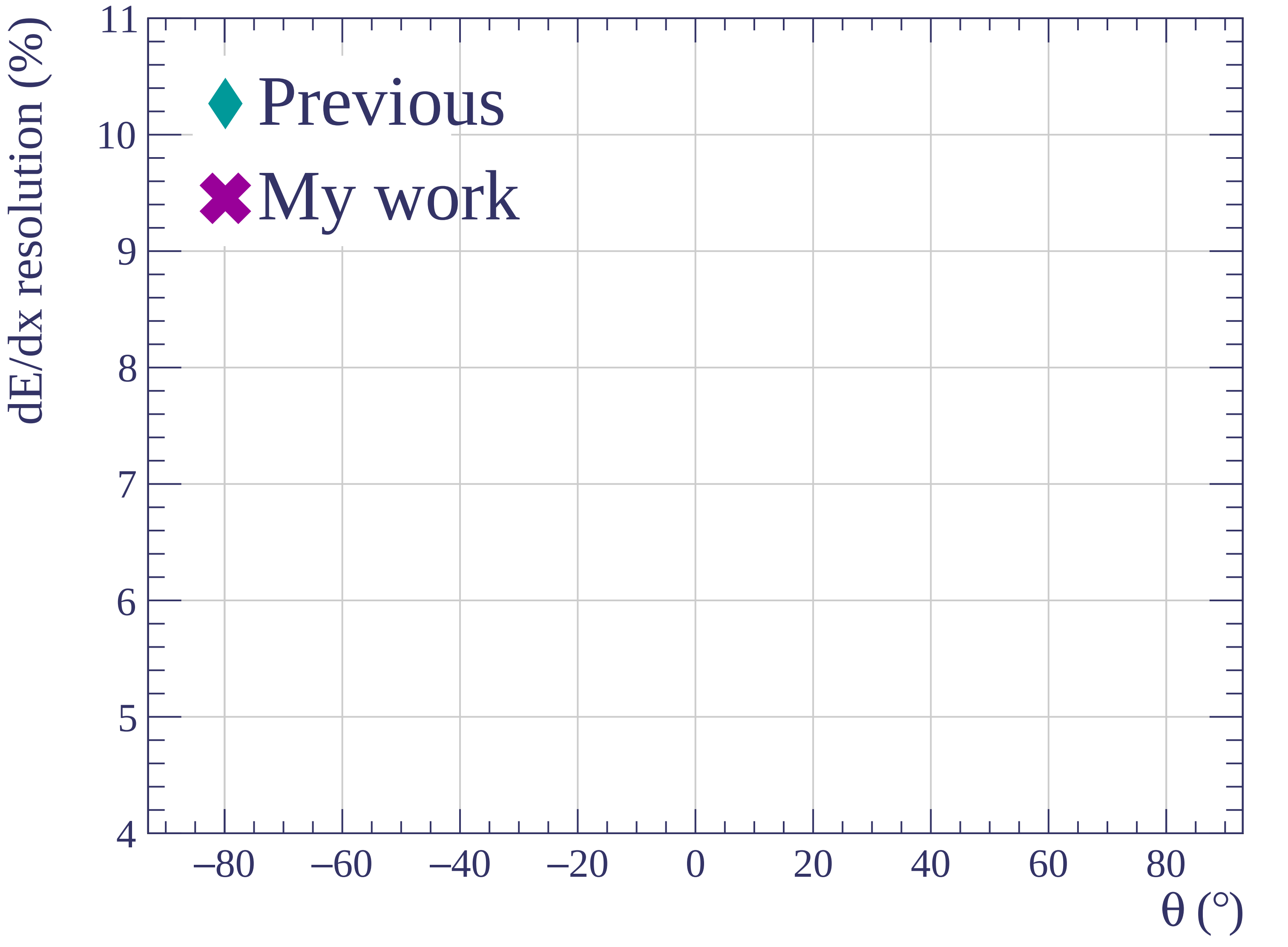


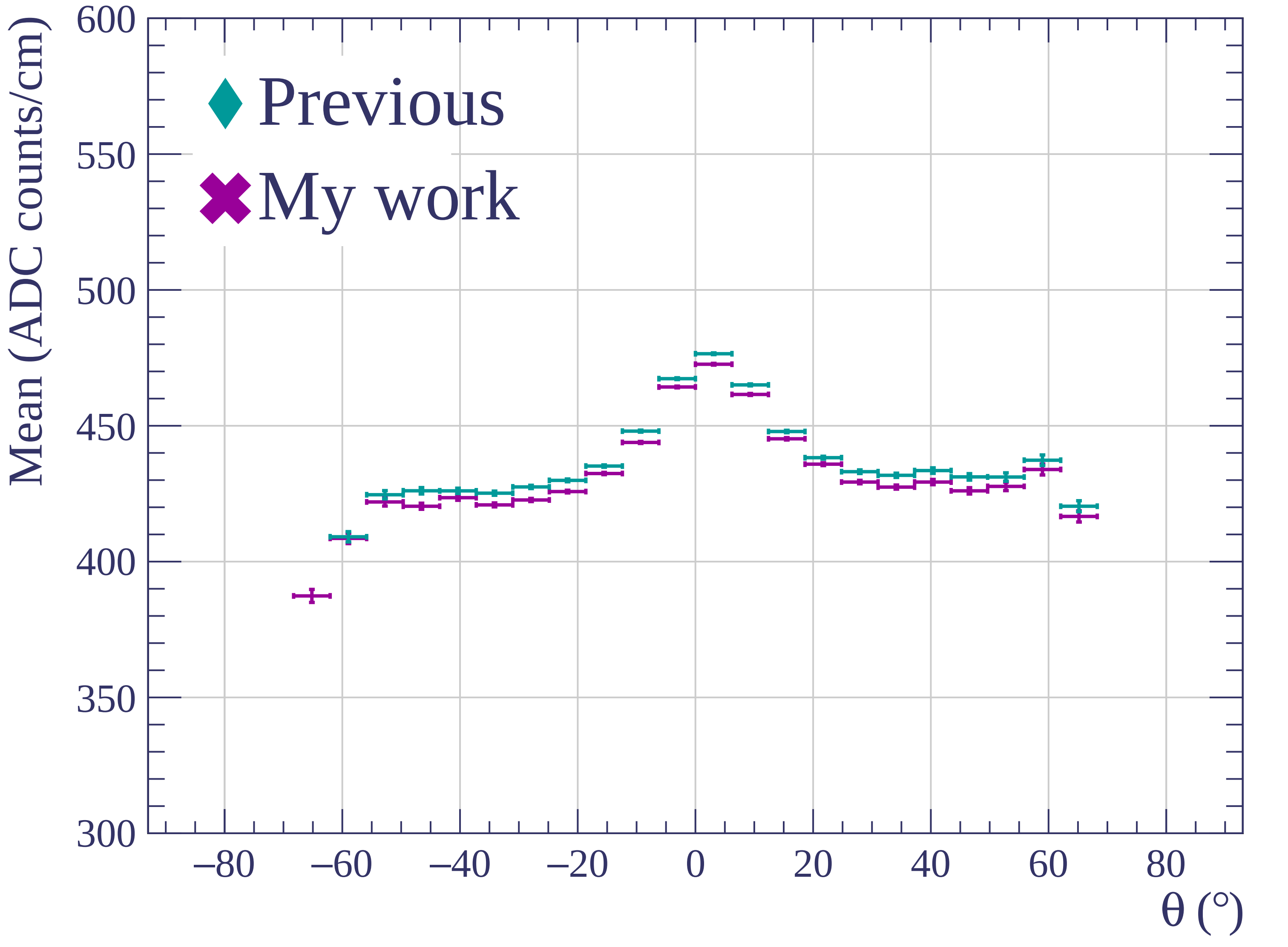
Count
×10
16
14
12
10
8
6
4
2
0
−80
−60
−40
−20
0
20
40
60
80
φ angle



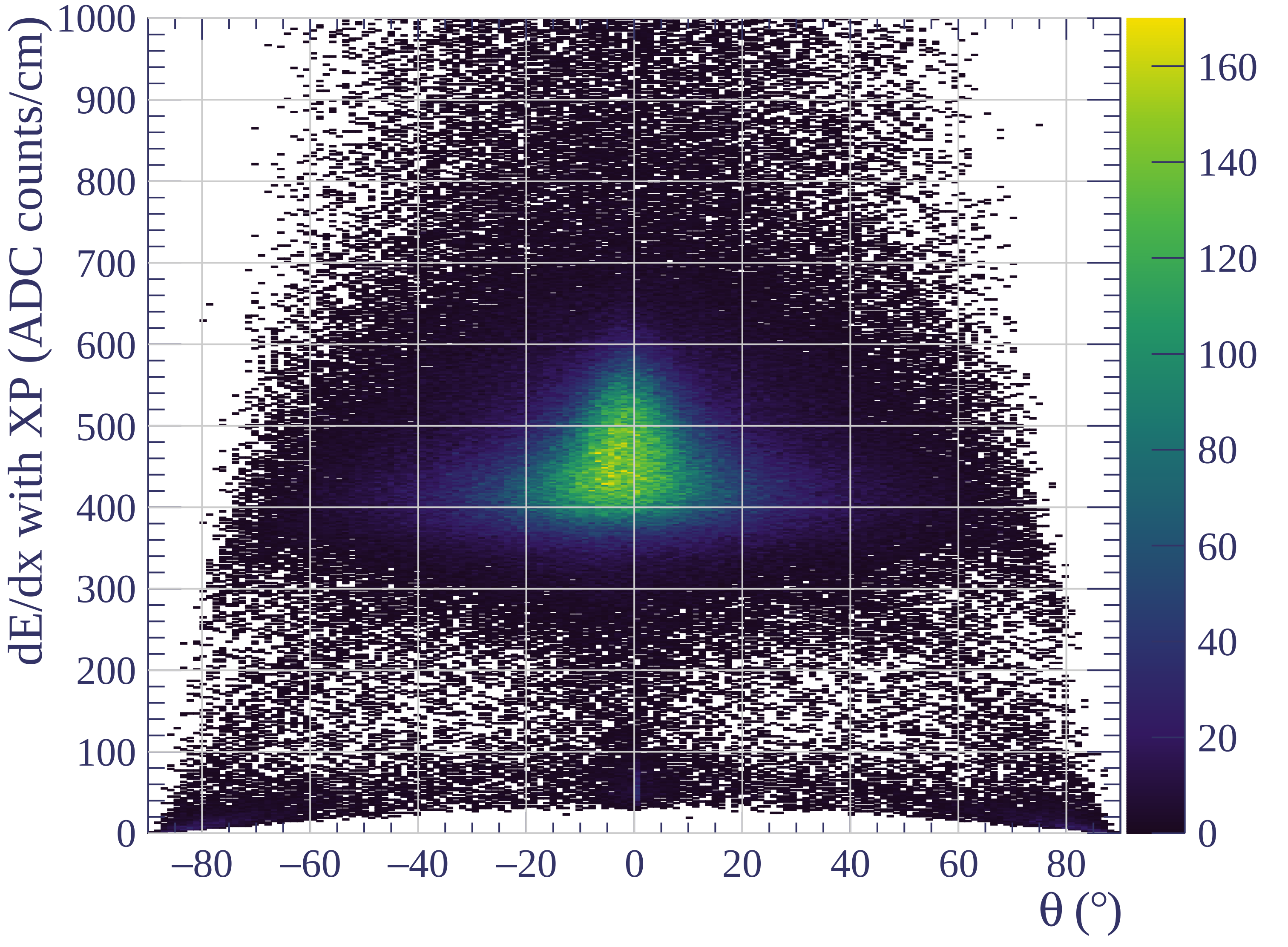
dE/dx resolution (%)
11
10
9
8
7
6
5
4
0
10
20
30
40
50
60
70
80
90
θ (°)
Previous
My work

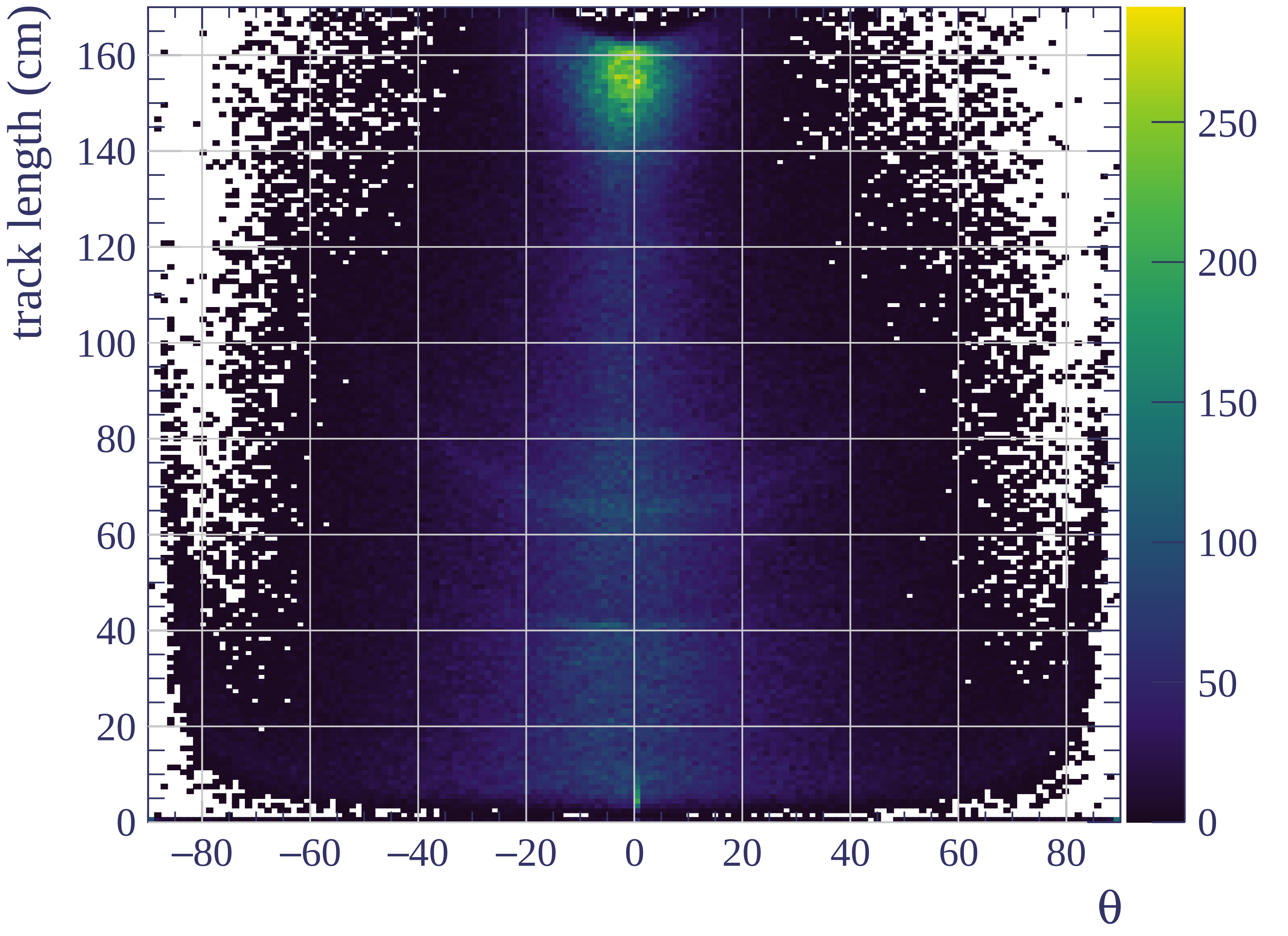

Previous
My work
Mean (ADC counts/cm)
600
550
500
450
400
350
300
0
10
20
30
40
50
60
70
80
90
θ (°)


Previous
My work
dE/dx resolution (%)
θ (°)
4
5
6
7
8
9
10
11
−80
−60
−40
−20
0
20
40
60
80

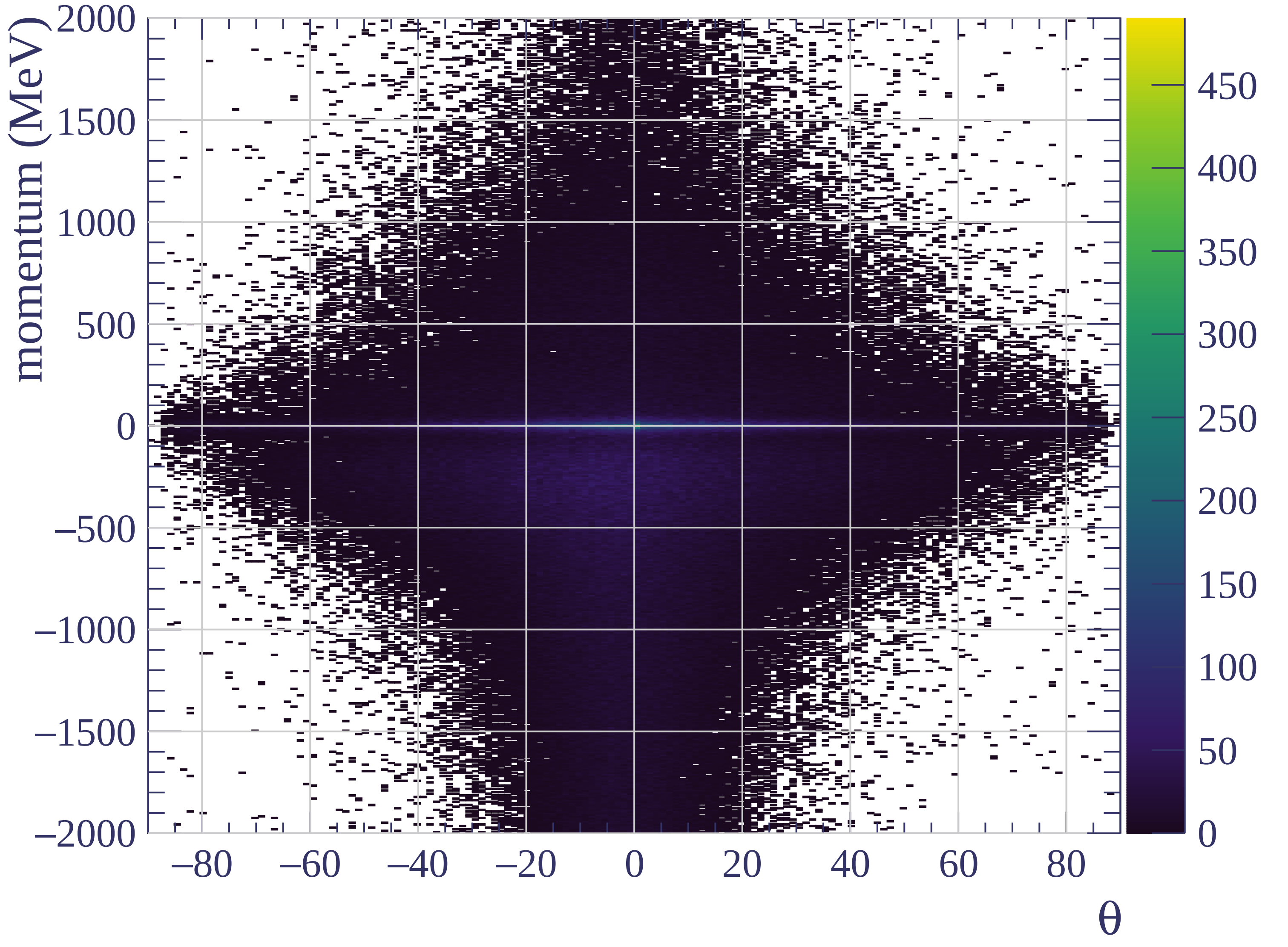


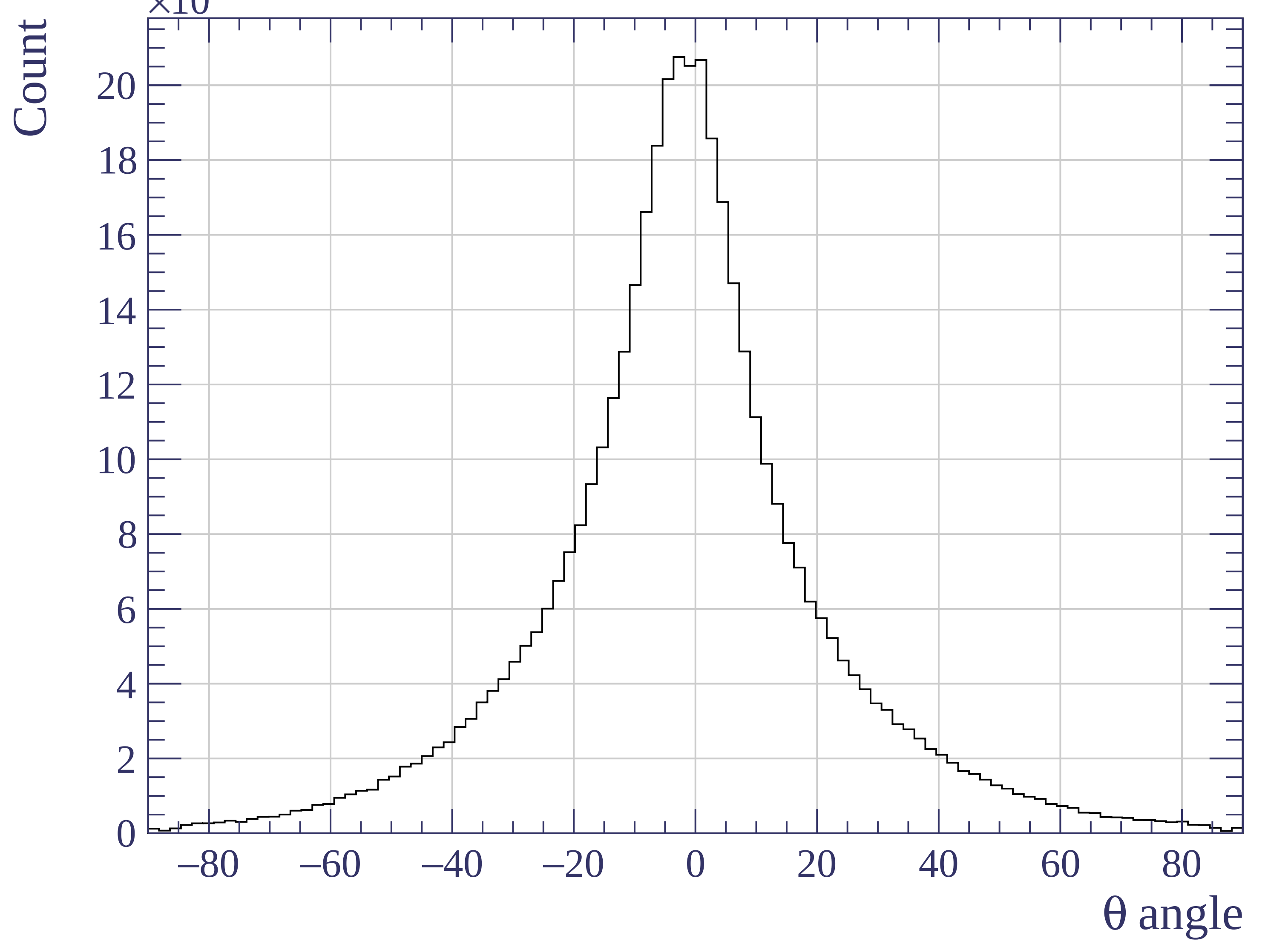
Previous
My work
Mean (ADC counts/cm)
600
550
500
450
400
350
300
−80
−60
−40
−20
0
20
40
60
80
θ (°)

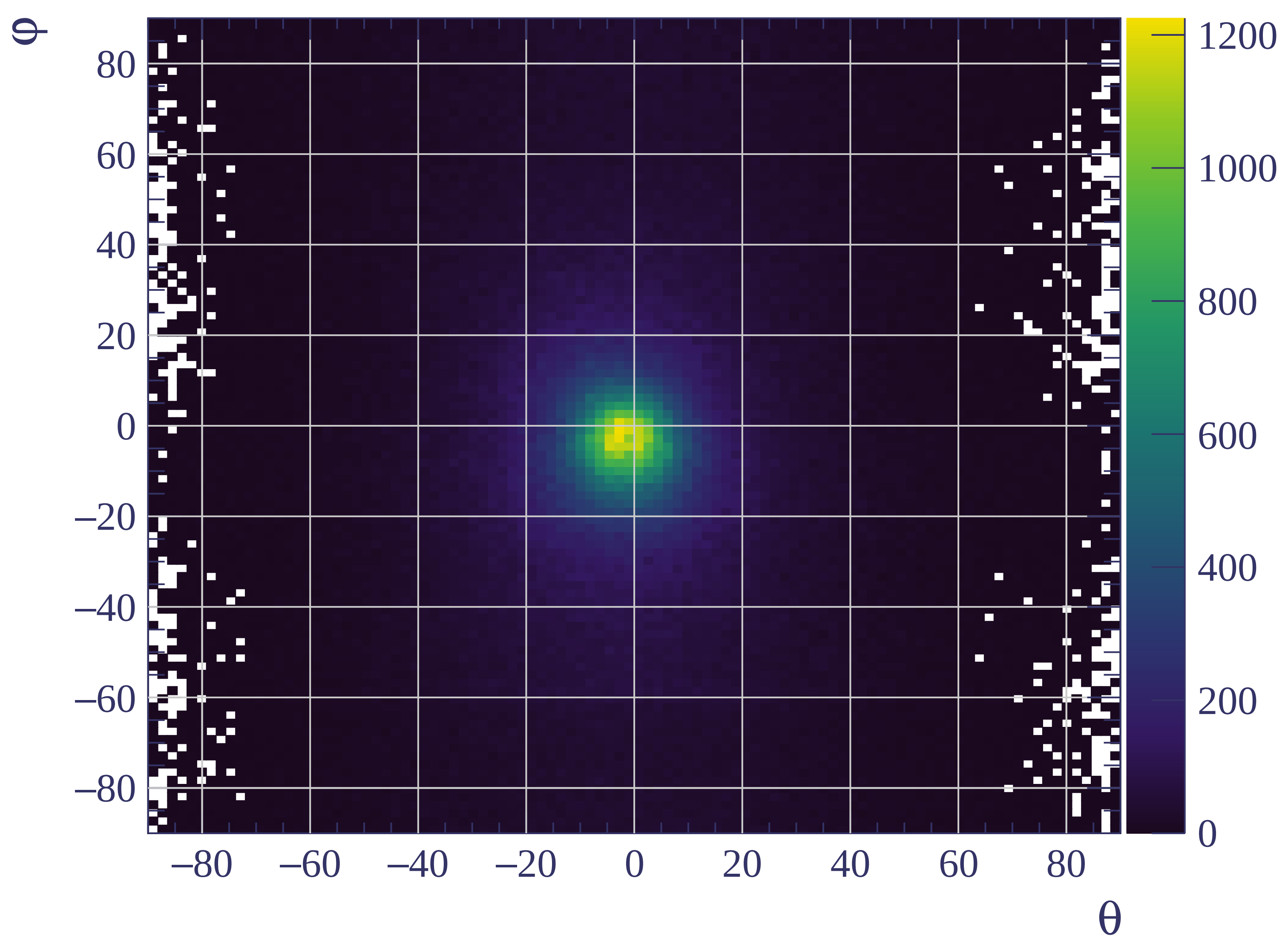

dE/dx with XP (ADC counts/cm)
1000
900
800
700
600
500
400
300
200
100
0
−80
−60
−40
−20
0
20
40
60
80
θ (°)
160
140
120
100
80
60
40
20
0



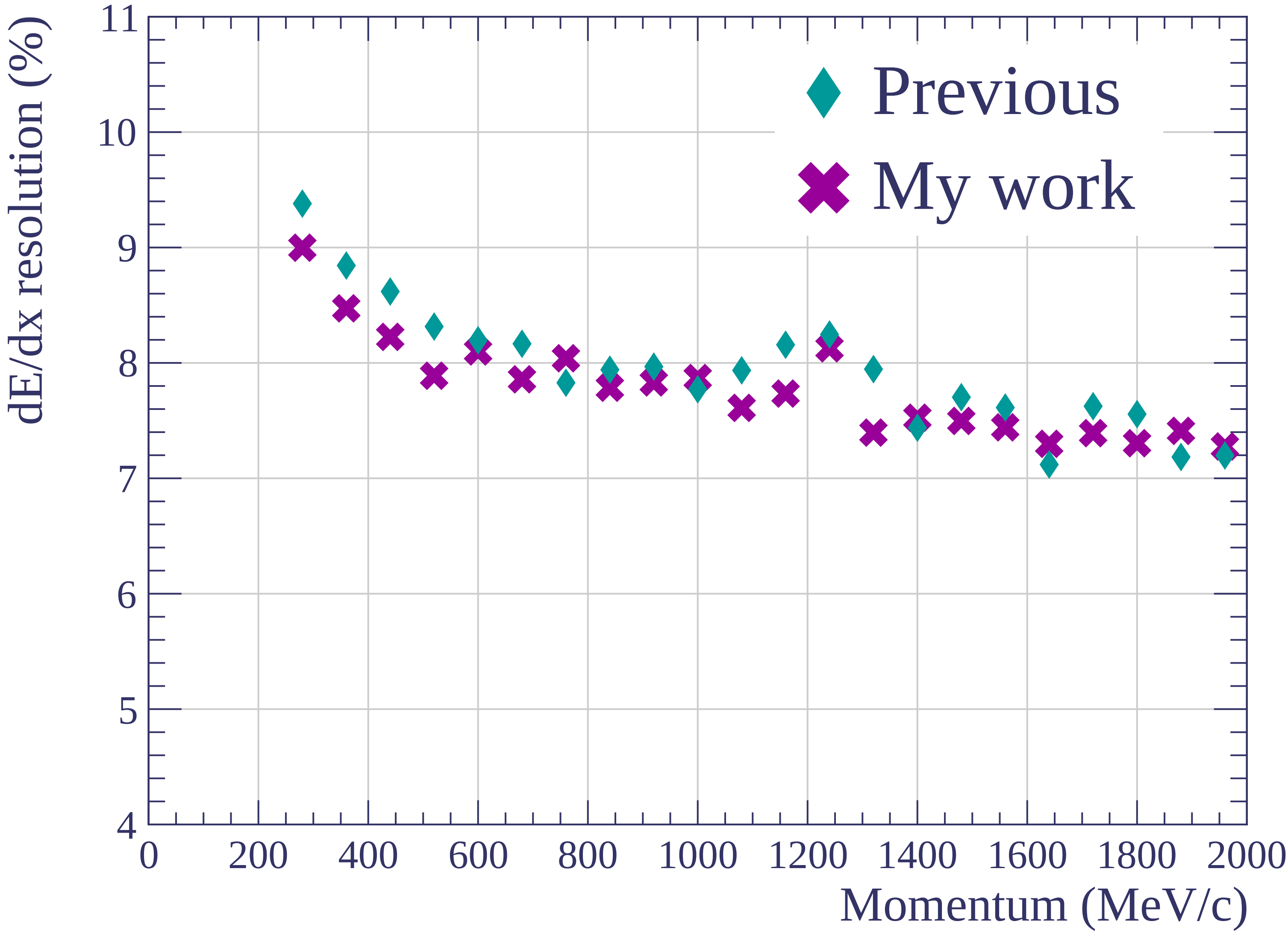
track length (cm)
160
140
120
100
80
60
40
20
0
−80
−60
−40
−20
0
20
40
60
80
θ
250
200
150
100
50
0


momentum (MeV)
2000
1500
1000
500
0
−500
−1000
−1500
−2000
−80
−60
−40
−20
0
20
40
60
80
θ
450
400
350
300
250
200
150
100
50
0

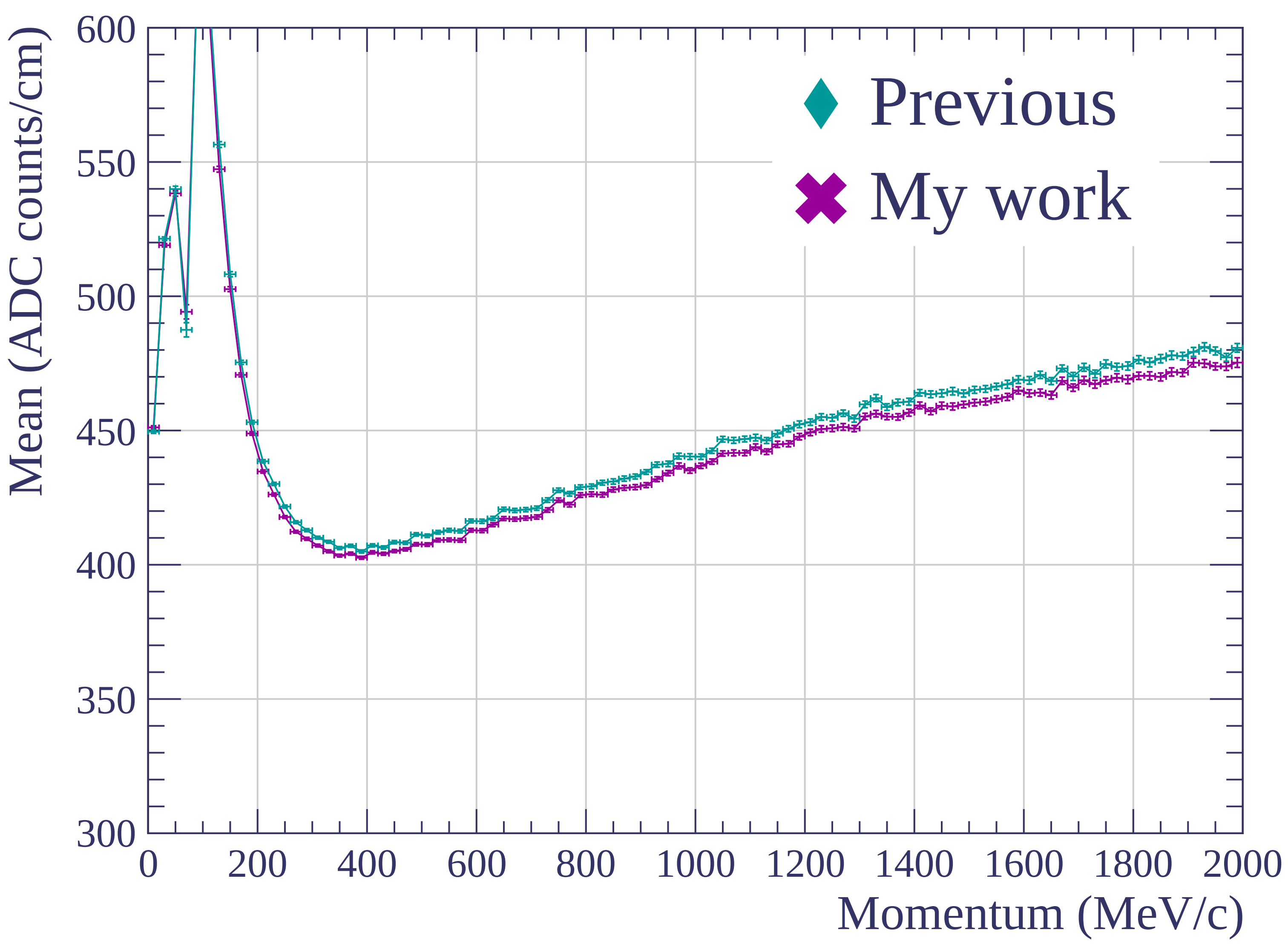

×10
Count
20
18
16
14
12
10
8
6
4
2
0
−80
−60
−40
−20
0
20
40
60
80
θ angle



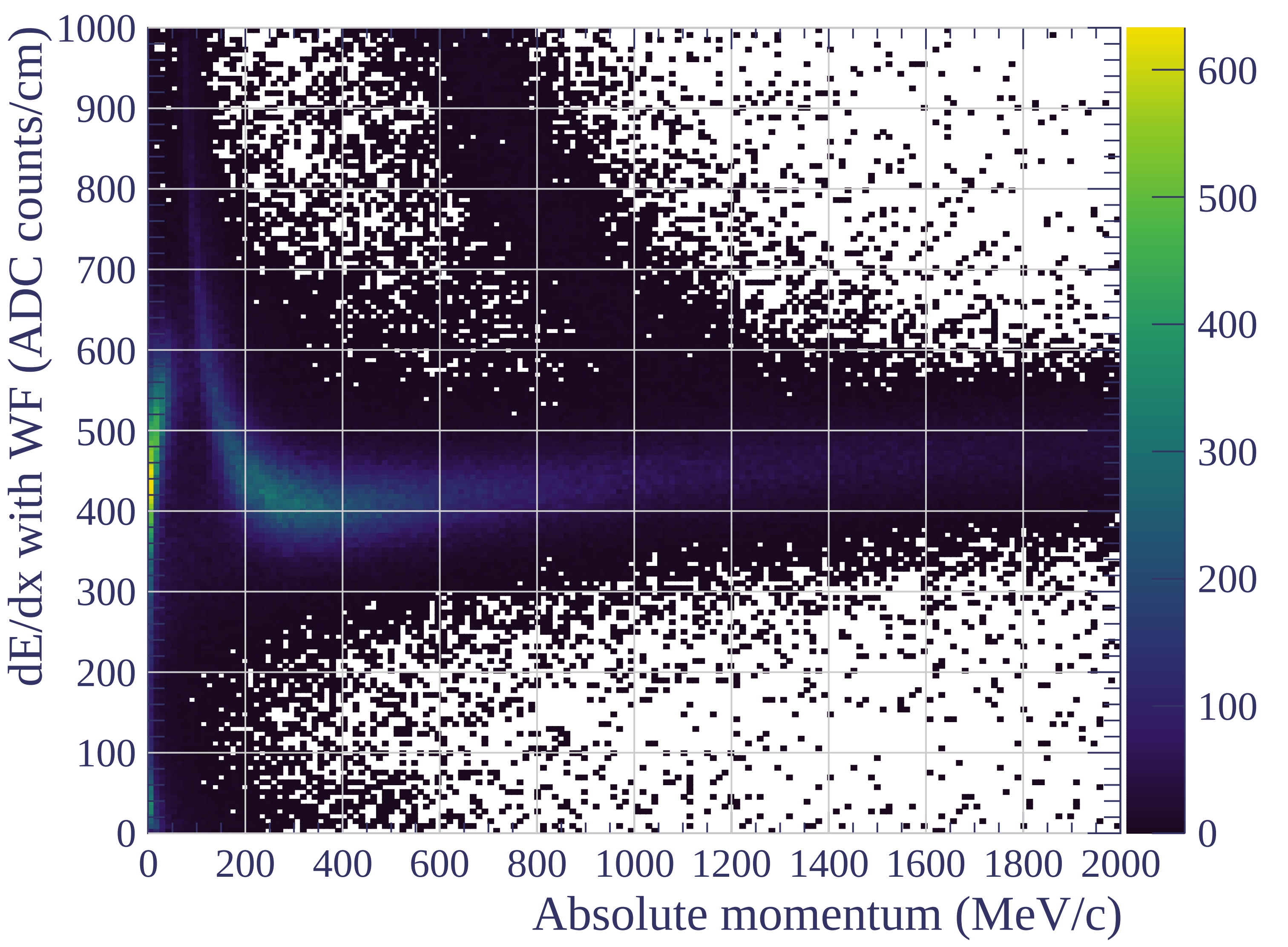
φ
80
60
40
20
0
−20
−40
−60
−80
−80
−60
−40
−20
0
20
40
60
80
θ
1200
1000
800
600
400
200
0

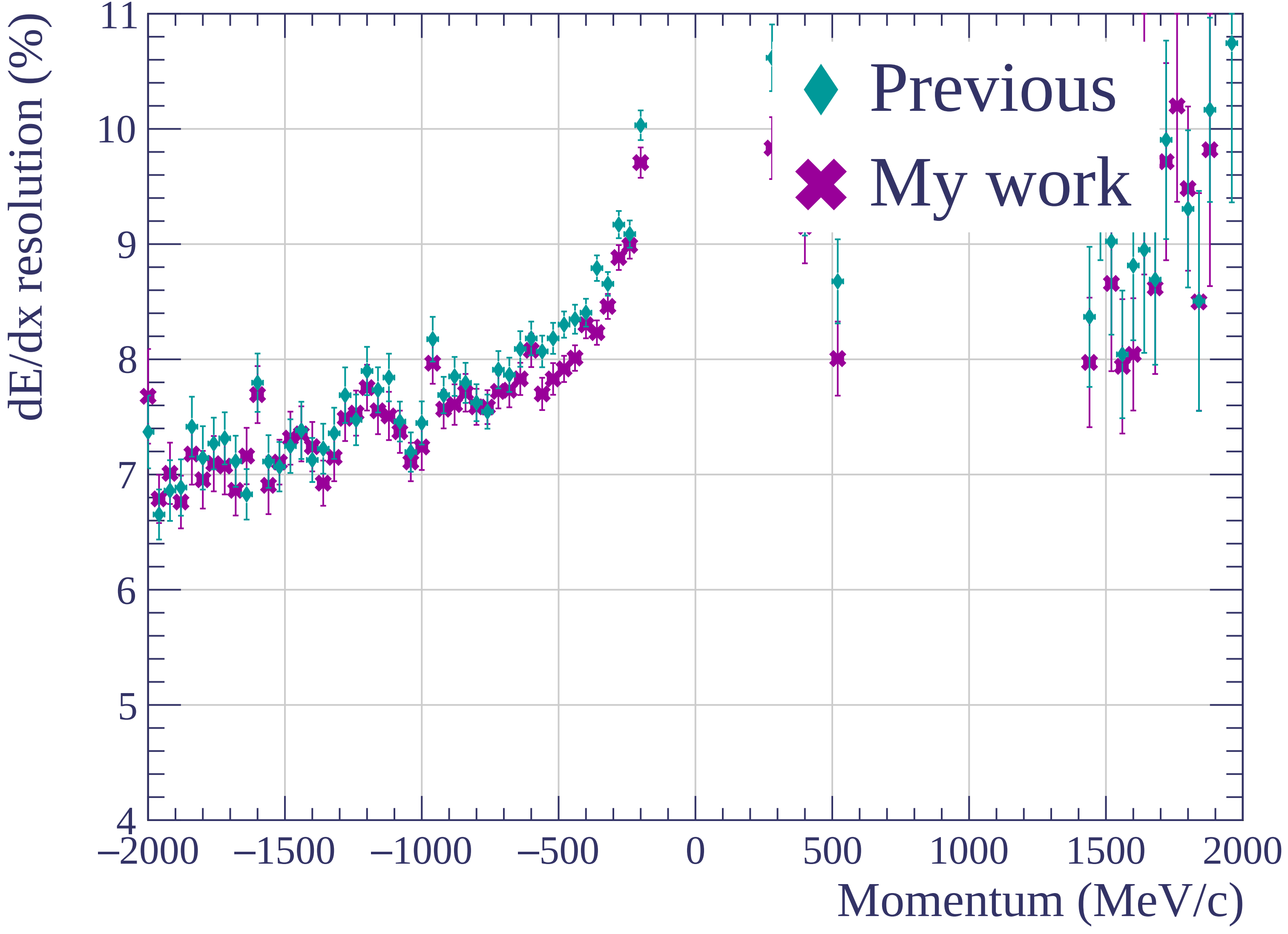


Previous
My work
dE/dx resolution (%)
Momentum (MeV/c)
4
5
6
7
8
9
10
11
0
200
400
600
800
1000
1200
1400
1600
1800
2000

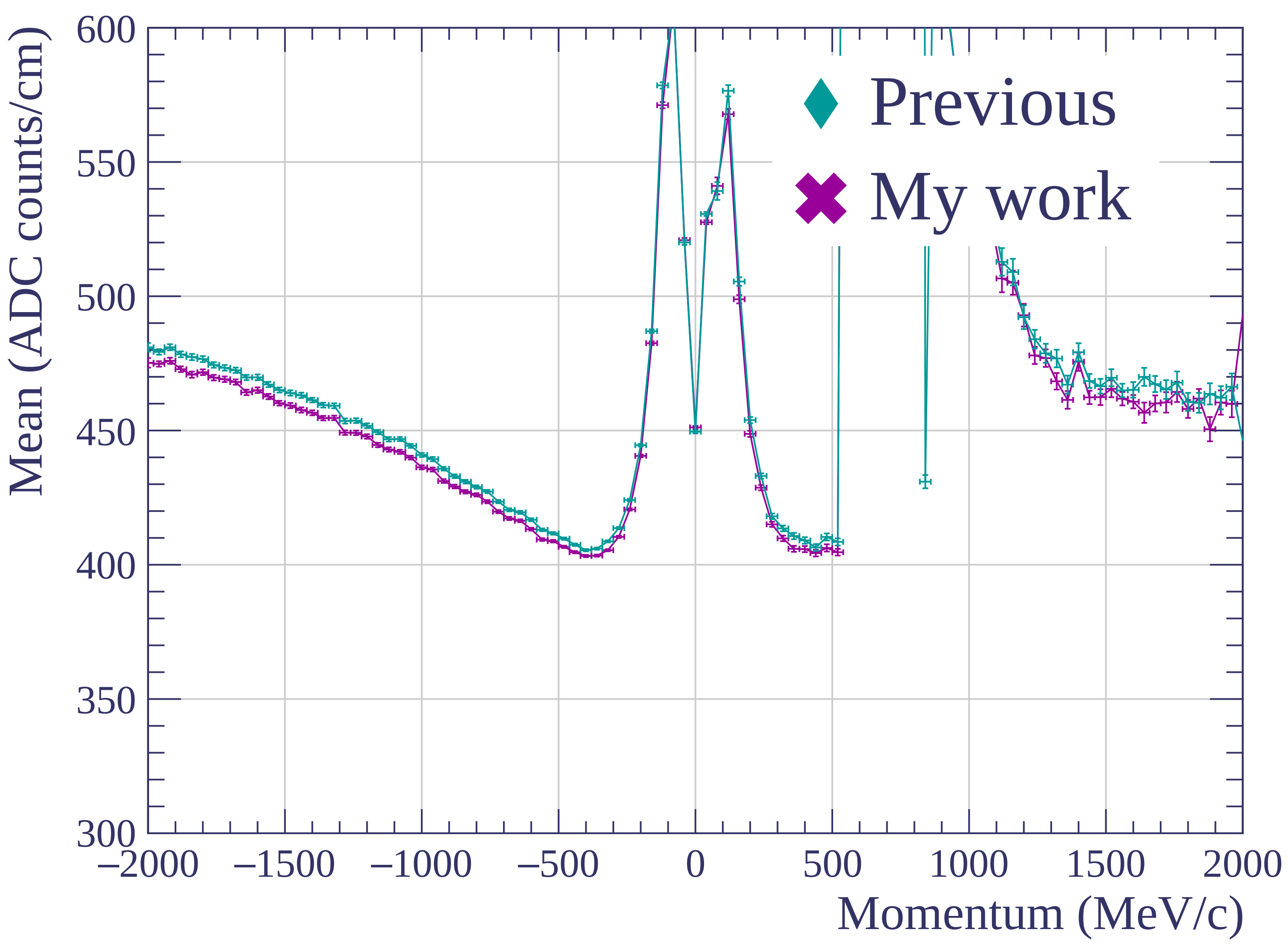


Previous
My work
Mean (ADC counts/cm)
Momentum (MeV/c)
600
550
500
450
400
350
300
0
200
400
600
800
1000
1200
1400
1600
1800
2000

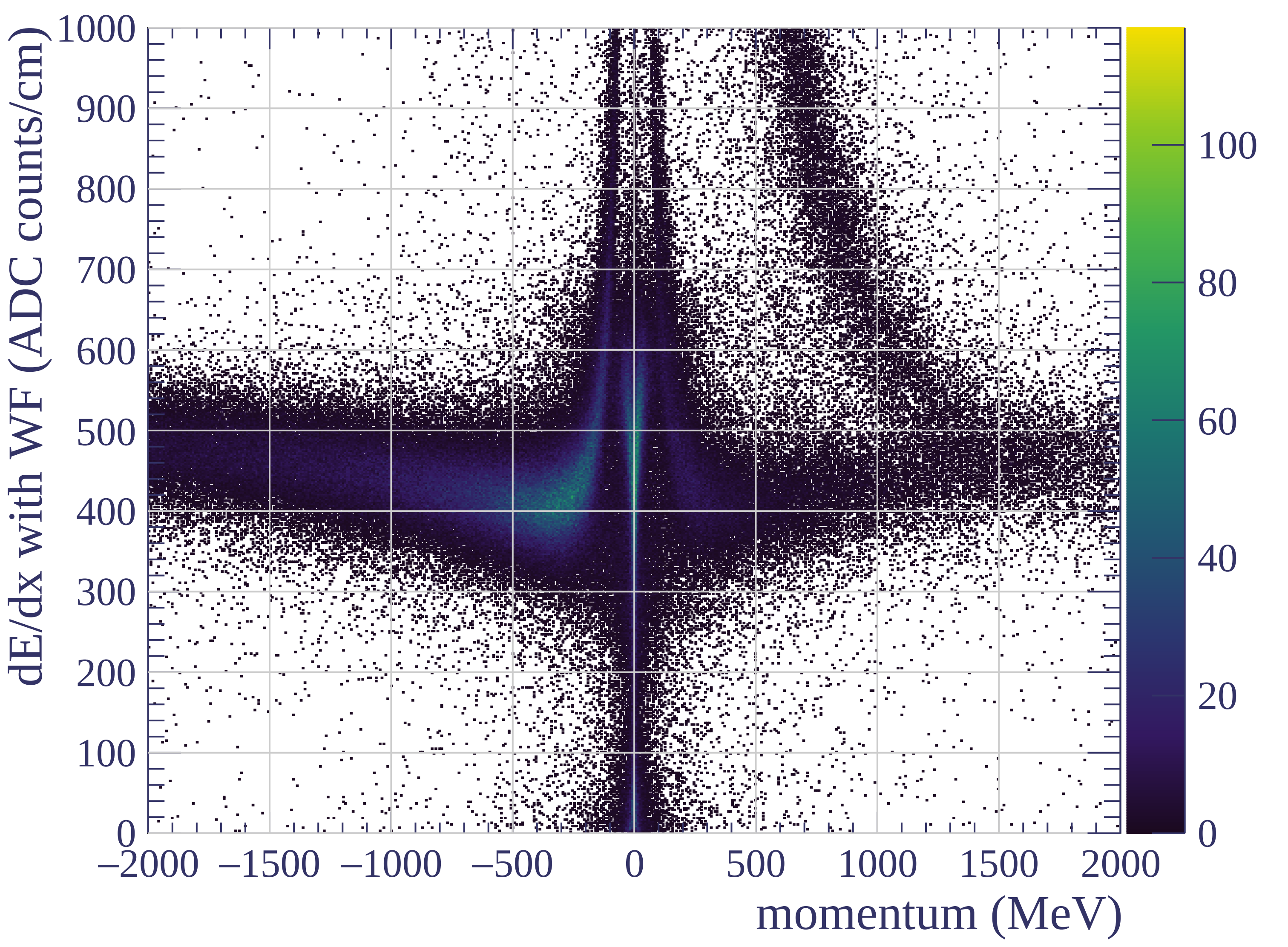

dE/dx with WF (ADC counts/cm)
Absolute momentum (MeV/c)
0
100
200
300
400
500
600
700
800
900
1000
0
200
400
600
800
1000
1200
1400
1600
1800
2000
0
100
200
300
400
500
600



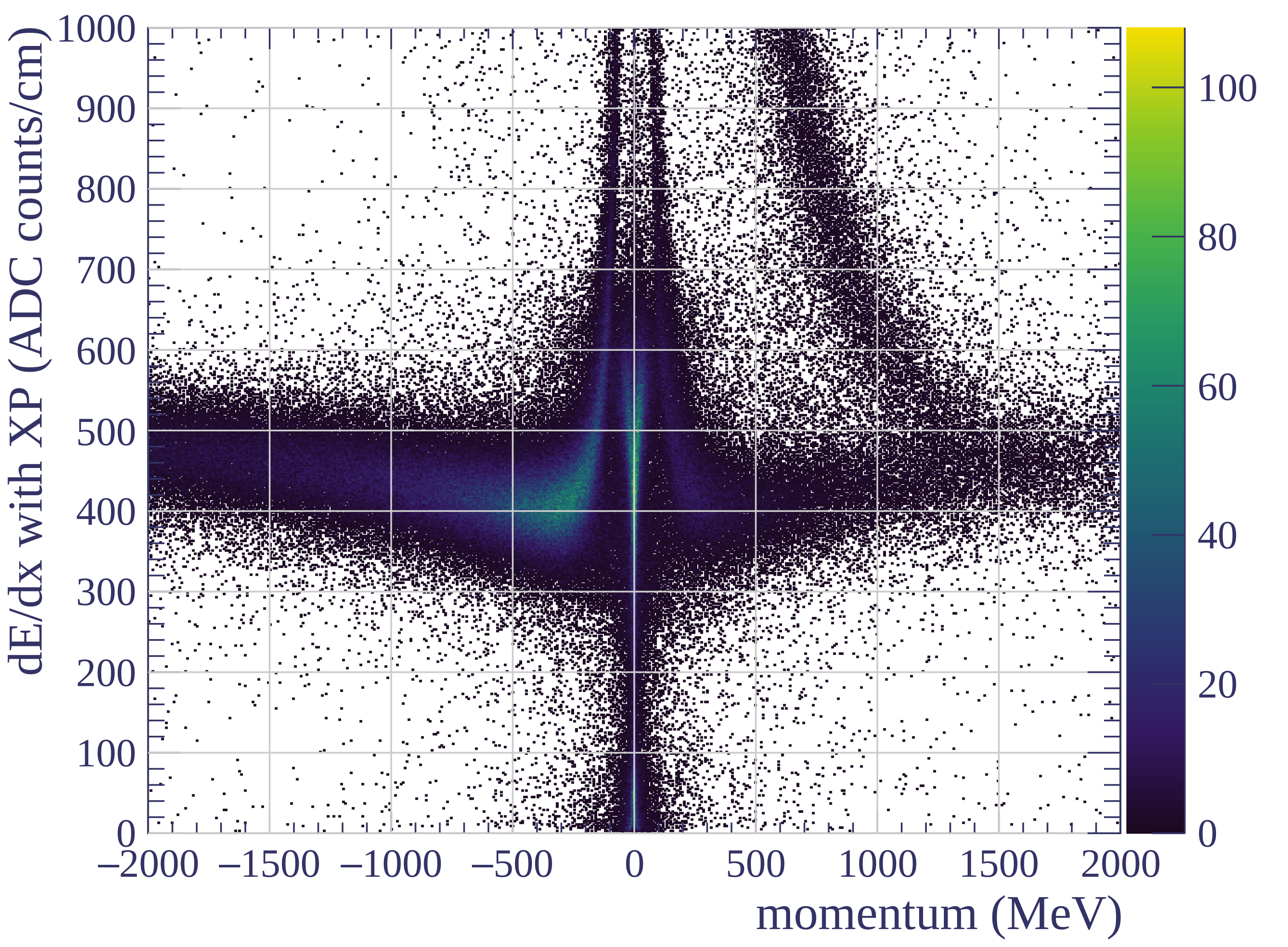
Previous
My work
dE/dx resolution (%)
Momentum (MeV/c)
4
5
6
7
8
9
10
11
−2000
−1500
−1000
−500
0
500
1000
1500
2000



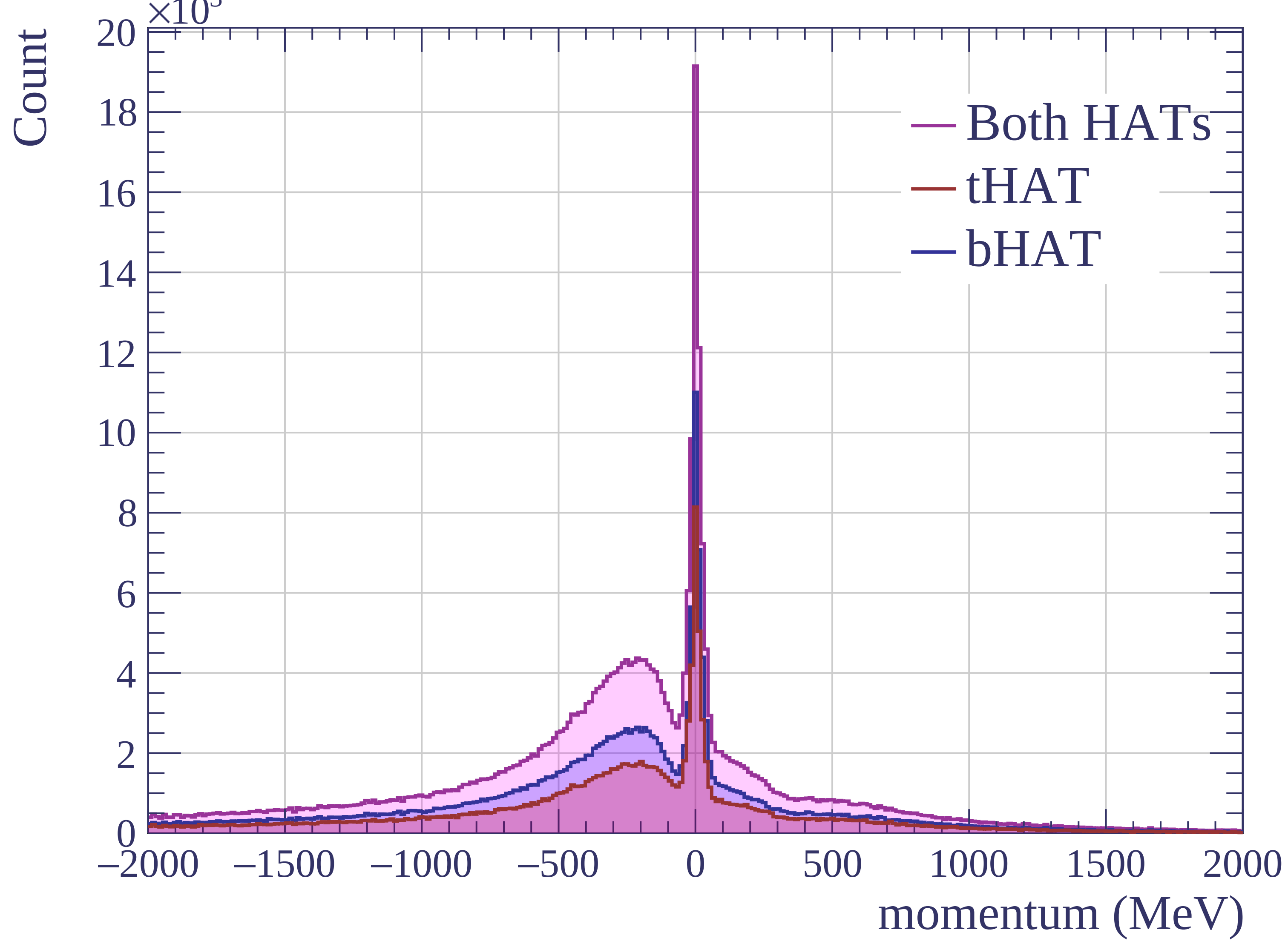
Mean (ADC counts/cm)
600
550
500
450
400
350
300
−2000
−1500
−1000
−500
0
500
1000
1500
2000
Momentum (MeV/c)
Previous
My work



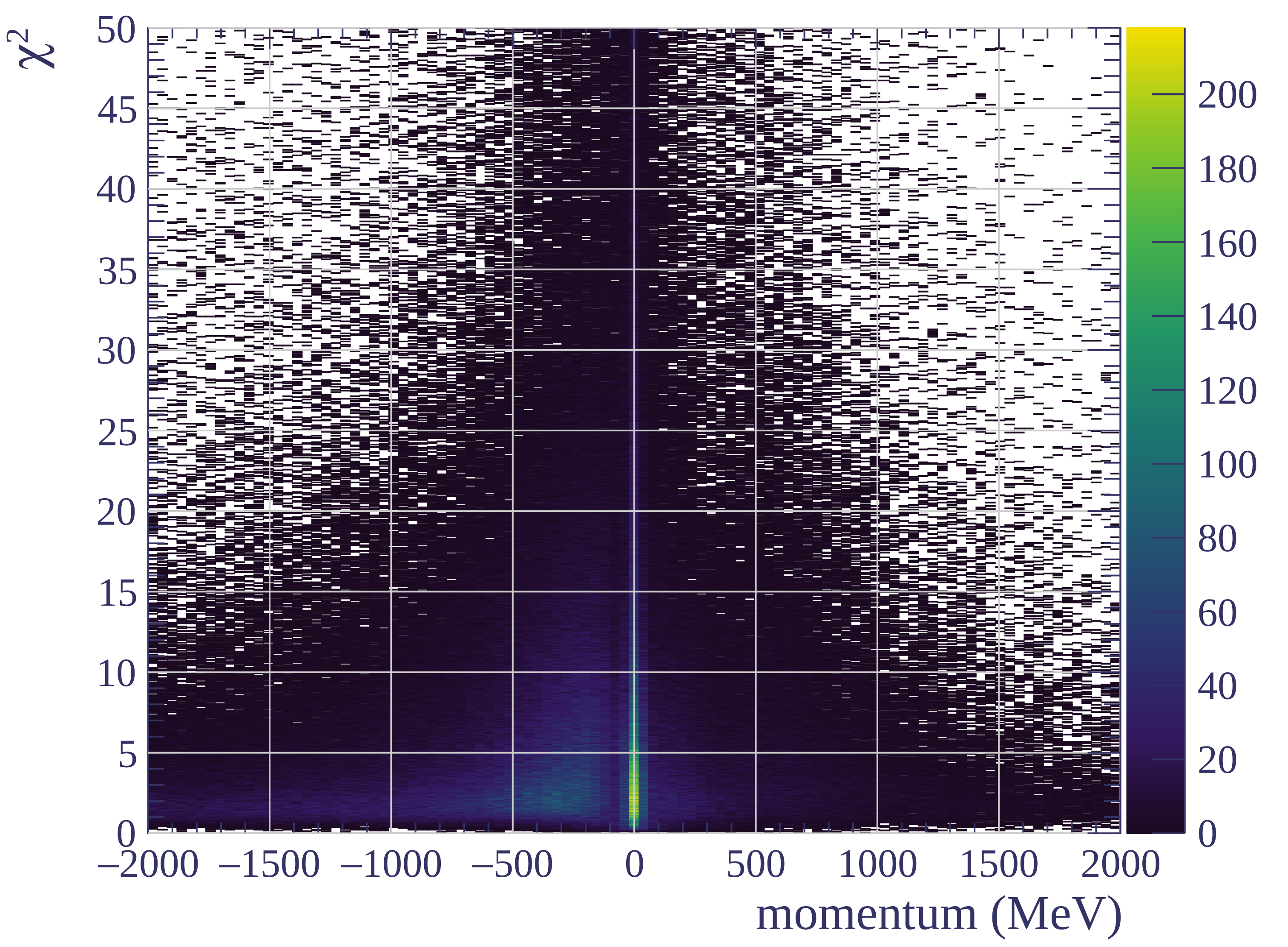
dE/dx with WF (ADC counts/cm)
1000
900
800
700
600
500
400
300
200
100
0
−2000
−1500
−1000
−500
0
500
1000
1500
2000
momentum (MeV)
100
80
60
40
20
0



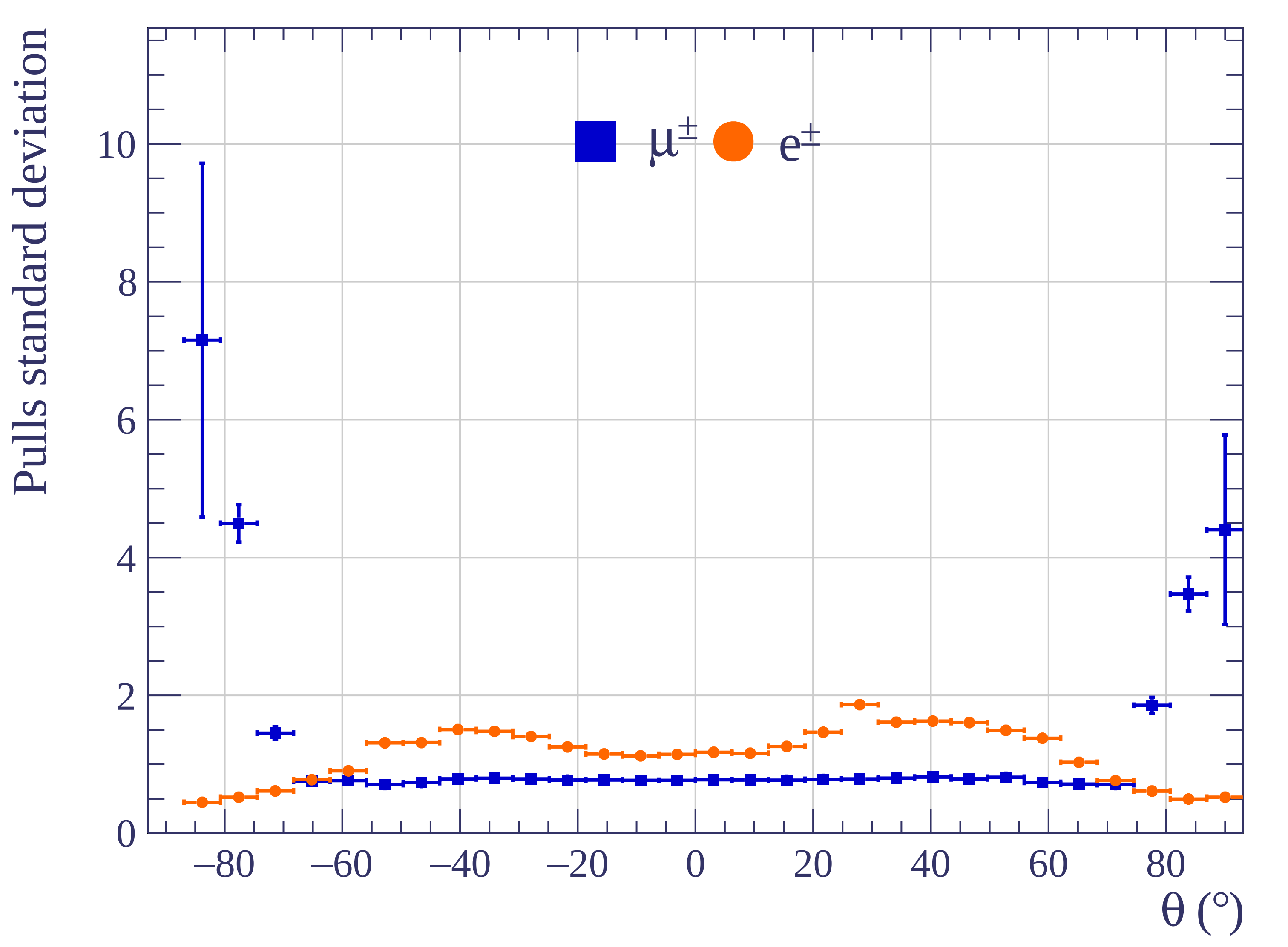
dE/dx with XP (ADC counts/cm)
1000
900
800
700
600
500
400
300
200
100
0
−2000
−1500
−1000
−500
0
500
1000
1500
2000
momentum (MeV)
100
80
60
40
20
0

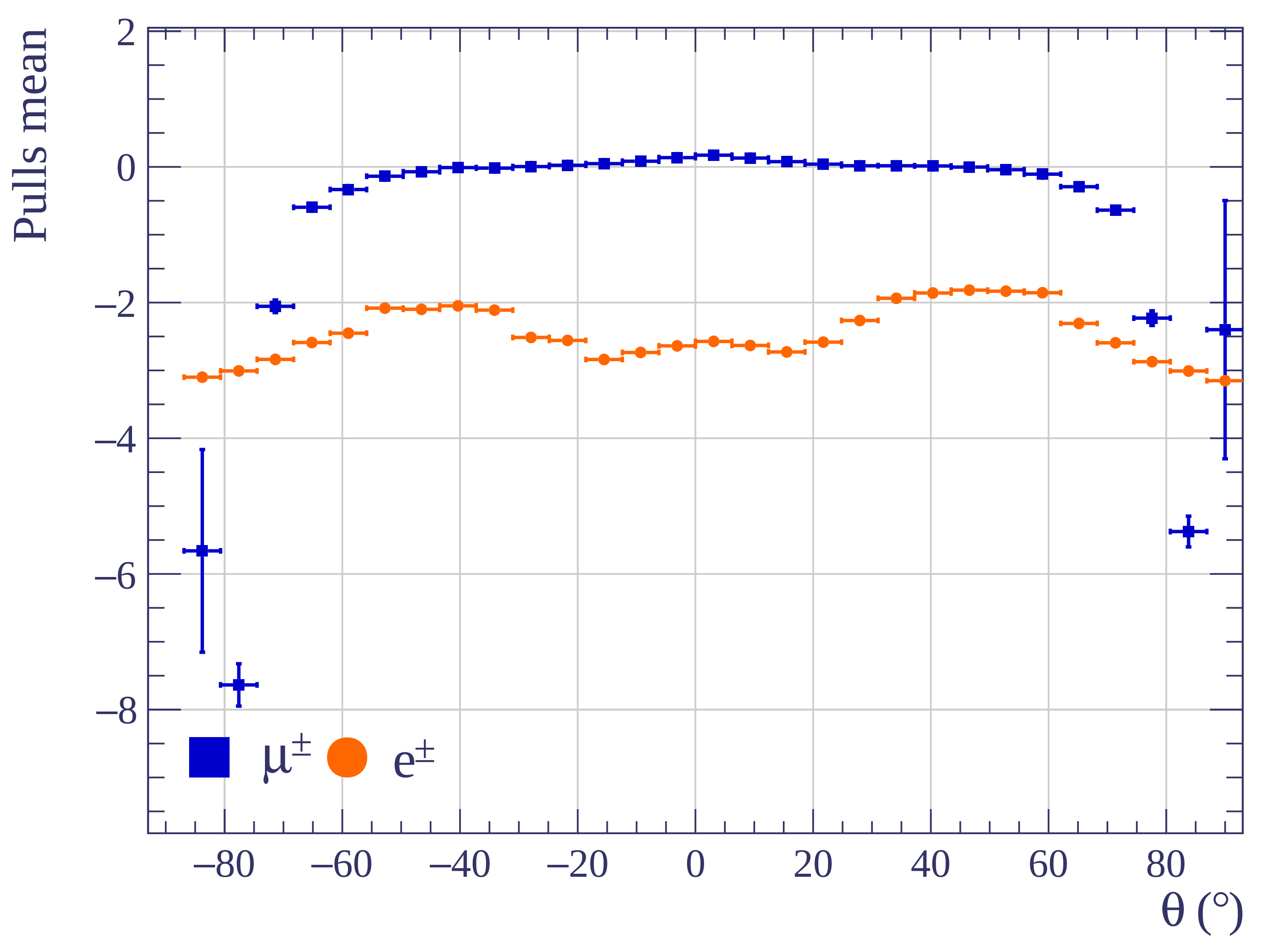


×10^3
Count
20
18
16
14
12
10
8
6
4
2
0
Both HATs
tHAT
bHAT
−2000
−1500
−1000
−500
0
500
1000
1500
2000
momentum (MeV)

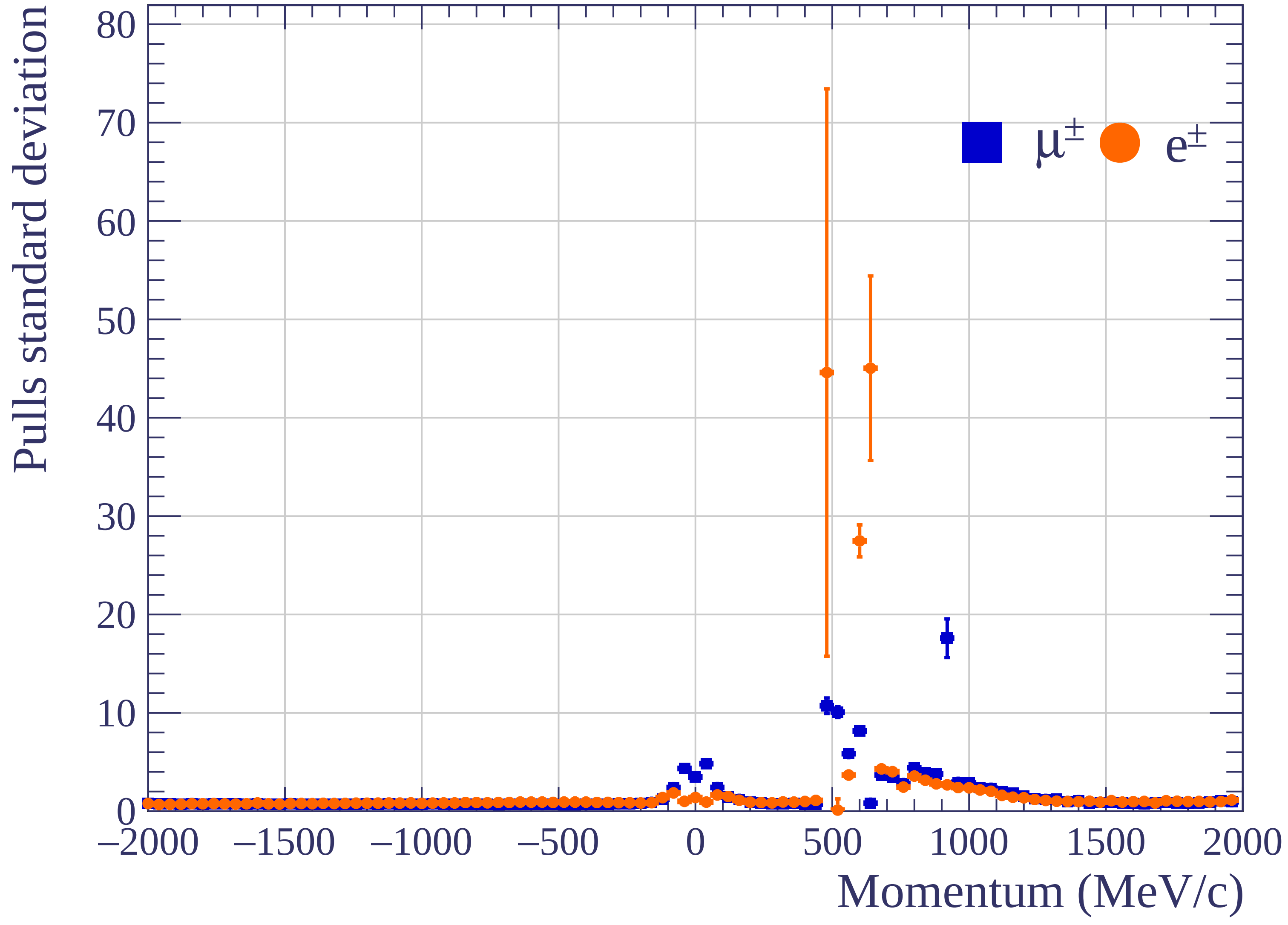


χ^2
50
45
40
35
30
25
20
15
10
5
0
−2000
−1500
−1000
−500
0
500
1000
1500
2000
momentum (MeV)
200
180
160
140
120
100
80
60
40
20
0

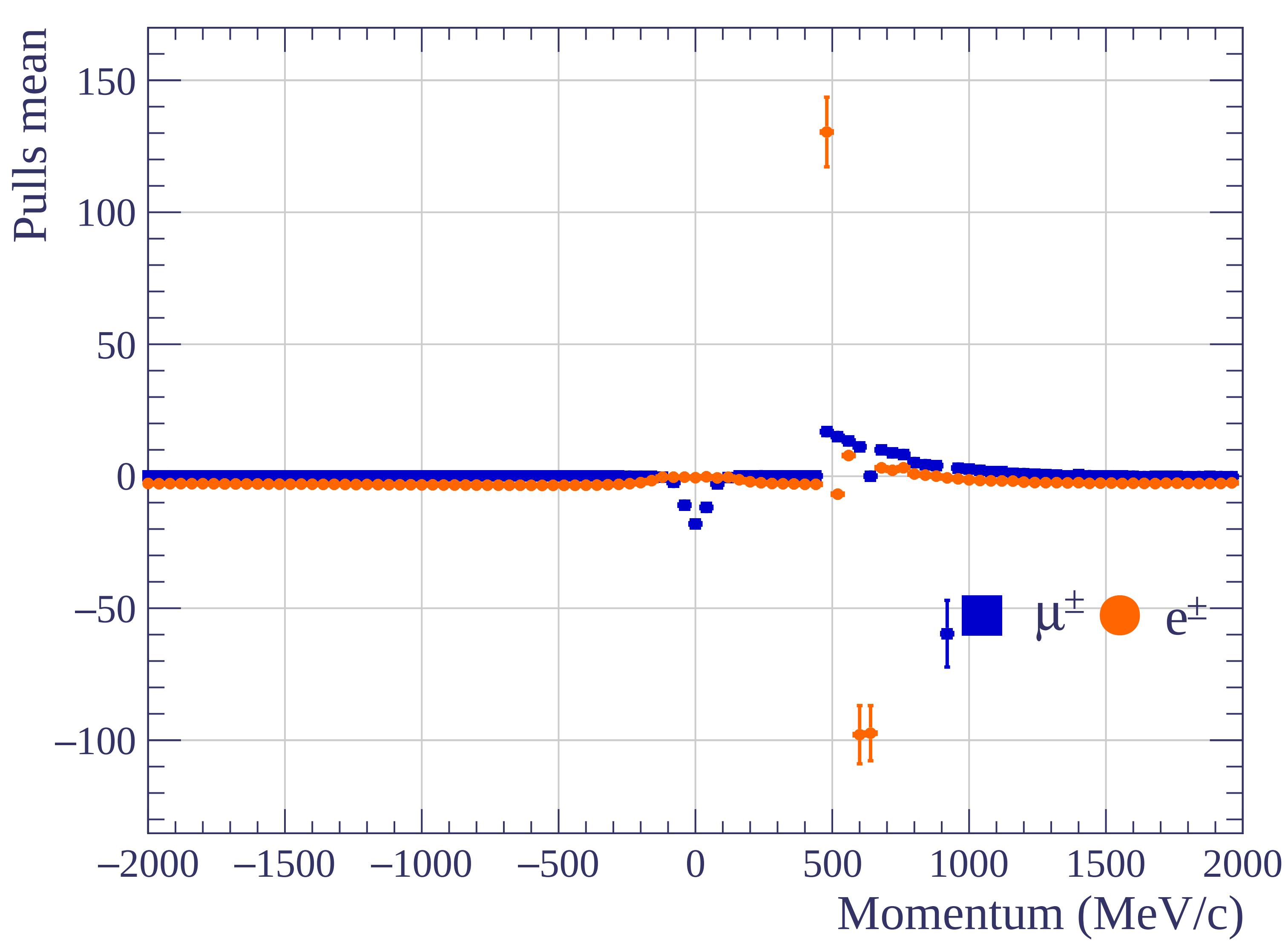

Pulls standard deviation
θ (°)
μ±
e±
0
2
4
6
8
10
−80
−60
−40
−20
0
20
40
60
80



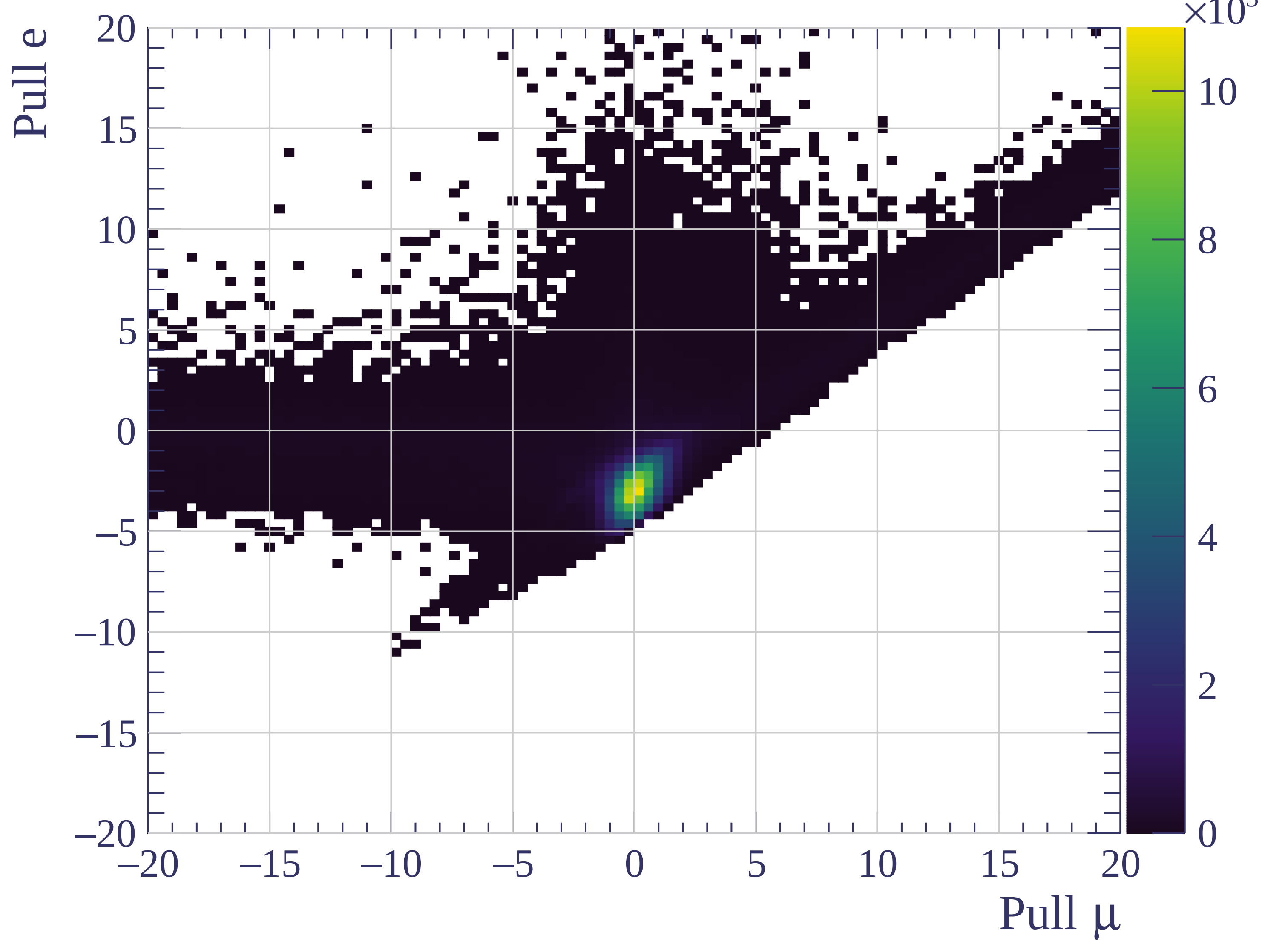
Pulls mean
2
0
−2
−4
−6
−8
−80
−60
−40
−20
0
20
40
60
80
θ (°)
μ±
e±



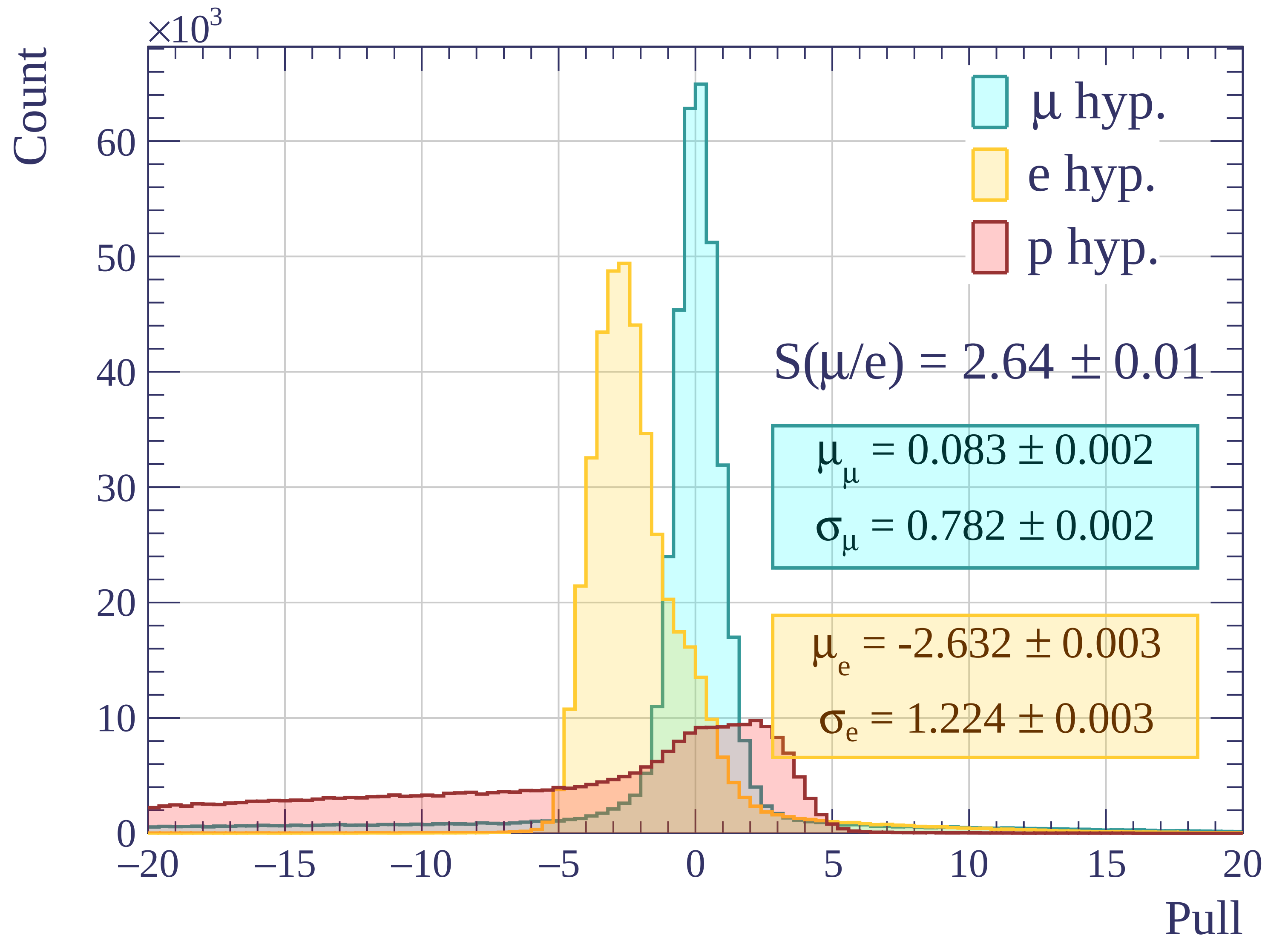
Pulls standard deviation
80
70
60
50
40
30
20
10
0
−2000
−1500
−1000
−500
0
500
1000
1500
2000
Momentum (MeV/c)
$\mu^{\pm}$
$e^{\pm}$



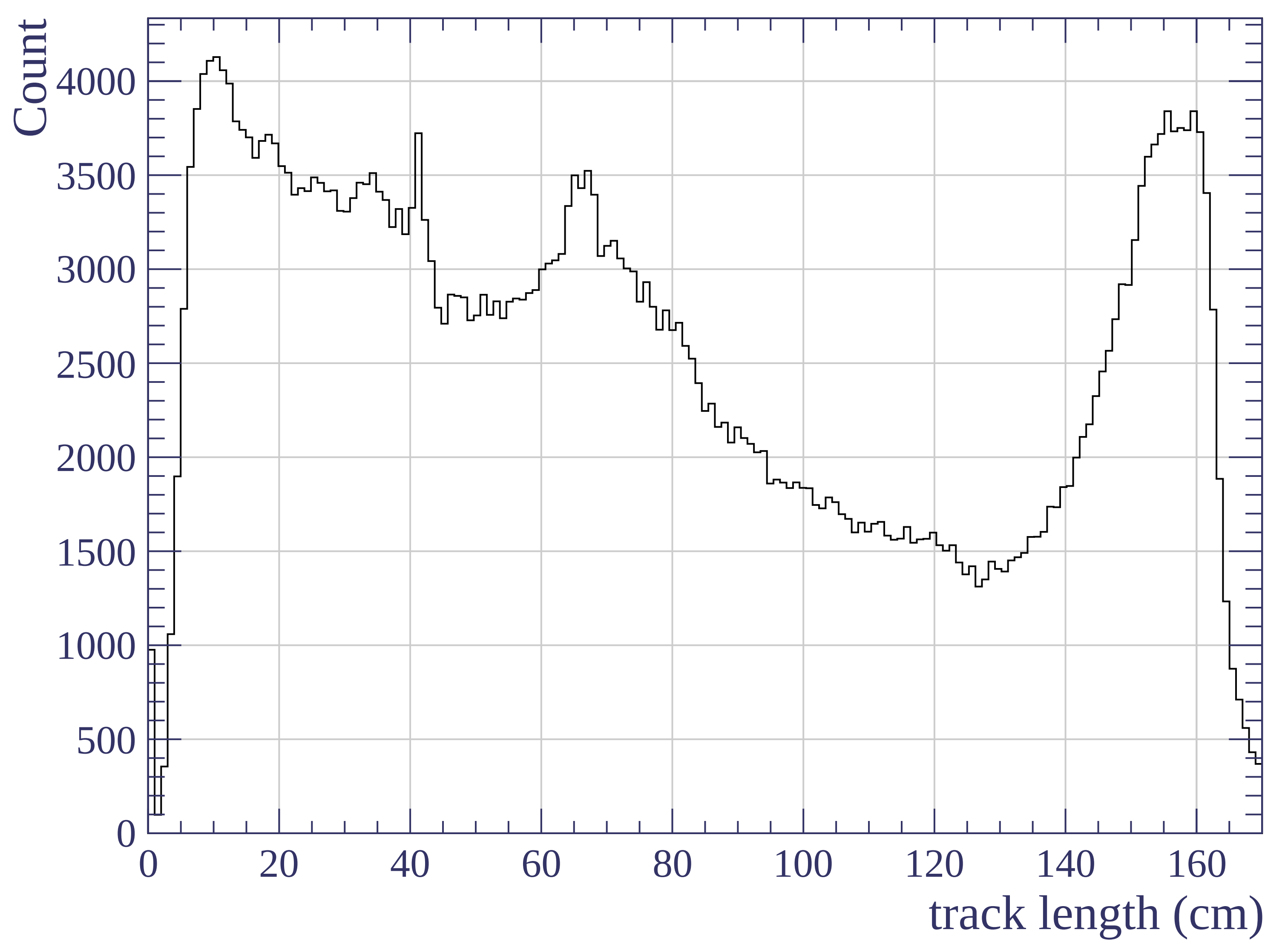
Pulls mean
150
100
50
0
−50
−100
−2000
−1500
−1000
−500
0
500
1000
1500
2000
Momentum (MeV/c)
$\mu^{\pm}$
$e^{\pm}$



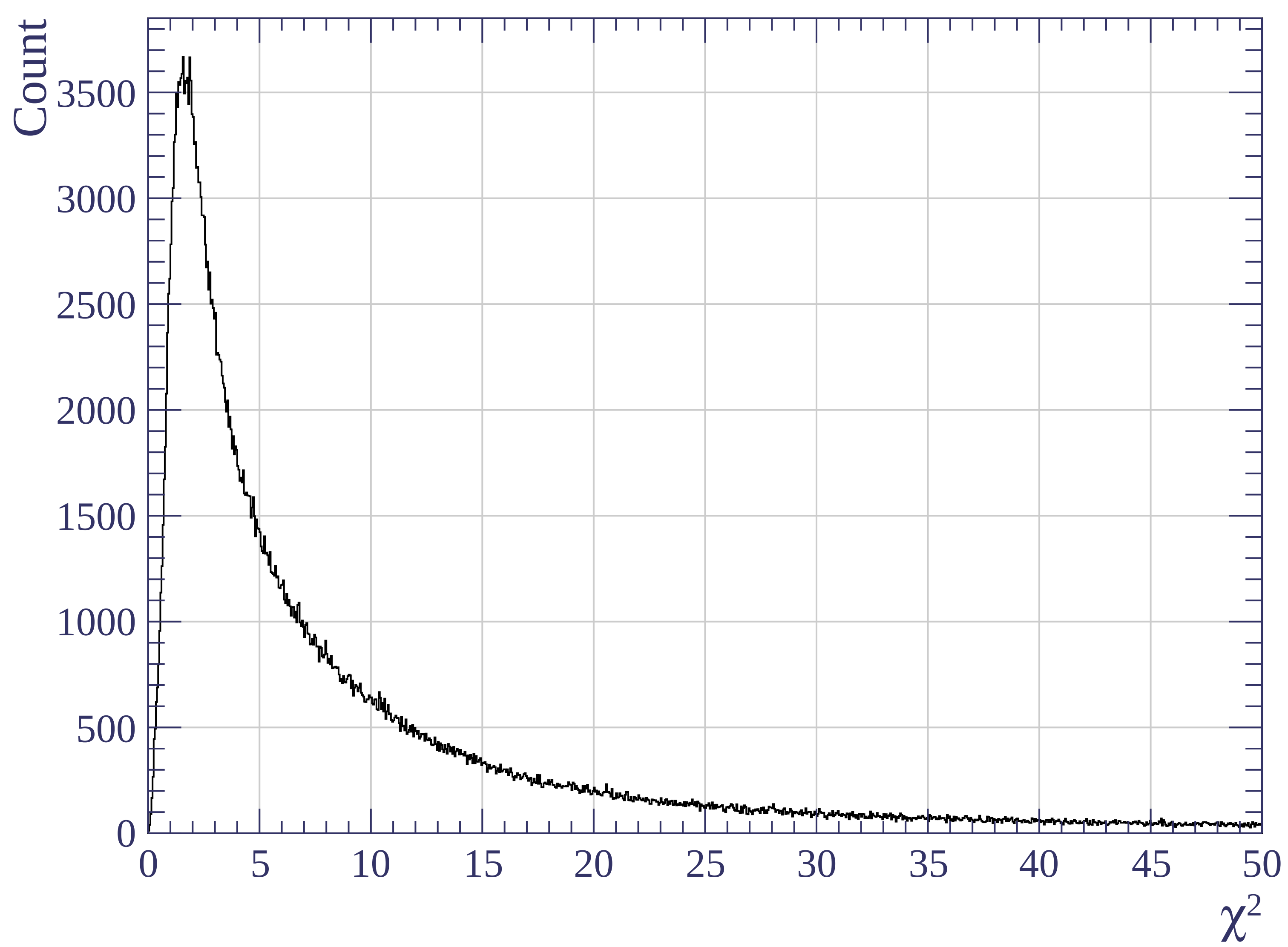
Pull e
Pull μ
×10³
20
15
10
5
0
−5
−10
−15
−20
−20
−15
−10
−5
0
5
10
15
20
10
8
6
4
2
0

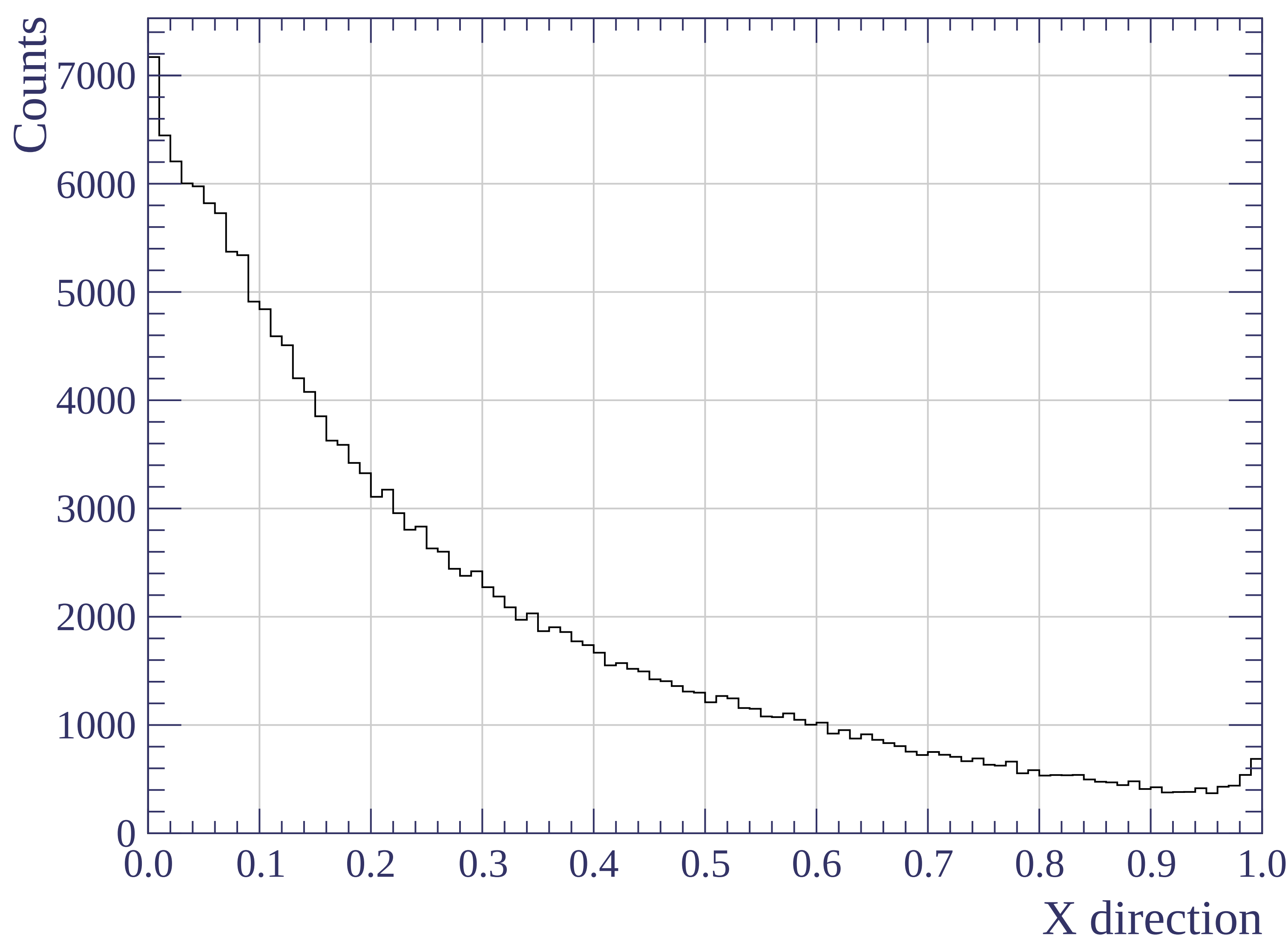


$\times 10^3$
Count
60
50
40
30
20
10
0
−20
−15
−10
−5
0
5
10
15
20
Pull
μ hyp.
e hyp.
p hyp.
S(μ/e) = 2.64 ± 0.01
μ_μ = 0.083 ± 0.002
σ_μ = 0.782 ± 0.002
μ_e = -2.632 ± 0.003
σ_e = 1.224 ± 0.003

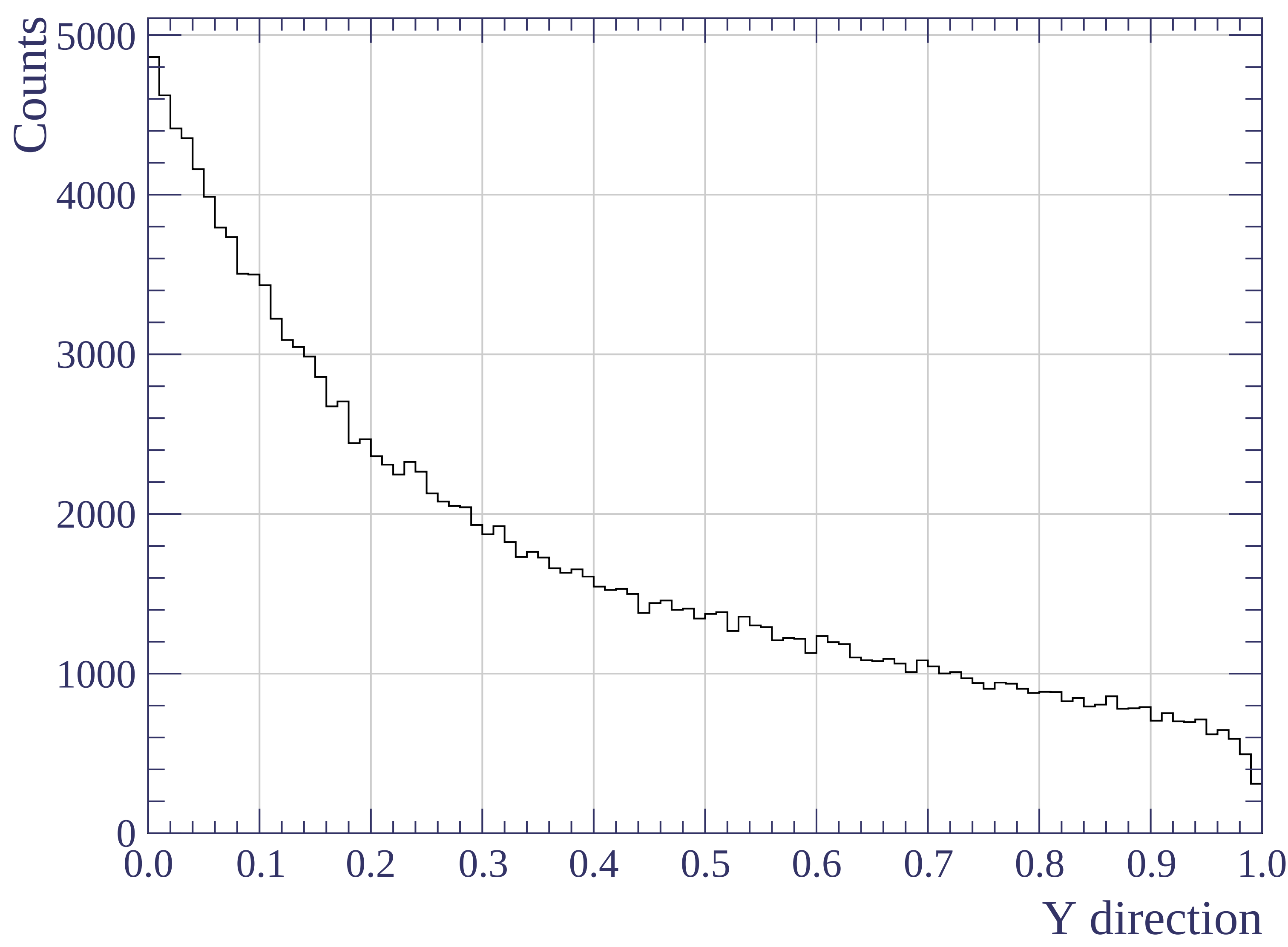


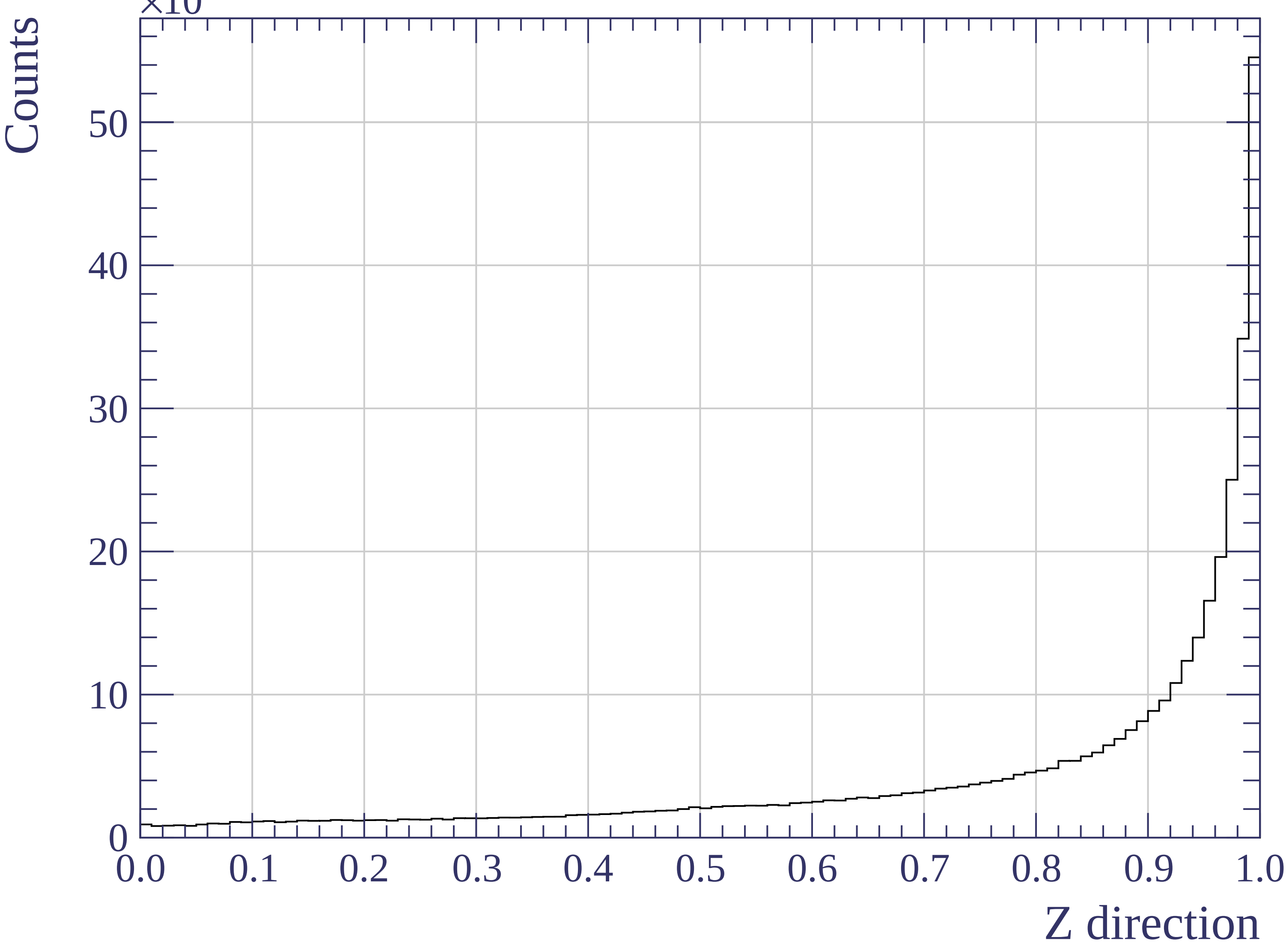
Count
0
500
1000
1500
2000
2500
3000
3500
4000
0
20
40
60
80
100
120
140
160
track length (cm)


Count
3500
3000
2500
2000
1500
1000
500
0
0
5
10
15
20
25
30
35
40
45
50
χ^2

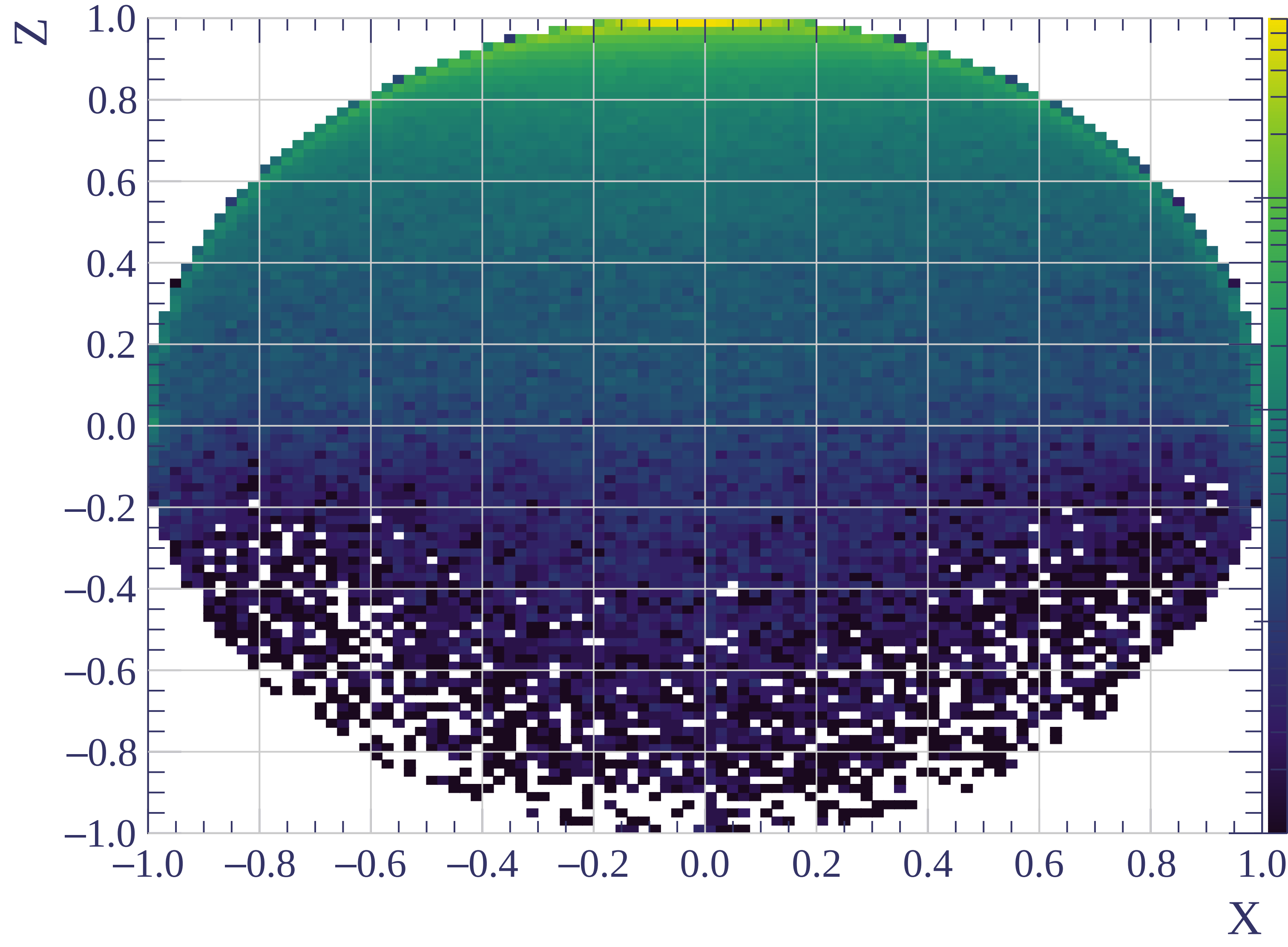


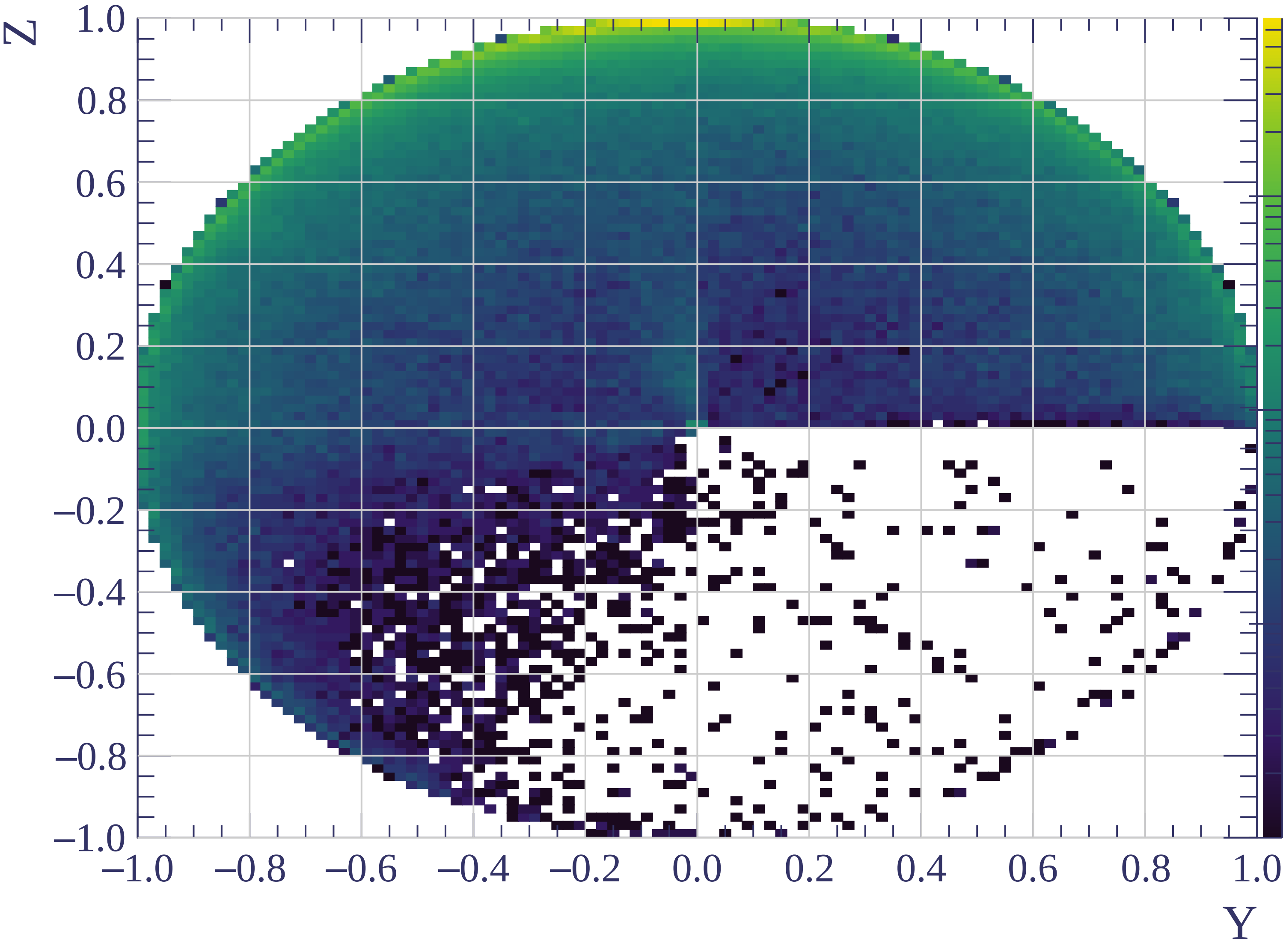
Counts
7000
6000
5000
4000
3000
2000
1000
0
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1.0
X direction

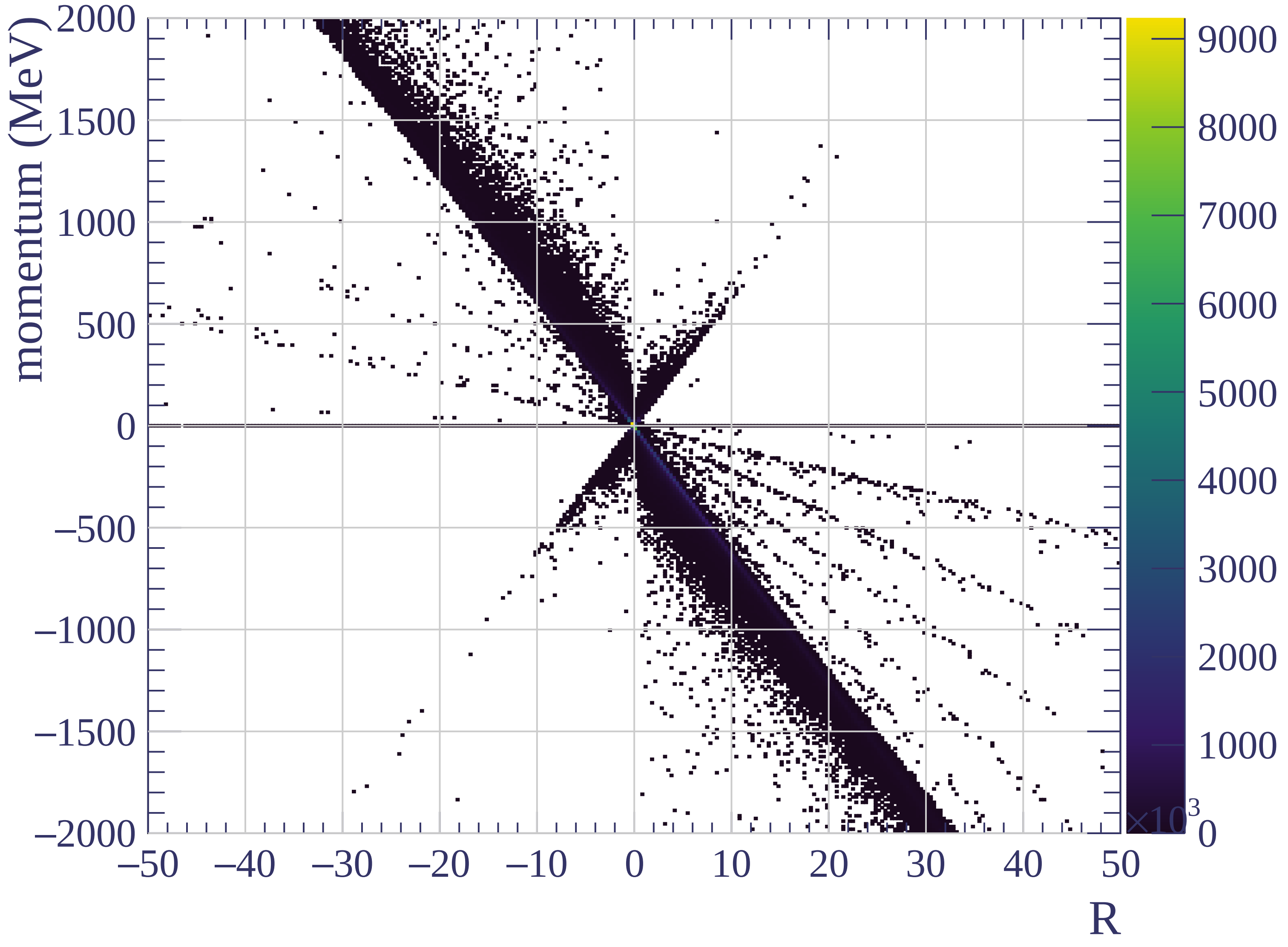

Counts
5000
4000
3000
2000
1000
0
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1.0
Y direction



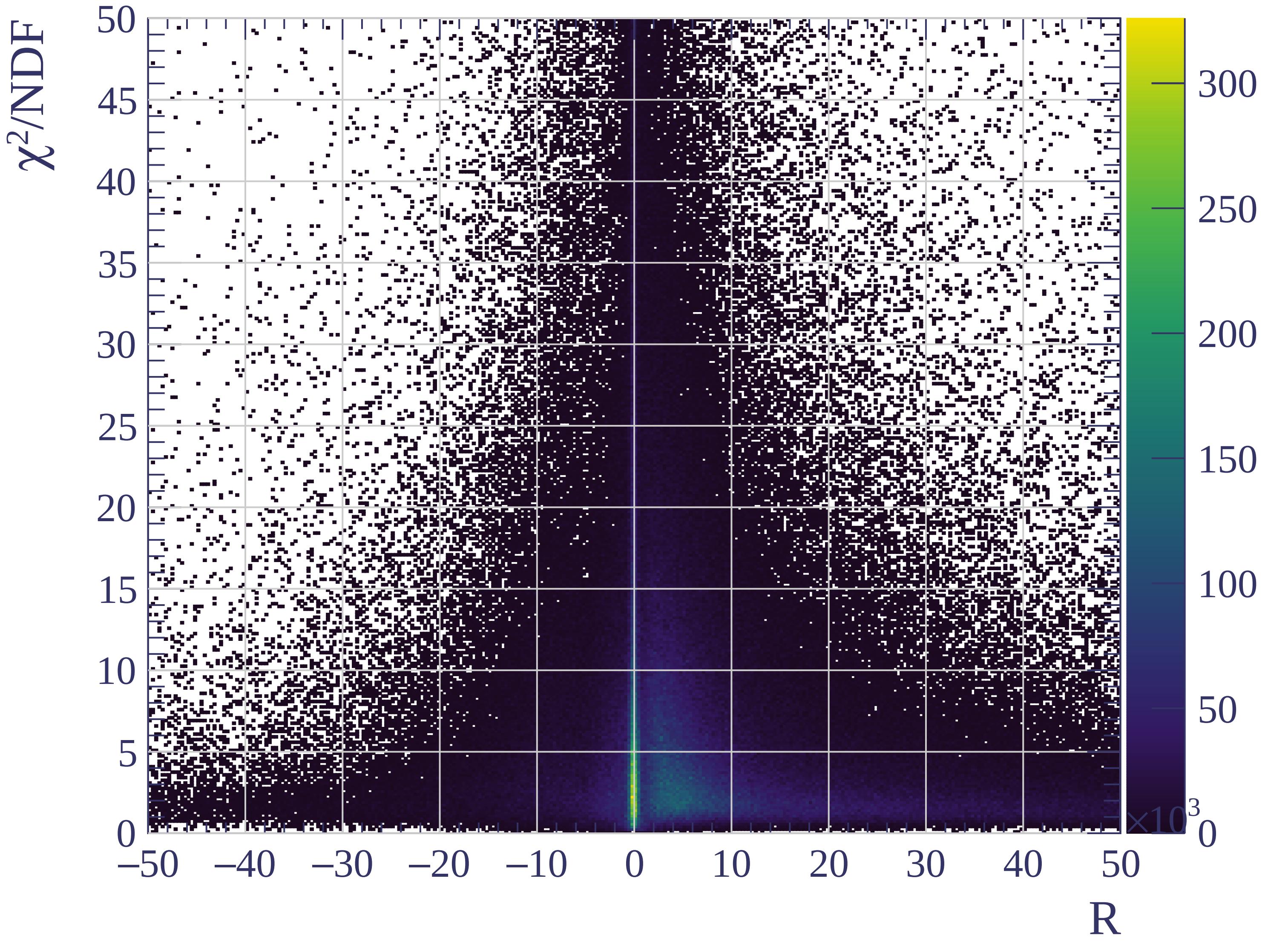
Counts
×10
50
40
30
20
10
0
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1.0
Z direction


Z
1.0
0.8
0.6
0.4
0.2
0.0
−0.2
−0.4
−0.6
−0.8
−1.0
−1.0
−0.8
−0.6
−0.4
−0.2
0.0
0.2
0.4
0.6
0.8
1.0
X

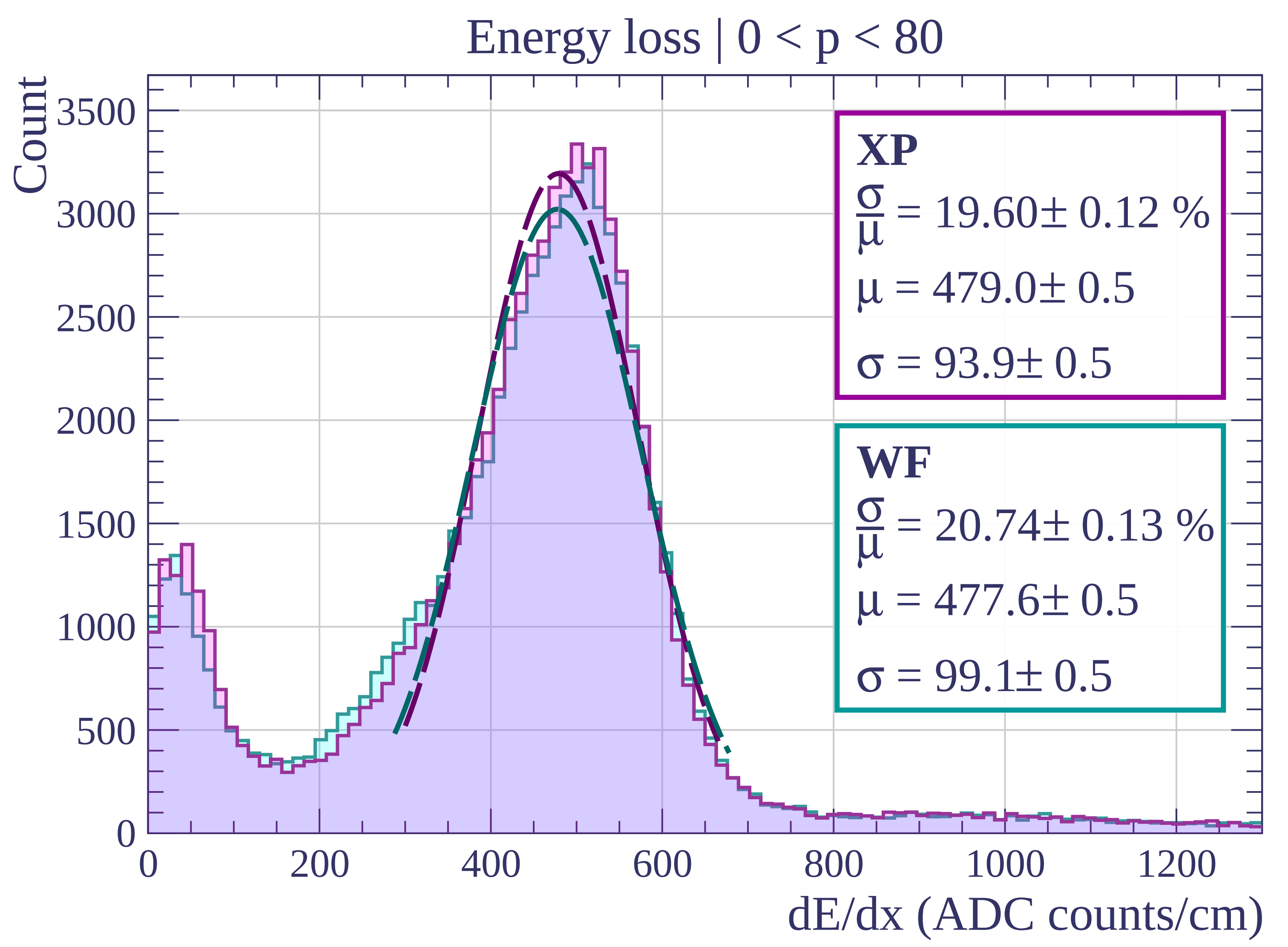

Z
1.0
0.8
0.6
0.4
0.2
0.0
−0.2
−0.4
−0.6
−0.8
−1.0
−1.0
−0.8
−0.6
−0.4
−0.2
0.0
0.2
0.4
0.6
0.8
1.0
Y



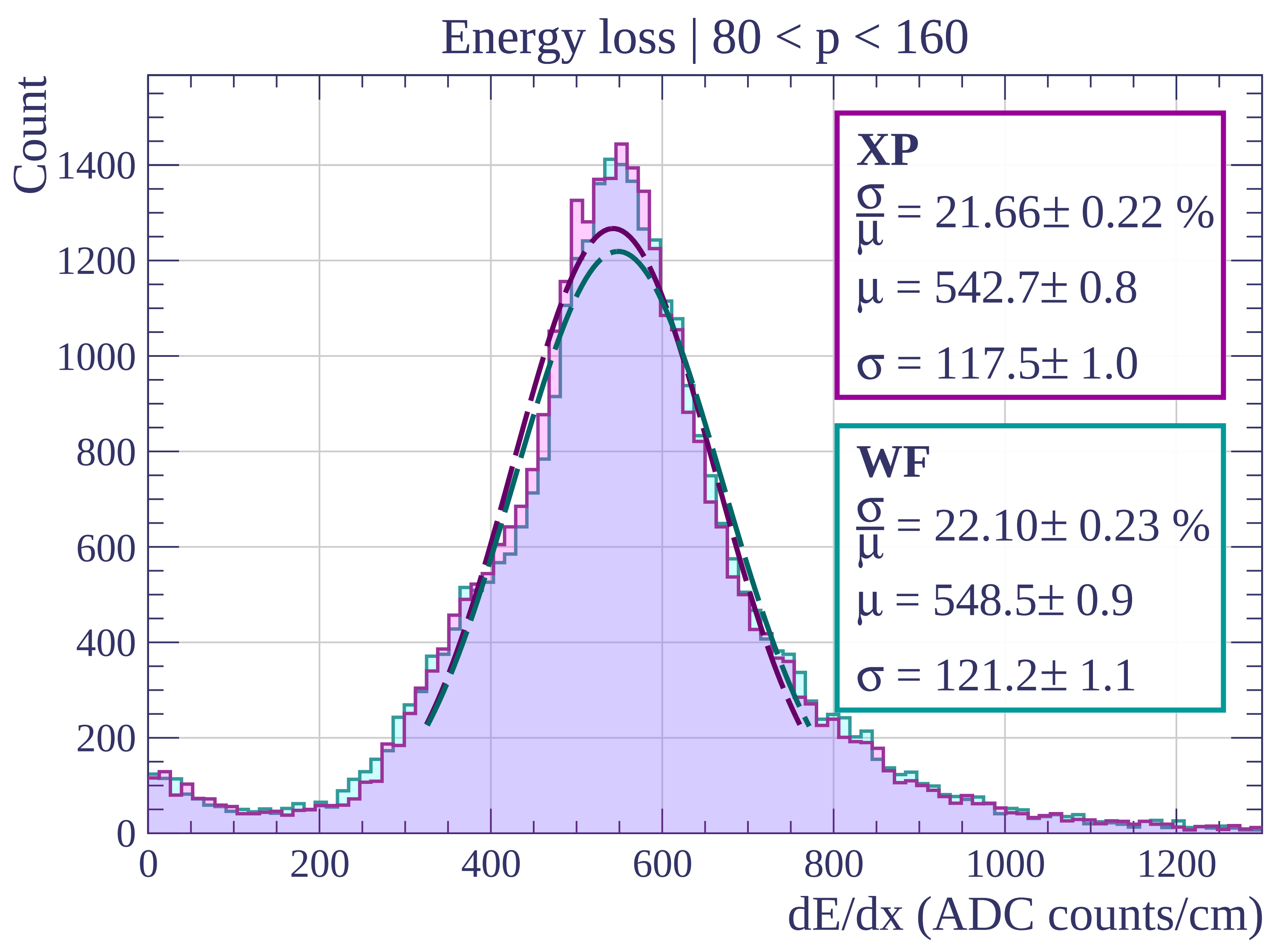
momentum (MeV)
2000
1500
1000
500
0
−500
−1000
−1500
−2000
−50
−40
−30
−20
−10
0
10
20
30
40
50
R
9000
8000
7000
6000
5000
4000
3000
2000
1000
0
×10³

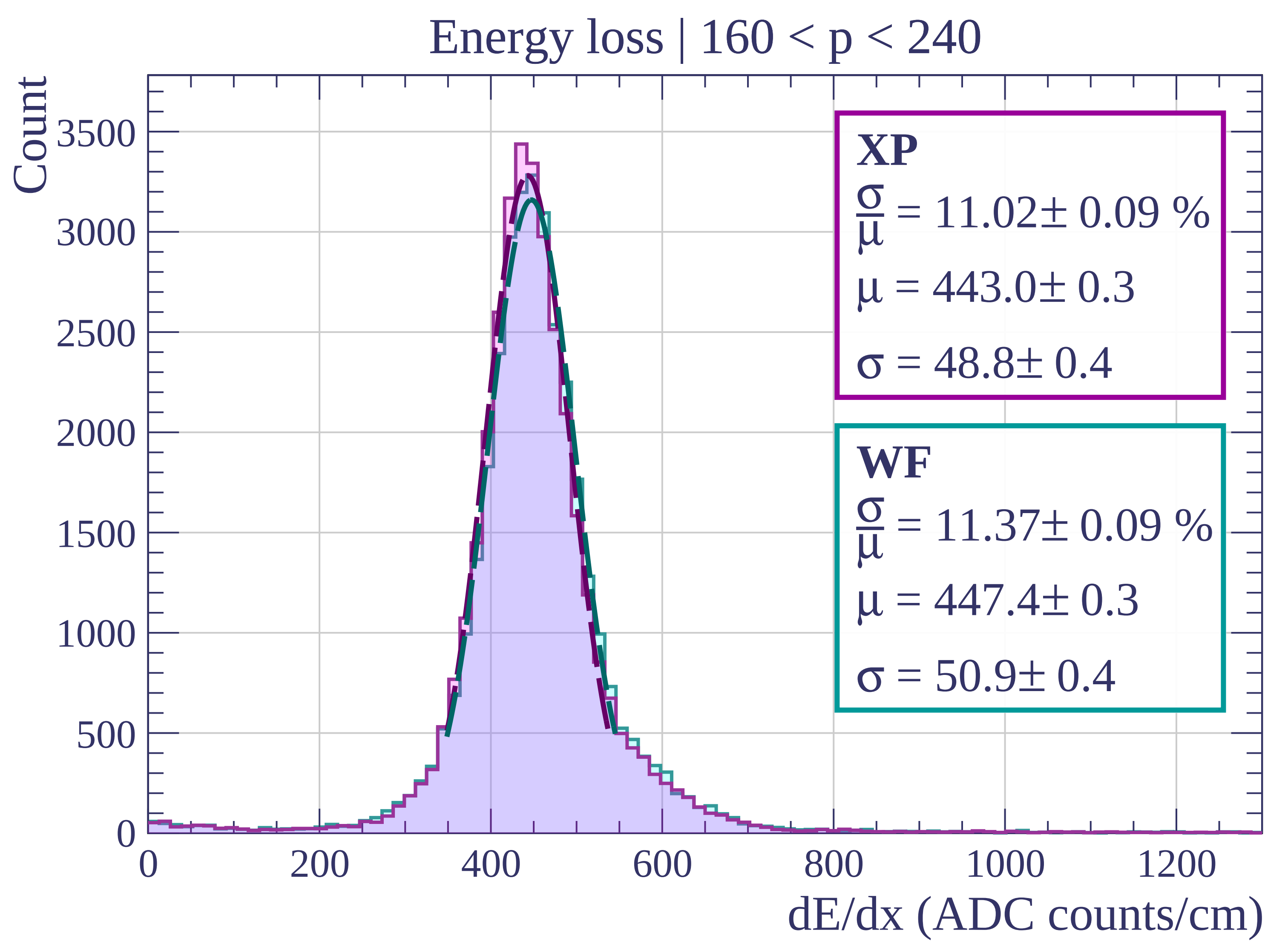


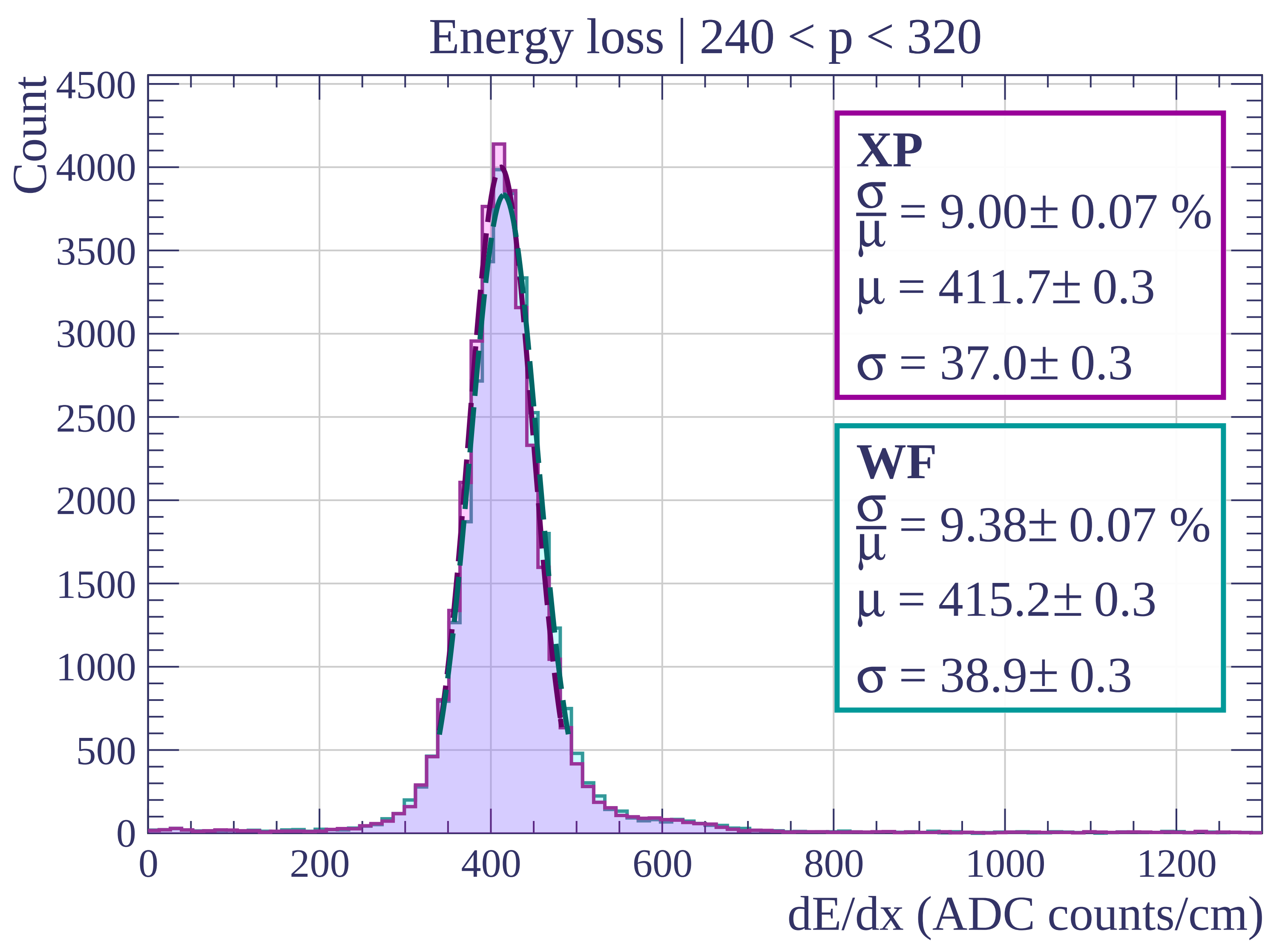
χ²/NDF
50
45
40
35
30
25
20
15
10
5
0
−50
−40
−30
−20
−10
0
10
20
30
40
50
R
300
250
200
150
100
50
0
×10³



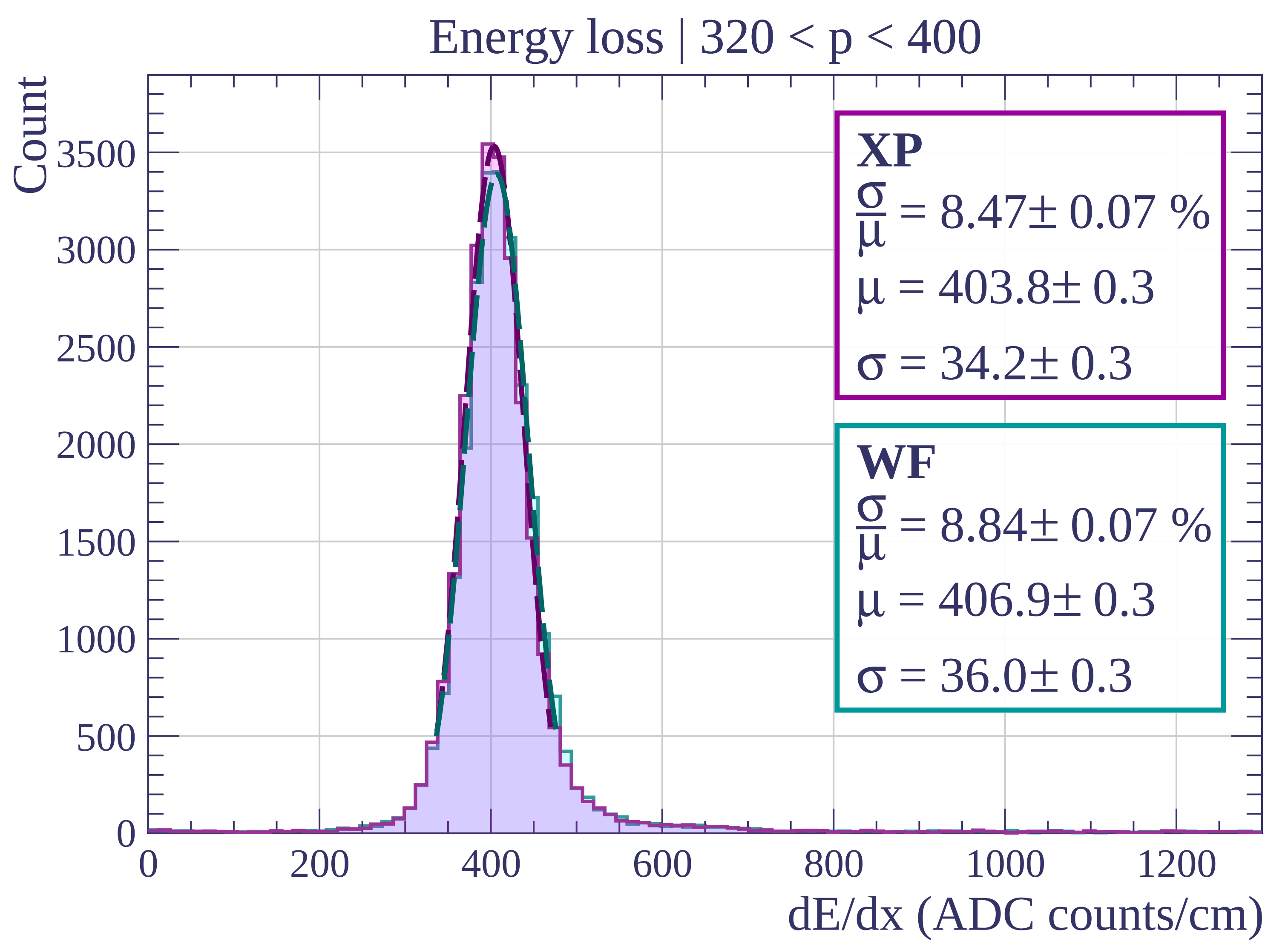
Energy loss | 0 < p < 80
Count
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu} = 19.60 \pm 0.12$ %
$\mu = 479.0 \pm 0.5$
$\sigma = 93.9 \pm 0.5$
WF
$\frac{\sigma}{\mu} = 20.74 \pm 0.13$ %
$\mu = 477.6 \pm 0.5$
$\sigma = 99.1 \pm 0.5$
0
500
1000
1500
2000
2500
3000
3500
0
200
400
600
800
1000
1200

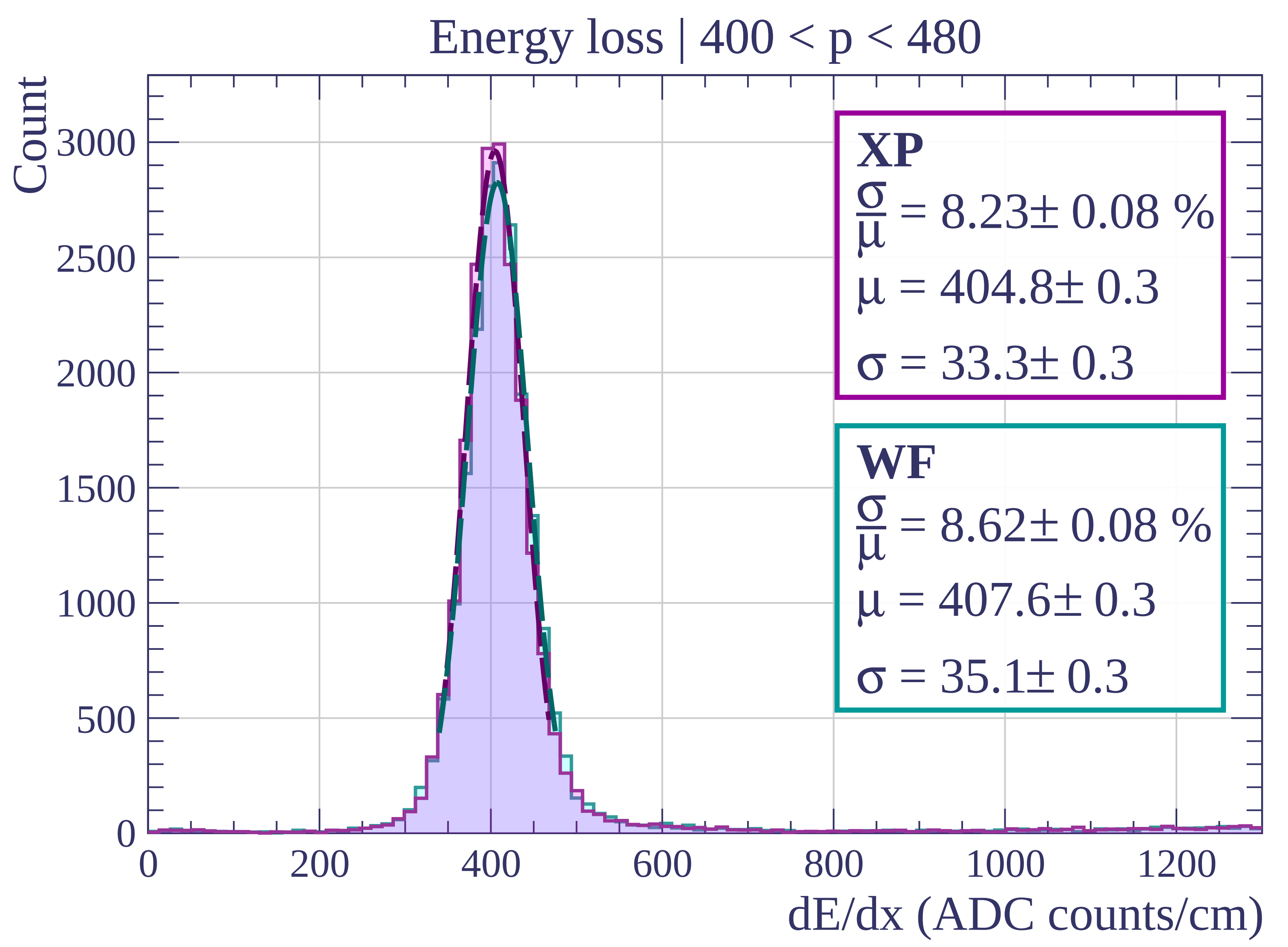

Energy loss | 80 < p < 160
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
1400
XP
$\frac{\sigma}{\mu}$ = 21.66± 0.22 %
μ = 542.7± 0.8
σ = 117.5± 1.0
WF
$\frac{\sigma}{\mu}$ = 22.10± 0.23 %
μ = 548.5± 0.9
σ = 121.2± 1.1


Energy loss | 160 < p < 240
Count
dE/dx (ADC counts/cm)
0
500
1000
1500
2000
2500
3000
3500
0
200
400
600
800
1000
1200
XP
σ/μ = 11.02± 0.09 %
μ = 443.0± 0.3
σ = 48.8± 0.4
WF
σ/μ = 11.37± 0.09 %
μ = 447.4± 0.3
σ = 50.9± 0.4

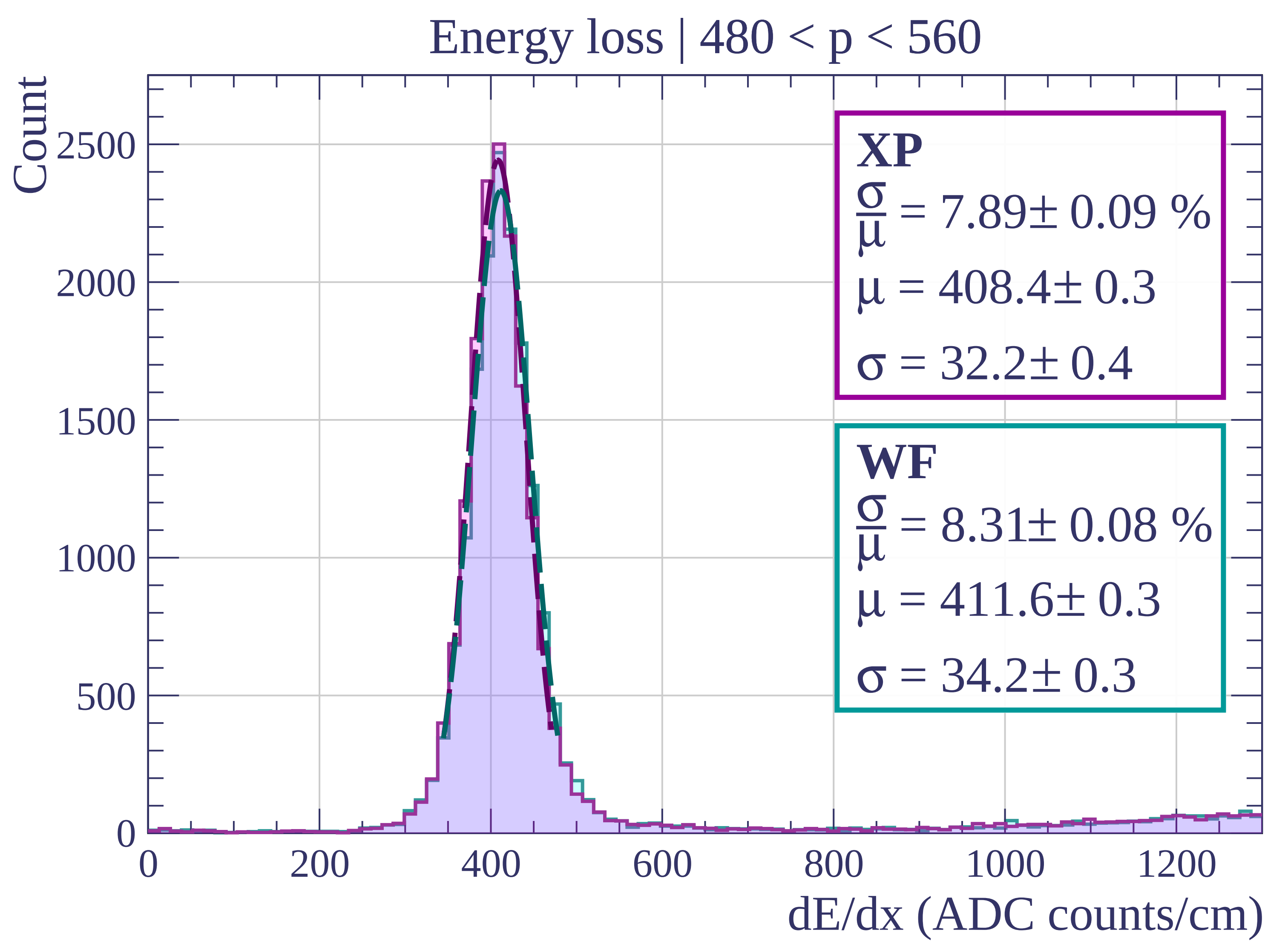


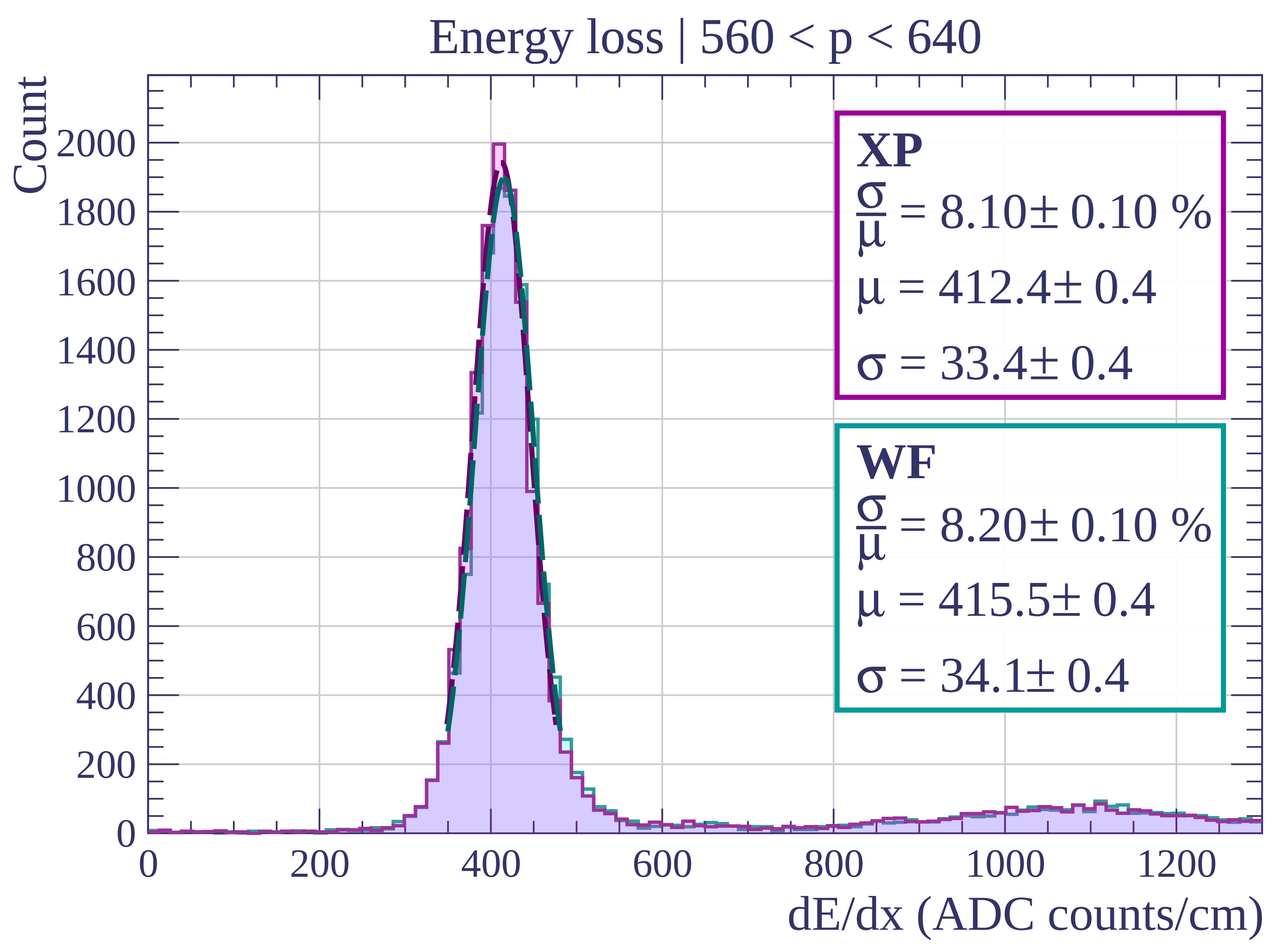
Energy loss | 240 < p < 320
Count
4500
4000
3500
3000
2500
2000
1500
1000
500
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 9.00± 0.07 %
μ = 411.7± 0.3
σ = 37.0± 0.3
WF
σ/μ = 9.38± 0.07 %
μ = 415.2± 0.3
σ = 38.9± 0.3

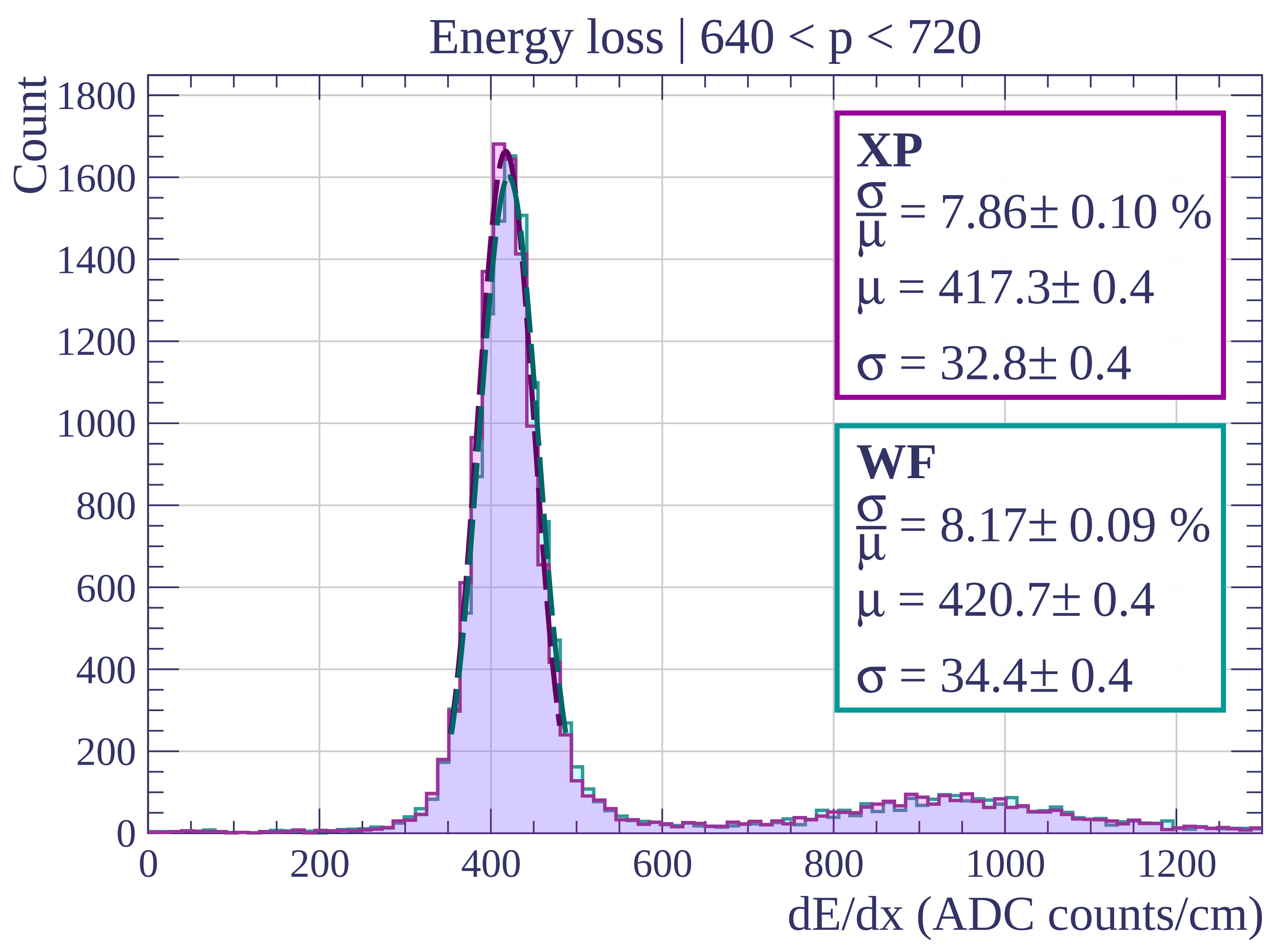

Energy loss | 320 < p < 400
Count
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 8.47± 0.07 %
μ = 403.8± 0.3
σ = 34.2± 0.3
WF
$\frac{\sigma}{\mu}$ = 8.84± 0.07 %
μ = 406.9± 0.3
σ = 36.0± 0.3

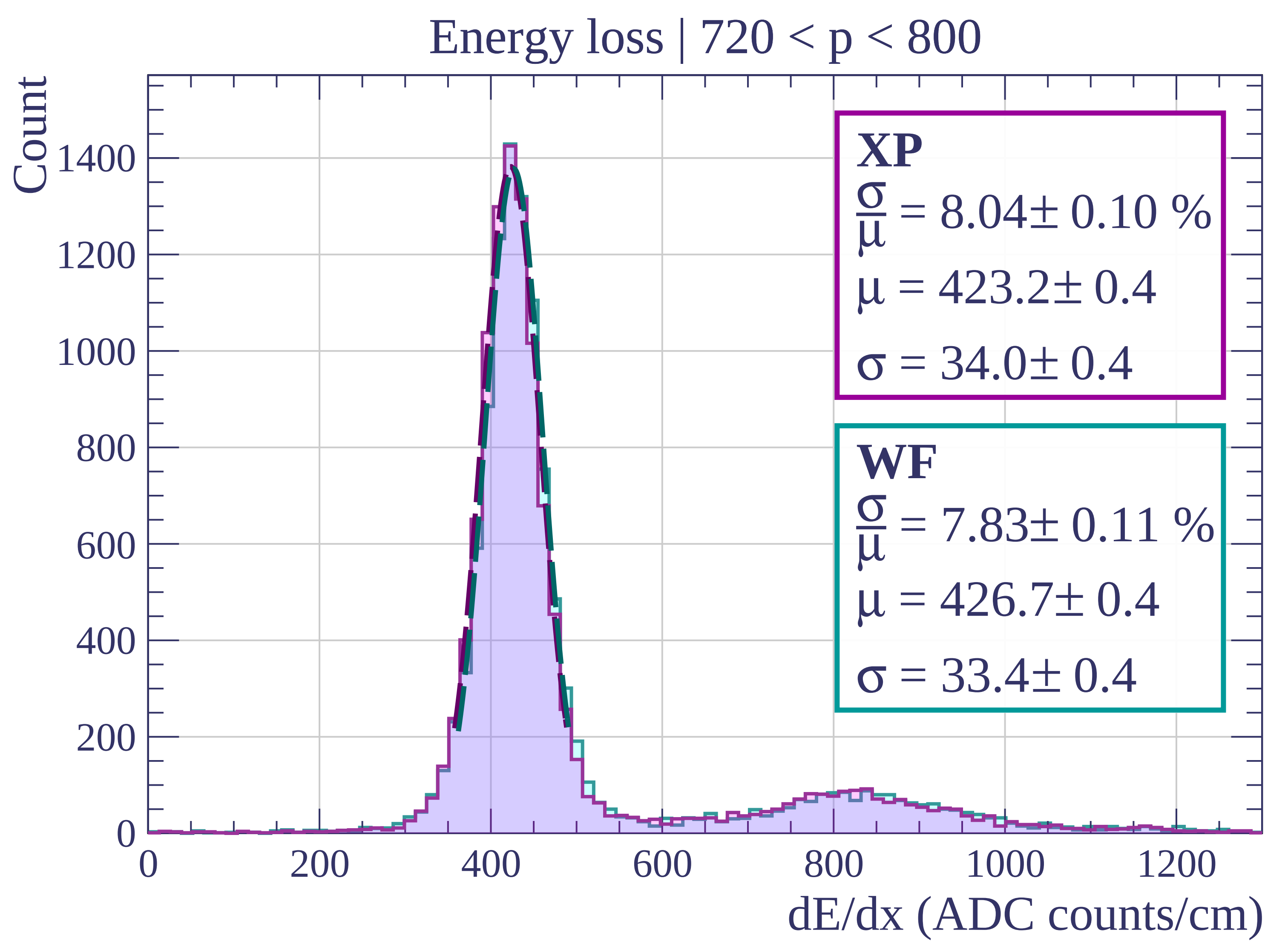


Energy loss | 400 < p < 480
Count
dE/dx (ADC counts/cm)
0
500
1000
1500
2000
2500
3000
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 8.23± 0.08 %
μ = 404.8± 0.3
σ = 33.3± 0.3
WF
$\frac{\sigma}{\mu}$ = 8.62± 0.08 %
μ = 407.6± 0.3
σ = 35.1± 0.3

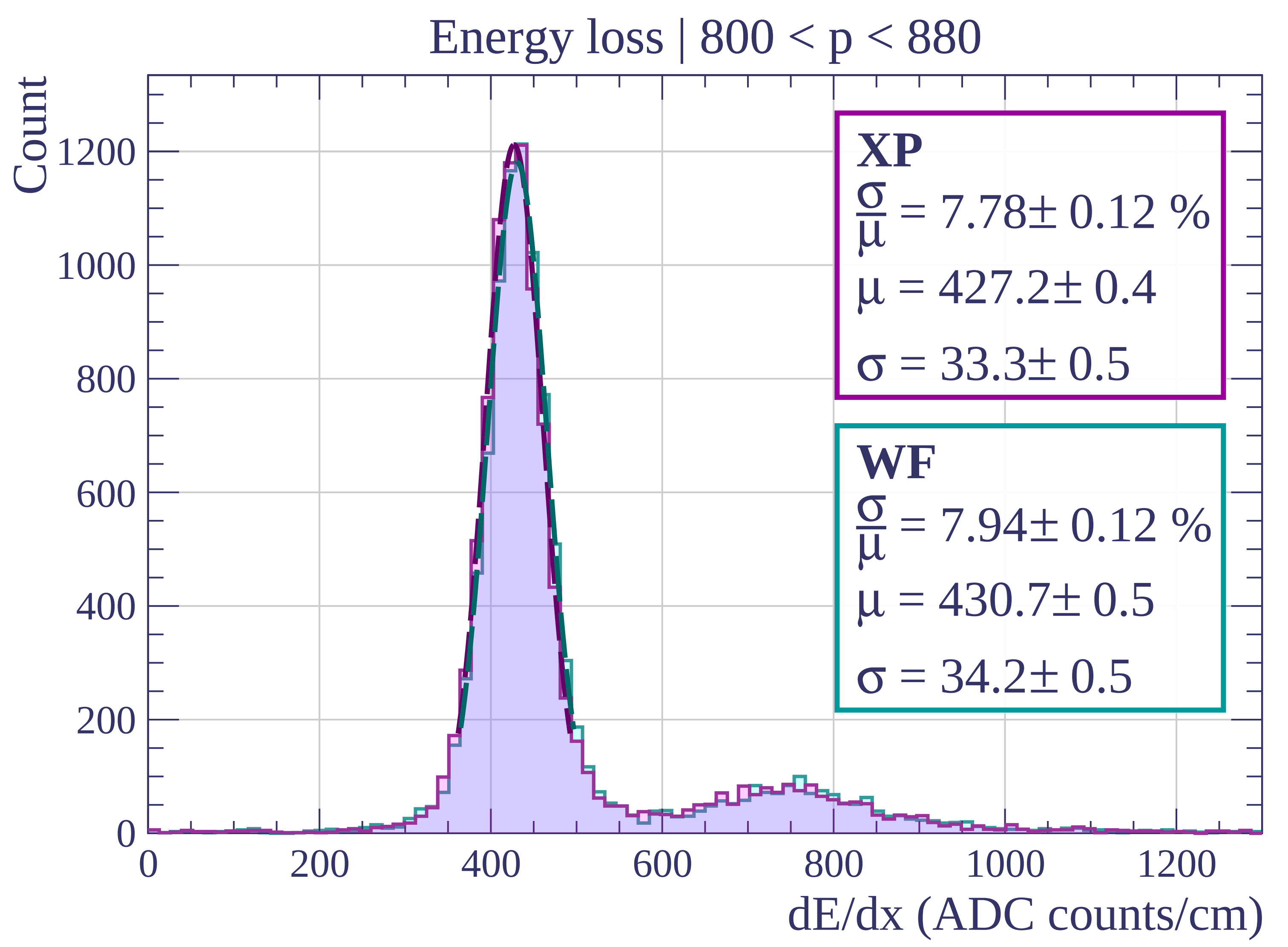

Energy loss | 480 < p < 560
Count
dE/dx (ADC counts/cm)
0
500
1000
1500
2000
2500
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 7.89± 0.09 %
μ = 408.4± 0.3
σ = 32.2± 0.4
WF
$\frac{\sigma}{\mu}$ = 8.31± 0.08 %
μ = 411.6± 0.3
σ = 34.2± 0.3


Energy loss | 560 < p < 640
Count
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 8.10± 0.10 %
μ = 412.4± 0.4
σ = 33.4± 0.4
WF
$\frac{\sigma}{\mu}$ = 8.20± 0.10 %
μ = 415.5± 0.4
σ = 34.1± 0.4
0
200
400
600
800
1000
1200
1400
1600
1800
2000

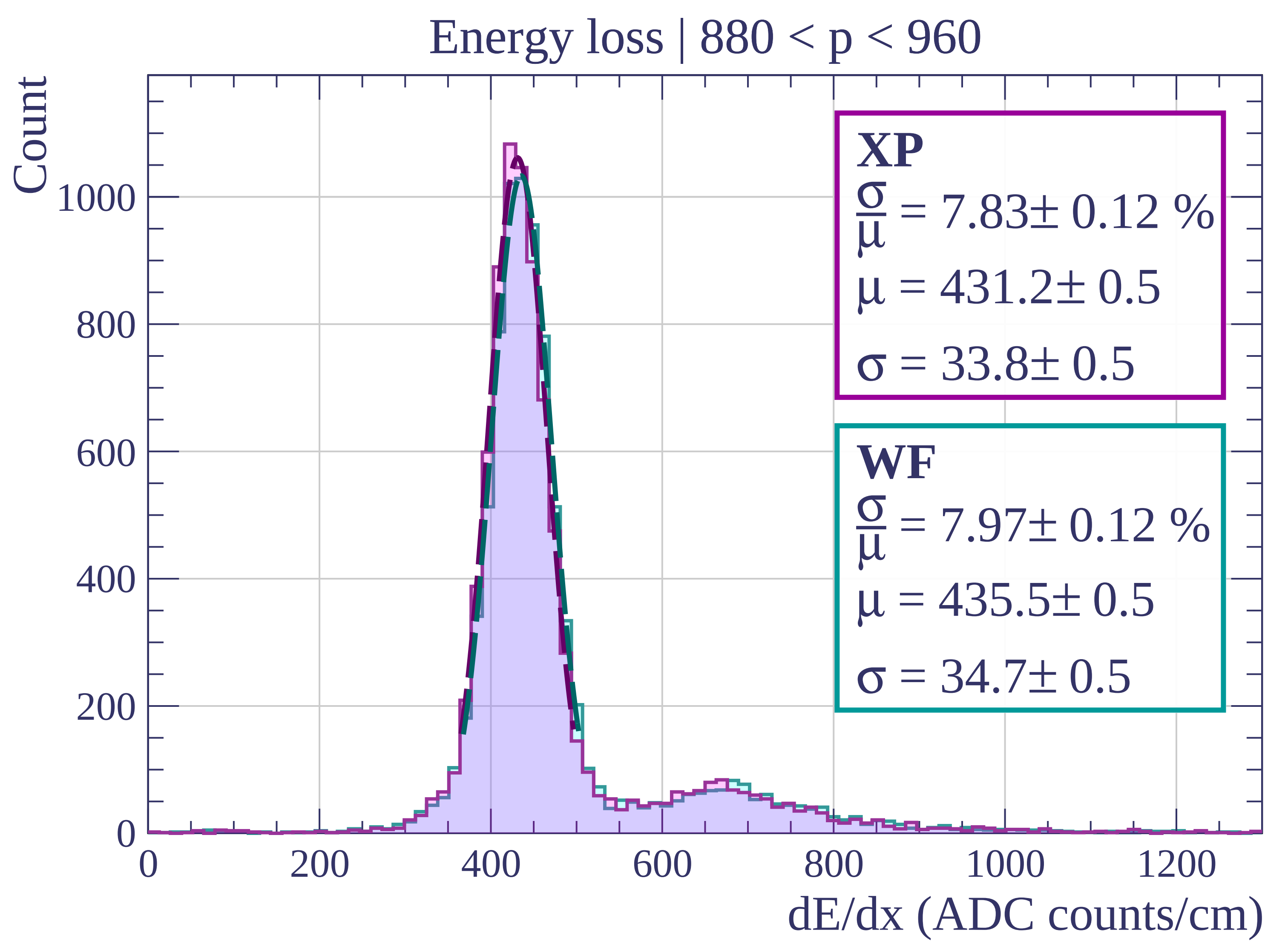

Energy loss | 640 < p < 720
Count
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 7.86± 0.10 %
μ = 417.3± 0.4
σ = 32.8± 0.4
WF
$\frac{\sigma}{\mu}$ = 8.17± 0.09 %
μ = 420.7± 0.4
σ = 34.4± 0.4
0
200
400
600
800
1000
1200
1400
1600
1800

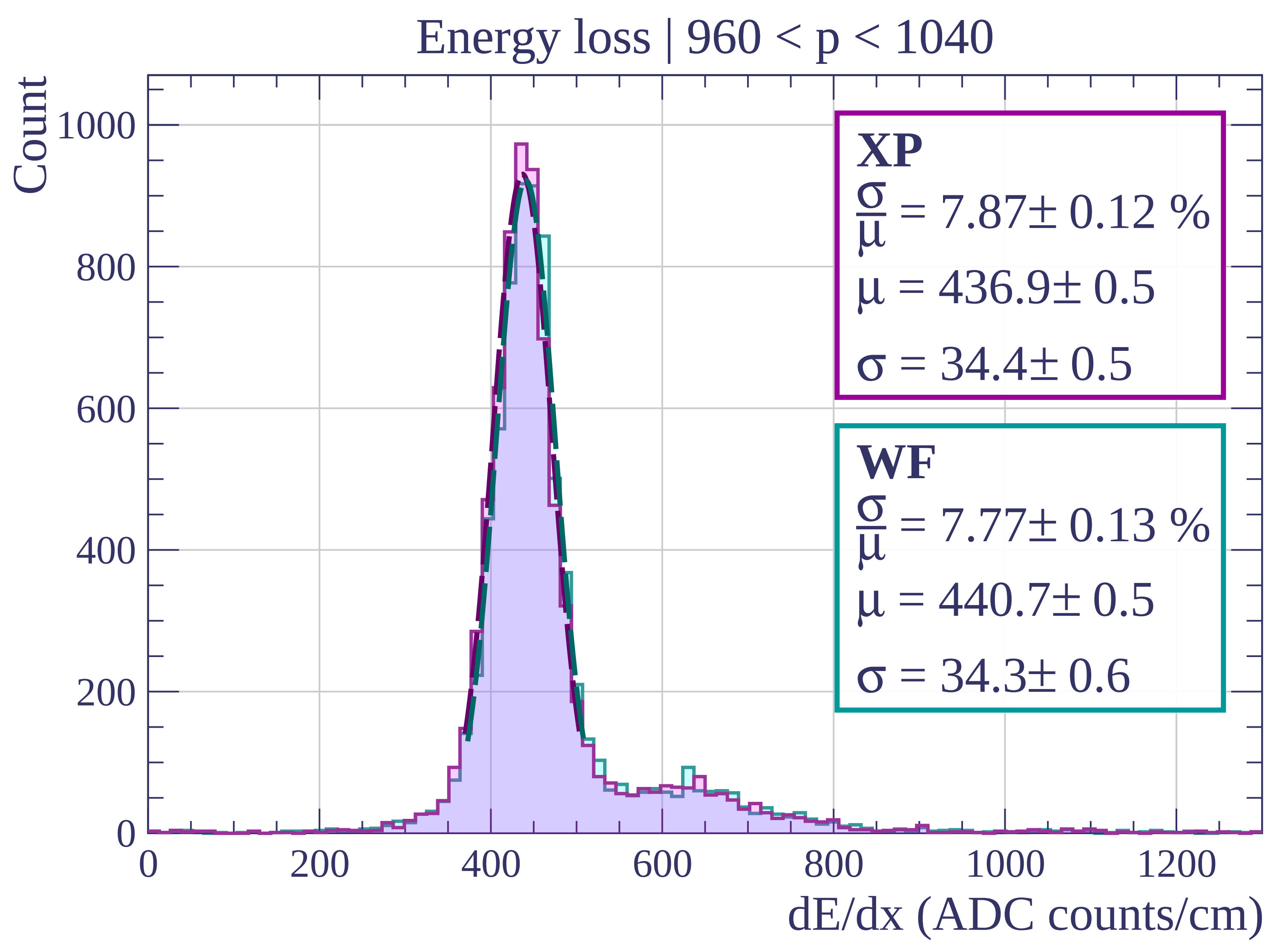


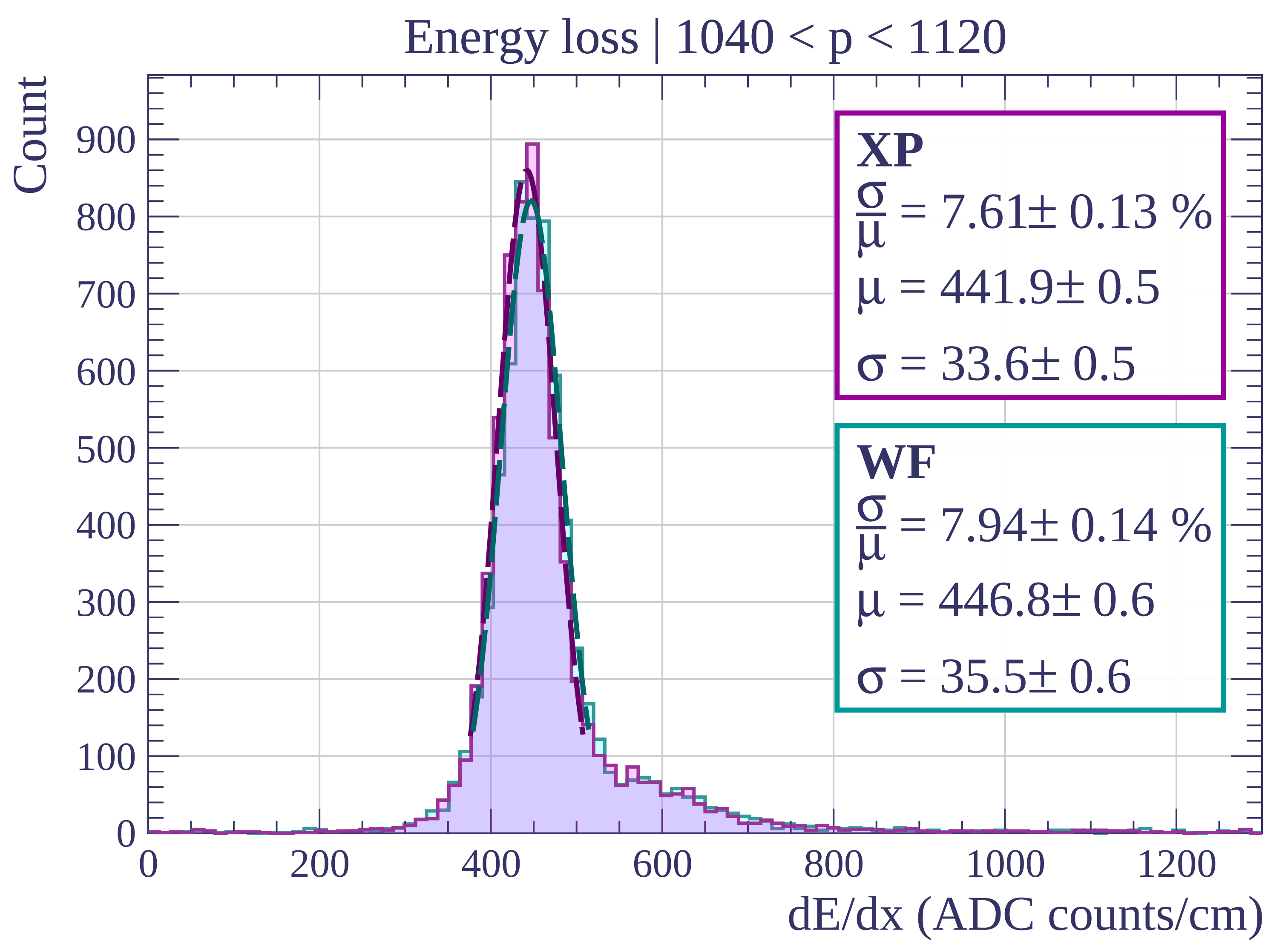
Energy loss | 720 < p < 800
Count
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 8.04± 0.10 %
μ = 423.2± 0.4
σ = 34.0± 0.4
WF
$\frac{\sigma}{\mu}$ = 7.83± 0.11 %
μ = 426.7± 0.4
σ = 33.4± 0.4
0
200
400
600
800
1000
1200
1400



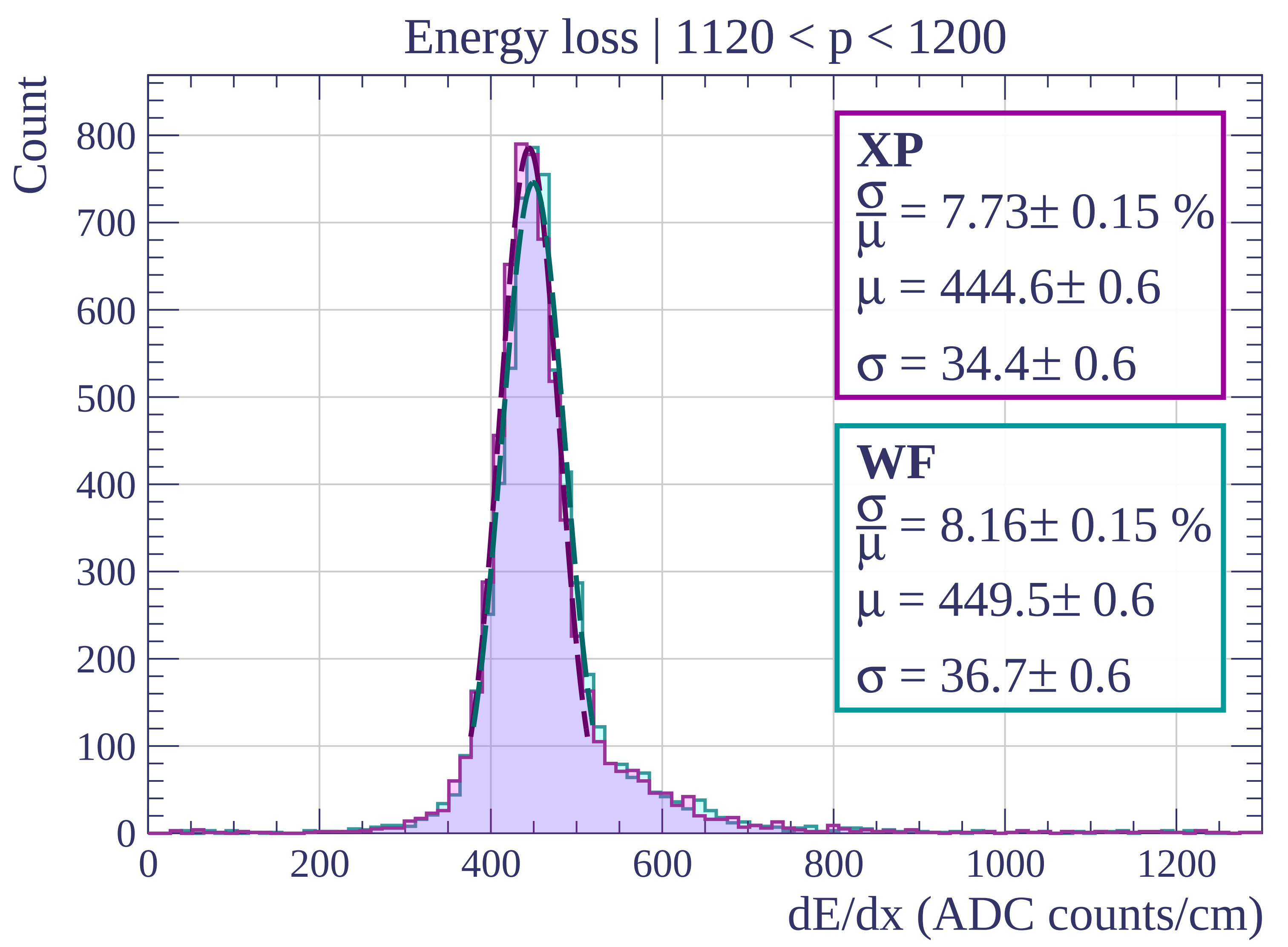
Energy loss | 800 < p < 880
Count
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 7.78± 0.12 %
μ = 427.2± 0.4
σ = 33.3± 0.5
WF
$\frac{\sigma}{\mu}$ = 7.94± 0.12 %
μ = 430.7± 0.5
σ = 34.2± 0.5
0
200
400
600
800
1000
1200


Energy loss | 880 < p < 960
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
XP
σ/μ = 7.83± 0.12 %
μ = 431.2± 0.5
σ = 33.8± 0.5
WF
σ/μ = 7.97± 0.12 %
μ = 435.5± 0.5
σ = 34.7± 0.5

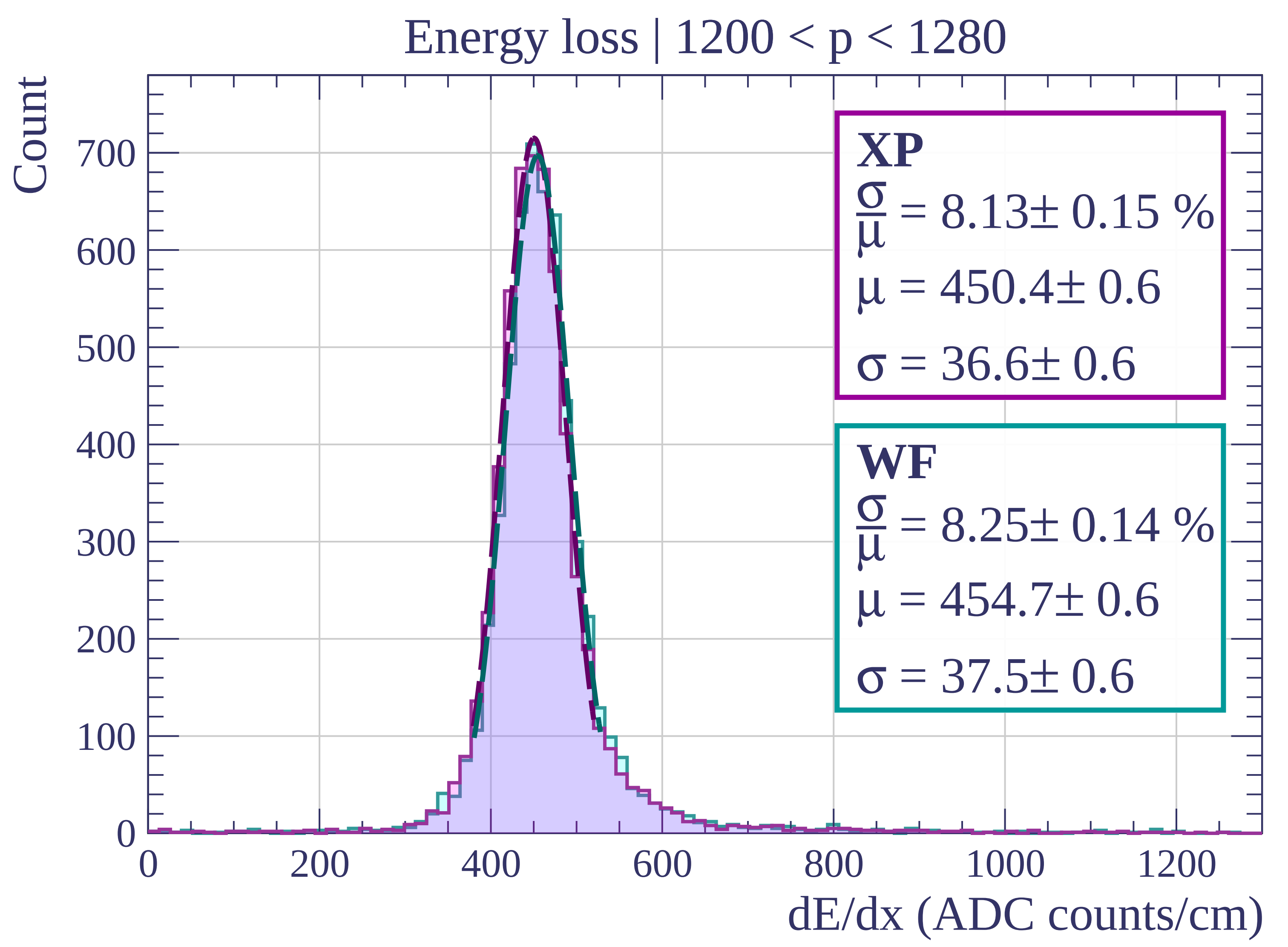

Energy loss | 960 < p < 1040
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu} = 7.87 \pm 0.12$ %
$\mu = 436.9 \pm 0.5$
$\sigma = 34.4 \pm 0.5$
WF
$\frac{\sigma}{\mu} = 7.77 \pm 0.13$ %
$\mu = 440.7 \pm 0.5$
$\sigma = 34.3 \pm 0.6$

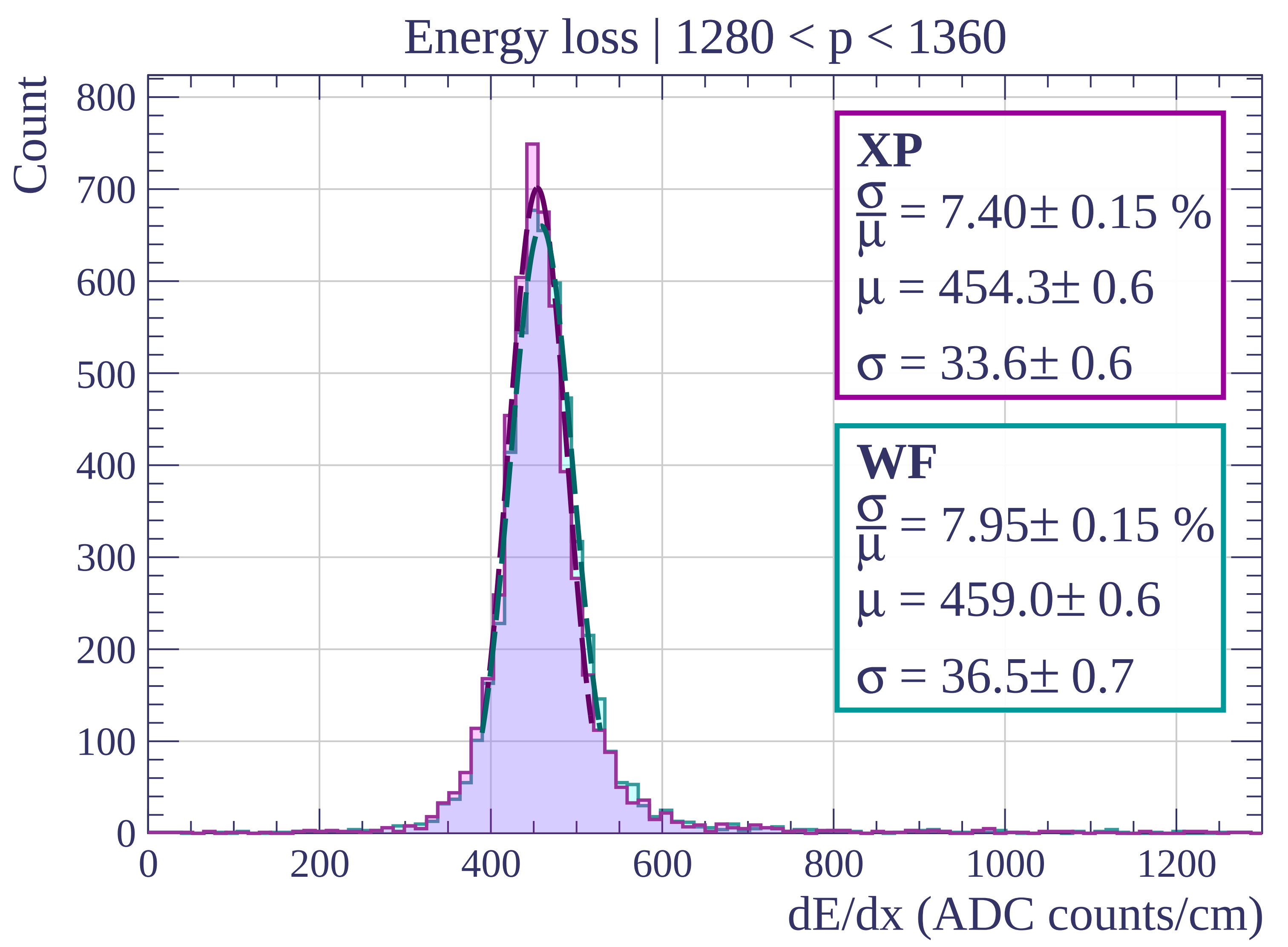


Energy loss | 1040 < p < 1120
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
700
800
900
0
200
400
600
800
1000
1200
XP
σ/μ = 7.61± 0.13 %
μ = 441.9± 0.5
σ = 33.6± 0.5
WF
σ/μ = 7.94± 0.14 %
μ = 446.8± 0.6
σ = 35.5± 0.6

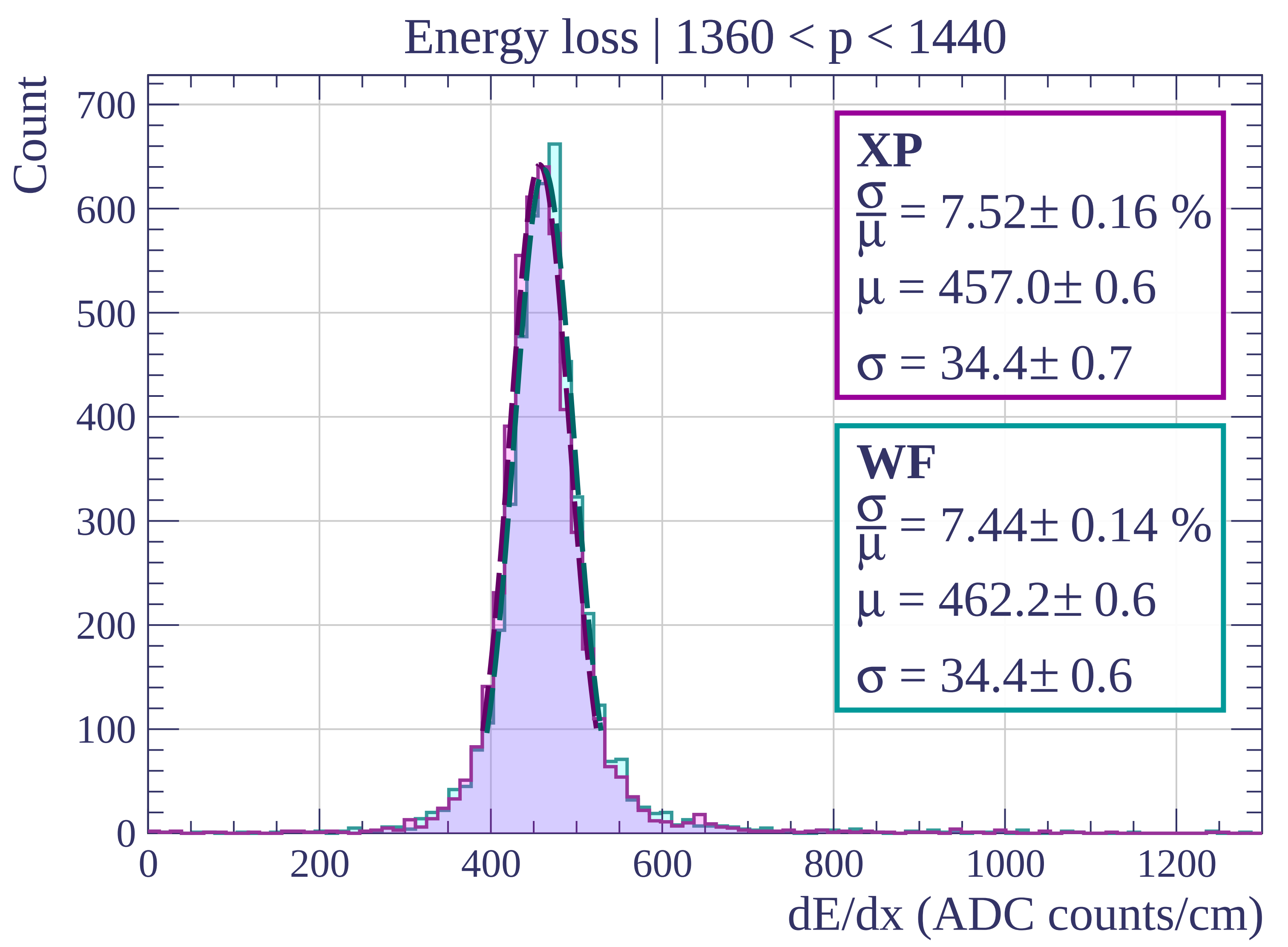


Energy loss | 1120 < p < 1200
Count
800
700
600
500
400
300
200
100
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 7.73± 0.15 %
μ = 444.6± 0.6
σ = 34.4± 0.6
WF
$\frac{\sigma}{\mu}$ = 8.16± 0.15 %
μ = 449.5± 0.6
σ = 36.7± 0.6

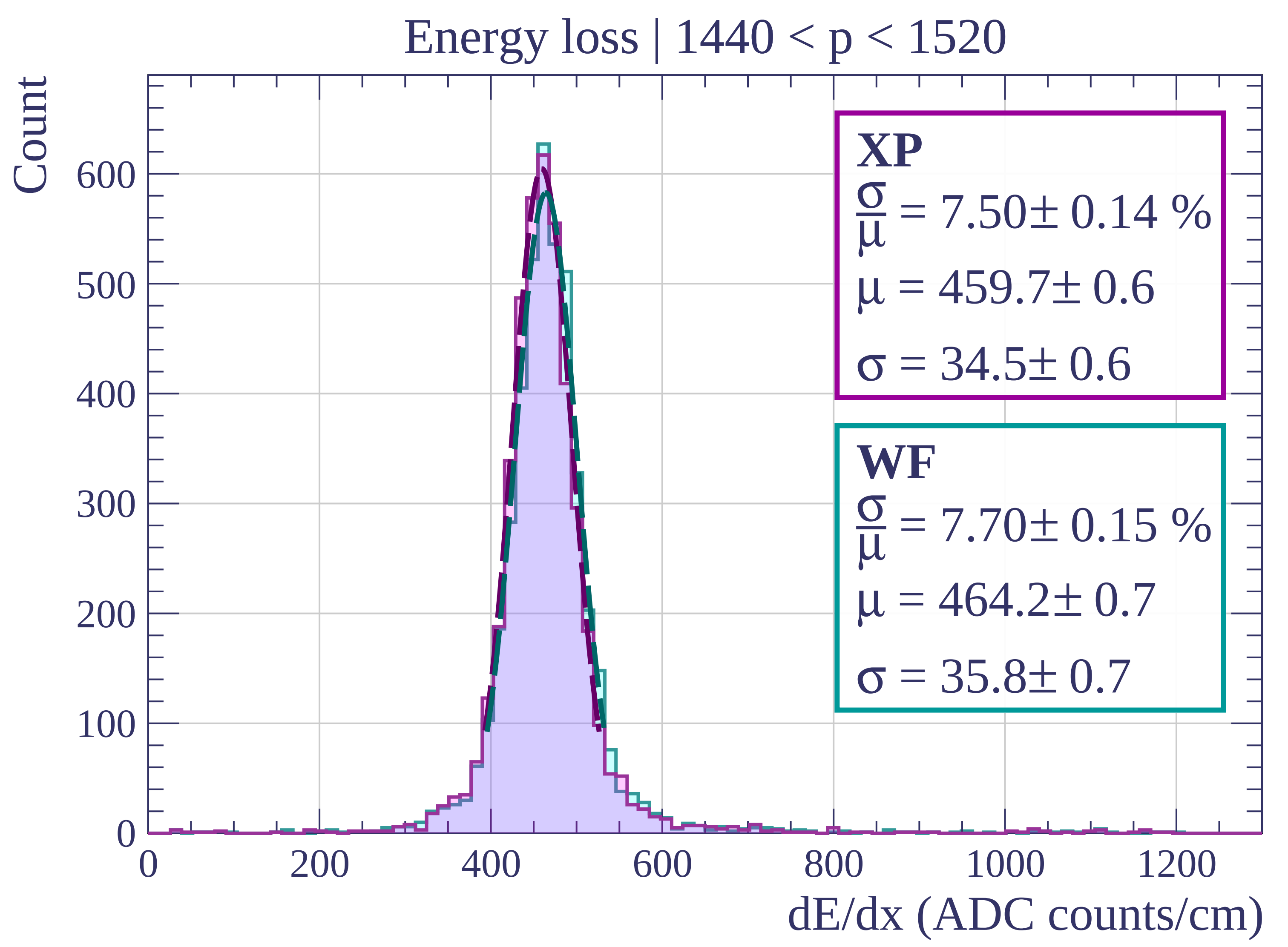

Energy loss | 1200 < p < 1280
Count
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu} = 8.13 \pm 0.15$ %
$\mu = 450.4 \pm 0.6$
$\sigma = 36.6 \pm 0.6$
WF
$\frac{\sigma}{\mu} = 8.25 \pm 0.14$ %
$\mu = 454.7 \pm 0.6$
$\sigma = 37.5 \pm 0.6$

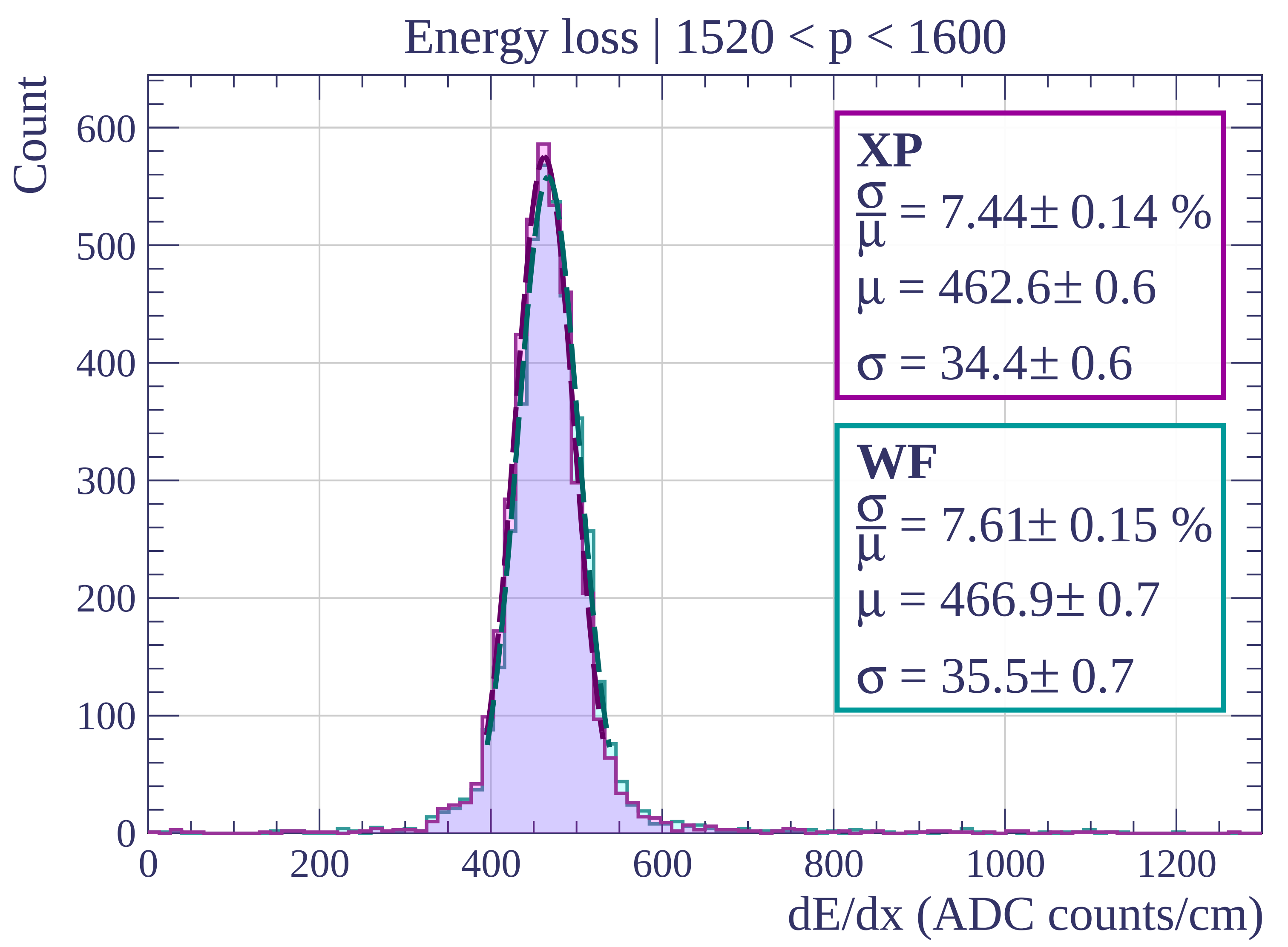


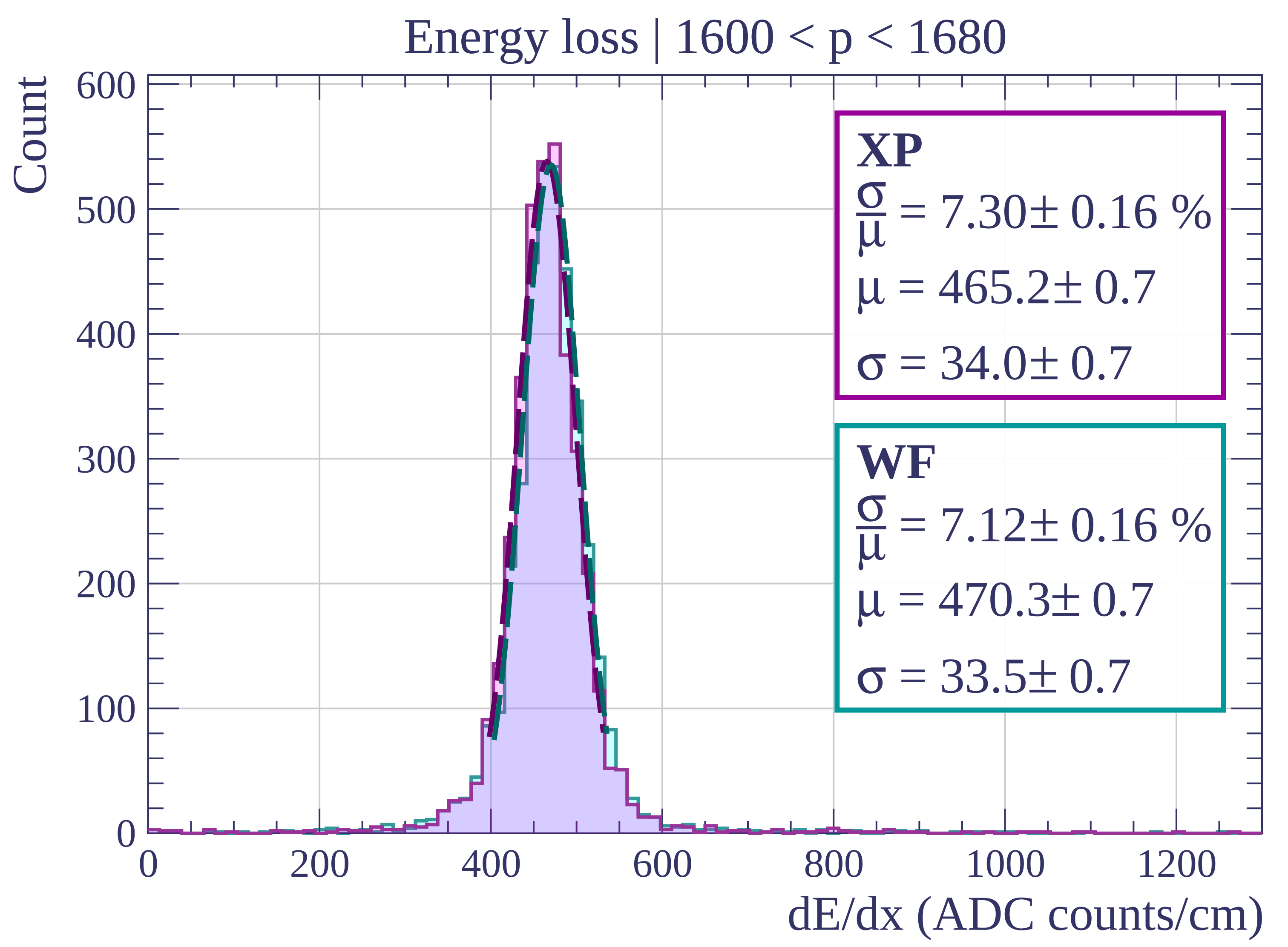
Energy loss | 1280 < p < 1360
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
700
800
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 7.40± 0.15 %
μ = 454.3± 0.6
σ = 33.6± 0.6
WF
$\frac{\sigma}{\mu}$ = 7.95± 0.15 %
μ = 459.0± 0.6
σ = 36.5± 0.7

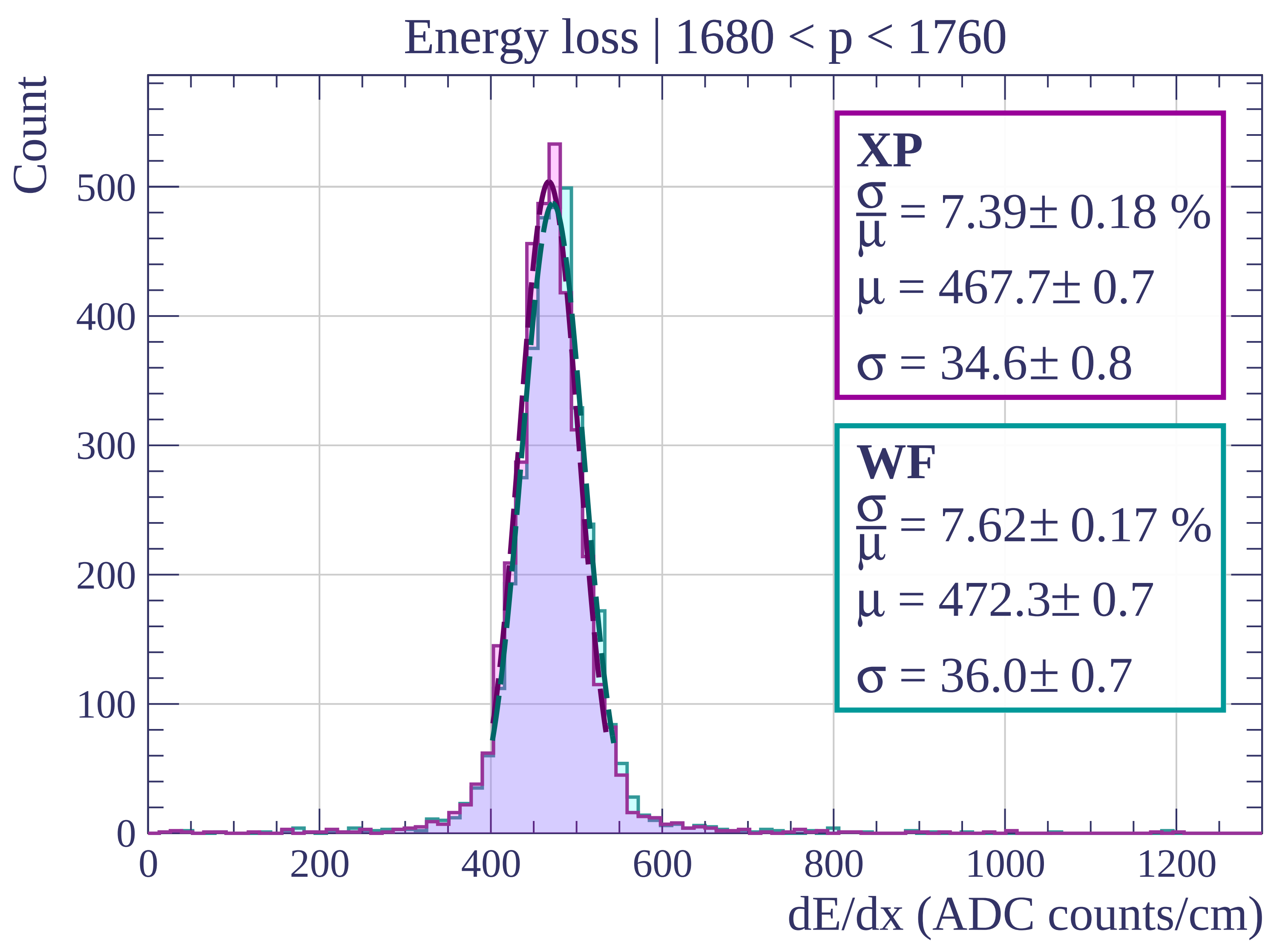


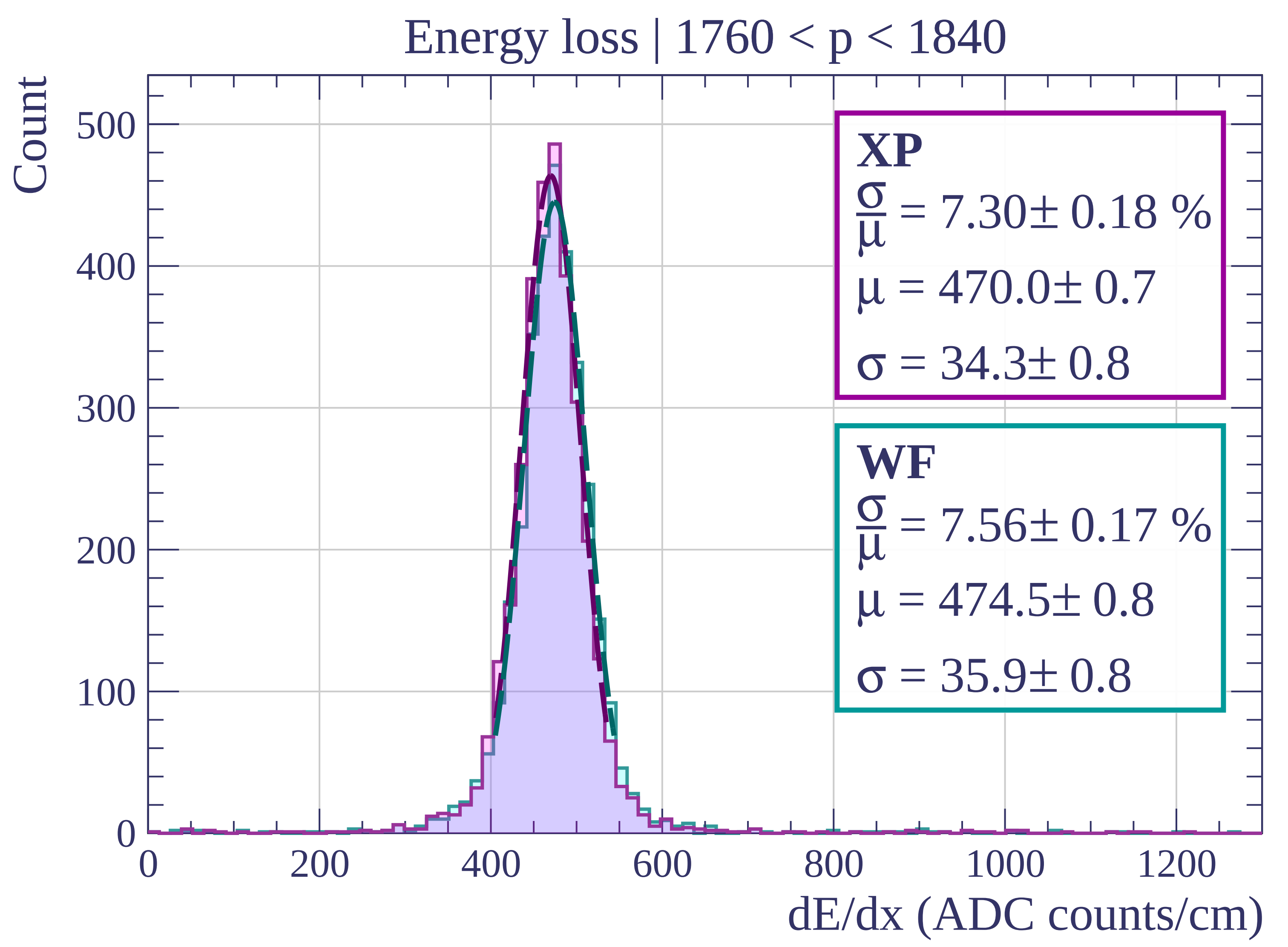
Energy loss | 1360 < p < 1440
Count
0
100
200
300
400
500
600
700
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu} = 7.52 \pm 0.16$ %
$\mu = 457.0 \pm 0.6$
$\sigma = 34.4 \pm 0.7$
WF
$\frac{\sigma}{\mu} = 7.44 \pm 0.14$ %
$\mu = 462.2 \pm 0.6$
$\sigma = 34.4 \pm 0.6$

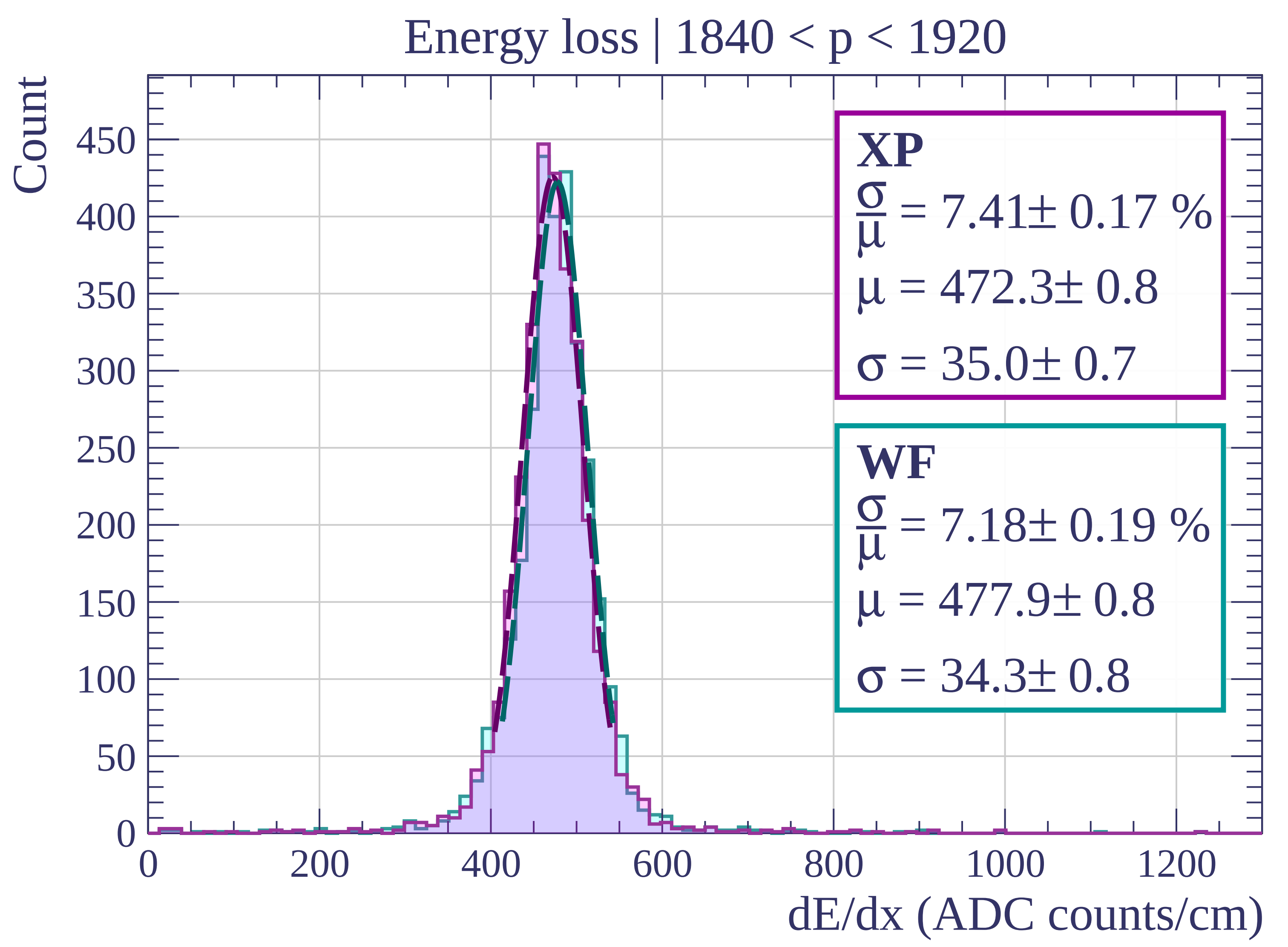

Energy loss | 1440 < p < 1520
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 7.50± 0.14 %
μ = 459.7± 0.6
σ = 34.5± 0.6
WF
$\frac{\sigma}{\mu}$ = 7.70± 0.15 %
μ = 464.2± 0.7
σ = 35.8± 0.7


Energy loss | 1520 < p < 1600
Count
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 7.44± 0.14 %
μ = 462.6± 0.6
σ = 34.4± 0.6
WF
$\frac{\sigma}{\mu}$ = 7.61± 0.15 %
μ = 466.9± 0.7
σ = 35.5± 0.7
0
100
200
300
400
500
600
0
200
400
600
800
1000
1200

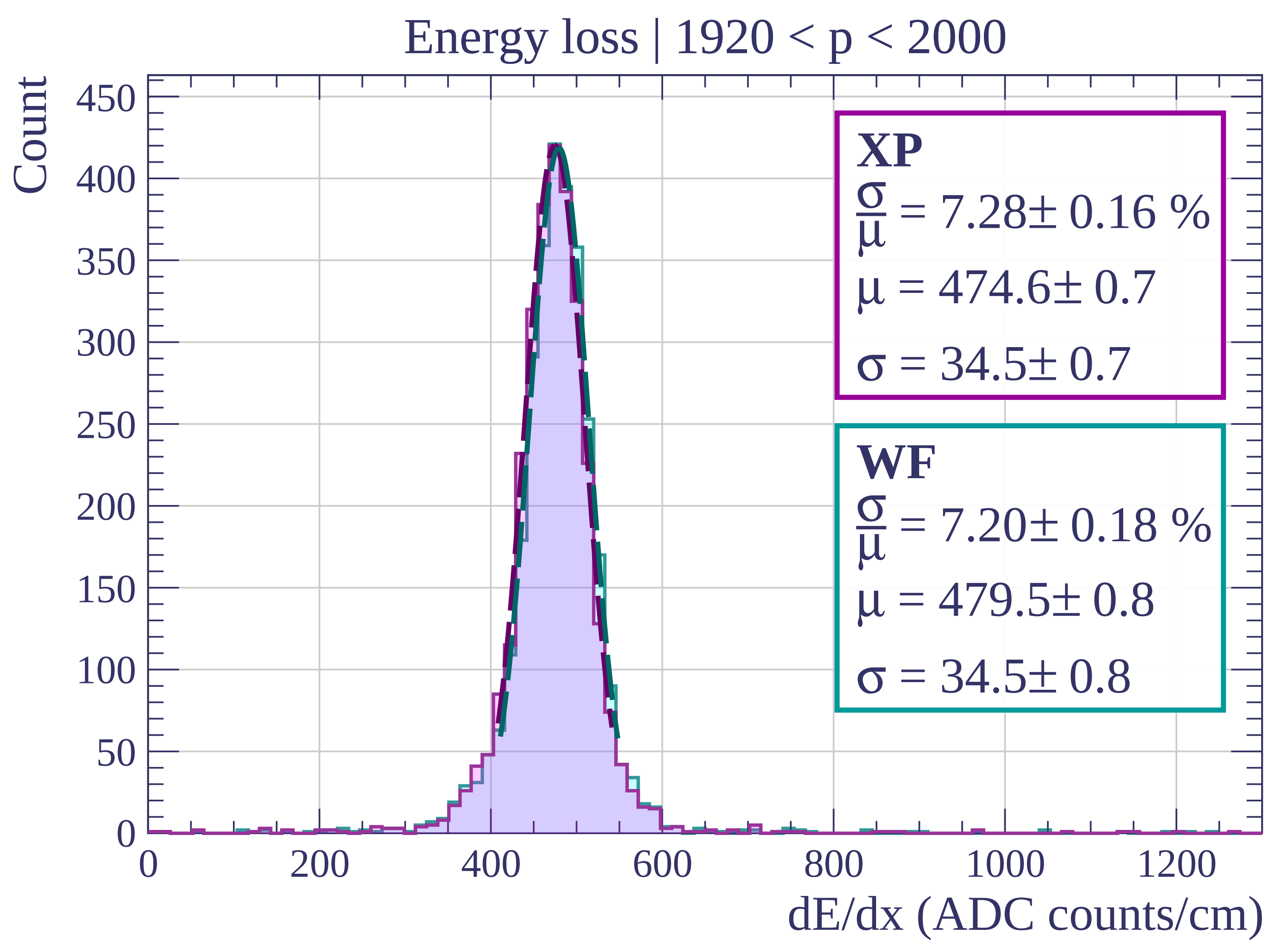

Energy loss | 1600 < p < 1680
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
0
200
400
600
800
1000
1200
XP
σ/μ = 7.30± 0.16 %
μ = 465.2± 0.7
σ = 34.0± 0.7
WF
σ/μ = 7.12± 0.16 %
μ = 470.3± 0.7
σ = 33.5± 0.7


Energy loss | 1680 < p < 1760
Count
500
400
300
200
100
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 7.39± 0.18 %
μ = 467.7± 0.7
σ = 34.6± 0.8
WF
σ/μ = 7.62± 0.17 %
μ = 472.3± 0.7
σ = 36.0± 0.7

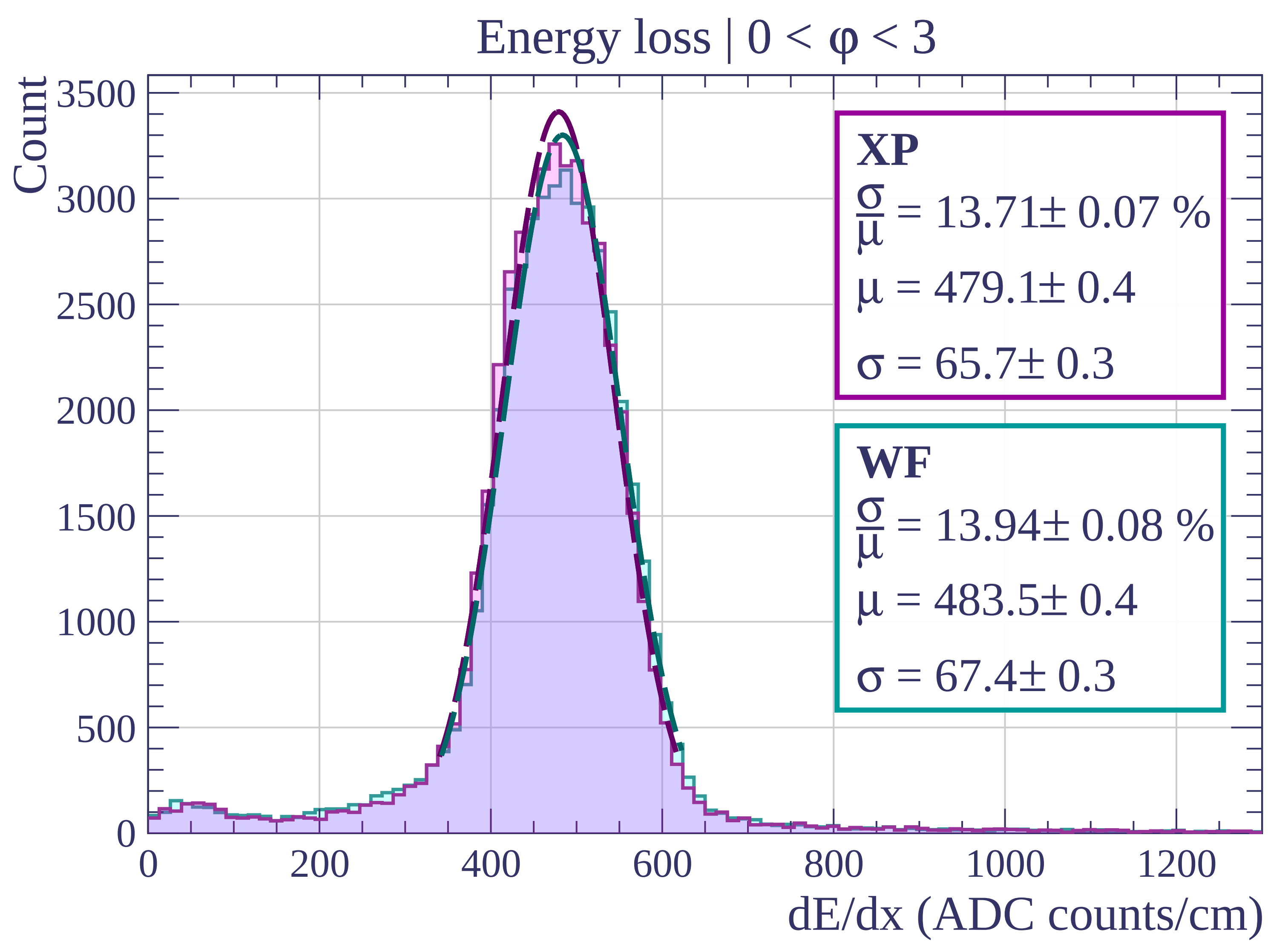

Energy loss | 1760 < p < 1840
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
100
300
500
XP
$\frac{\sigma}{\mu}$ = 7.30± 0.18 %
μ = 470.0± 0.7
σ = 34.3± 0.8
WF
$\frac{\sigma}{\mu}$ = 7.56± 0.17 %
μ = 474.5± 0.8
σ = 35.9± 0.8



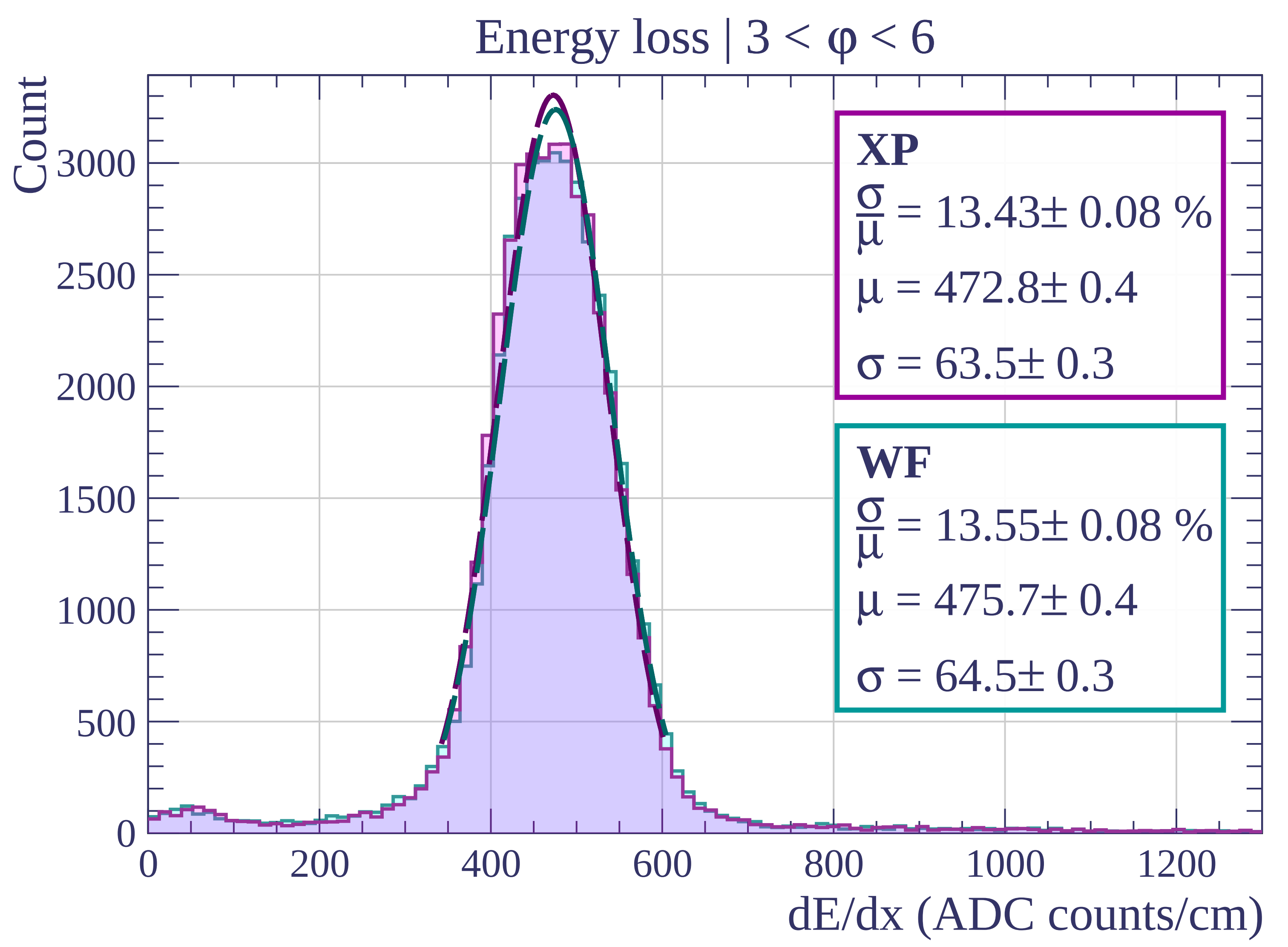
Energy loss | 1840 < p < 1920
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
400
450
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 7.41± 0.17 %
μ = 472.3± 0.8
σ = 35.0± 0.7
WF
$\frac{\sigma}{\mu}$ = 7.18± 0.19 %
μ = 477.9± 0.8
σ = 34.3± 0.8



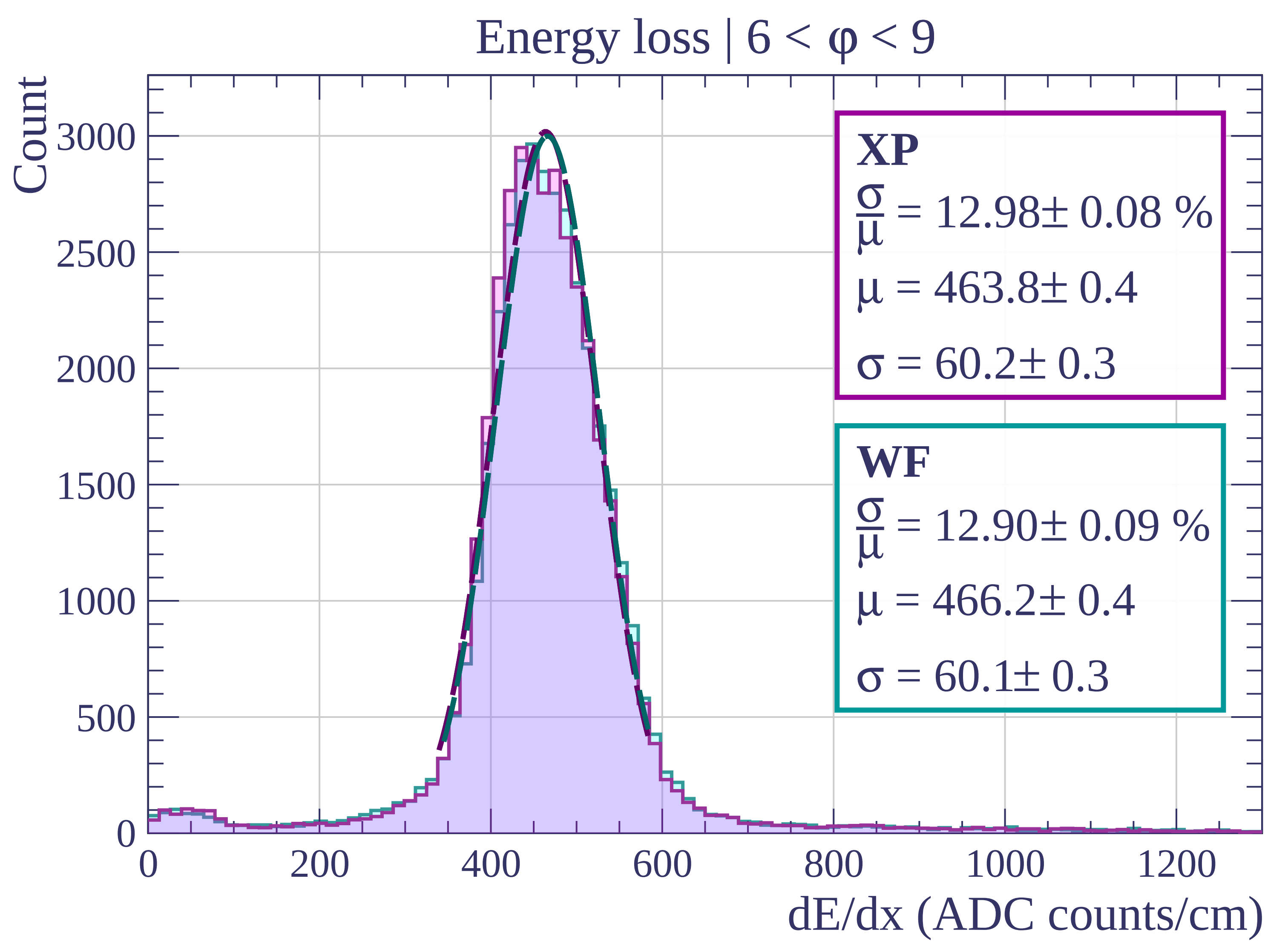
Energy loss | 1920 < p < 2000
Count
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu} = 7.28 \pm 0.16$ %
$\mu = 474.6 \pm 0.7$
$\sigma = 34.5 \pm 0.7$
WF
$\frac{\sigma}{\mu} = 7.20 \pm 0.18$ %
$\mu = 479.5 \pm 0.8$
$\sigma = 34.5 \pm 0.8$



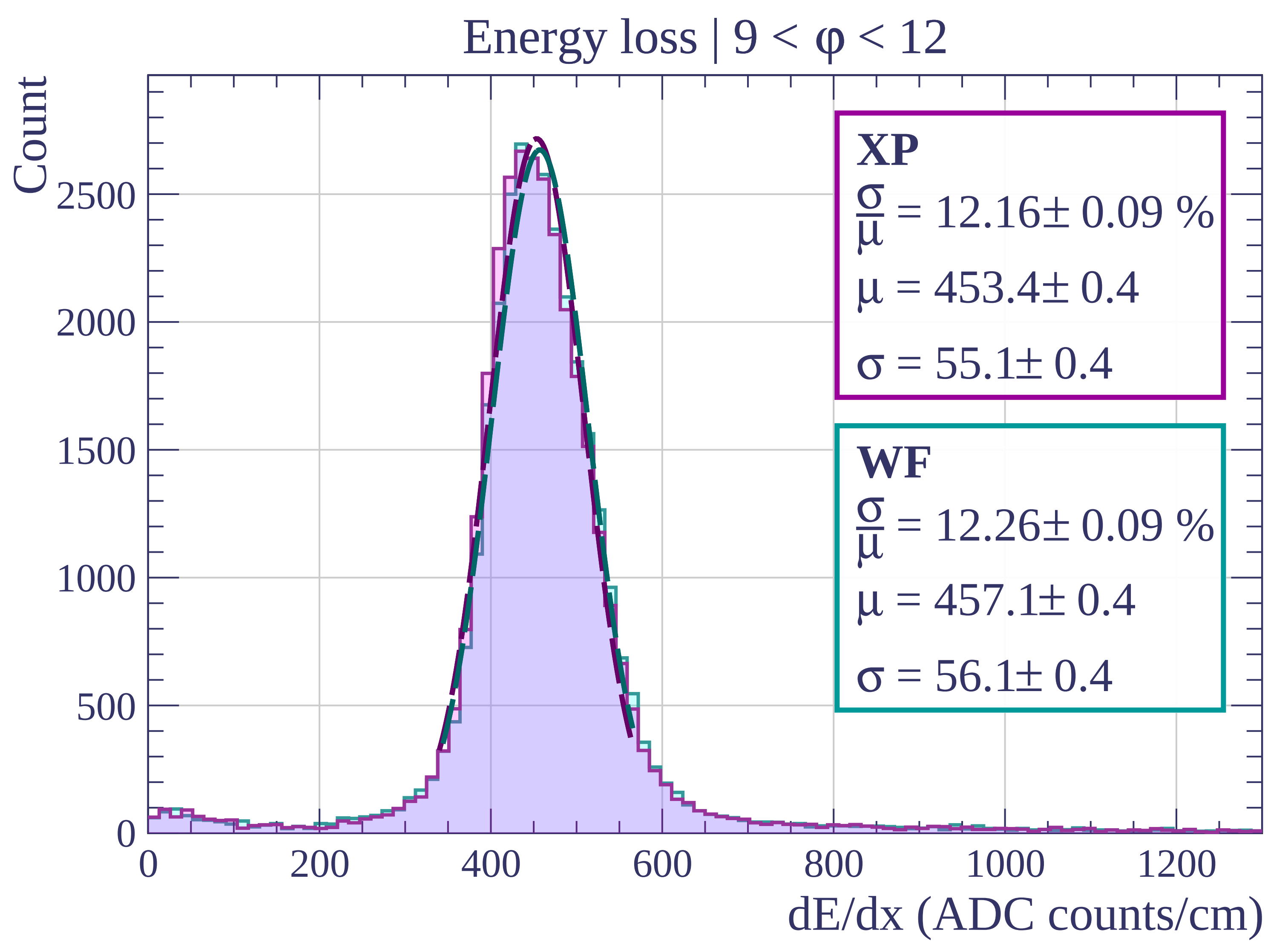
Energy loss | 0 < φ < 3
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 13.71± 0.07 %
μ = 479.1± 0.4
σ = 65.7± 0.3
WF
σ/μ = 13.94± 0.08 %
μ = 483.5± 0.4
σ = 67.4± 0.3

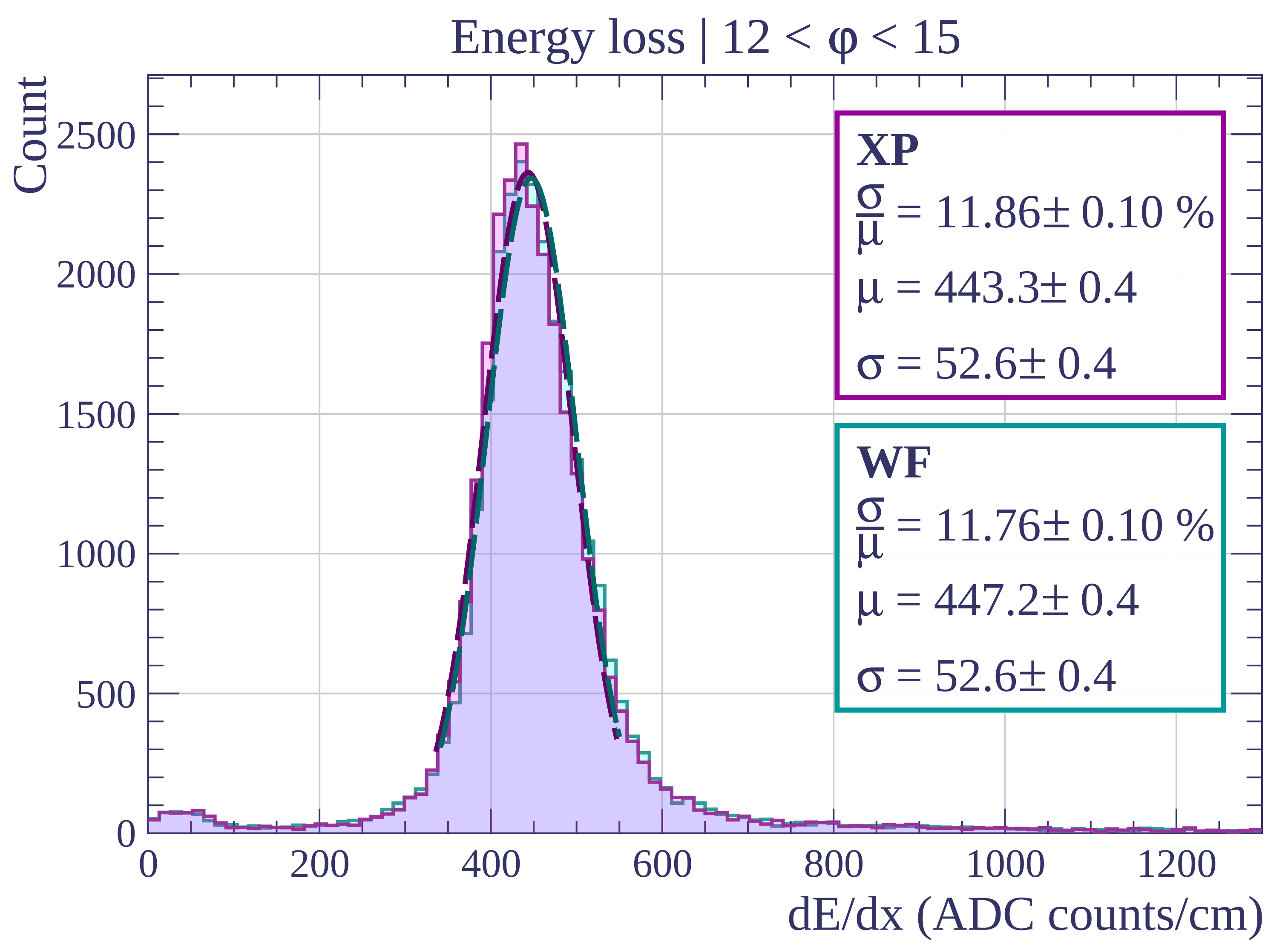


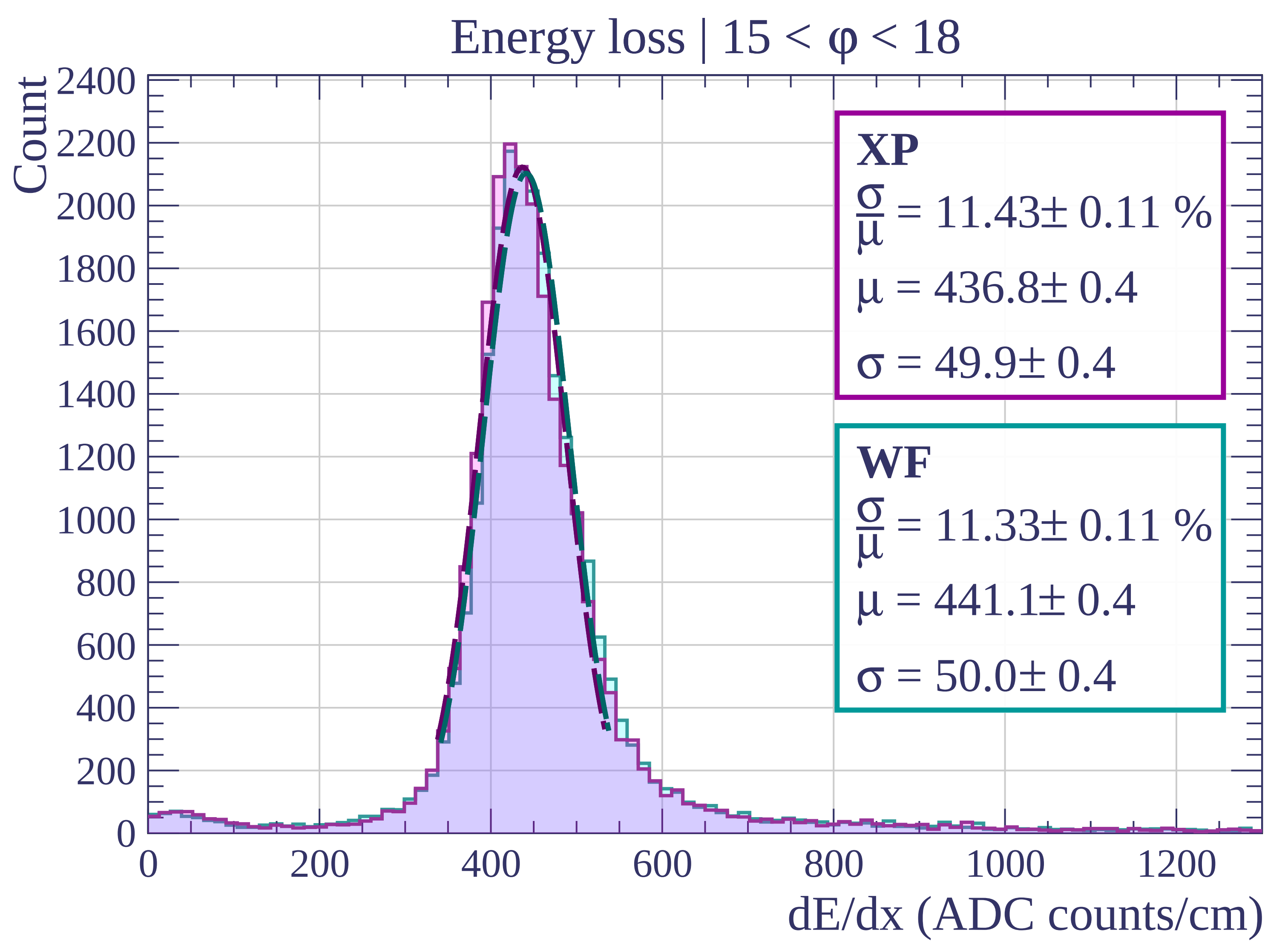
Energy loss | 3 < φ < 6
Count
3000
2500
2000
1500
1000
500
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 13.43± 0.08 %
μ = 472.8± 0.4
σ = 63.5± 0.3
WF
σ/μ = 13.55± 0.08 %
μ = 475.7± 0.4
σ = 64.5± 0.3



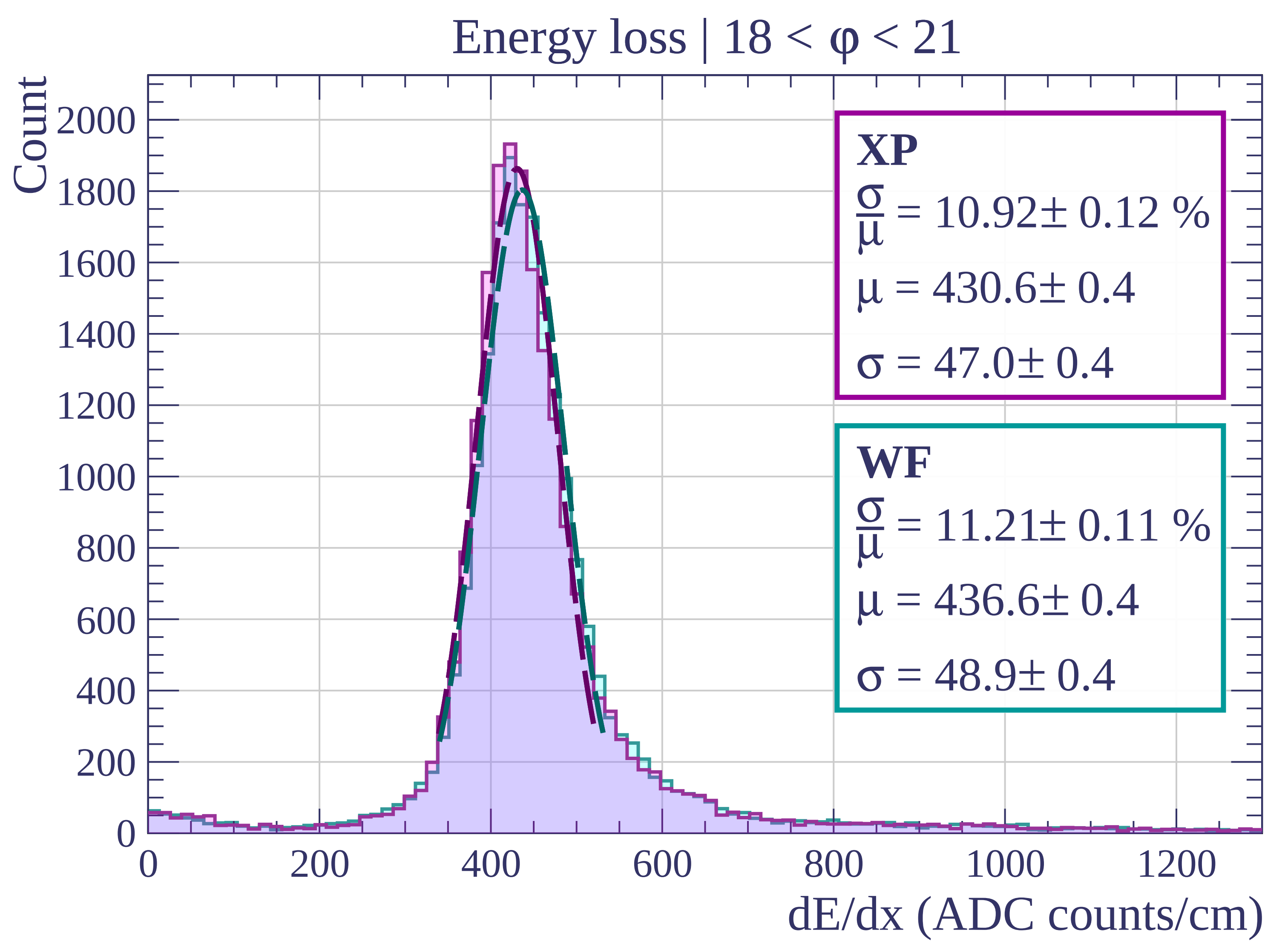
Energy loss | 6 < φ < 9
Count
3000
2500
2000
1500
1000
500
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 12.98± 0.08 %
μ = 463.8± 0.4
σ = 60.2± 0.3
WF
$\frac{\sigma}{\mu}$ = 12.90± 0.09 %
μ = 466.2± 0.4
σ = 60.1± 0.3


Energy loss | 9 < φ < 12
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
500
1000
1500
2000
2500
XP
$\frac{\sigma}{\mu}$ = 12.16± 0.09 %
μ = 453.4± 0.4
σ = 55.1± 0.4
WF
$\frac{\sigma}{\mu}$ = 12.26± 0.09 %
μ = 457.1± 0.4
σ = 56.1± 0.4

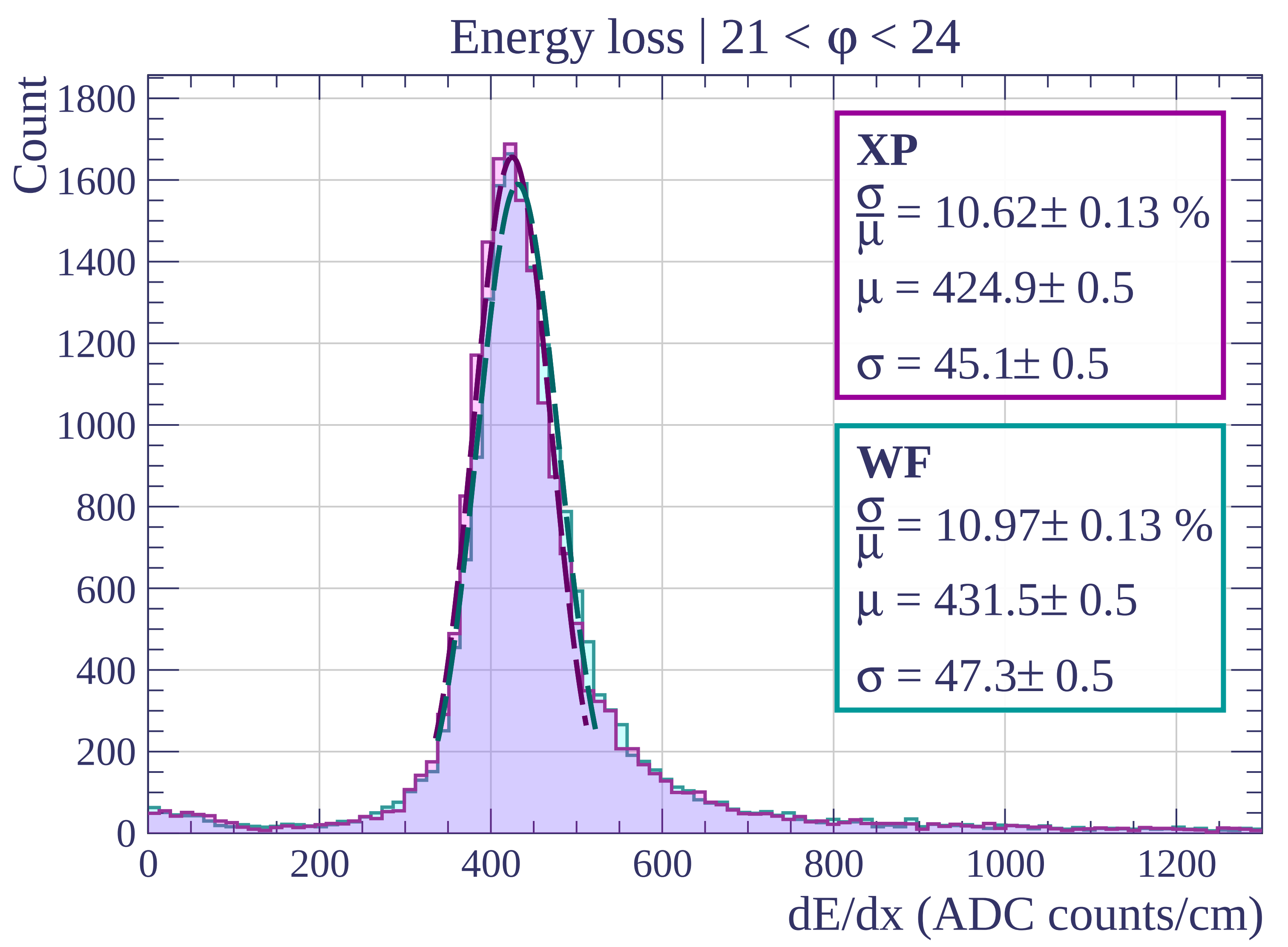

Energy loss | 12 < φ < 15
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 11.86± 0.10 %
μ = 443.3± 0.4
σ = 52.6± 0.4
WF
σ/μ = 11.76± 0.10 %
μ = 447.2± 0.4
σ = 52.6± 0.4



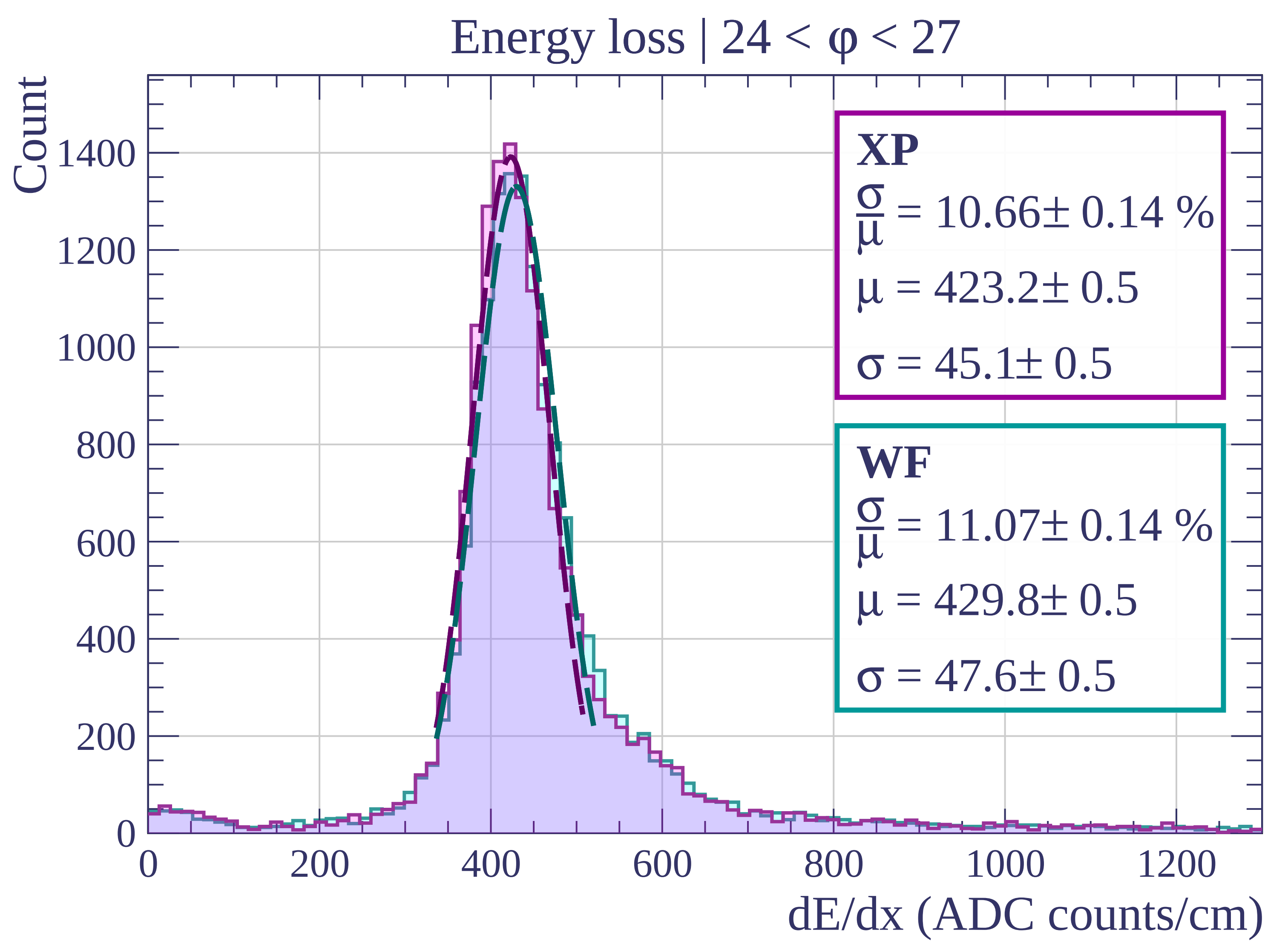
Energy loss | 15 < φ < 18
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 11.43± 0.11 %
μ = 436.8± 0.4
σ = 49.9± 0.4
WF
σ/μ = 11.33± 0.11 %
μ = 441.1± 0.4
σ = 50.0± 0.4



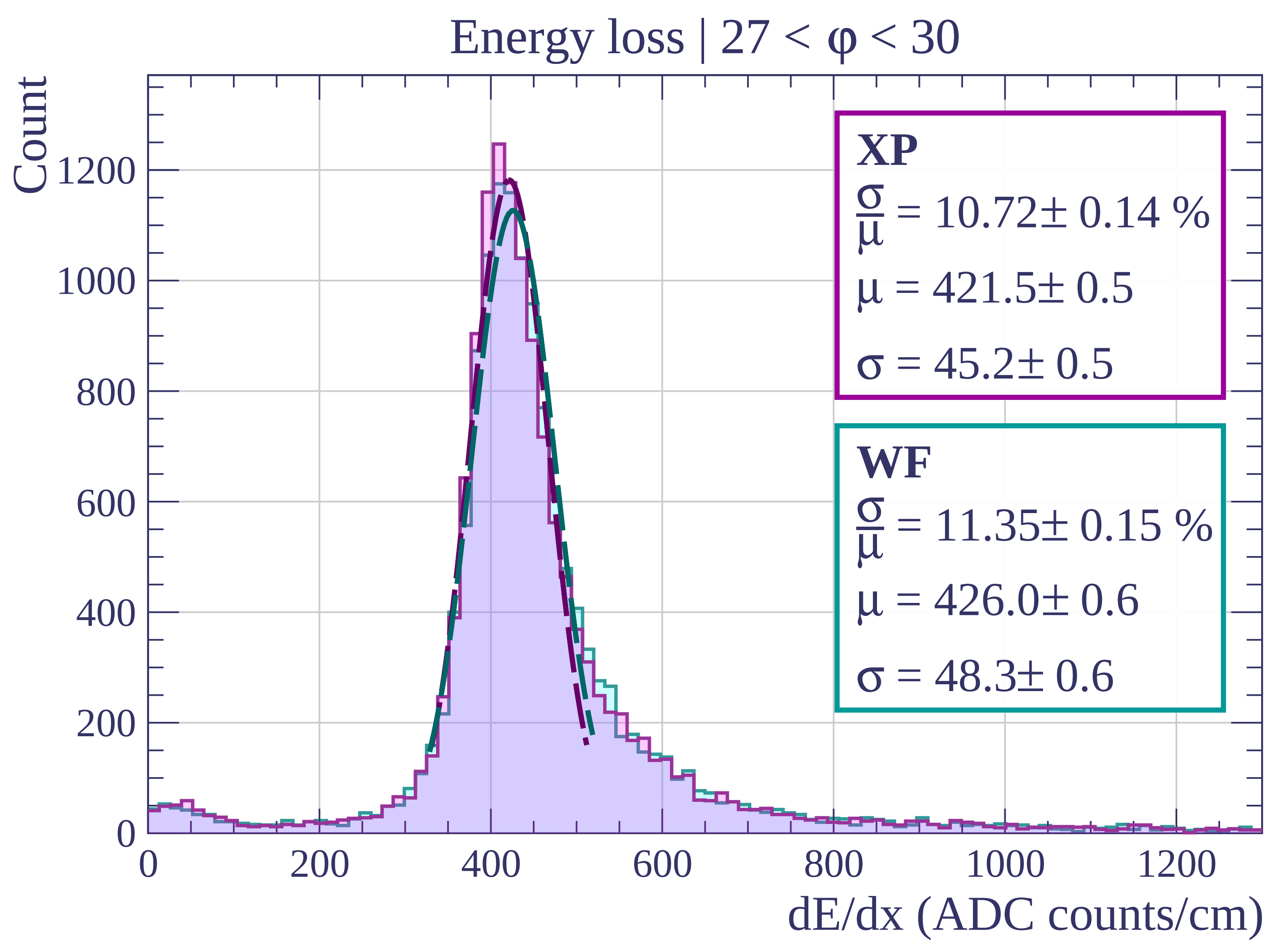
Energy loss | 18 < φ < 21
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 10.92± 0.12 %
μ = 430.6± 0.4
σ = 47.0± 0.4
WF
σ/μ = 11.21± 0.11 %
μ = 436.6± 0.4
σ = 48.9± 0.4



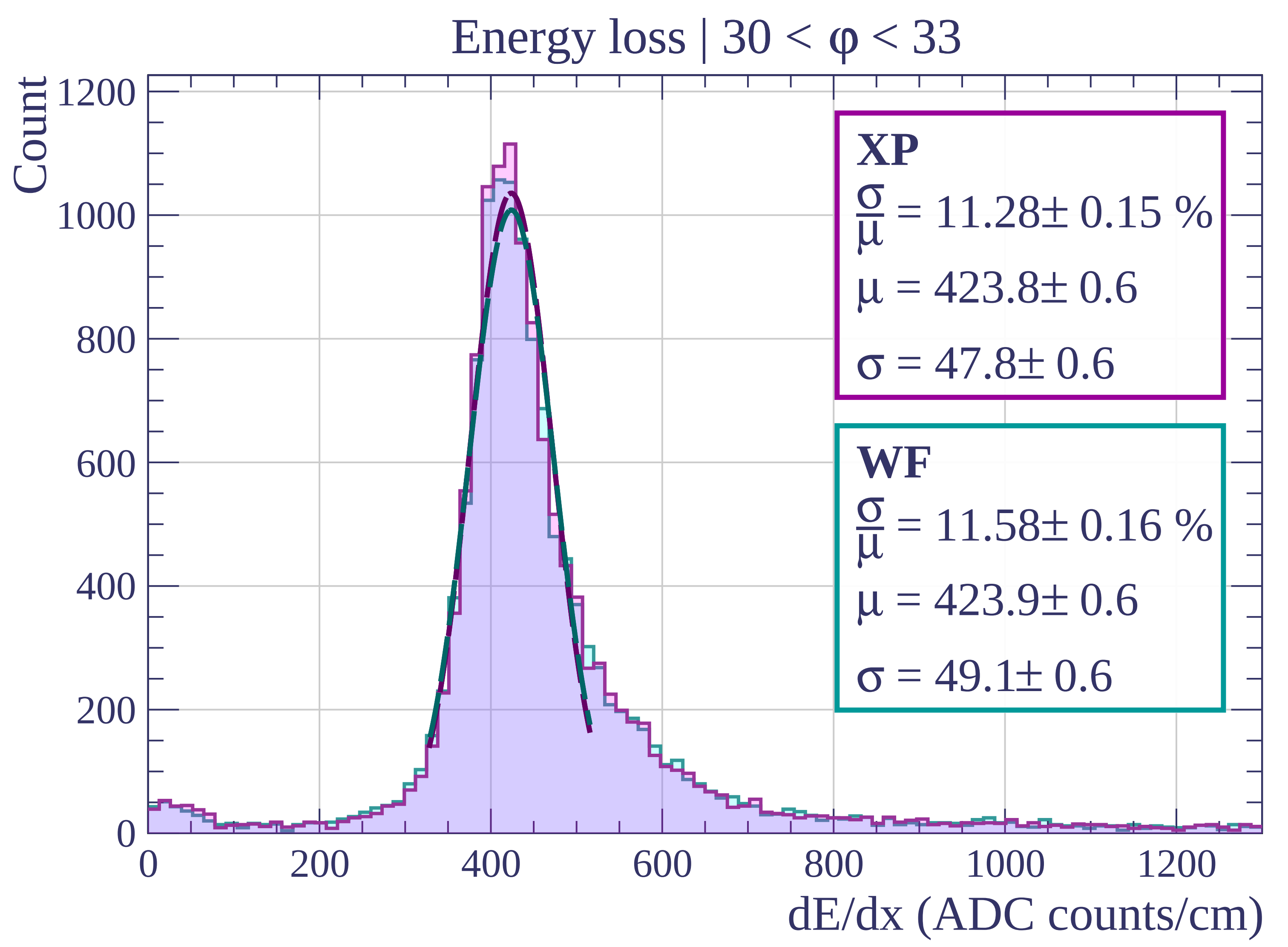
Energy loss | 21 < φ < 24
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
1400
1600
1800
XP
σ/μ = 10.62± 0.13 %
μ = 424.9± 0.5
σ = 45.1± 0.5
WF
σ/μ = 10.97± 0.13 %
μ = 431.5± 0.5
σ = 47.3± 0.5

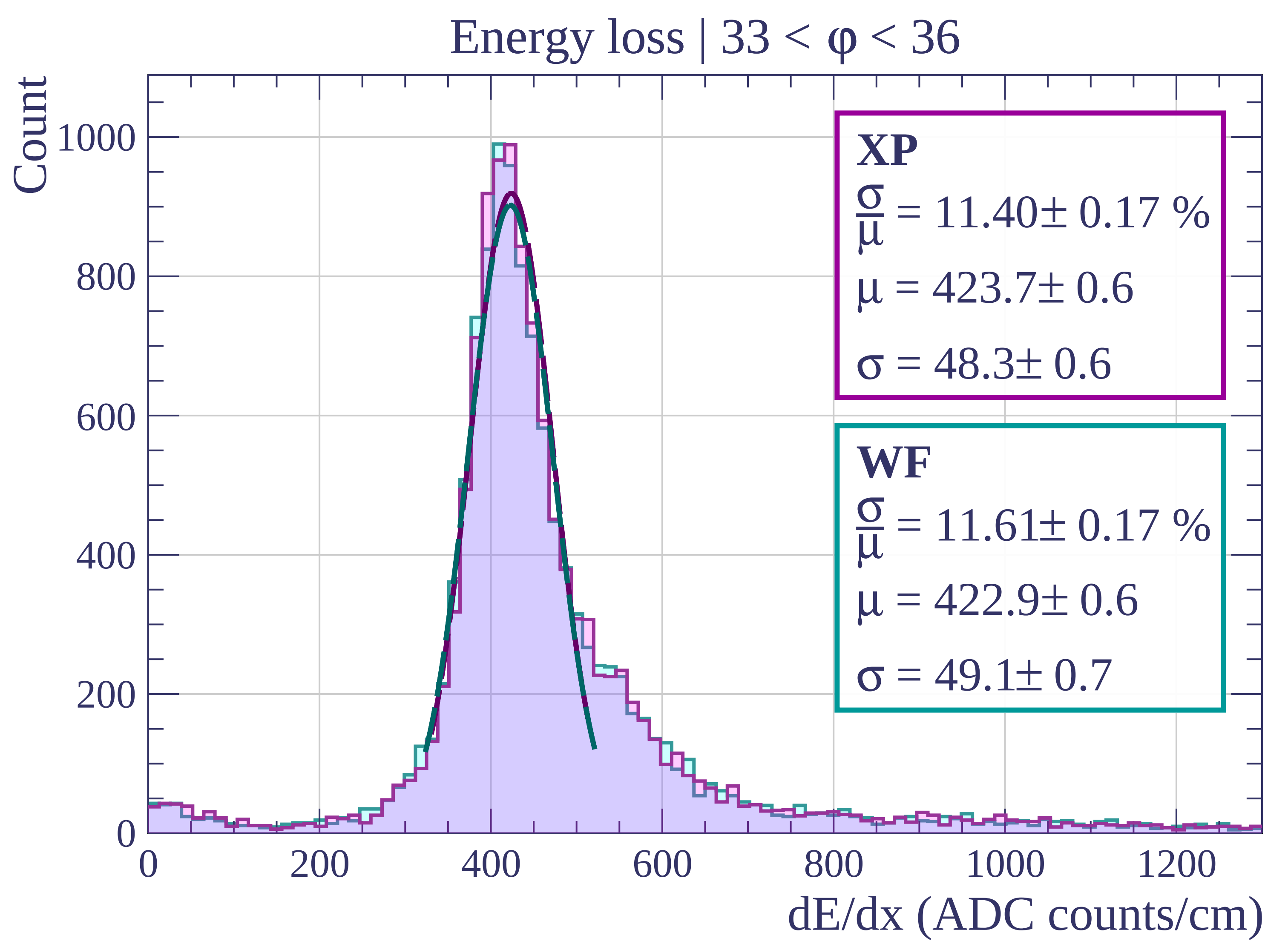


Energy loss | 24 < φ < 27
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 10.66± 0.14 %
μ = 423.2± 0.5
σ = 45.1± 0.5
WF
σ/μ = 11.07± 0.14 %
μ = 429.8± 0.5
σ = 47.6± 0.5

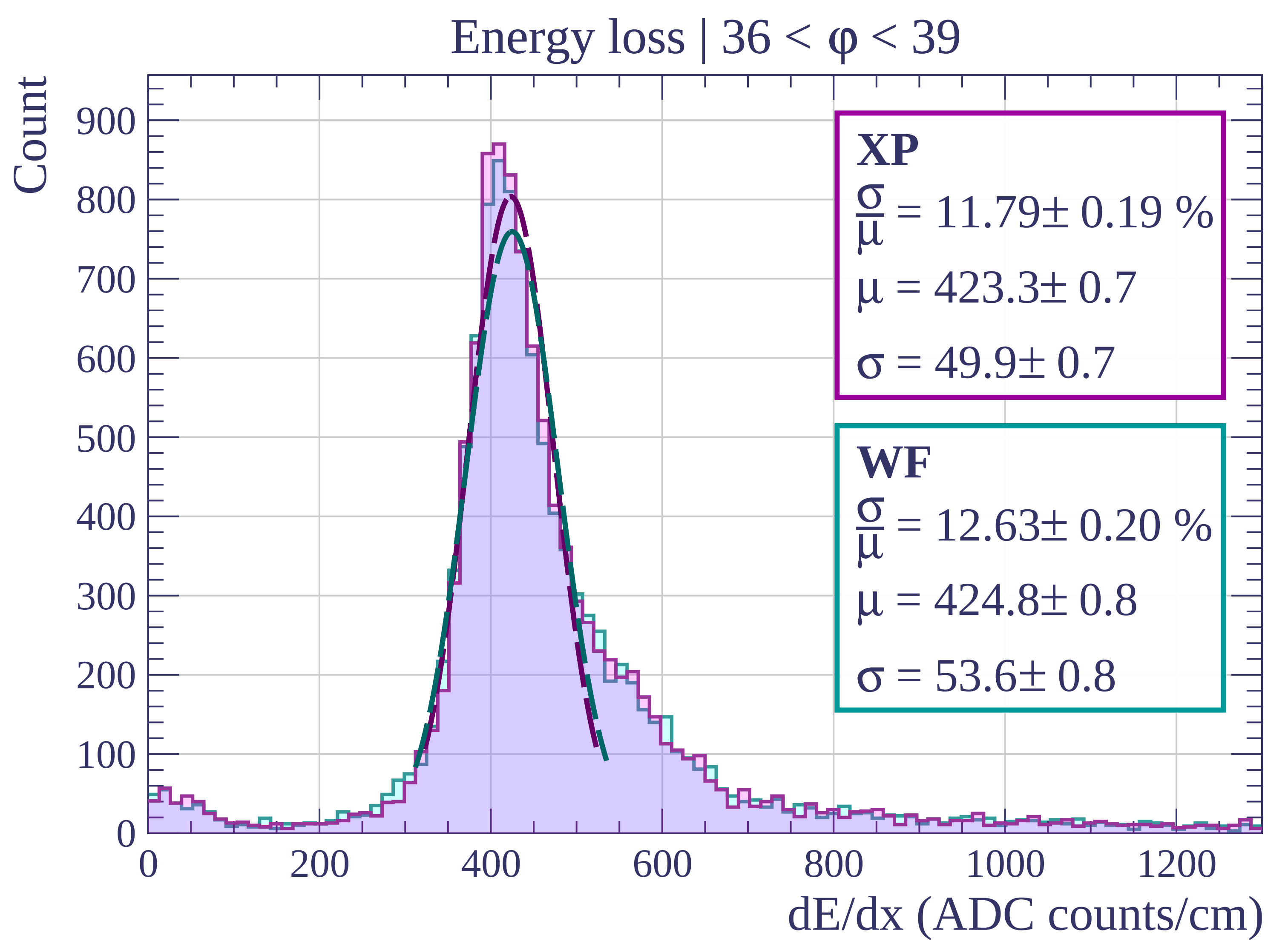


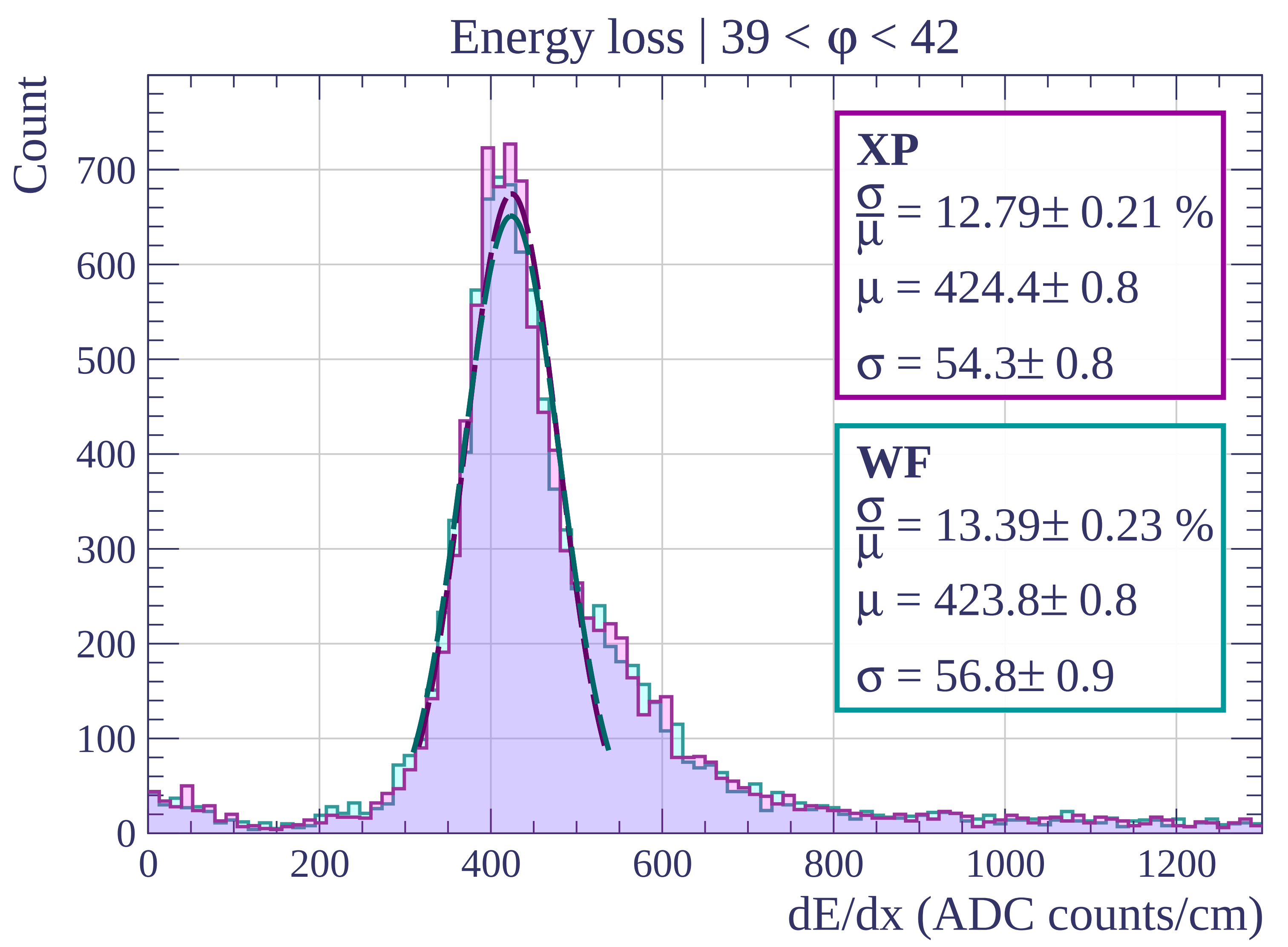
Energy loss | 27 < φ < 30
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 10.72 ± 0.14 %
μ = 421.5 ± 0.5
σ = 45.2 ± 0.5
WF
σ/μ = 11.35 ± 0.15 %
μ = 426.0 ± 0.6
σ = 48.3 ± 0.6


Energy loss | 30 < φ < 33
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
XP
σ/μ = 11.28± 0.15 %
μ = 423.8± 0.6
σ = 47.8± 0.6
WF
σ/μ = 11.58± 0.16 %
μ = 423.9± 0.6
σ = 49.1± 0.6

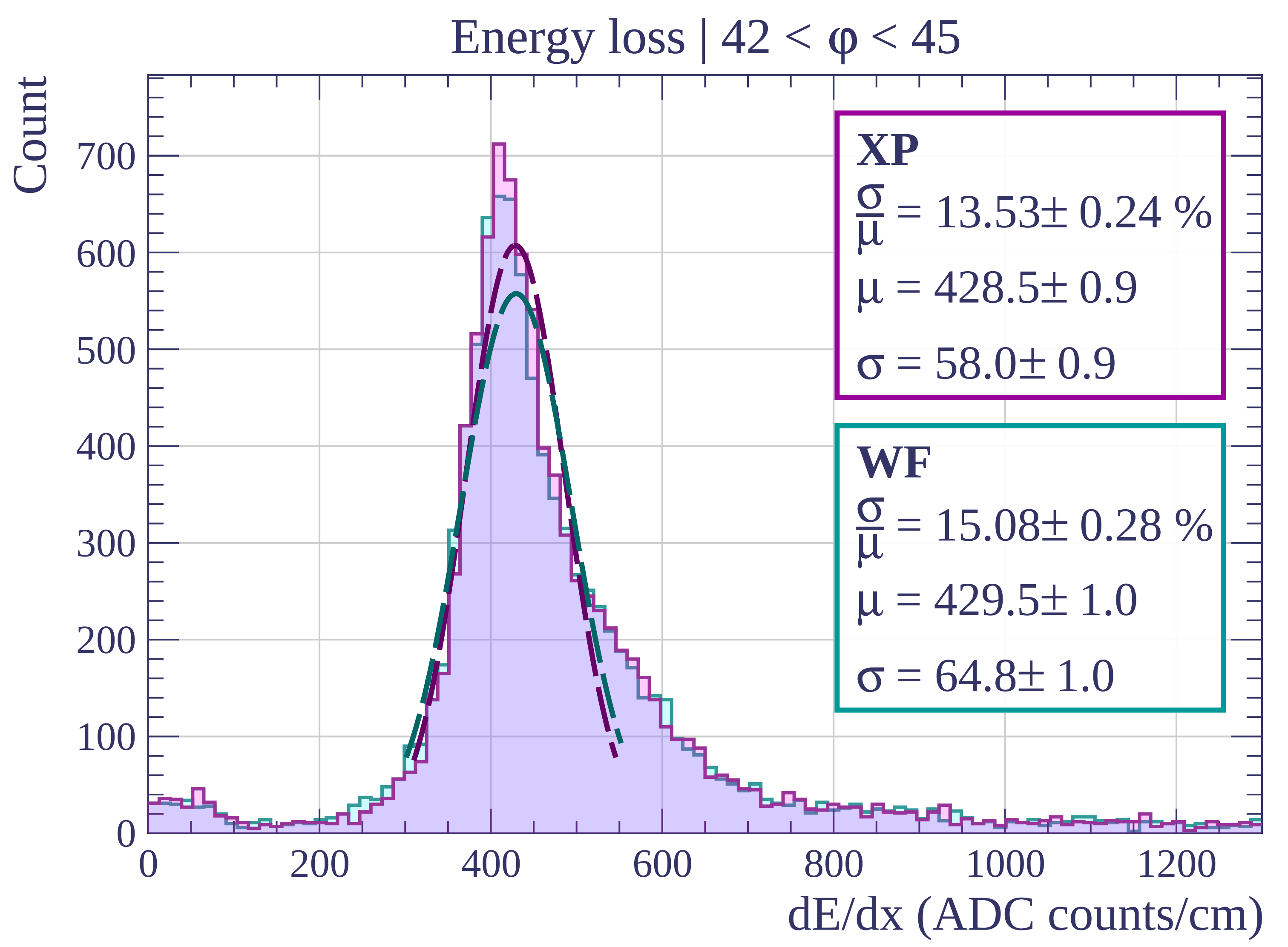


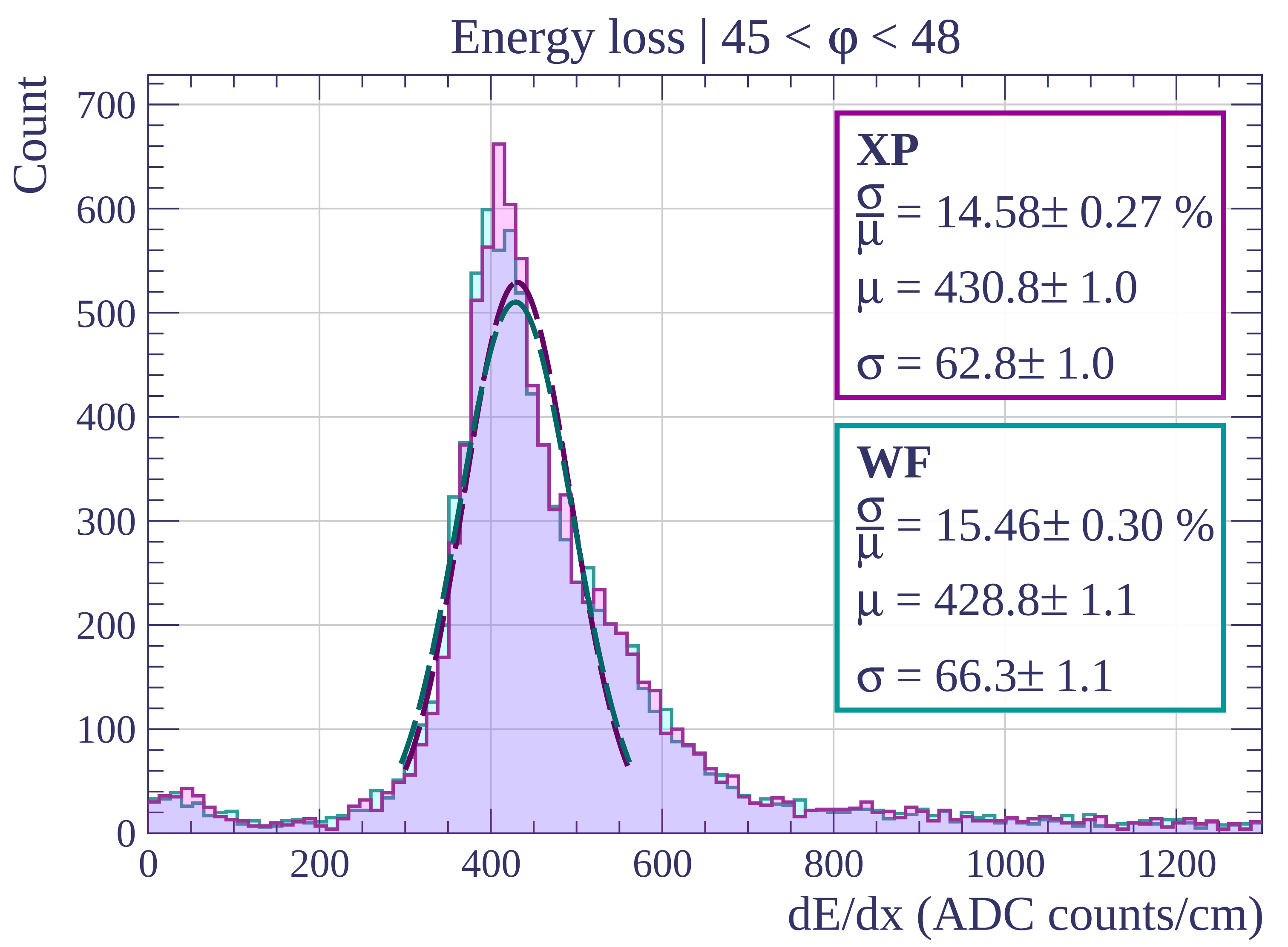
Energy loss | 33 < φ < 36
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 11.40± 0.17 %
μ = 423.7± 0.6
σ = 48.3± 0.6
WF
$\frac{\sigma}{\mu}$ = 11.61± 0.17 %
μ = 422.9± 0.6
σ = 49.1± 0.7

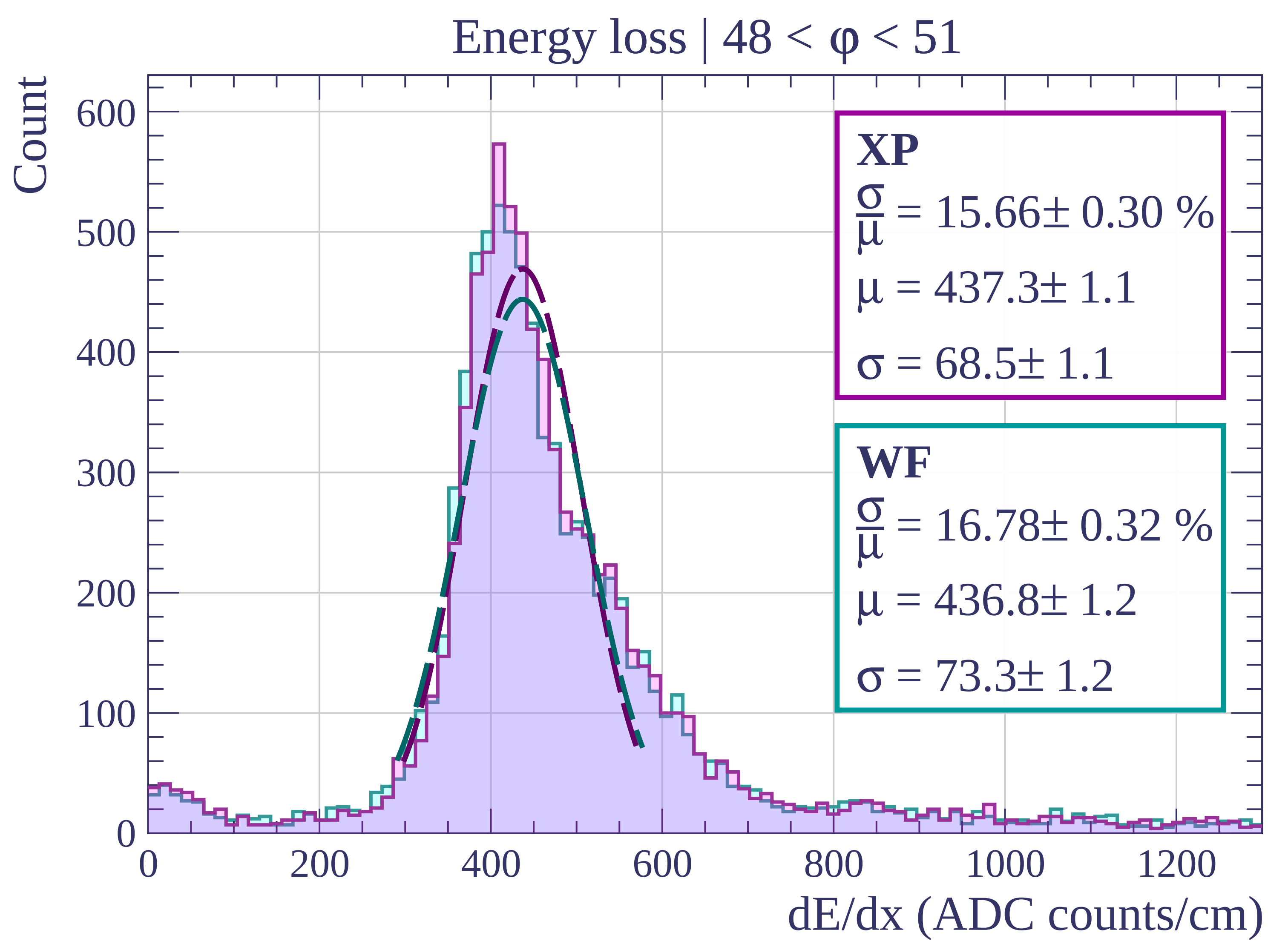

Energy loss | 36 < φ < 39
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
700
800
900
0
200
400
600
800
1000
1200
XP
σ/μ = 11.79± 0.19 %
μ = 423.3± 0.7
σ = 49.9± 0.7
WF
σ/μ = 12.63± 0.20 %
μ = 424.8± 0.8
σ = 53.6± 0.8



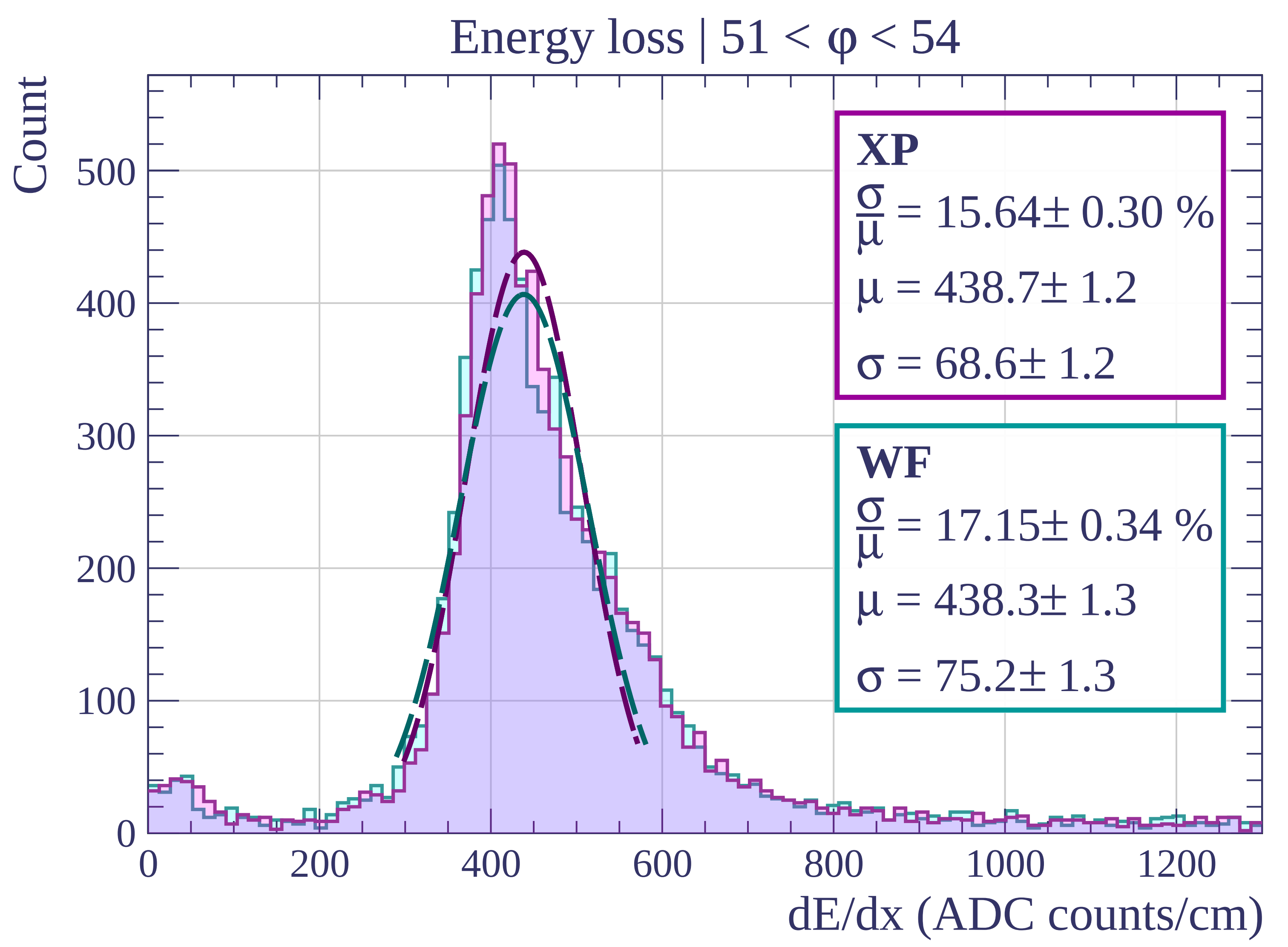
Energy loss | 39 < φ < 42
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
700
0
200
400
600
800
1000
1200
XP
σ/μ = 12.79± 0.21 %
μ = 424.4± 0.8
σ = 54.3± 0.8
WF
σ/μ = 13.39± 0.23 %
μ = 423.8± 0.8
σ = 56.8± 0.9


Energy loss | 42 < φ < 45
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 13.53± 0.24 %
μ = 428.5± 0.9
σ = 58.0± 0.9
WF
σ/μ = 15.08± 0.28 %
μ = 429.5± 1.0
σ = 64.8± 1.0
0
100
200
300
400
500
600
700
0
200
400
600
800
1000
1200

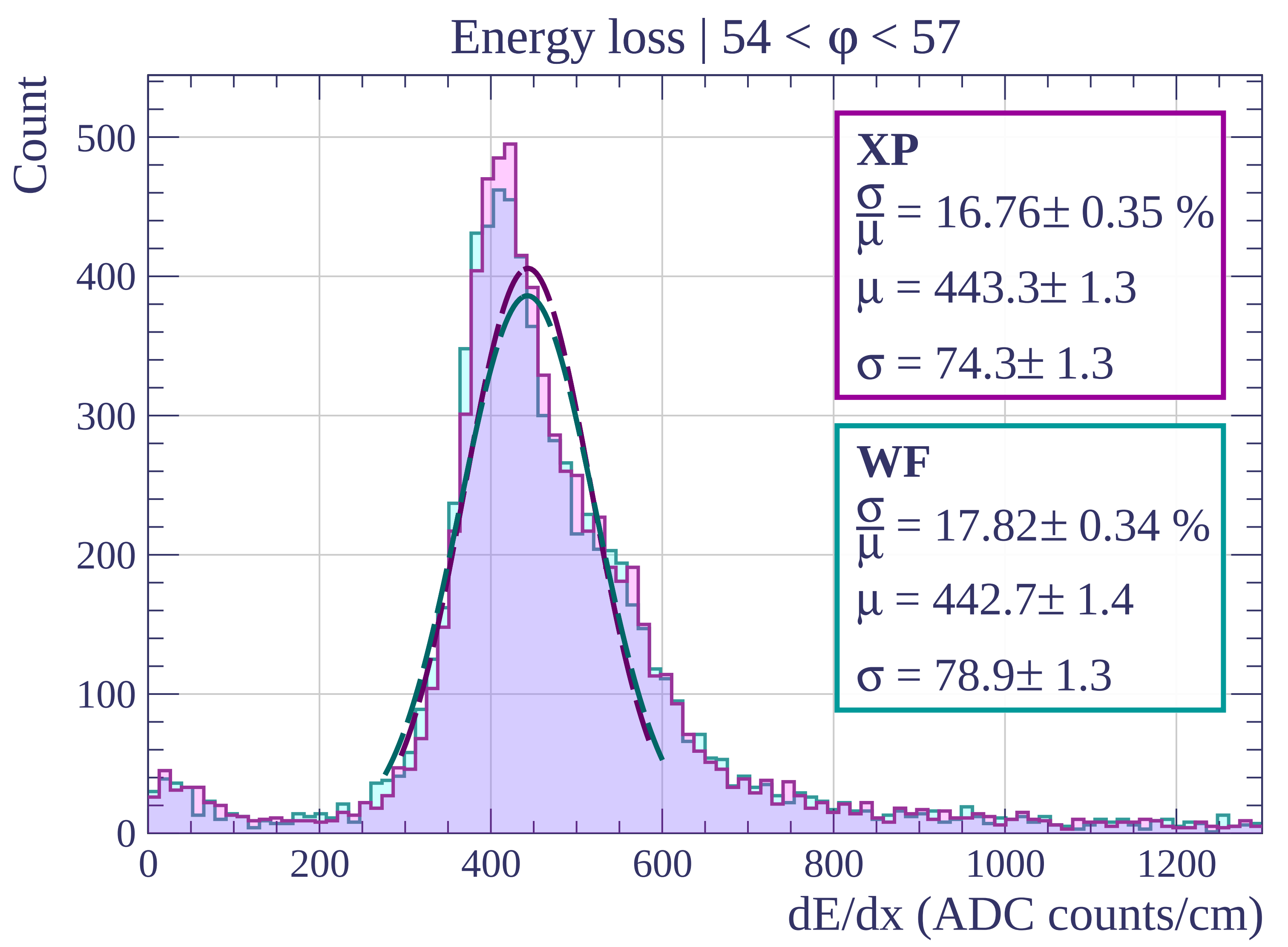

Energy loss | 45 < φ < 48
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
700
0
200
400
600
800
1000
1200
XP
σ/μ = 14.58± 0.27 %
μ = 430.8± 1.0
σ = 62.8± 1.0
WF
σ/μ = 15.46± 0.30 %
μ = 428.8± 1.1
σ = 66.3± 1.1

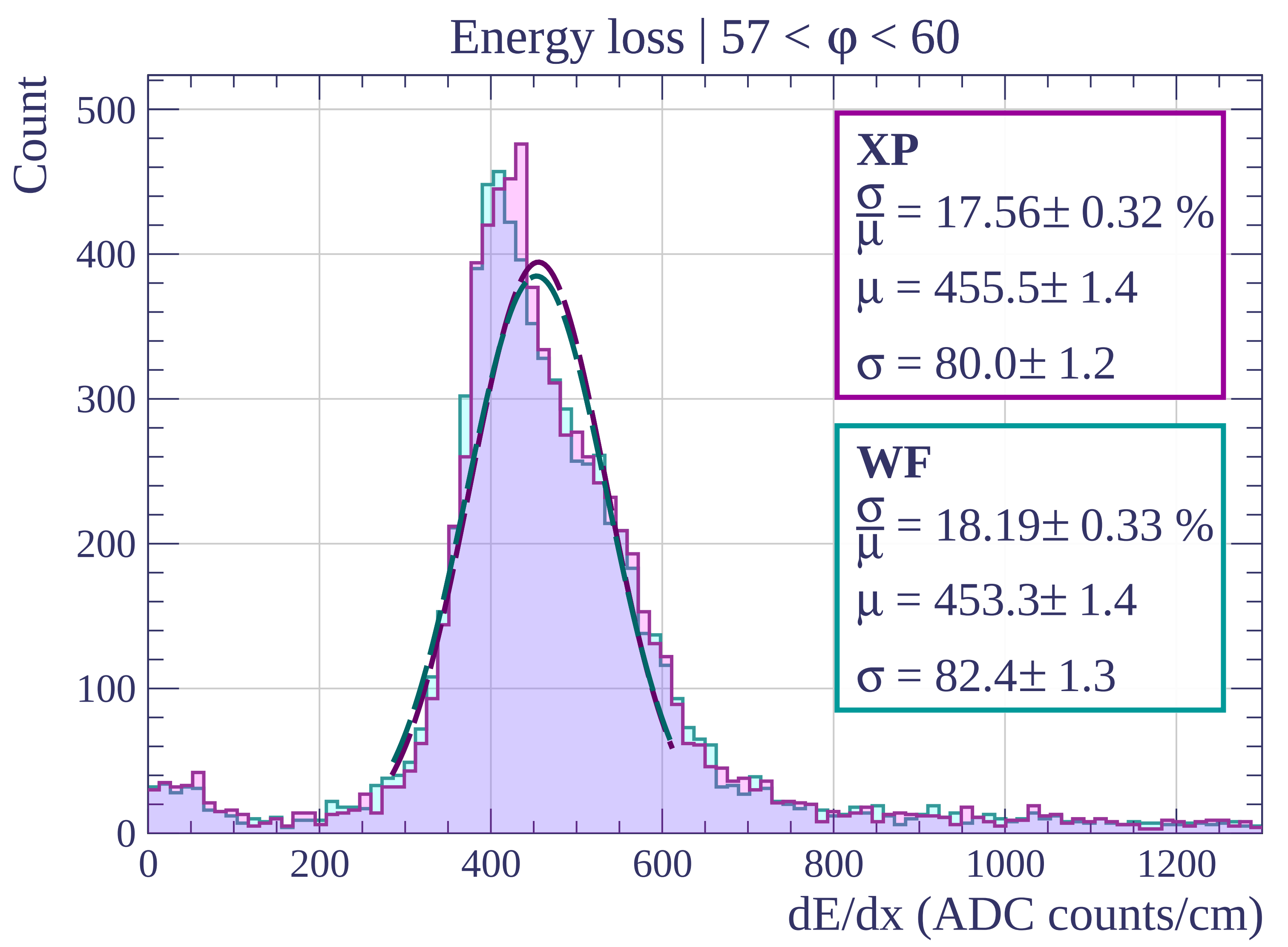


Energy loss | 48 < φ < 51
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
100
300
500
XP
$\frac{\sigma}{\mu}$ = 15.66± 0.30 %
μ = 437.3± 1.1
σ = 68.5± 1.1
WF
$\frac{\sigma}{\mu}$ = 16.78± 0.32 %
μ = 436.8± 1.2
σ = 73.3± 1.2

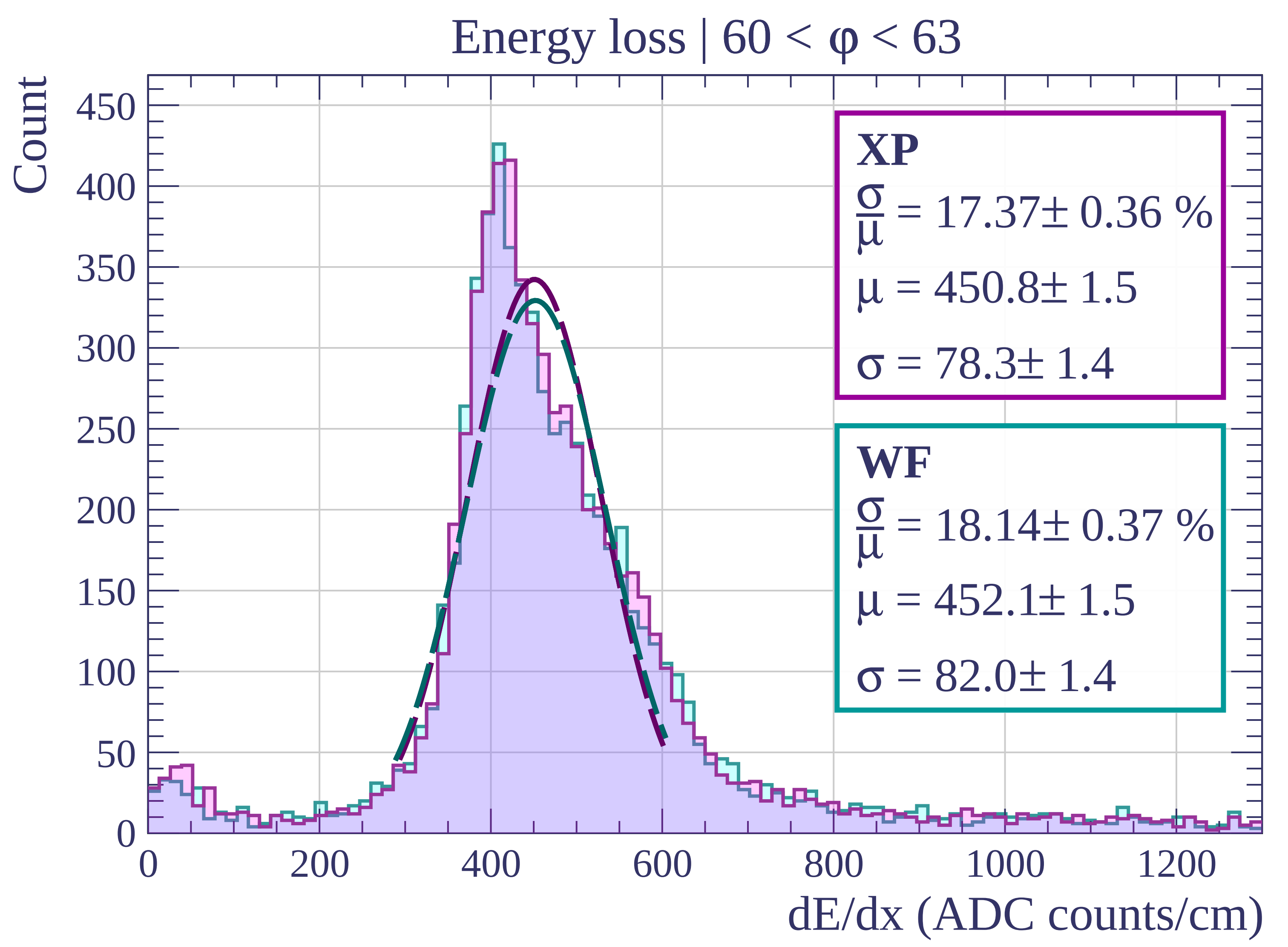


Energy loss | 51 < φ < 54
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
0
200
400
600
800
1000
1200
XP
σ/μ = 15.64± 0.30 %
μ = 438.7± 1.2
σ = 68.6± 1.2
WF
σ/μ = 17.15± 0.34 %
μ = 438.3± 1.3
σ = 75.2± 1.3

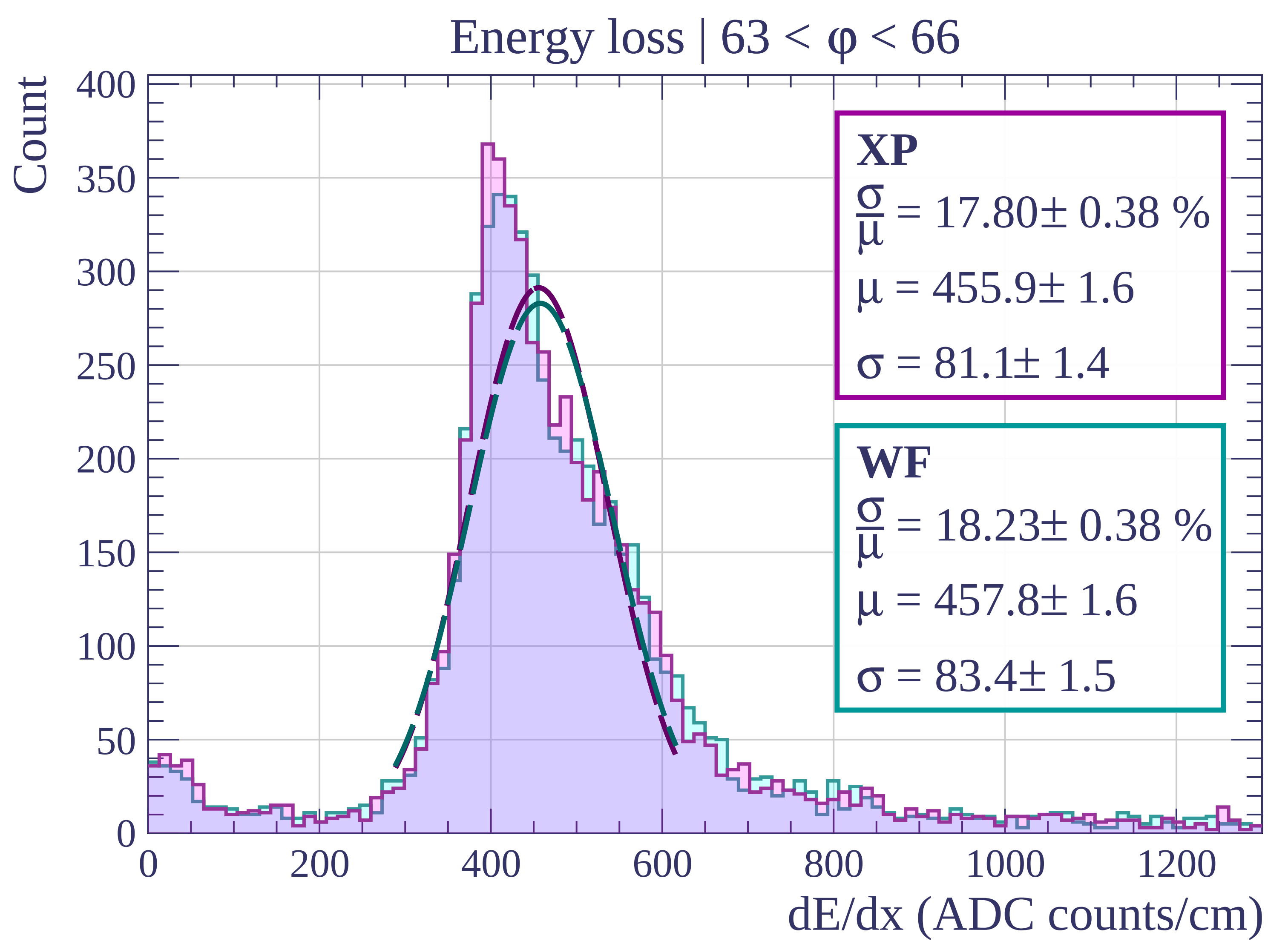

Energy loss | 54 < φ < 57
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
0
200
400
600
800
1000
1200
XP
σ/μ = 16.76± 0.35 %
μ = 443.3± 1.3
σ = 74.3± 1.3
WF
σ/μ = 17.82± 0.34 %
μ = 442.7± 1.4
σ = 78.9± 1.3

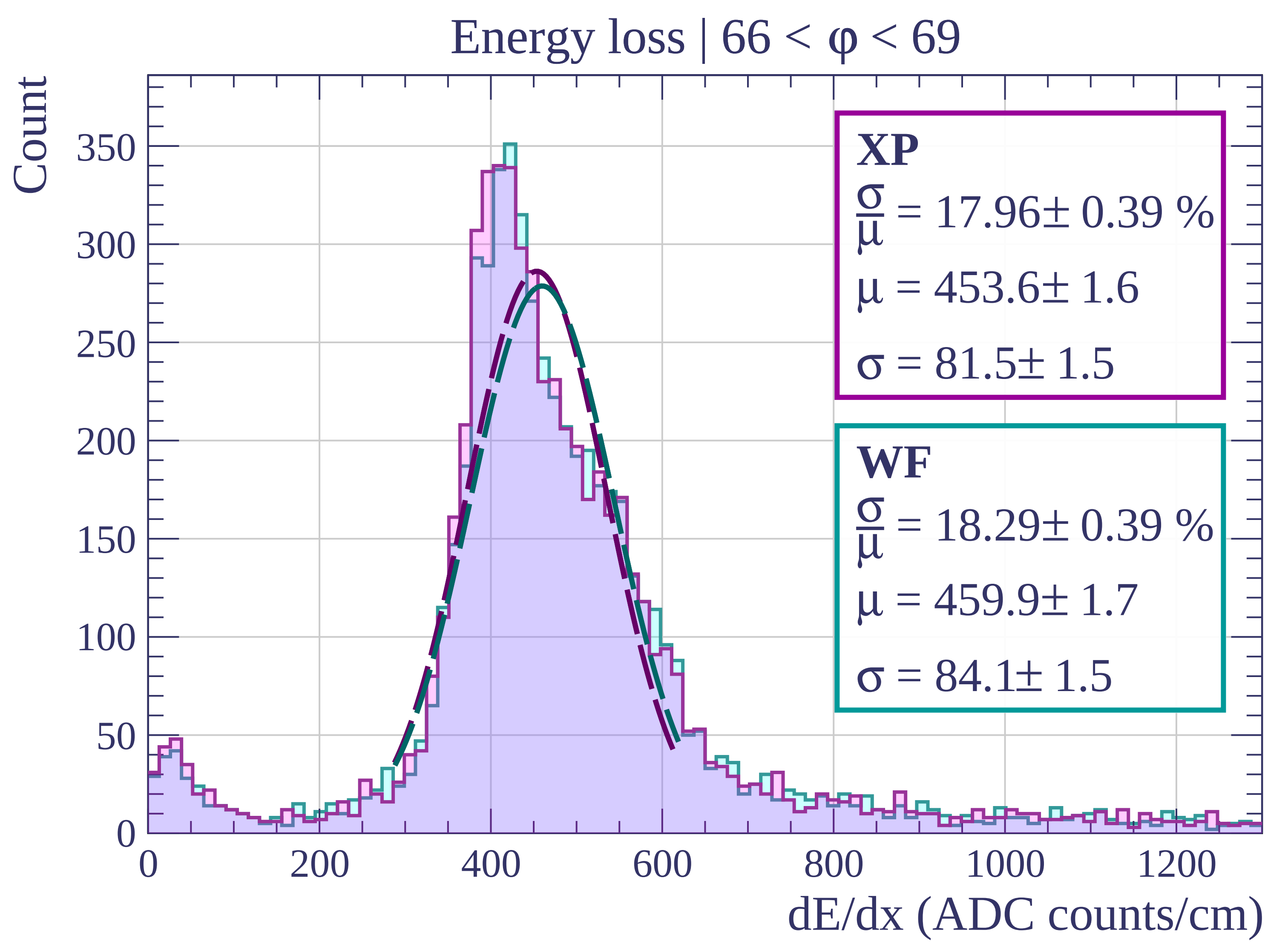


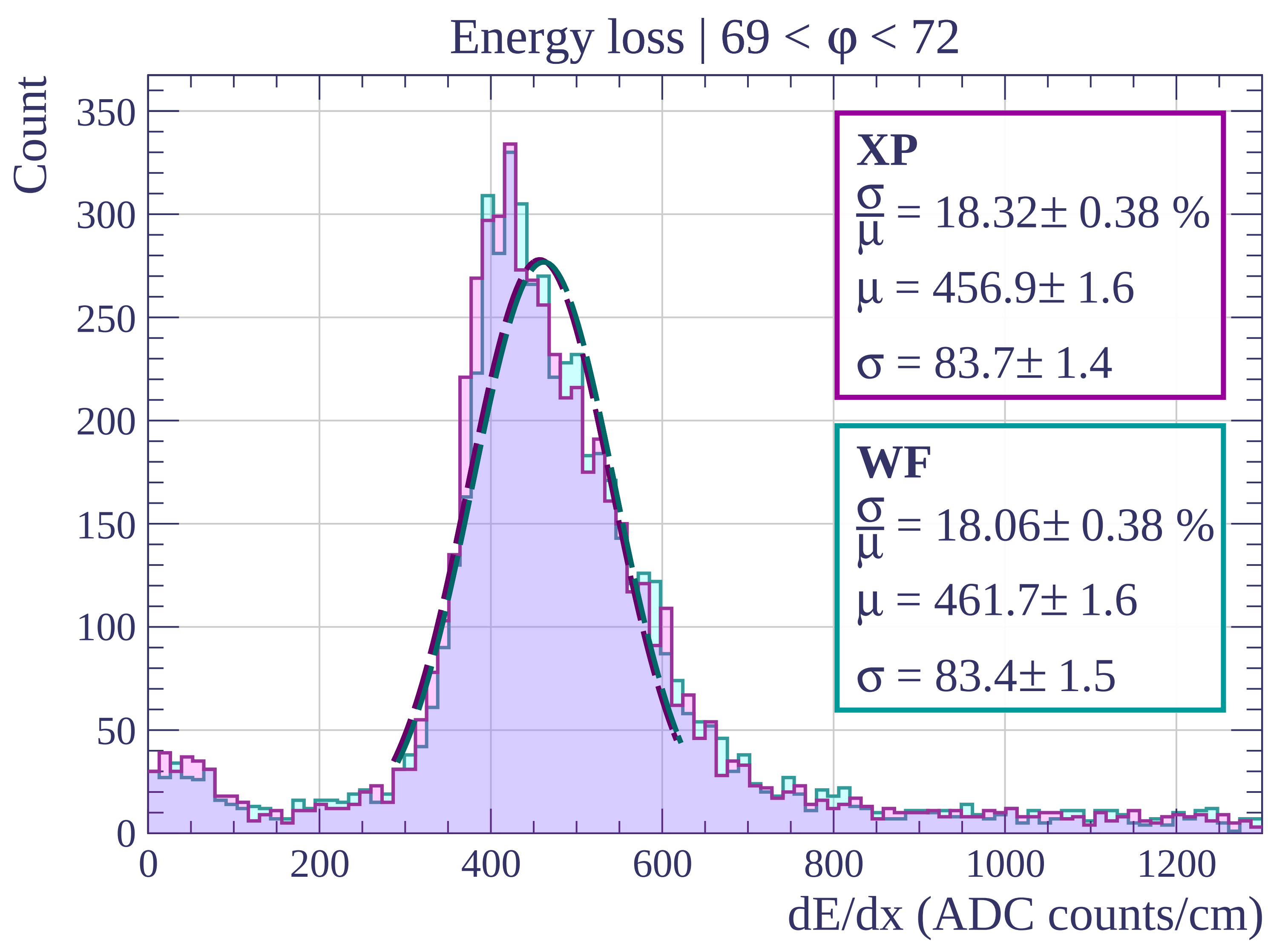
Energy loss | 57 < φ < 60
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
0
200
400
600
800
1000
1200
XP
σ/μ = 17.56± 0.32 %
μ = 455.5± 1.4
σ = 80.0± 1.2
WF
σ/μ = 18.19± 0.33 %
μ = 453.3± 1.4
σ = 82.4± 1.3



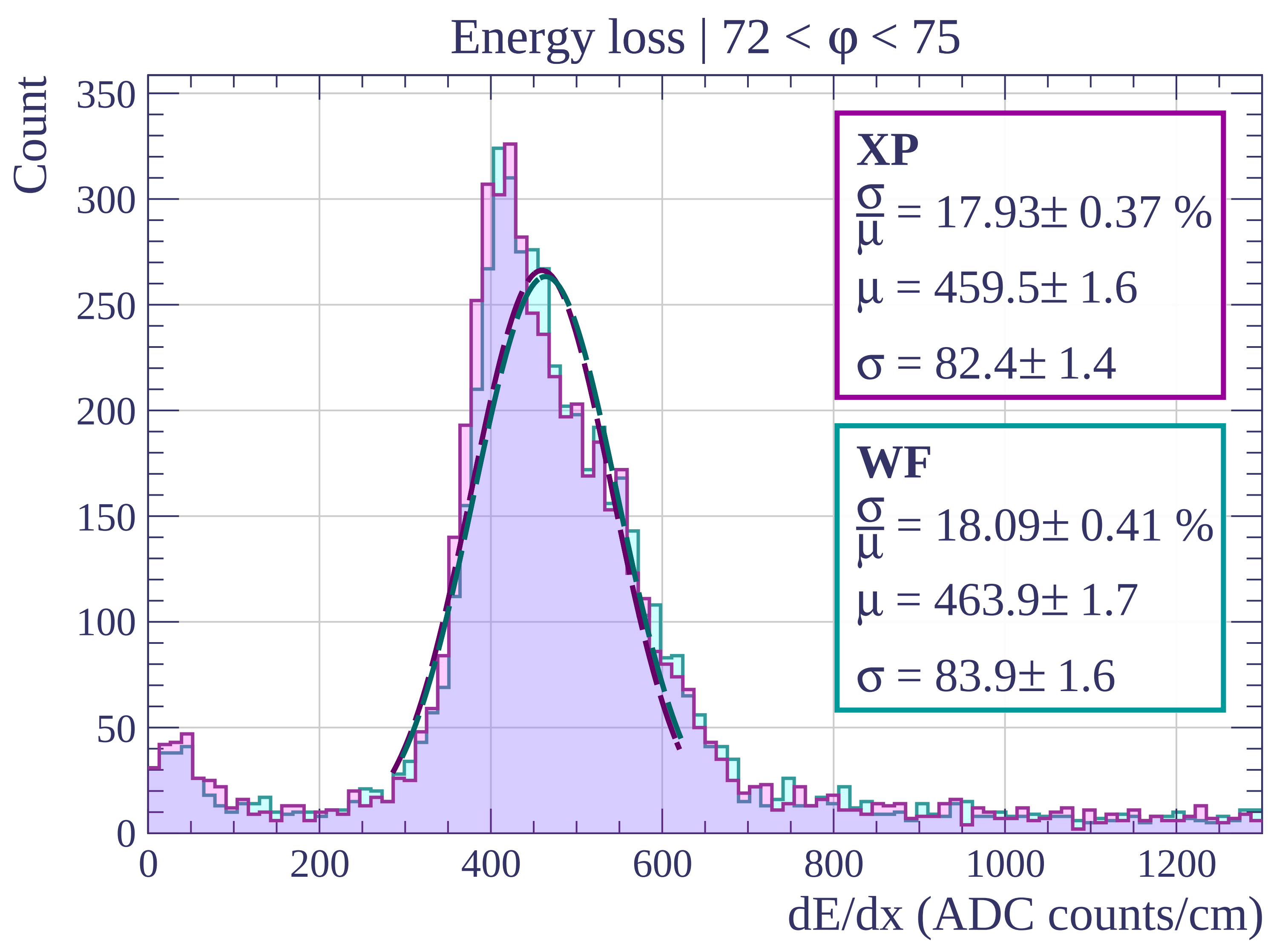
Energy loss | 60 < φ < 63
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
400
450
0
200
400
600
800
1000
1200
XP
σ/μ = 17.37± 0.36 %
μ = 450.8± 1.5
σ = 78.3± 1.4
WF
σ/μ = 18.14± 0.37 %
μ = 452.1± 1.5
σ = 82.0± 1.4

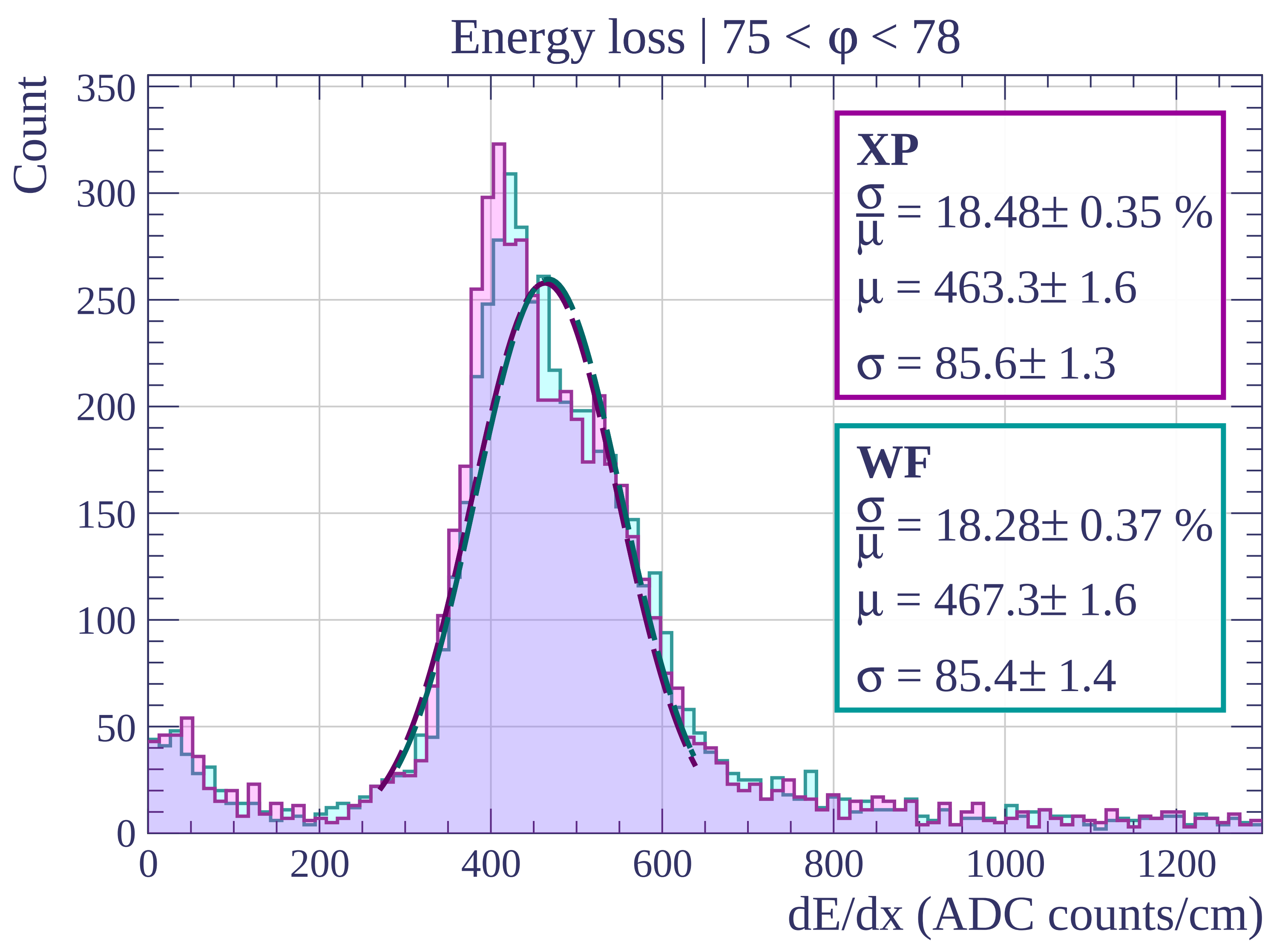

Energy loss | 63 < φ < 66
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
400
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 17.80± 0.38 %
μ = 455.9± 1.6
σ = 81.1± 1.4
WF
$\frac{\sigma}{\mu}$ = 18.23± 0.38 %
μ = 457.8± 1.6
σ = 83.4± 1.5


Energy loss | 66 < φ < 69
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 17.96± 0.39 %
μ = 453.6± 1.6
σ = 81.5± 1.5
WF
σ/μ = 18.29± 0.39 %
μ = 459.9± 1.7
σ = 84.1± 1.5
0
50
100
150
200
250
300
350
0
200
400
600
800
1000
1200

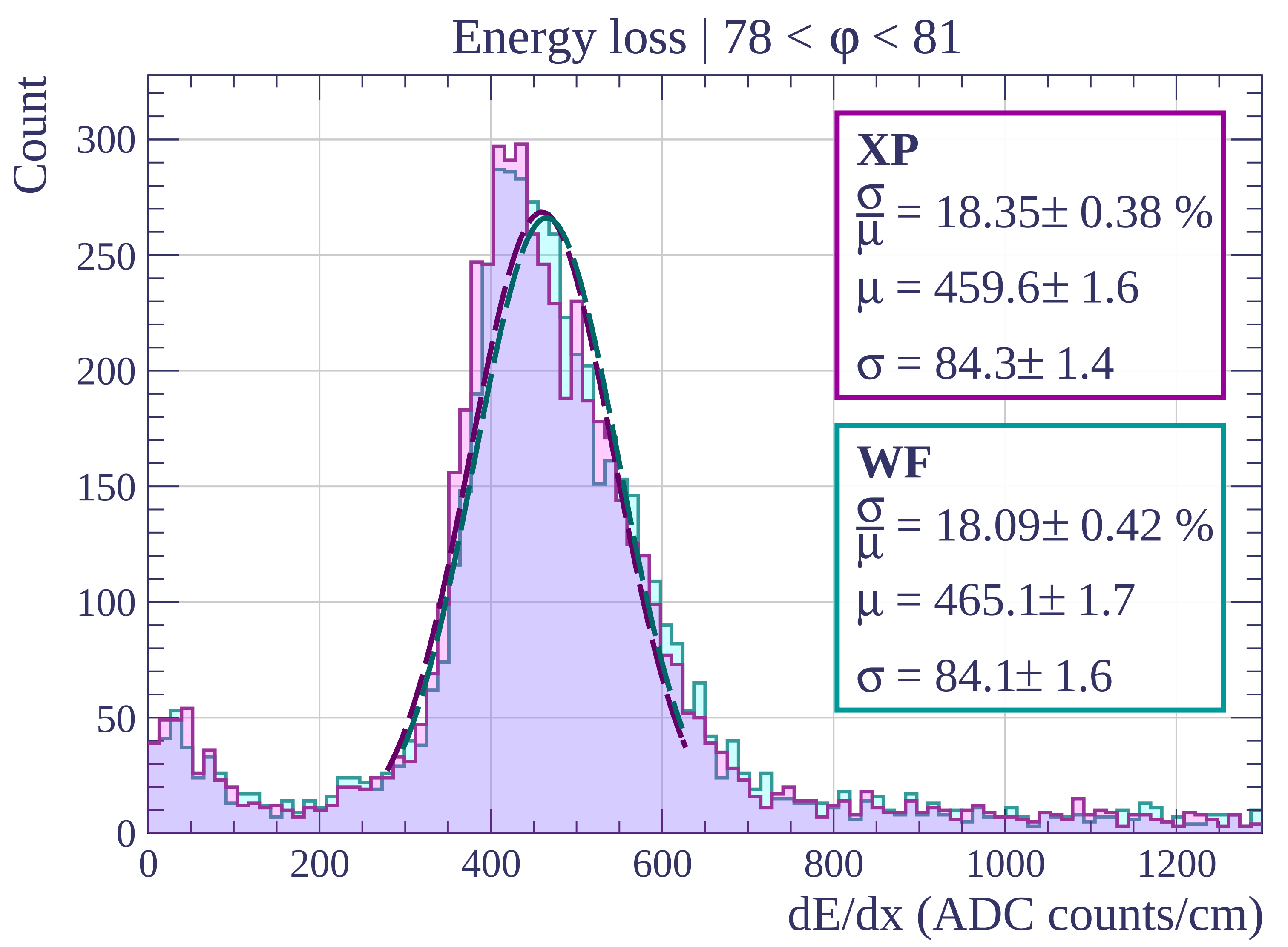


Energy loss | 69 < φ < 72
Count
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 18.32± 0.38 %
μ = 456.9± 1.6
σ = 83.7± 1.4
WF
$\frac{\sigma}{\mu}$ = 18.06± 0.38 %
μ = 461.7± 1.6
σ = 83.4± 1.5

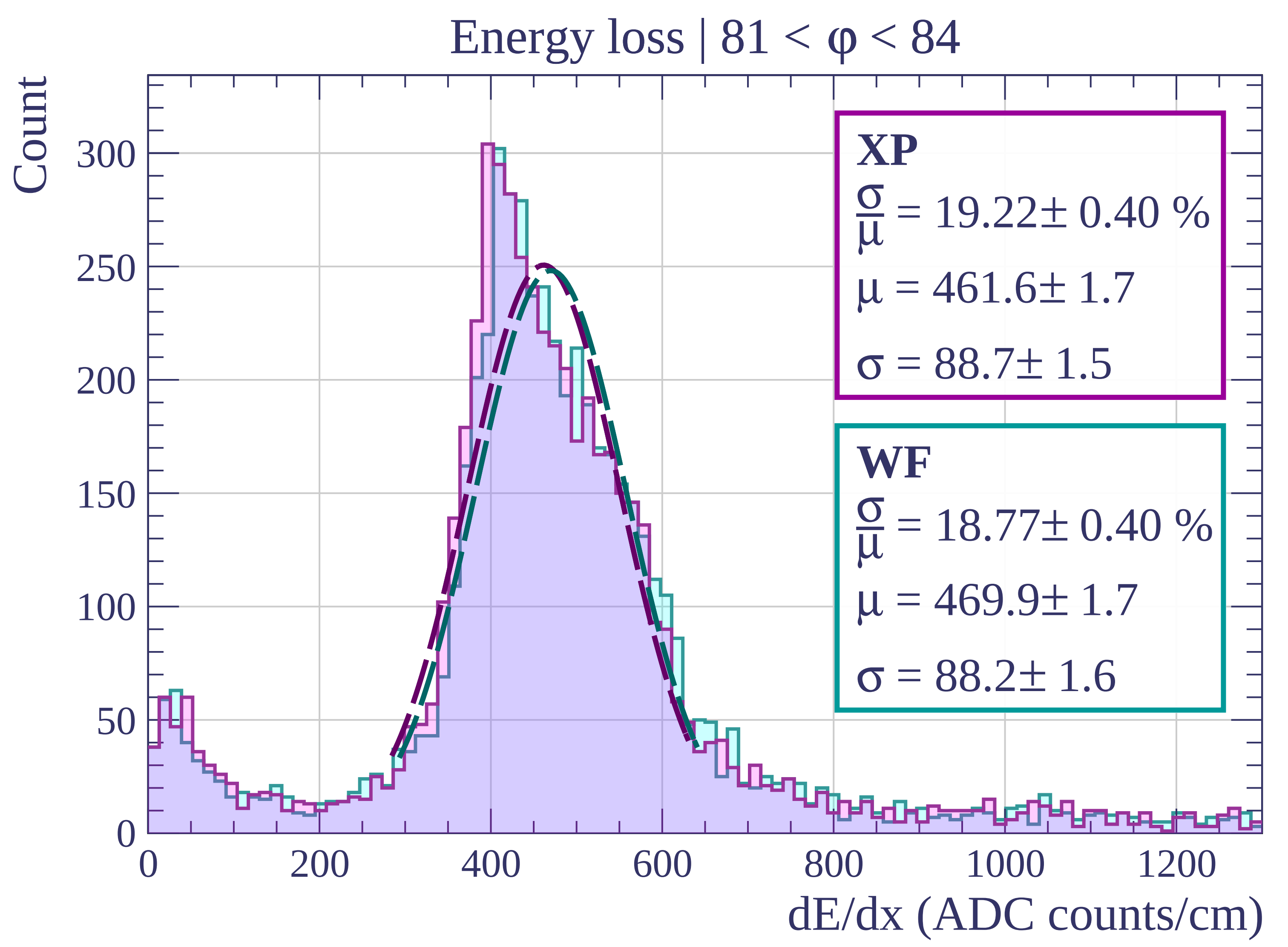

Energy loss | 72 < φ < 75
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
0
200
400
600
800
1000
1200
XP
σ/μ = 17.93± 0.37 %
μ = 459.5± 1.6
σ = 82.4± 1.4
WF
σ/μ = 18.09± 0.41 %
μ = 463.9± 1.7
σ = 83.9± 1.6

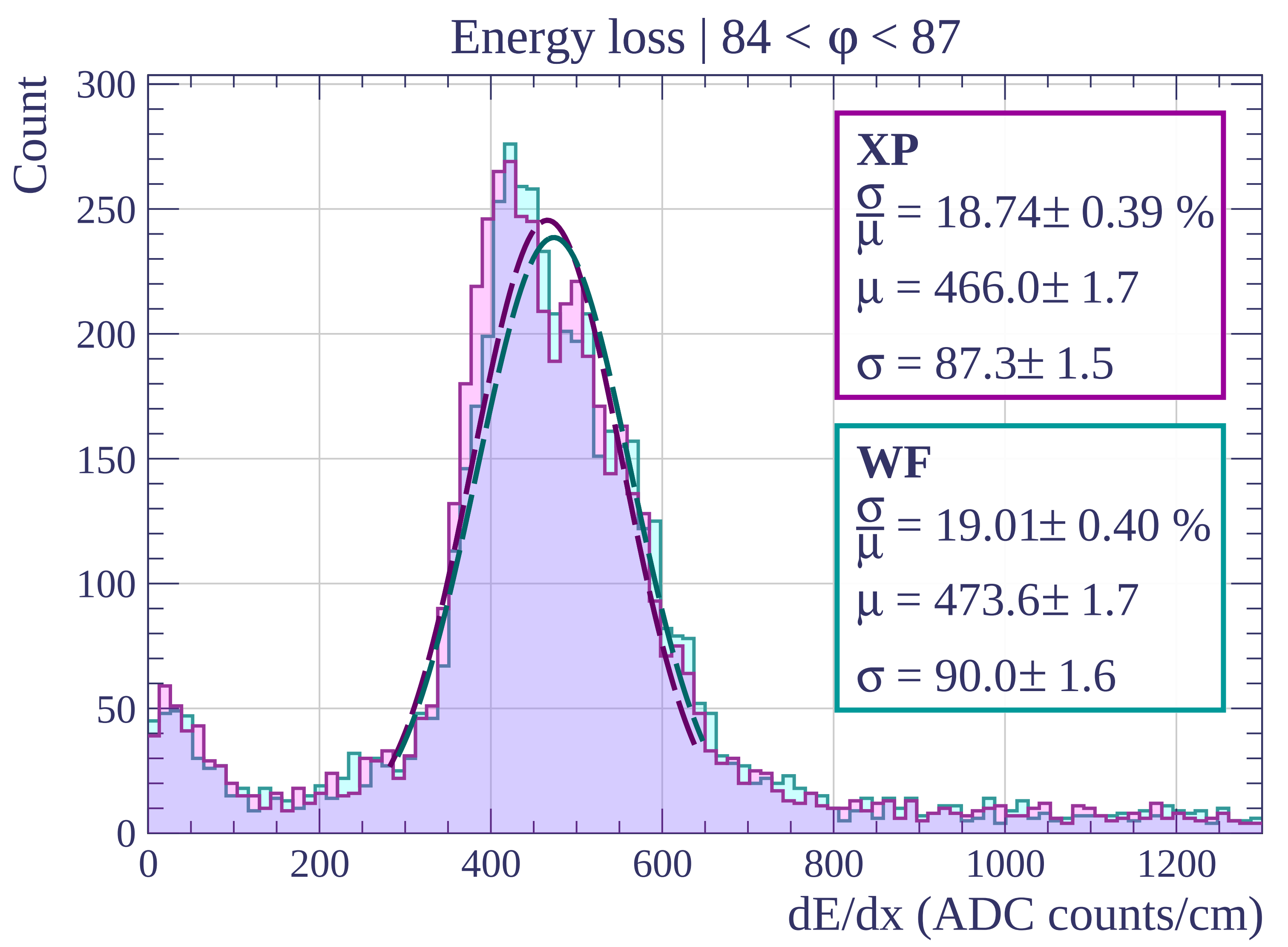


Energy loss | 75 < φ < 78
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 18.48± 0.35 %
μ = 463.3± 1.6
σ = 85.6± 1.3
WF
σ/μ = 18.28± 0.37 %
μ = 467.3± 1.6
σ = 85.4± 1.4
0
50
100
150
200
250
300
350
0
200
400
600
800
1000
1200

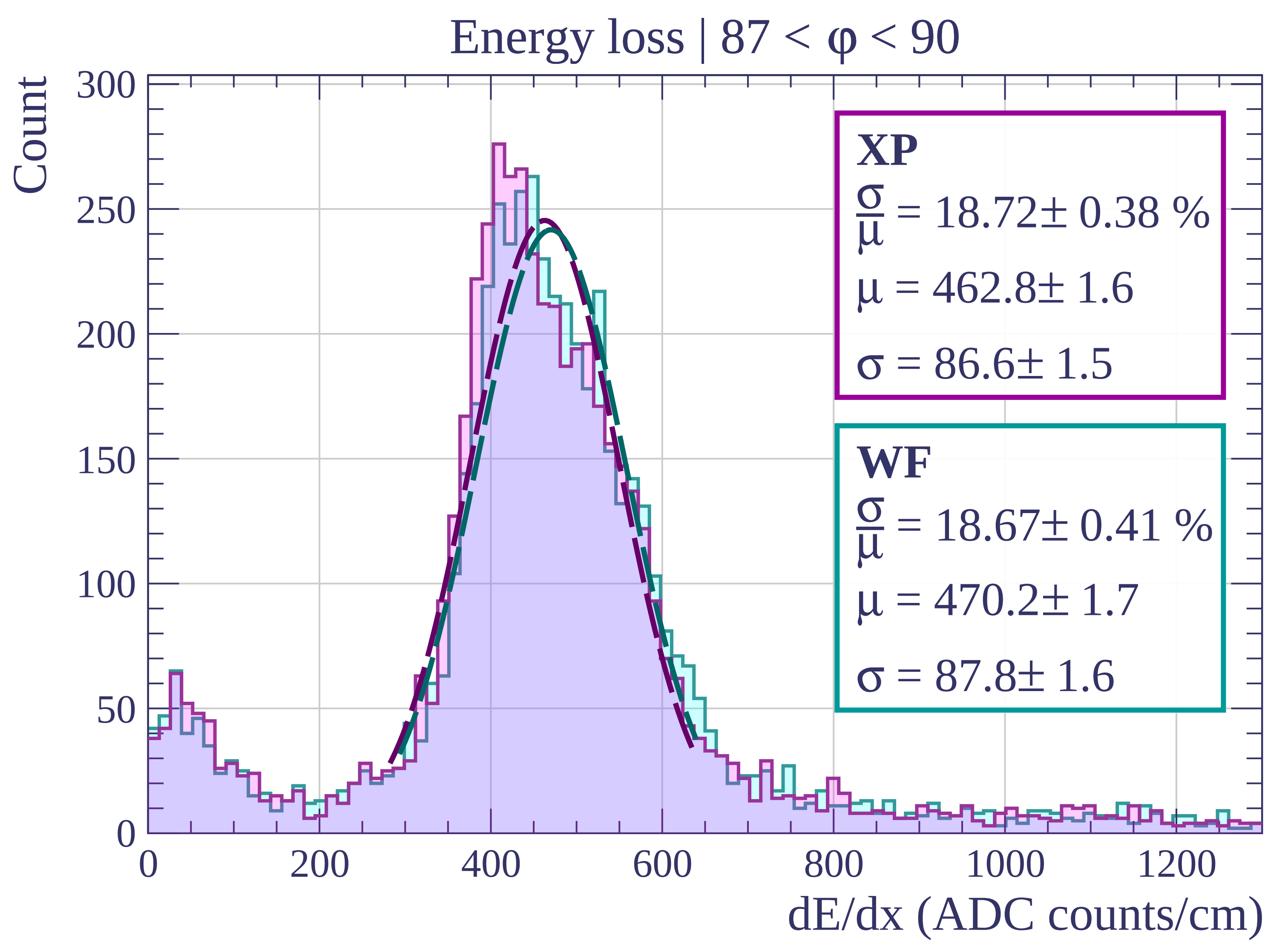


Energy loss | 78 < φ < 81
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
0
200
400
600
800
1000
1200
XP
σ/μ = 18.35± 0.38 %
μ = 459.6± 1.6
σ = 84.3± 1.4
WF
σ/μ = 18.09± 0.42 %
μ = 465.1± 1.7
σ = 84.1± 1.6

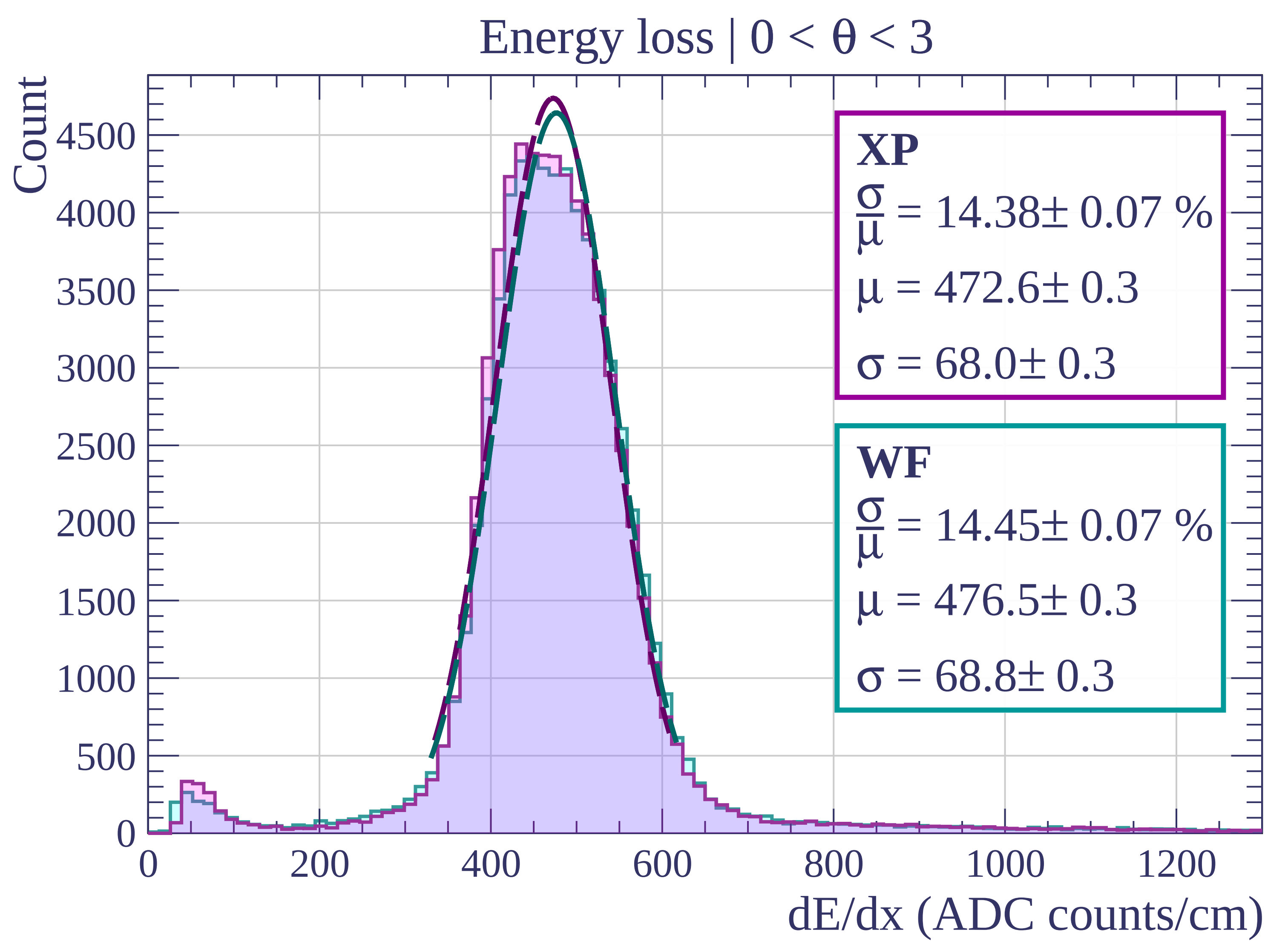

Energy loss | 81 < φ < 84
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
0
200
400
600
800
1000
1200
XP
σ/μ = 19.22± 0.40 %
μ = 461.6± 1.7
σ = 88.7± 1.5
WF
σ/μ = 18.77± 0.40 %
μ = 469.9± 1.7
σ = 88.2± 1.6



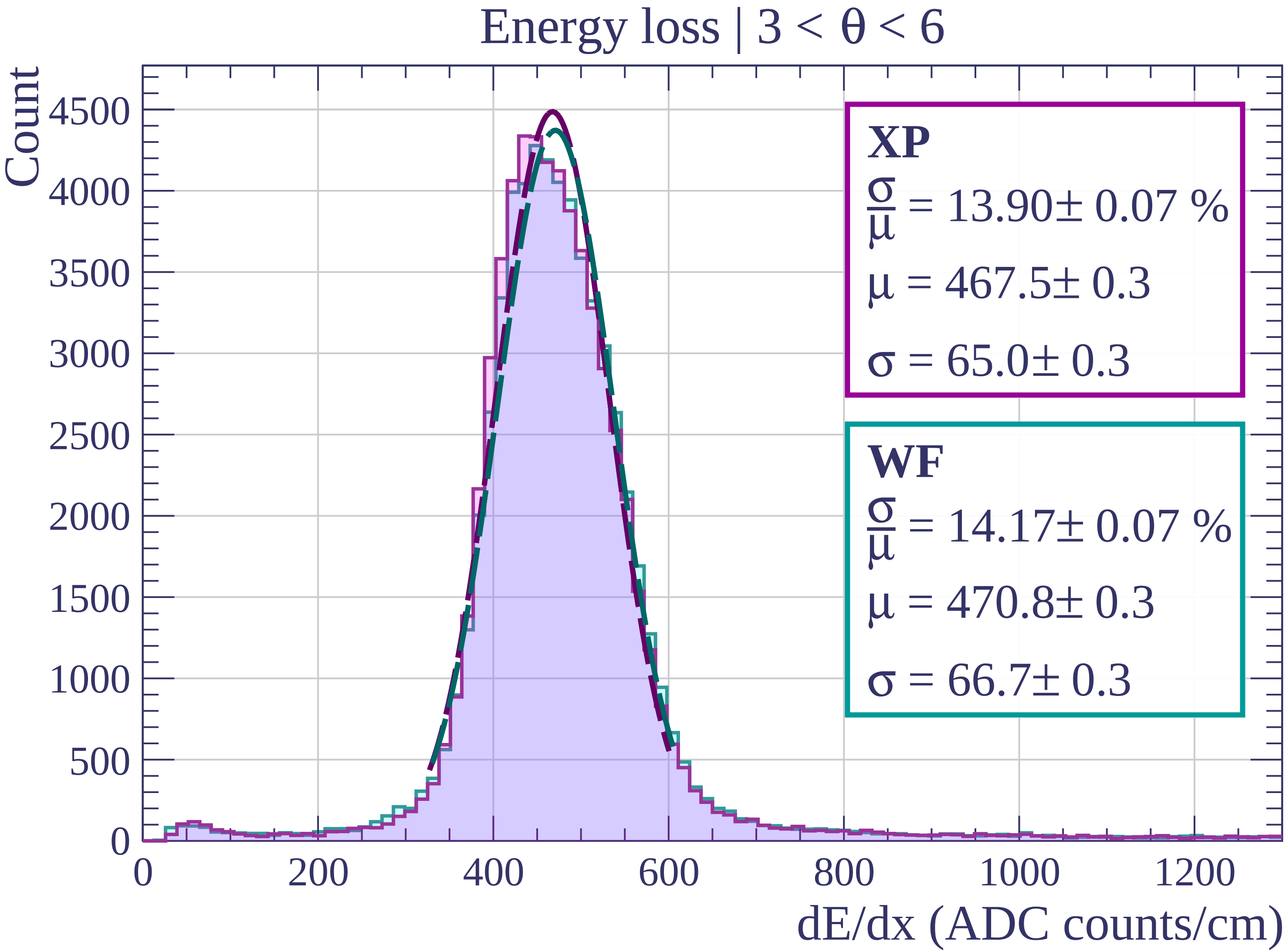
Energy loss | 84 < φ < 87
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
0
200
400
600
800
1000
1200
XP
σ/μ = 18.74± 0.39 %
μ = 466.0± 1.7
σ = 87.3± 1.5
WF
σ/μ = 19.01± 0.40 %
μ = 473.6± 1.7
σ = 90.0± 1.6



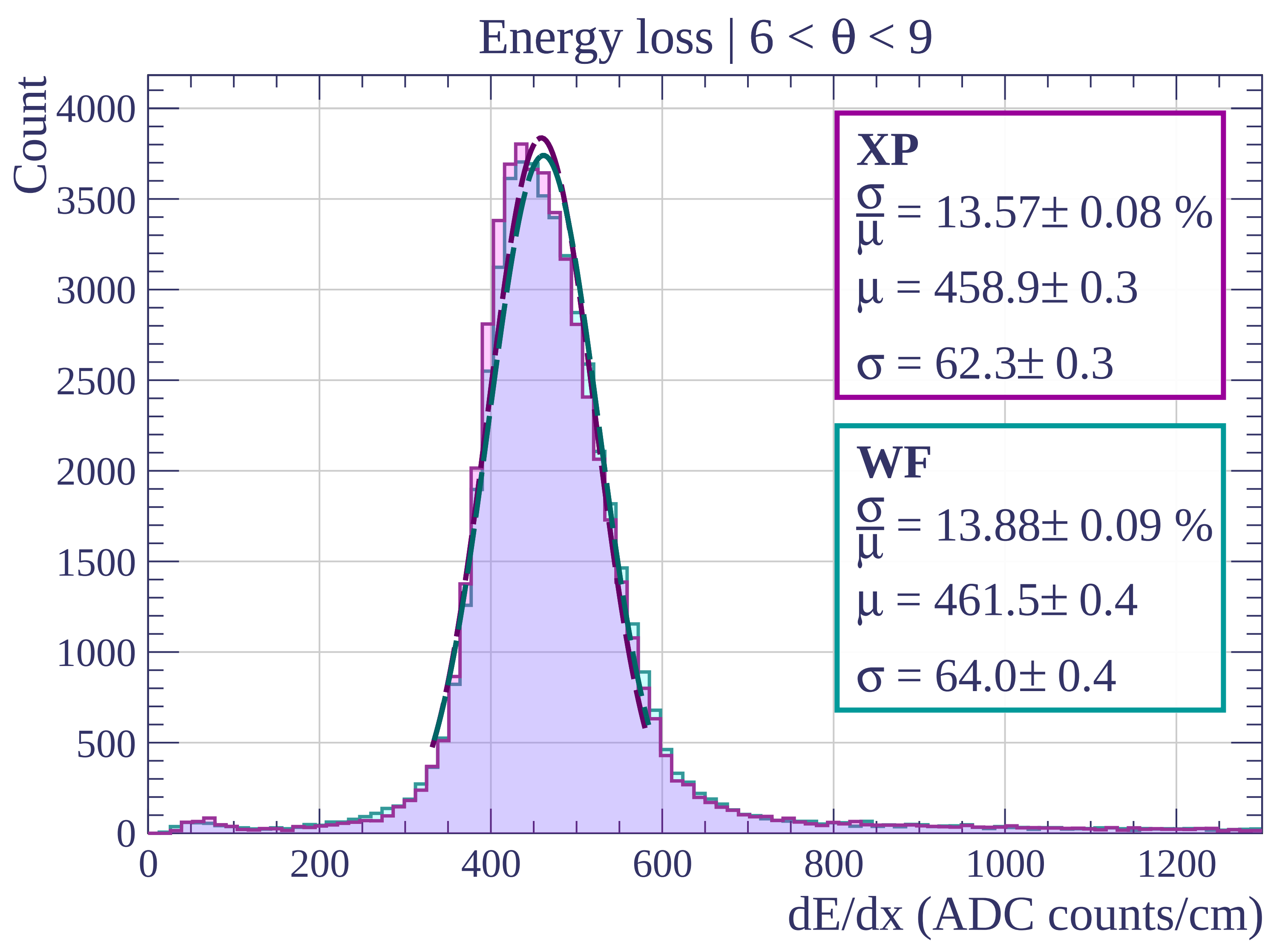
Energy loss | 87 < φ < 90
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 18.72± 0.38 %
μ = 462.8± 1.6
σ = 86.6± 1.5
WF
$\frac{\sigma}{\mu}$ = 18.67± 0.41 %
μ = 470.2± 1.7
σ = 87.8± 1.6



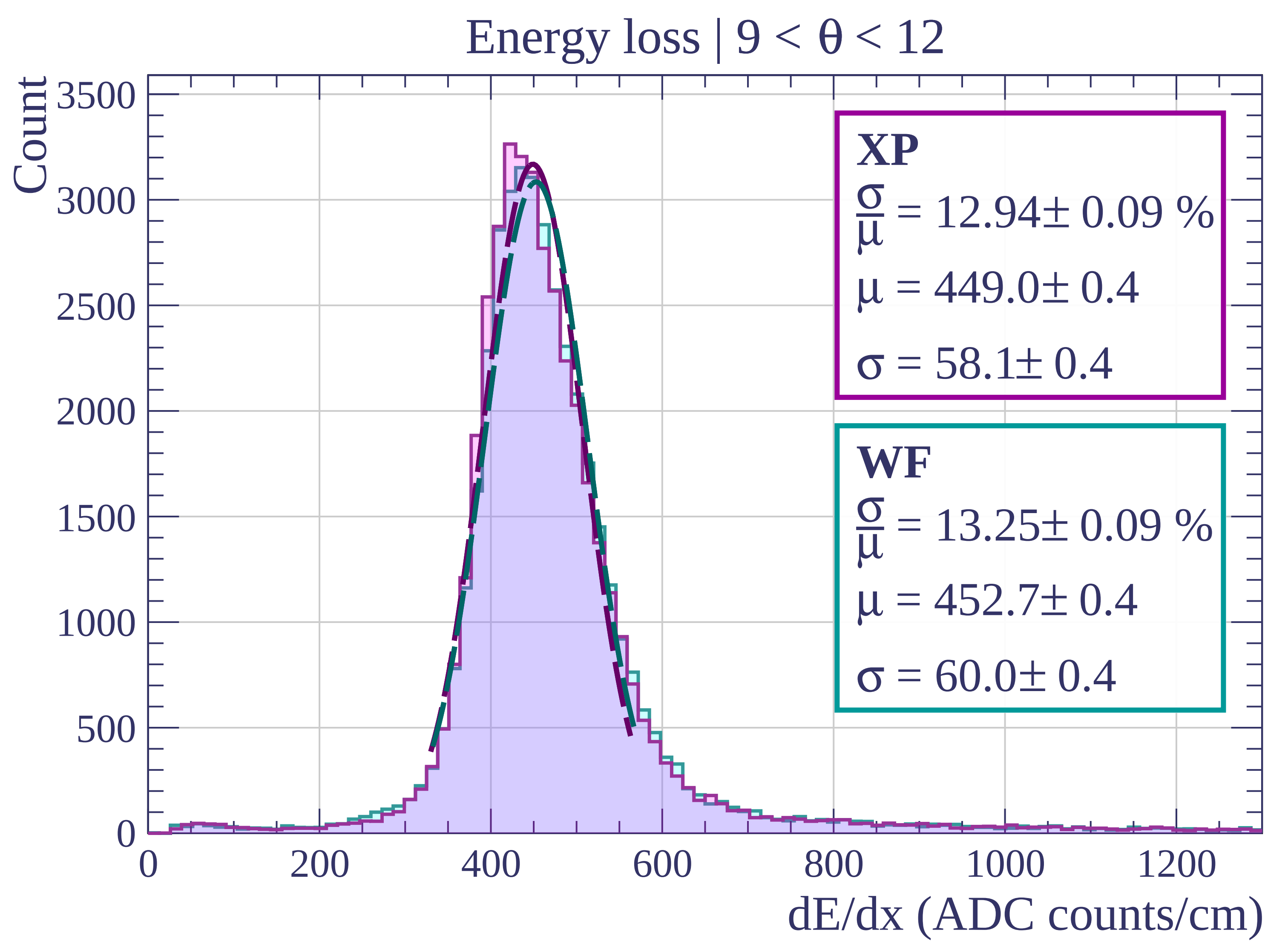
Energy loss | 0 < θ < 3
Count
dE/dx (ADC counts/cm)
0
500
1000
1500
2000
2500
3000
3500
4000
4500
0
200
400
600
800
1000
1200
XP
σ/μ = 14.38± 0.07 %
μ = 472.6± 0.3
σ = 68.0± 0.3
WF
σ/μ = 14.45± 0.07 %
μ = 476.5± 0.3
σ = 68.8± 0.3

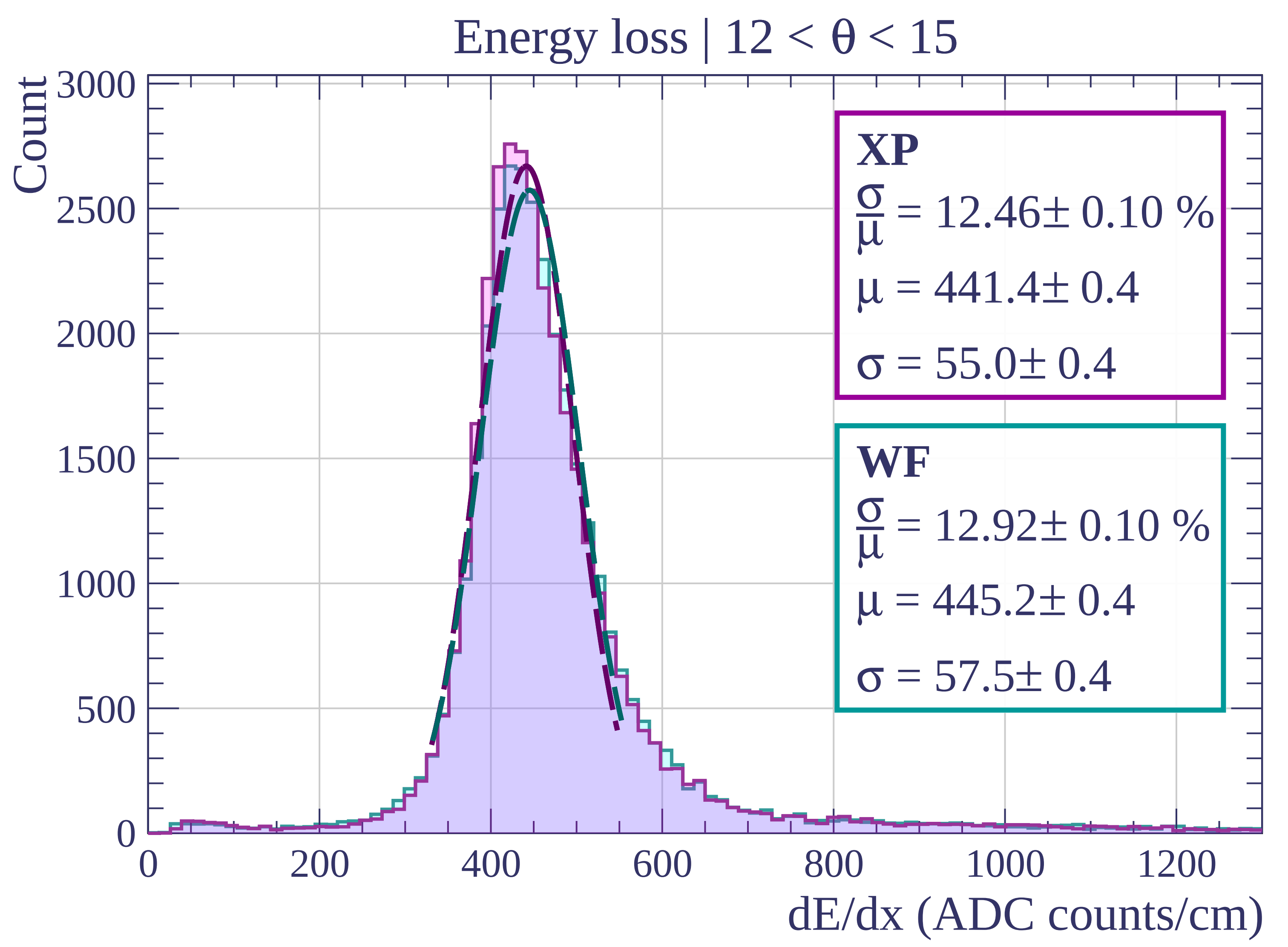


Energy loss | 3 < θ < 6
Count
4500
4000
3500
3000
2500
2000
1500
1000
500
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 13.90± 0.07 %
μ = 467.5± 0.3
σ = 65.0± 0.3
WF
σ/μ = 14.17± 0.07 %
μ = 470.8± 0.3
σ = 66.7± 0.3

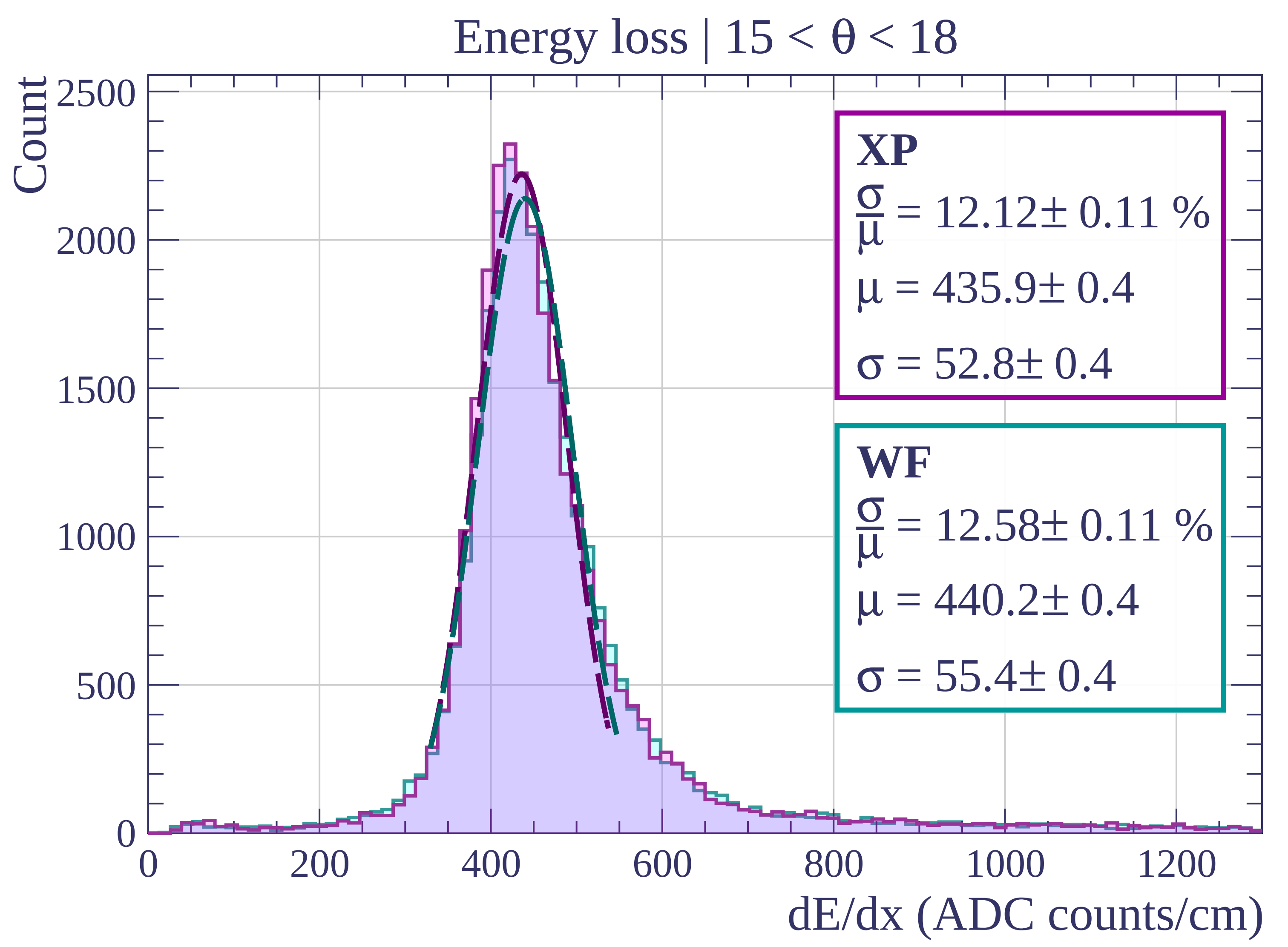


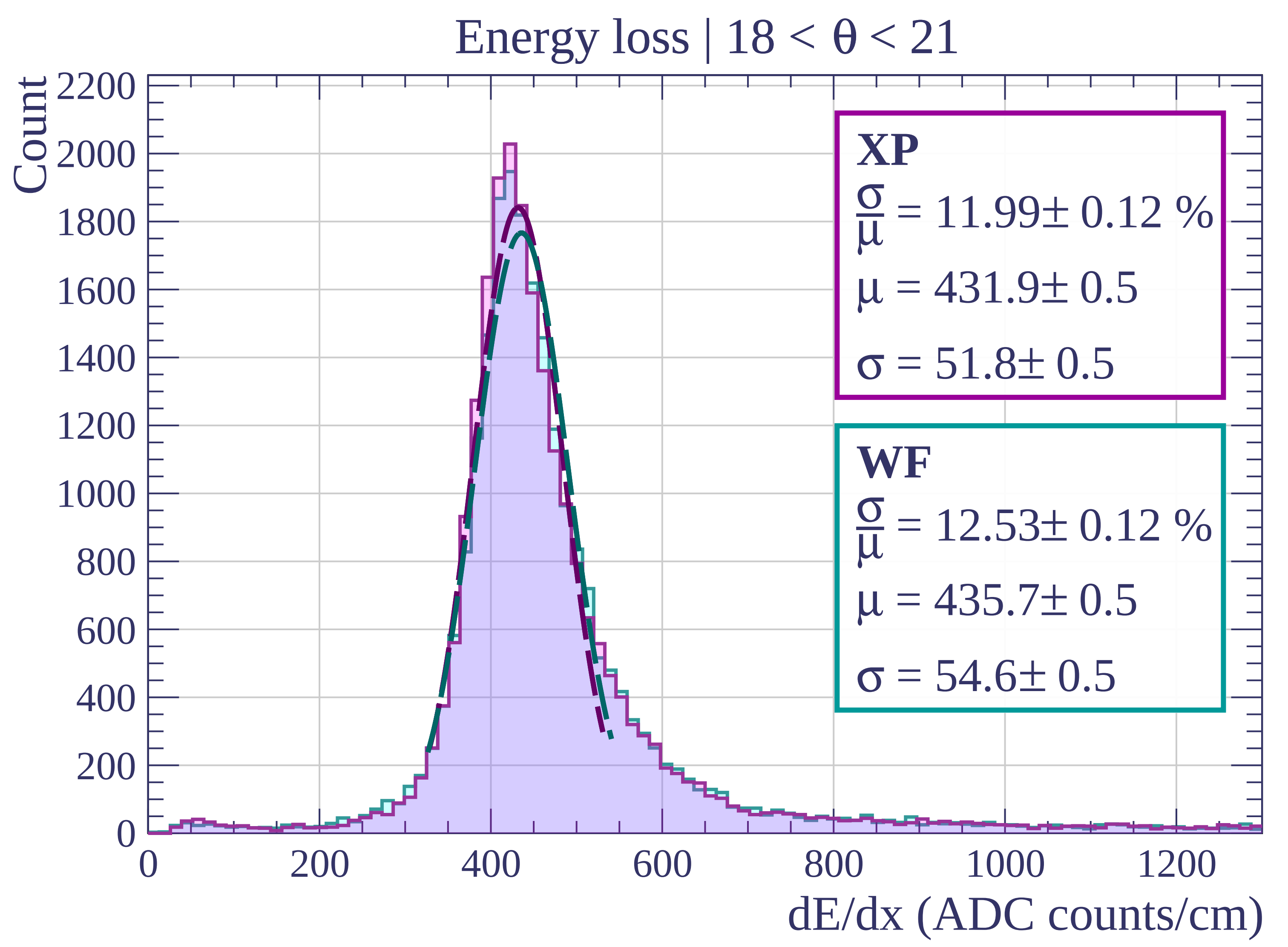
Energy loss | 6 < θ < 9
Count
4000
3500
3000
2500
2000
1500
1000
500
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 13.57± 0.08 %
μ = 458.9± 0.3
σ = 62.3± 0.3
WF
σ/μ = 13.88± 0.09 %
μ = 461.5± 0.4
σ = 64.0± 0.4

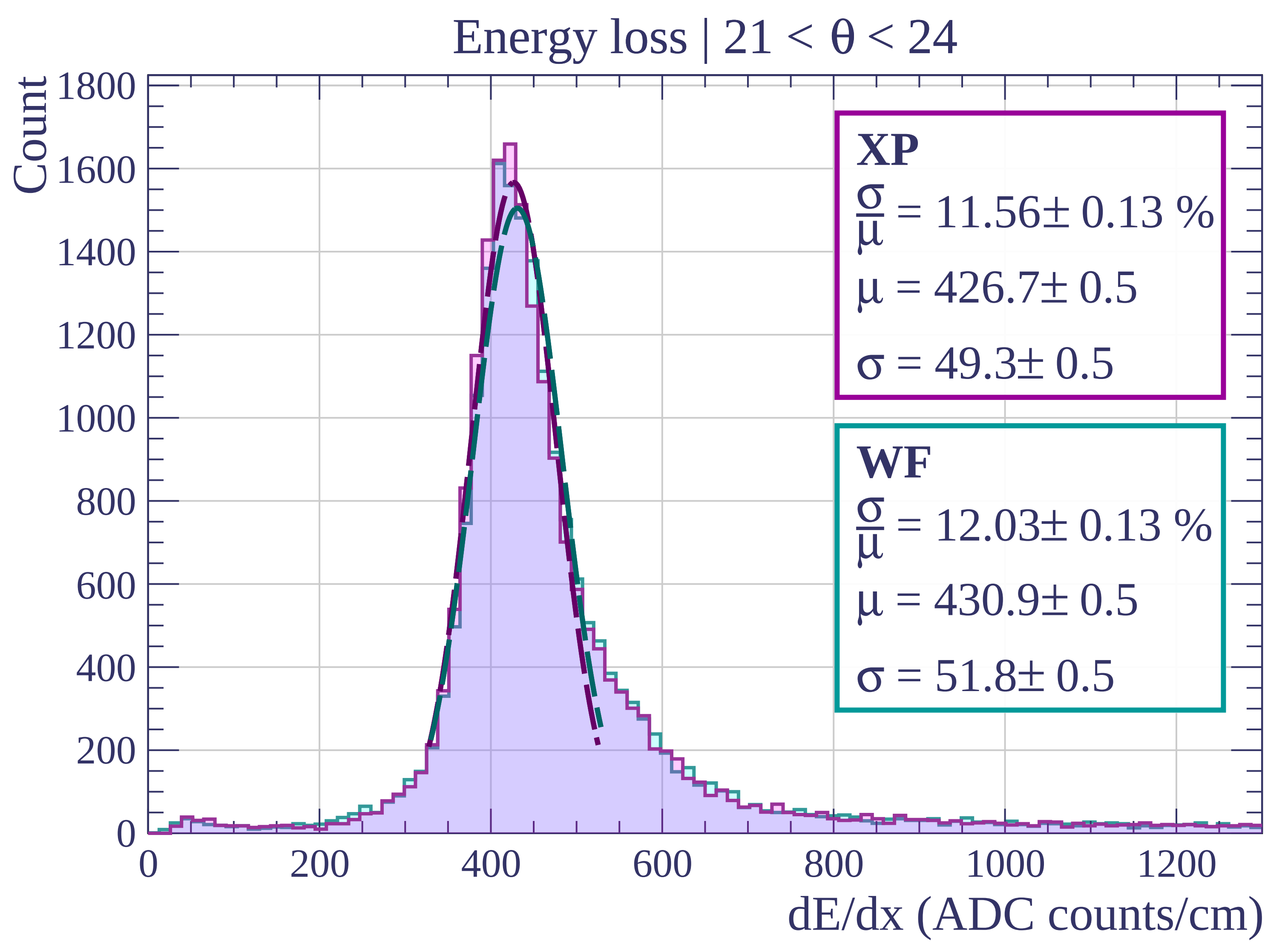

Energy loss | 9 < θ < 12
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
500
1000
1500
2000
2500
3000
3500
XP
$\frac{\sigma}{\mu}$ = 12.94± 0.09 %
μ = 449.0± 0.4
σ = 58.1± 0.4
WF
$\frac{\sigma}{\mu}$ = 13.25± 0.09 %
μ = 452.7± 0.4
σ = 60.0± 0.4

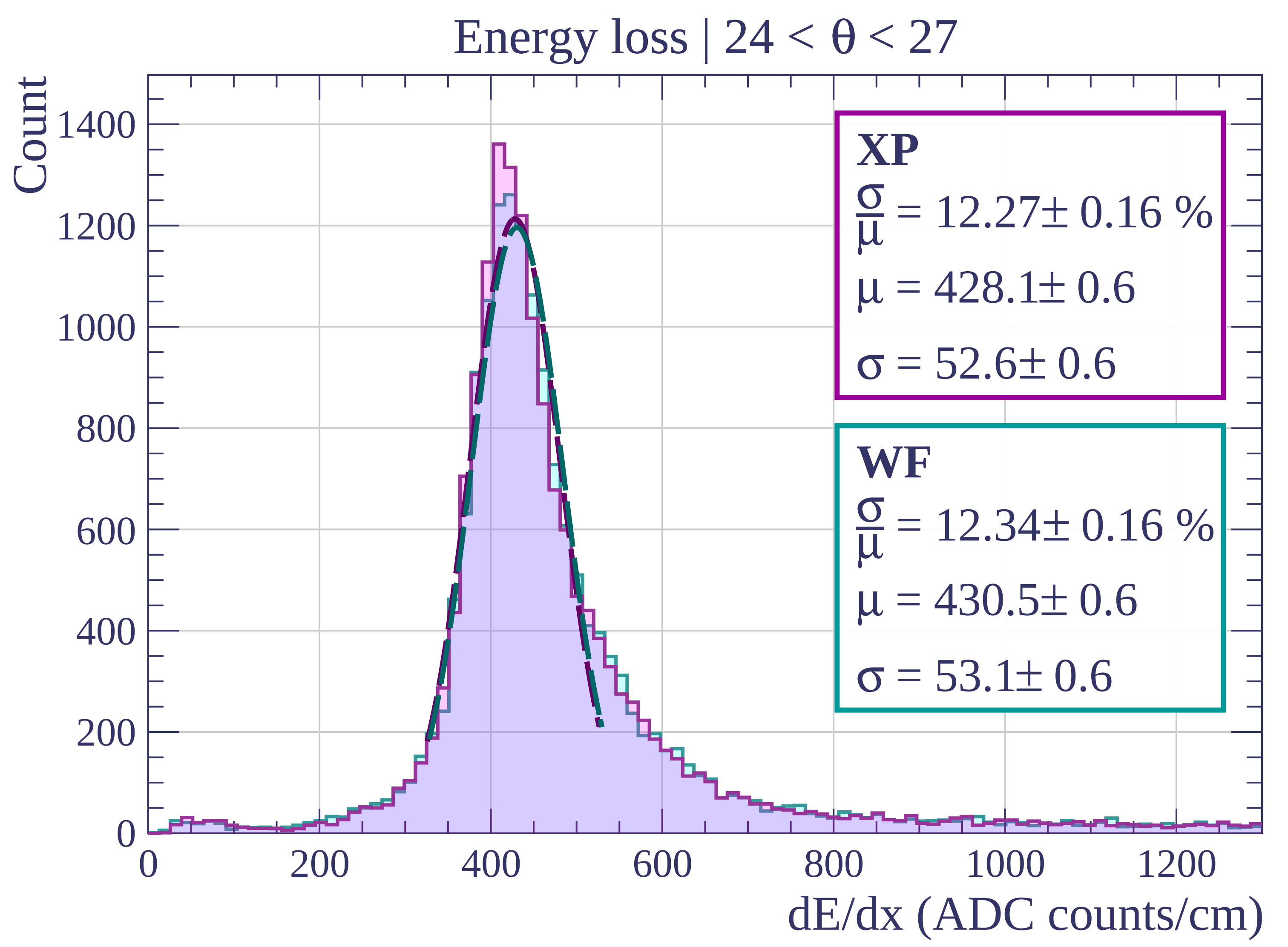


Energy loss | 12 < θ < 15
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 12.46± 0.10 %
μ = 441.4± 0.4
σ = 55.0± 0.4
WF
σ/μ = 12.92± 0.10 %
μ = 445.2± 0.4
σ = 57.5± 0.4

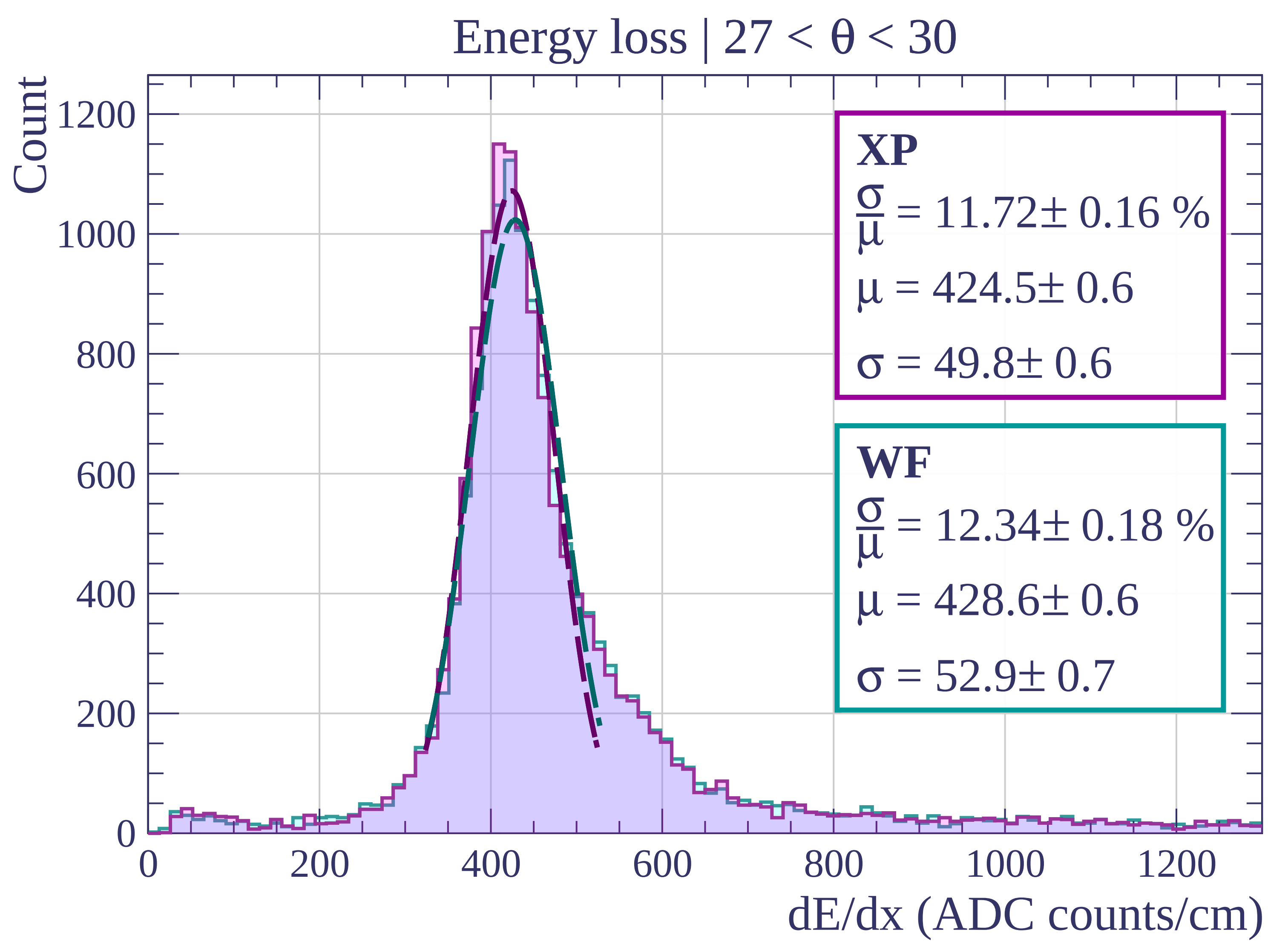

Energy loss | 15 < θ < 18
Count
2500
2000
1500
1000
500
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 12.12± 0.11 %
μ = 435.9± 0.4
σ = 52.8± 0.4
WF
σ/μ = 12.58± 0.11 %
μ = 440.2± 0.4
σ = 55.4± 0.4


Energy loss | 18 < θ < 21
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
1400
1600
1800
2000
2200
0
200
400
600
800
1000
1200
XP
σ/μ = 11.99± 0.12 %
μ = 431.9± 0.5
σ = 51.8± 0.5
WF
σ/μ = 12.53± 0.12 %
μ = 435.7± 0.5
σ = 54.6± 0.5

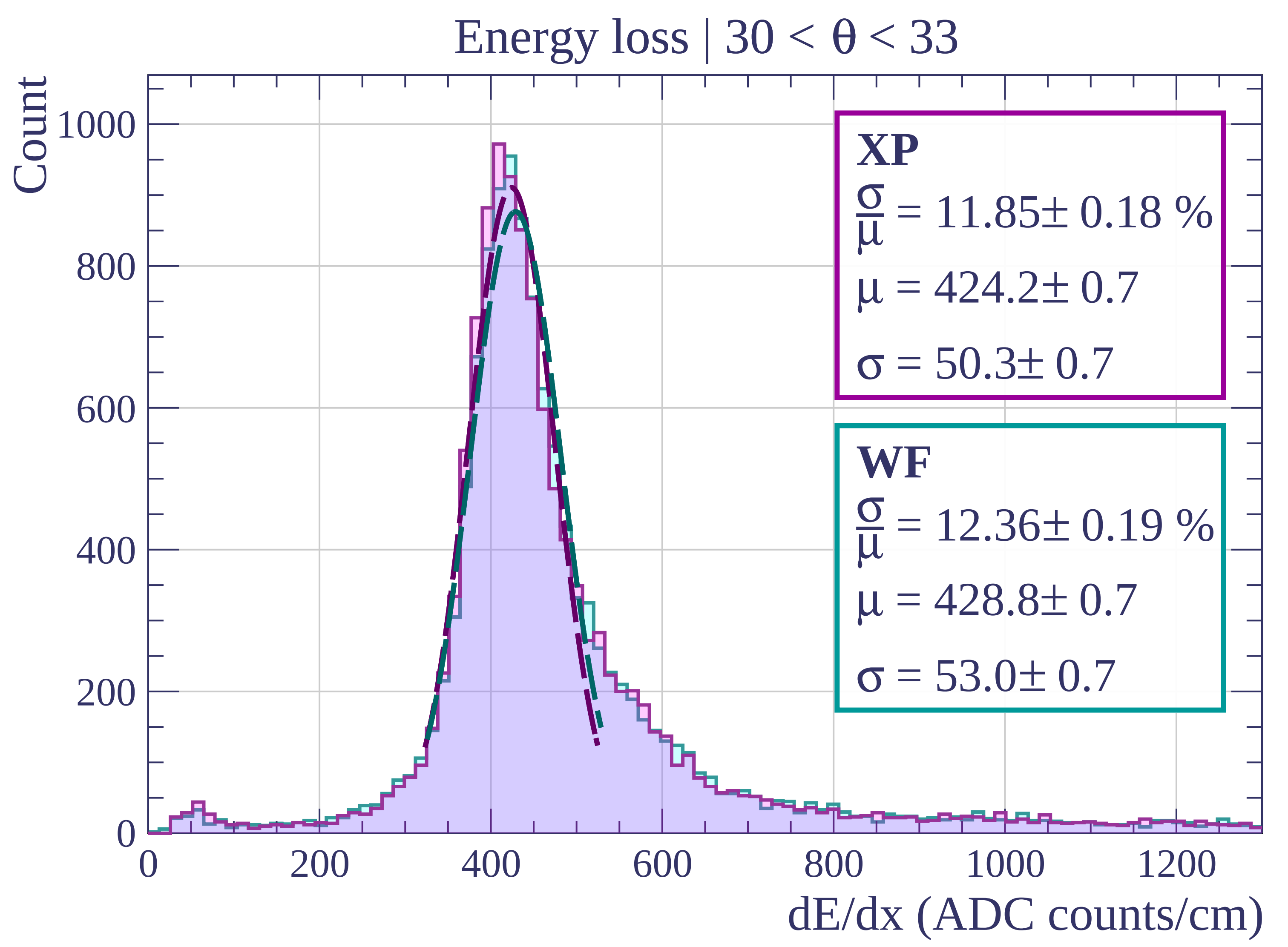

Energy loss | 21 < θ < 24
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
1400
1600
1800
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 11.56± 0.13 %
μ = 426.7± 0.5
σ = 49.3± 0.5
WF
$\frac{\sigma}{\mu}$ = 12.03± 0.13 %
μ = 430.9± 0.5
σ = 51.8± 0.5



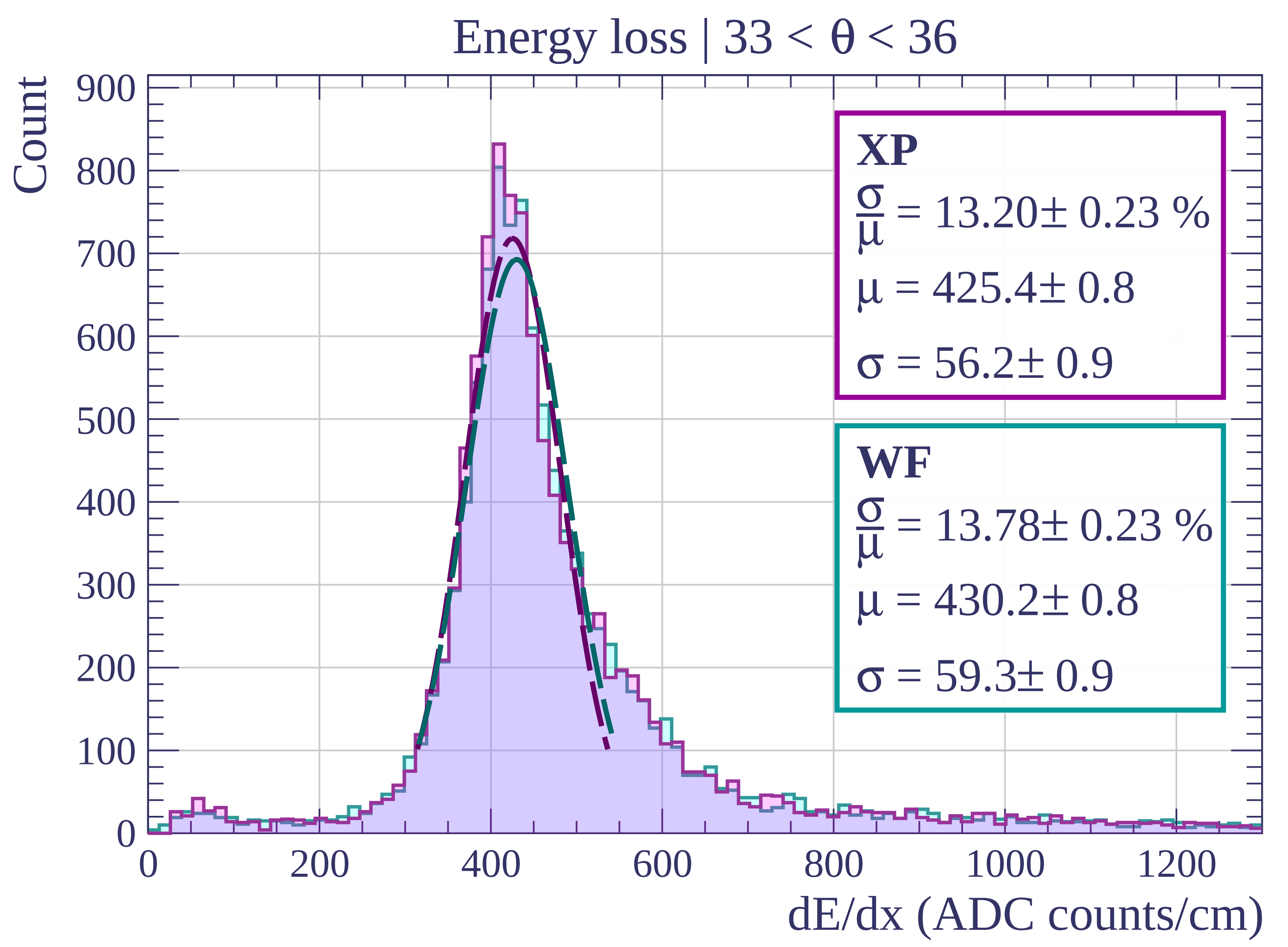
Energy loss | 24 < θ < 27
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 12.27± 0.16 %
μ = 428.1± 0.6
σ = 52.6± 0.6
WF
σ/μ = 12.34± 0.16 %
μ = 430.5± 0.6
σ = 53.1± 0.6

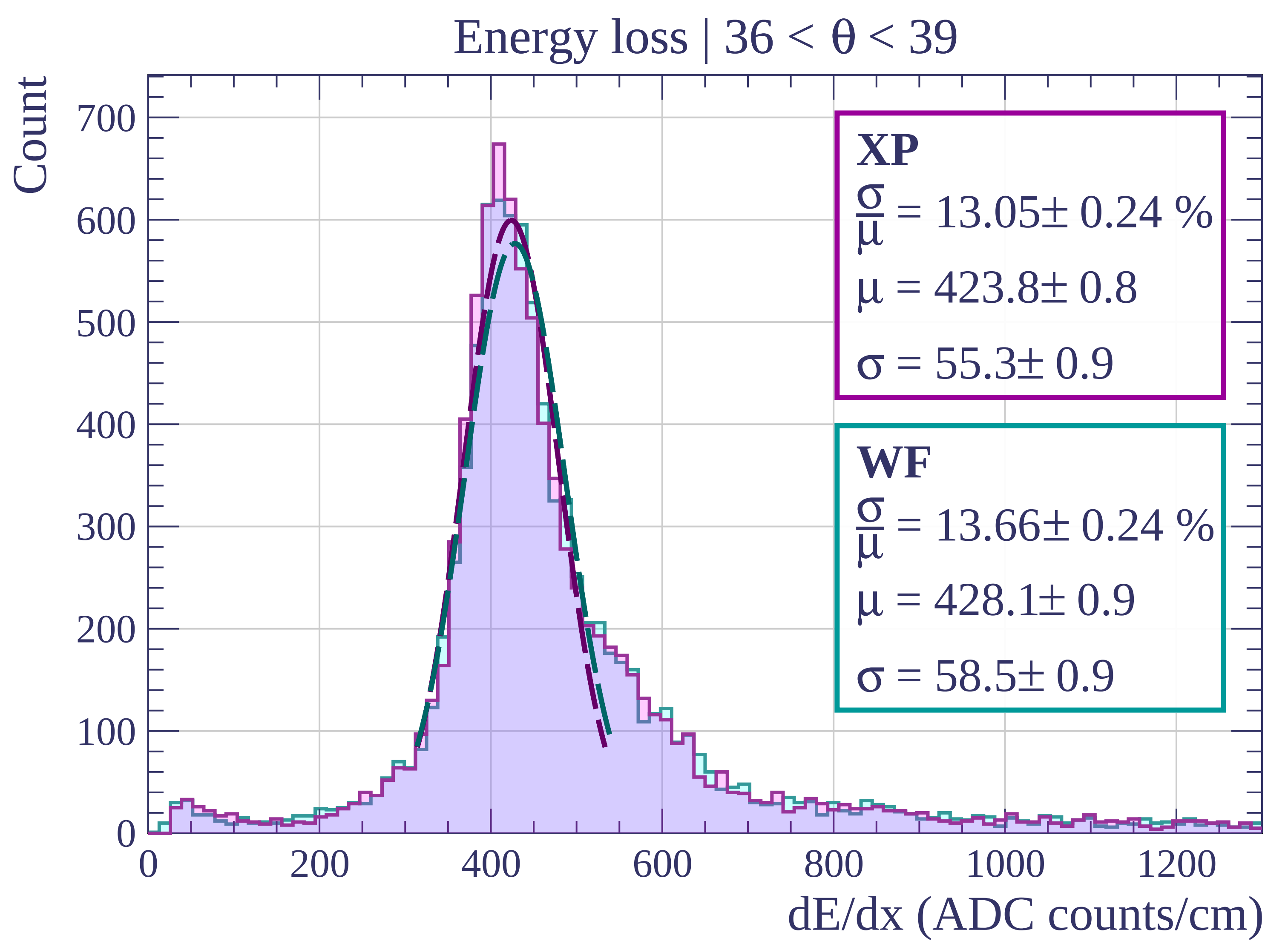


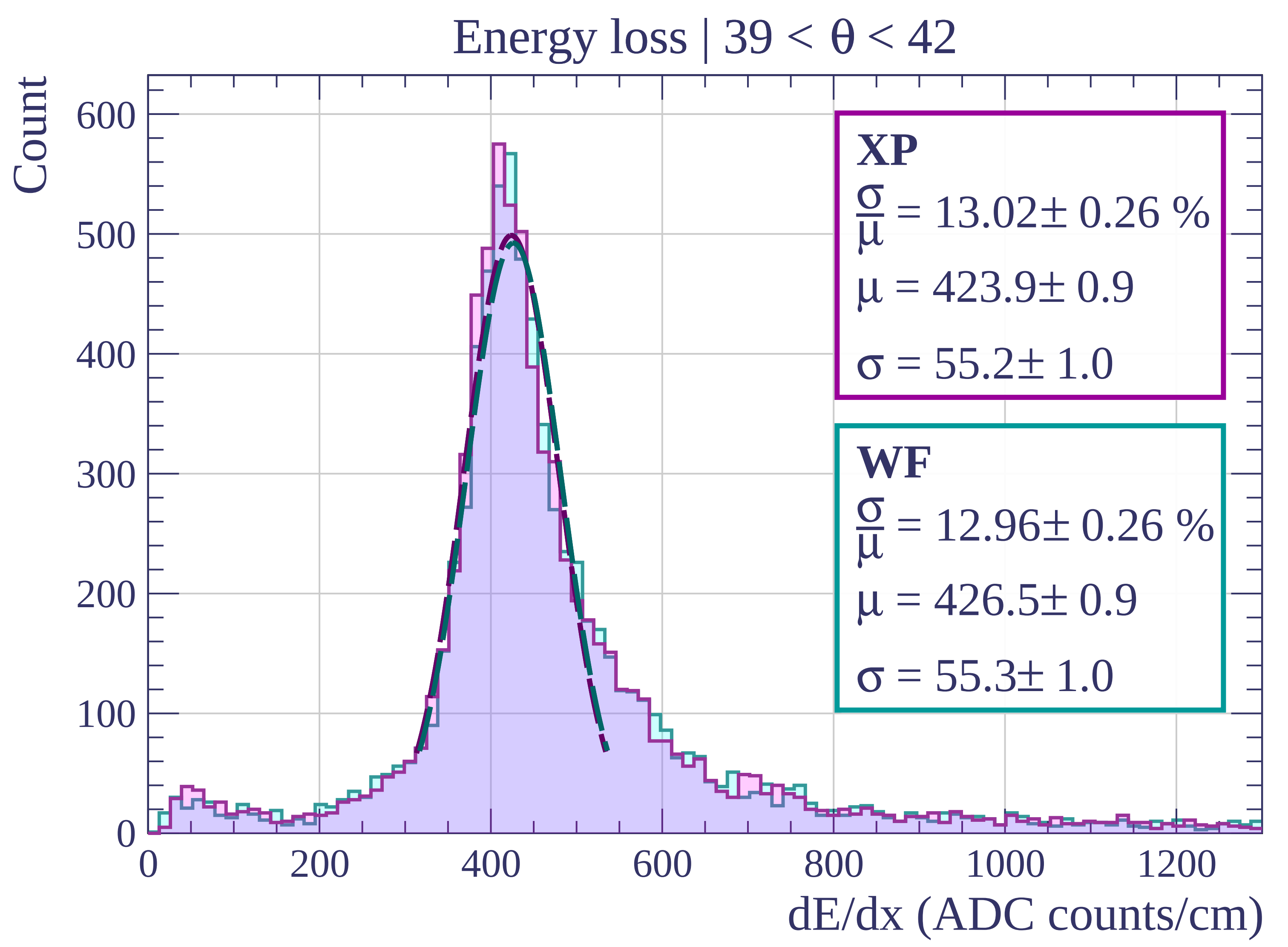
Energy loss | 27 < θ < 30
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 11.72± 0.16 %
μ = 424.5± 0.6
σ = 49.8± 0.6
WF
σ/μ = 12.34± 0.18 %
μ = 428.6± 0.6
σ = 52.9± 0.7



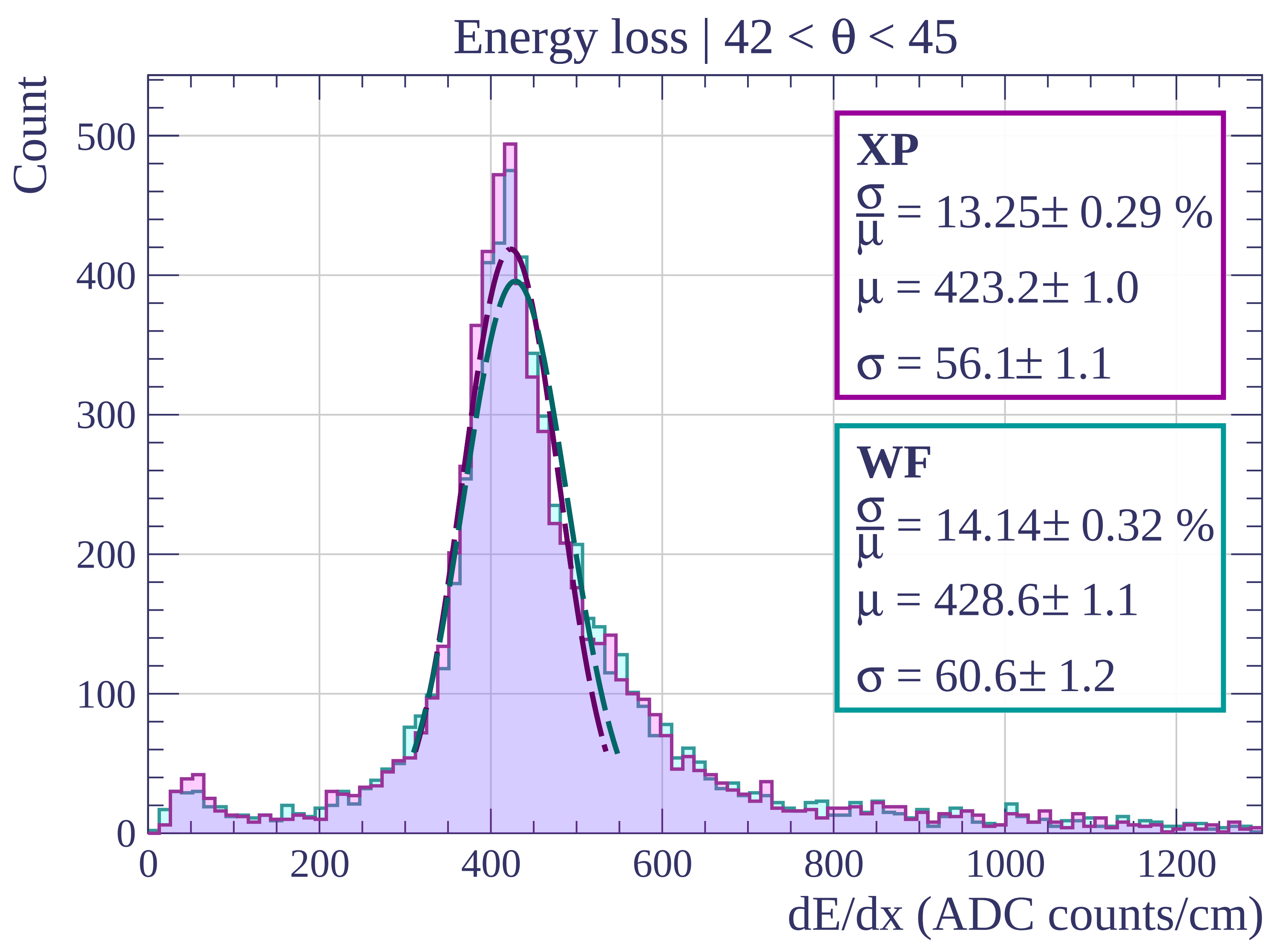
Energy loss | 30 < θ < 33
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 11.85± 0.18 %
μ = 424.2± 0.7
σ = 50.3± 0.7
WF
σ/μ = 12.36± 0.19 %
μ = 428.8± 0.7
σ = 53.0± 0.7
0
200
400
600
800
1000
1200


Energy loss | 33 < θ < 36
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 13.20± 0.23 %
μ = 425.4± 0.8
σ = 56.2± 0.9
WF
σ/μ = 13.78± 0.23 %
μ = 430.2± 0.8
σ = 59.3± 0.9
0
100
200
300
400
500
600
700
800
900
1000
1200

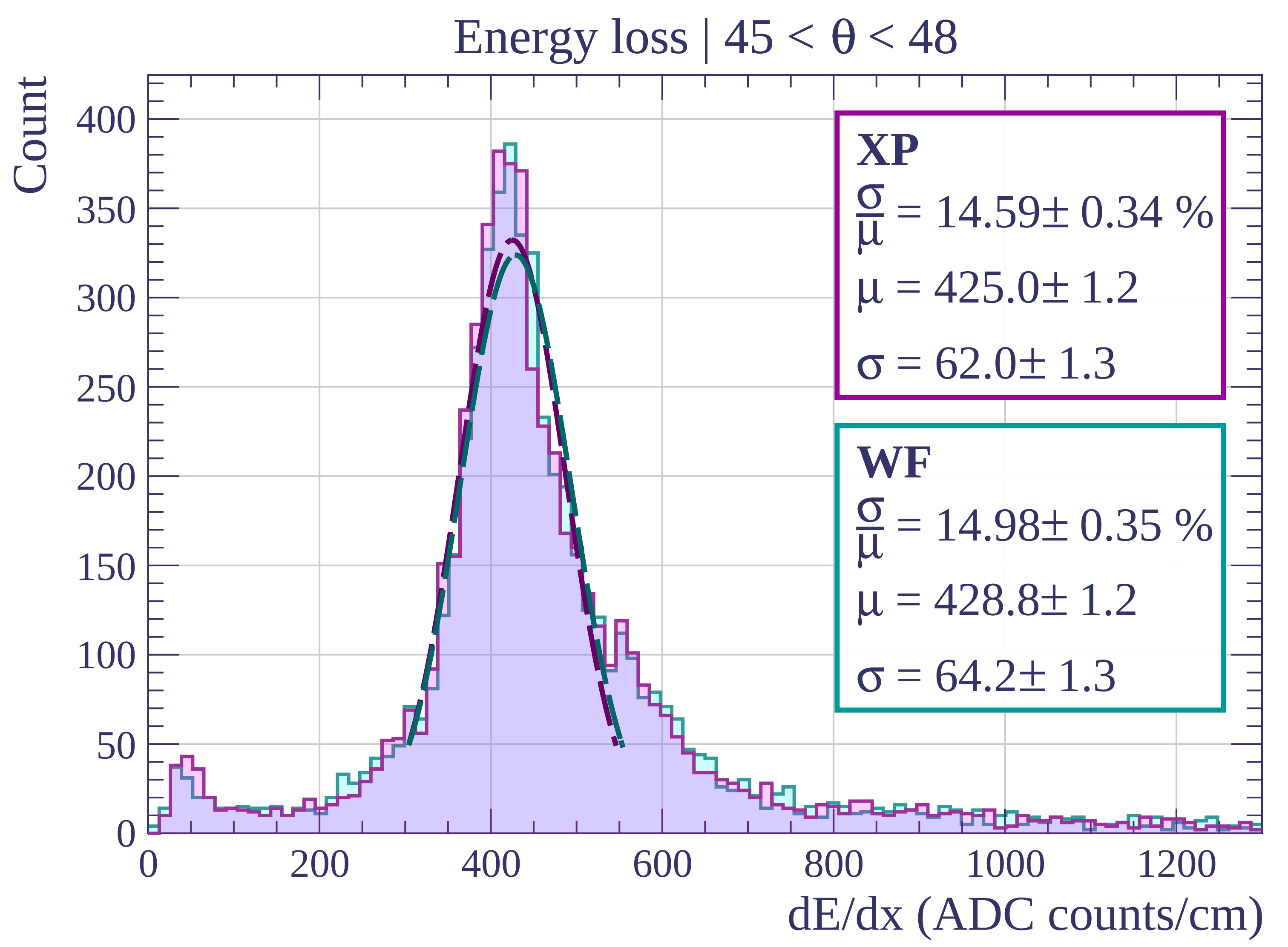

Energy loss | 36 < θ < 39
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
700
0
200
400
600
800
1000
1200
XP
σ/μ = 13.05± 0.24 %
μ = 423.8± 0.8
σ = 55.3± 0.9
WF
σ/μ = 13.66± 0.24 %
μ = 428.1± 0.9
σ = 58.5± 0.9

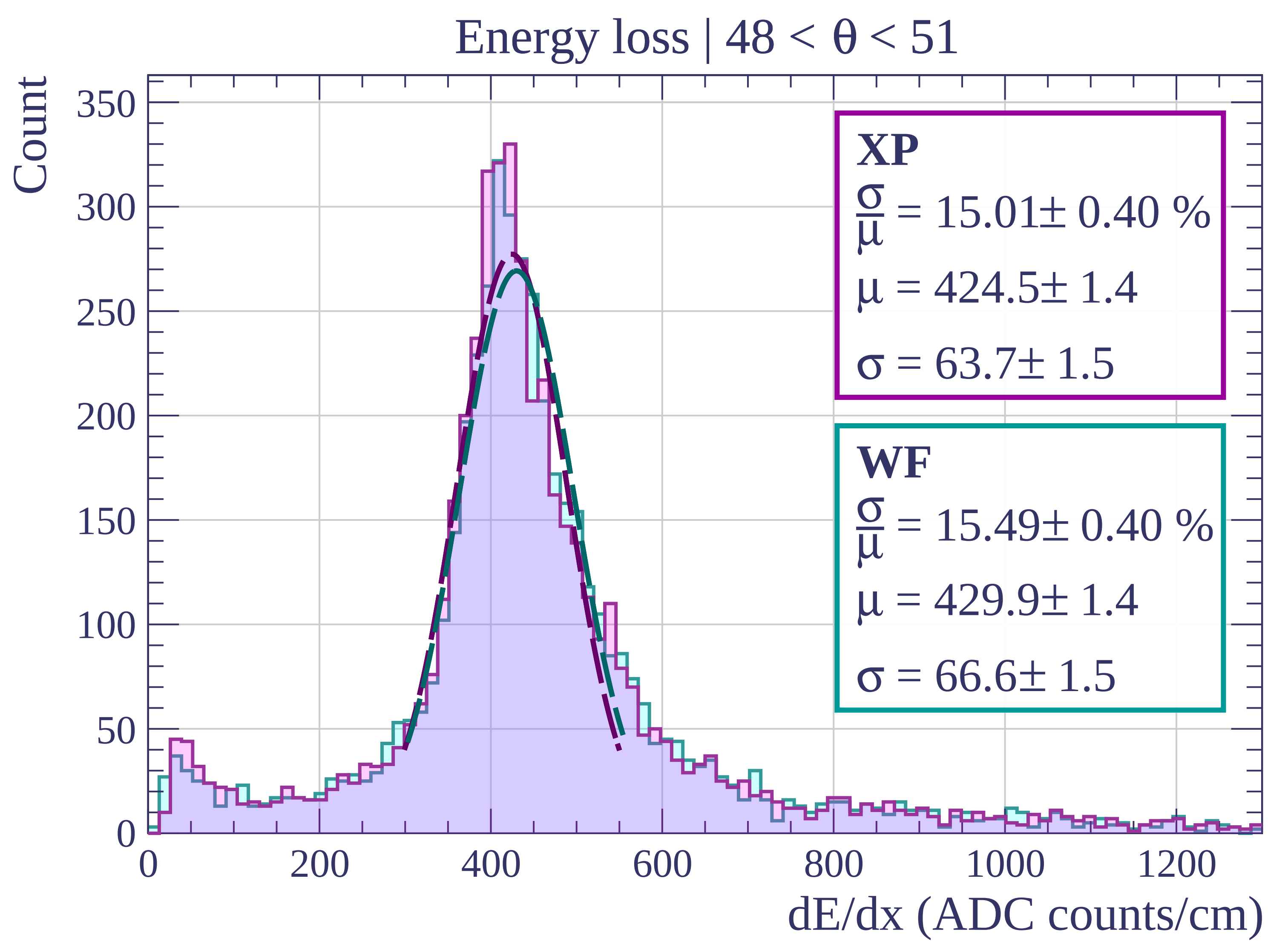


Energy loss | 39 < θ < 42
Count
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 13.02± 0.26 %
μ = 423.9± 0.9
σ = 55.2± 1.0
WF
$\frac{\sigma}{\mu}$ = 12.96± 0.26 %
μ = 426.5± 0.9
σ = 55.3± 1.0

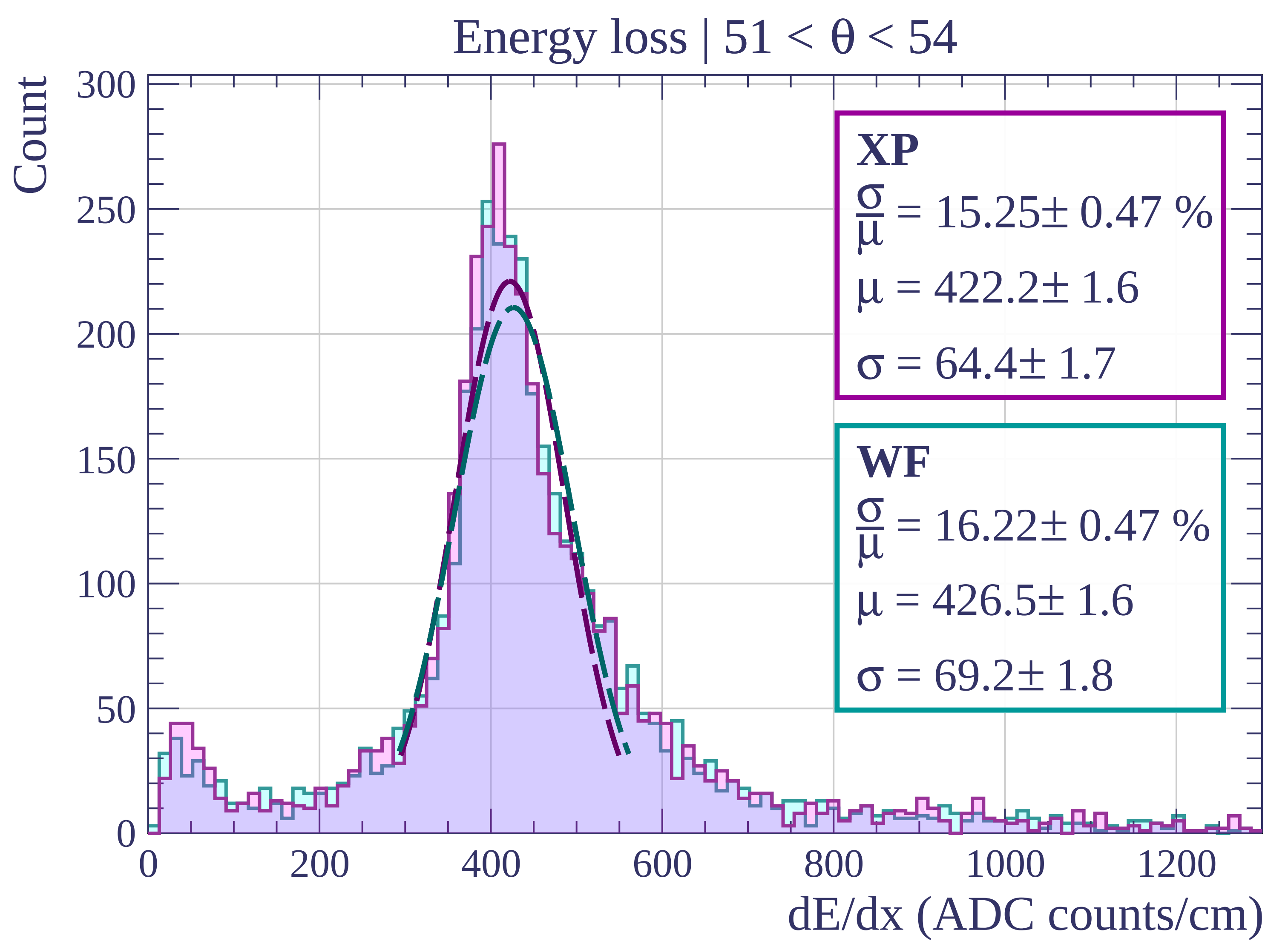


Energy loss | 42 < θ < 45
Count
XP
$\frac{\sigma}{\mu}$ = 13.25± 0.29 %
μ = 423.2± 1.0
σ = 56.1± 1.1
WF
$\frac{\sigma}{\mu}$ = 14.14± 0.32 %
μ = 428.6± 1.1
σ = 60.6± 1.2
0
100
200
300
400
500
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)

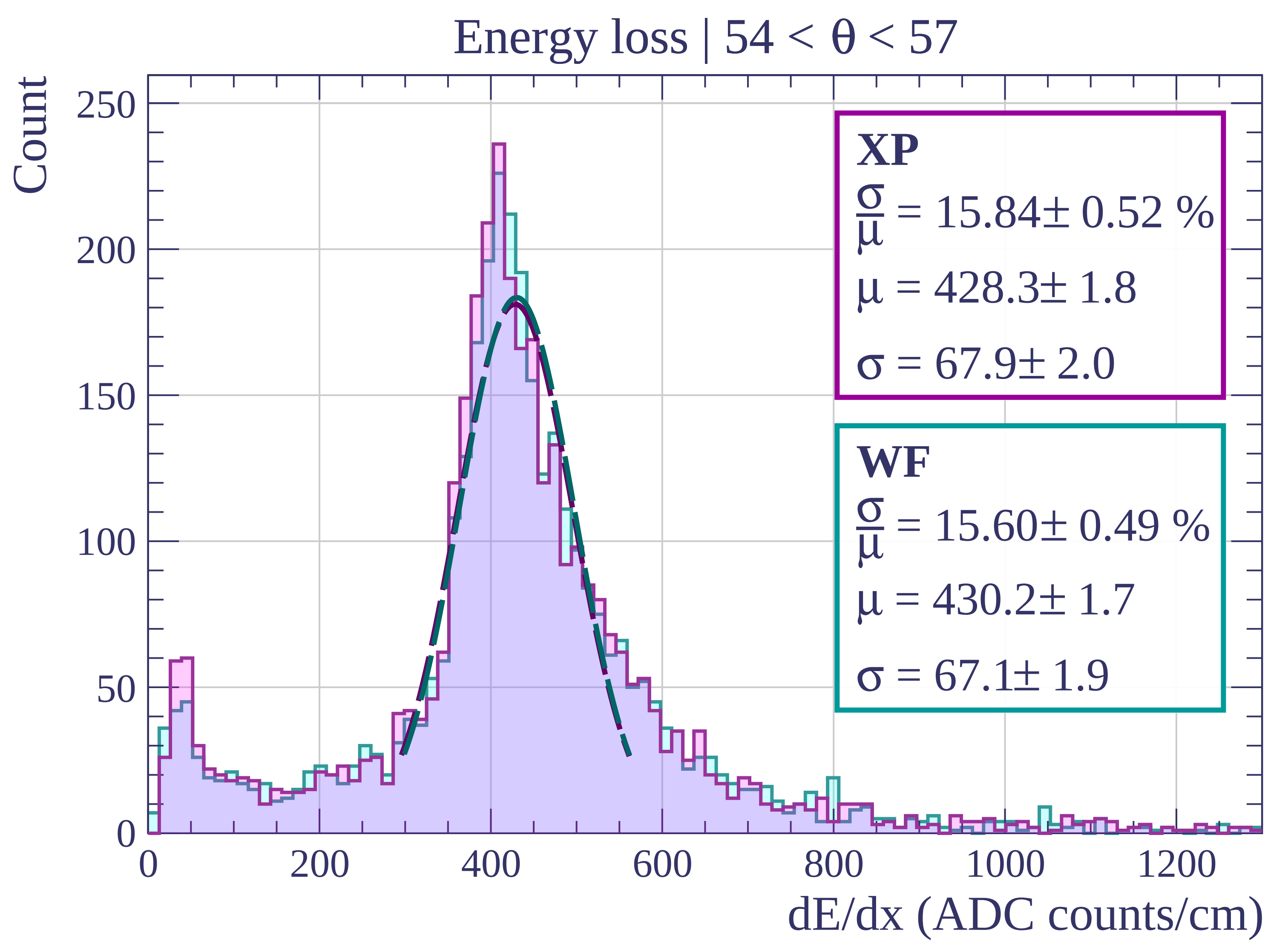

Energy loss | 45 < θ < 48
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
400
0
200
400
600
800
1000
1200
XP
σ/μ = 14.59± 0.34 %
μ = 425.0± 1.2
σ = 62.0± 1.3
WF
σ/μ = 14.98± 0.35 %
μ = 428.8± 1.2
σ = 64.2± 1.3



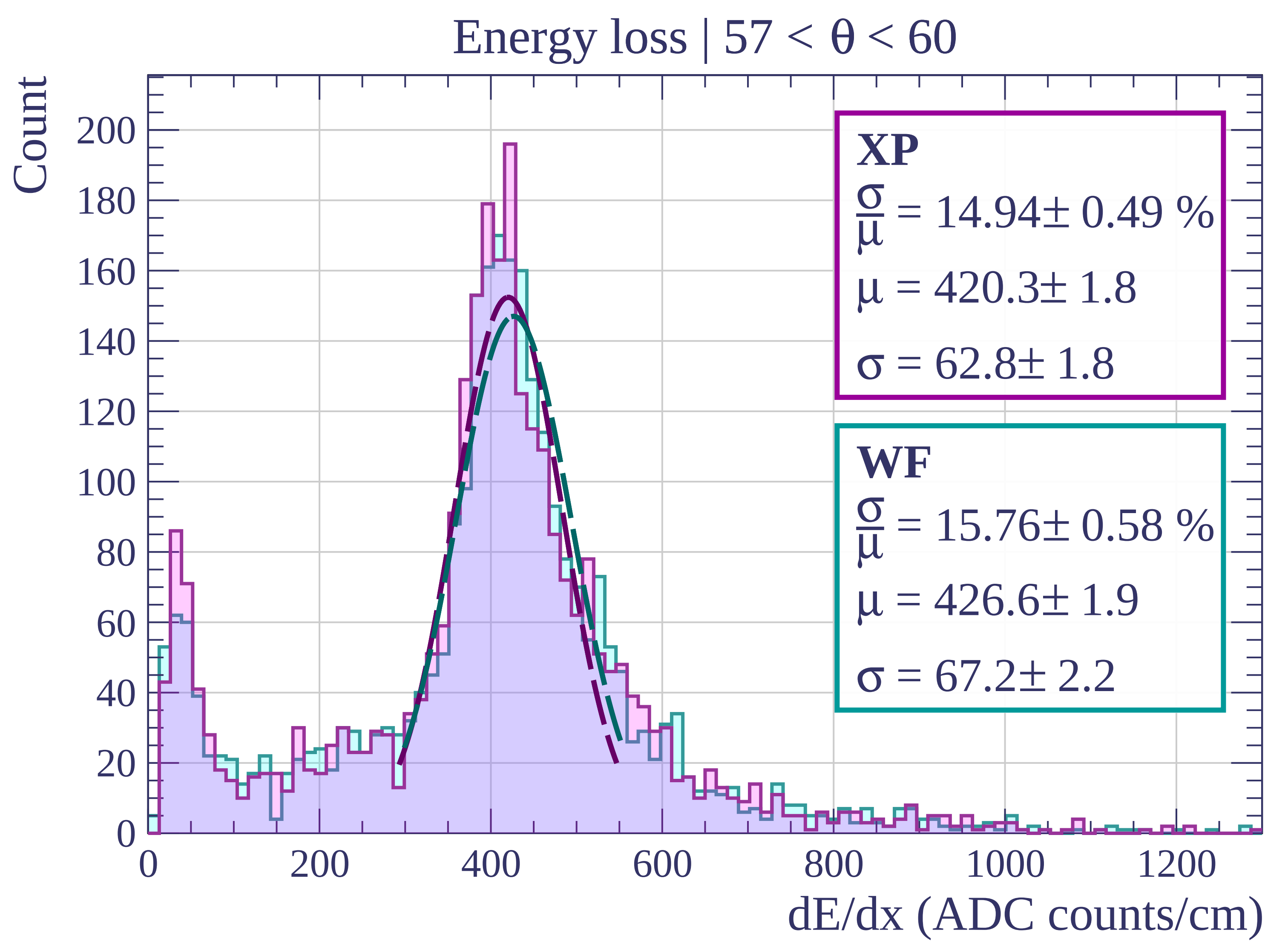
Energy loss | 48 < θ < 51
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 15.01± 0.40 %
μ = 424.5± 1.4
σ = 63.7± 1.5
WF
$\frac{\sigma}{\mu}$ = 15.49± 0.40 %
μ = 429.9± 1.4
σ = 66.6± 1.5



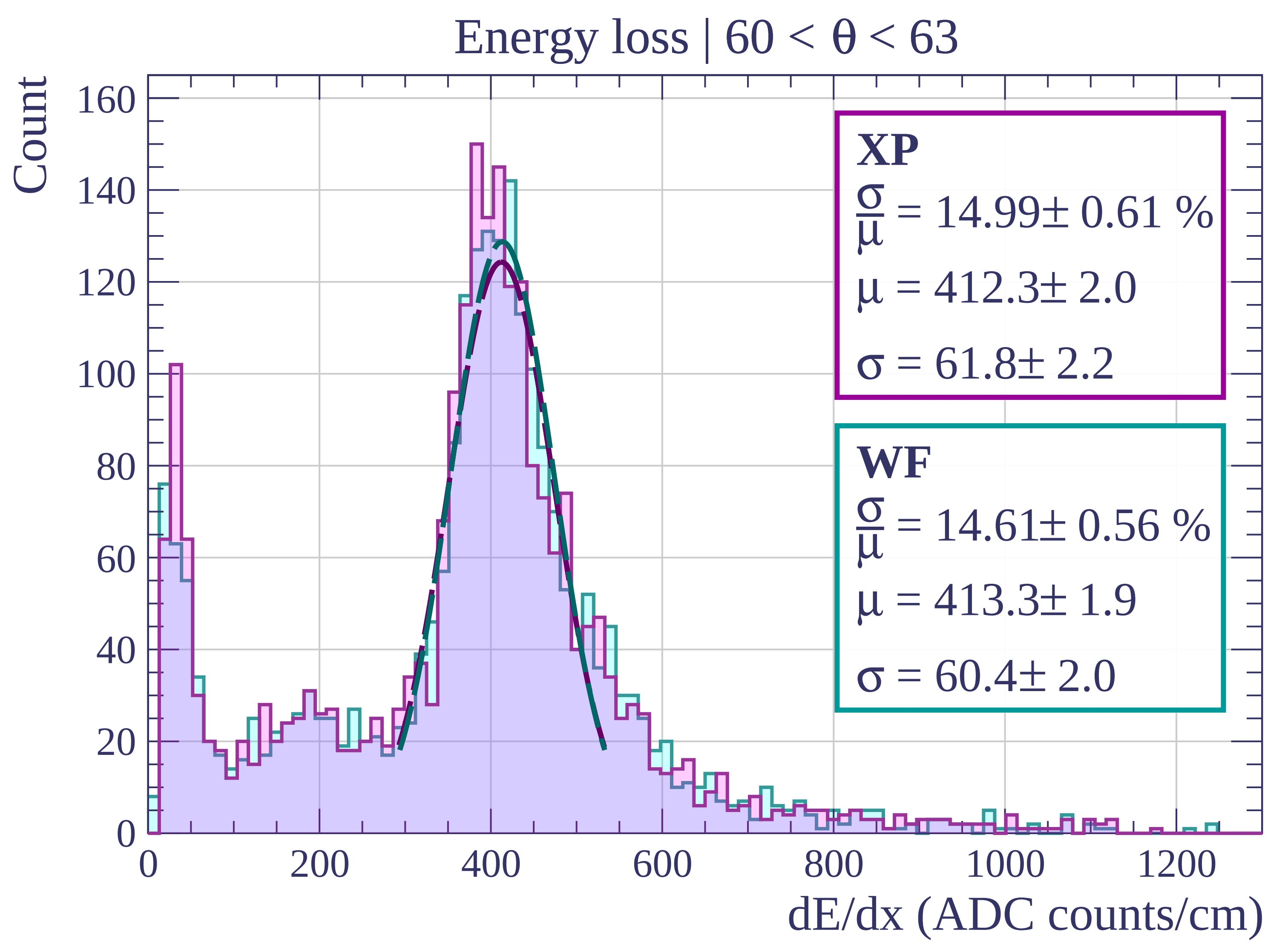
Energy loss | 51 < θ < 54
Count
300
250
200
150
100
50
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 15.25± 0.47 %
μ = 422.2± 1.6
σ = 64.4± 1.7
WF
σ/μ = 16.22± 0.47 %
μ = 426.5± 1.6
σ = 69.2± 1.8



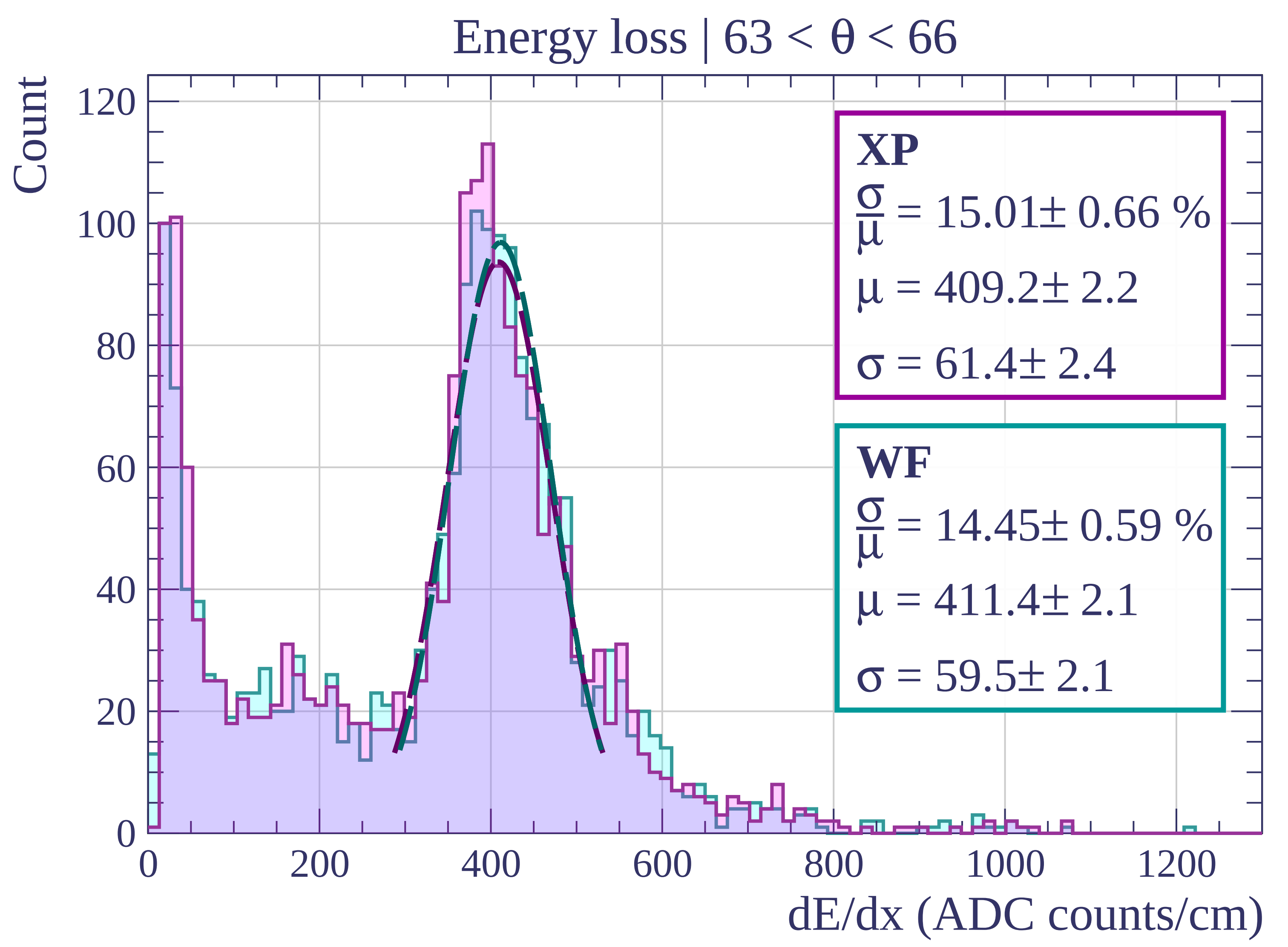
Energy loss | 54 < θ < 57
Count
250
200
150
100
50
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 15.84± 0.52 %
μ = 428.3± 1.8
σ = 67.9± 2.0
WF
$\frac{\sigma}{\mu}$ = 15.60± 0.49 %
μ = 430.2± 1.7
σ = 67.1± 1.9

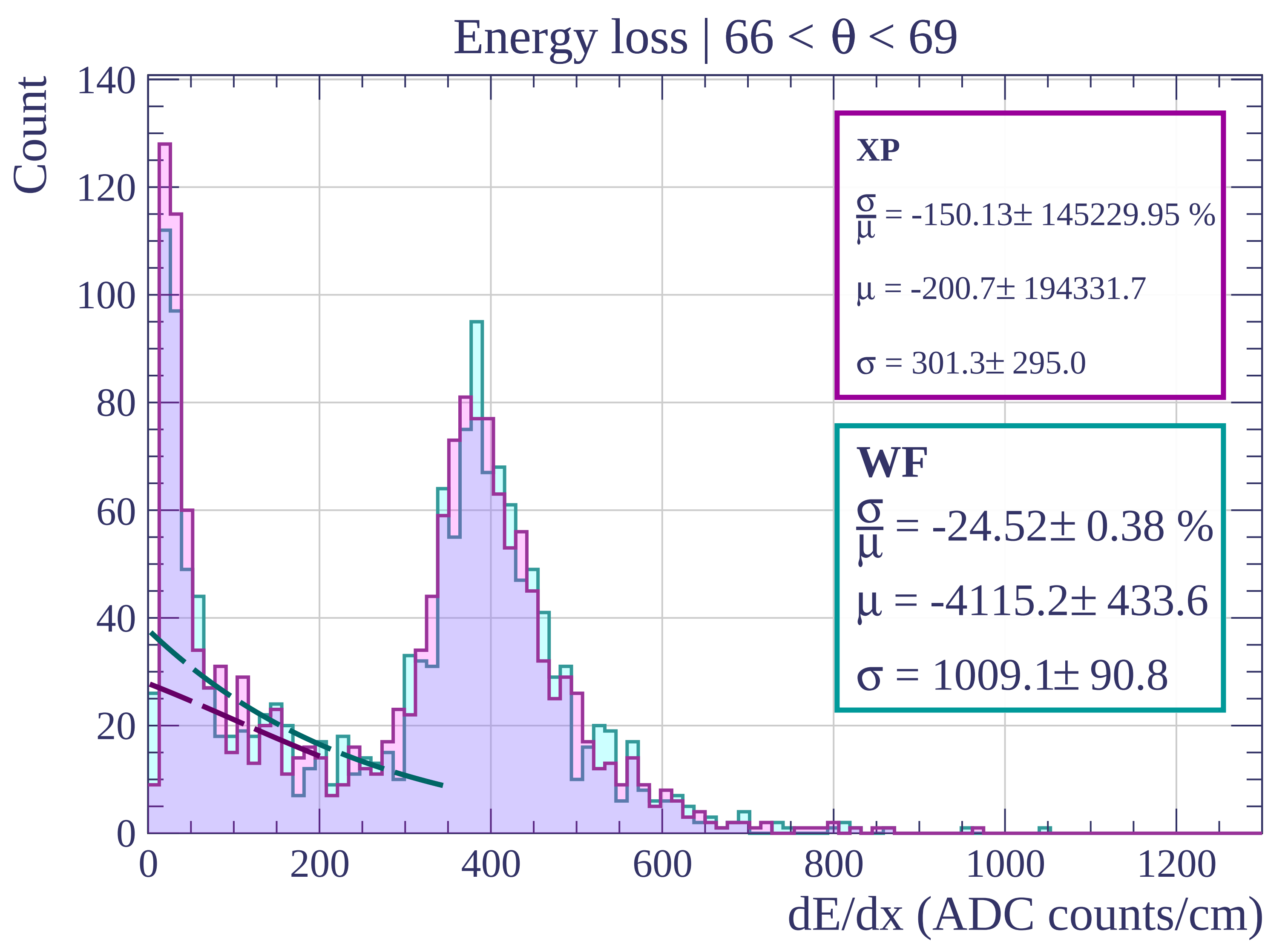


Energy loss | 57 < θ < 60
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
140
160
180
200
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 14.94± 0.49 %
μ = 420.3± 1.8
σ = 62.8± 1.8
WF
$\frac{\sigma}{\mu}$ = 15.76± 0.58 %
μ = 426.6± 1.9
σ = 67.2± 2.2

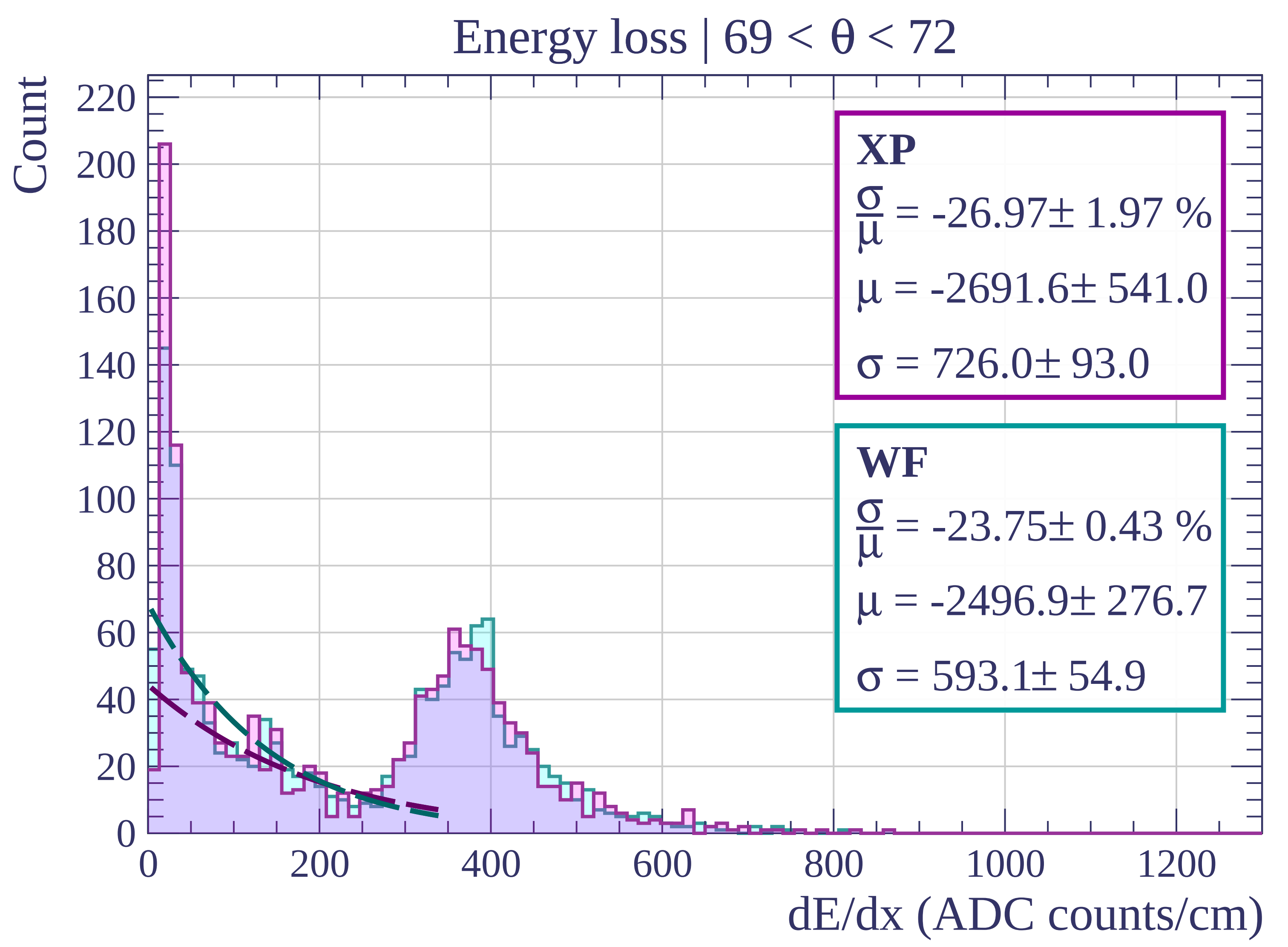


Energy loss | 60 < θ < 63
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
140
160
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 14.99± 0.61 %
μ = 412.3± 2.0
σ = 61.8± 2.2
WF
$\frac{\sigma}{\mu}$ = 14.61± 0.56 %
μ = 413.3± 1.9
σ = 60.4± 2.0

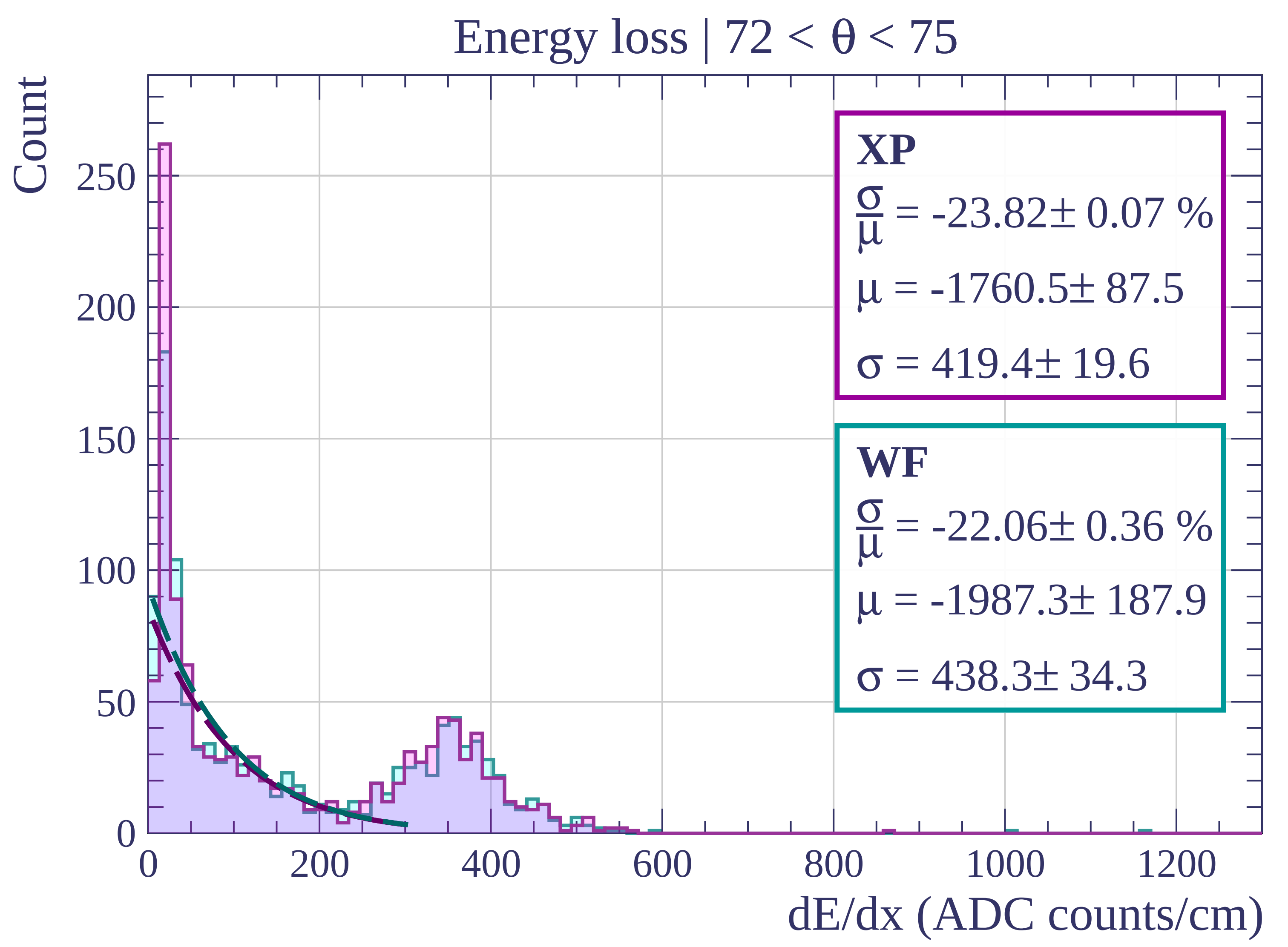

Energy loss | 63 < θ < 66
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 15.01± 0.66 %
μ = 409.2± 2.2
σ = 61.4± 2.4
WF
$\frac{\sigma}{\mu}$ = 14.45± 0.59 %
μ = 411.4± 2.1
σ = 59.5± 2.1



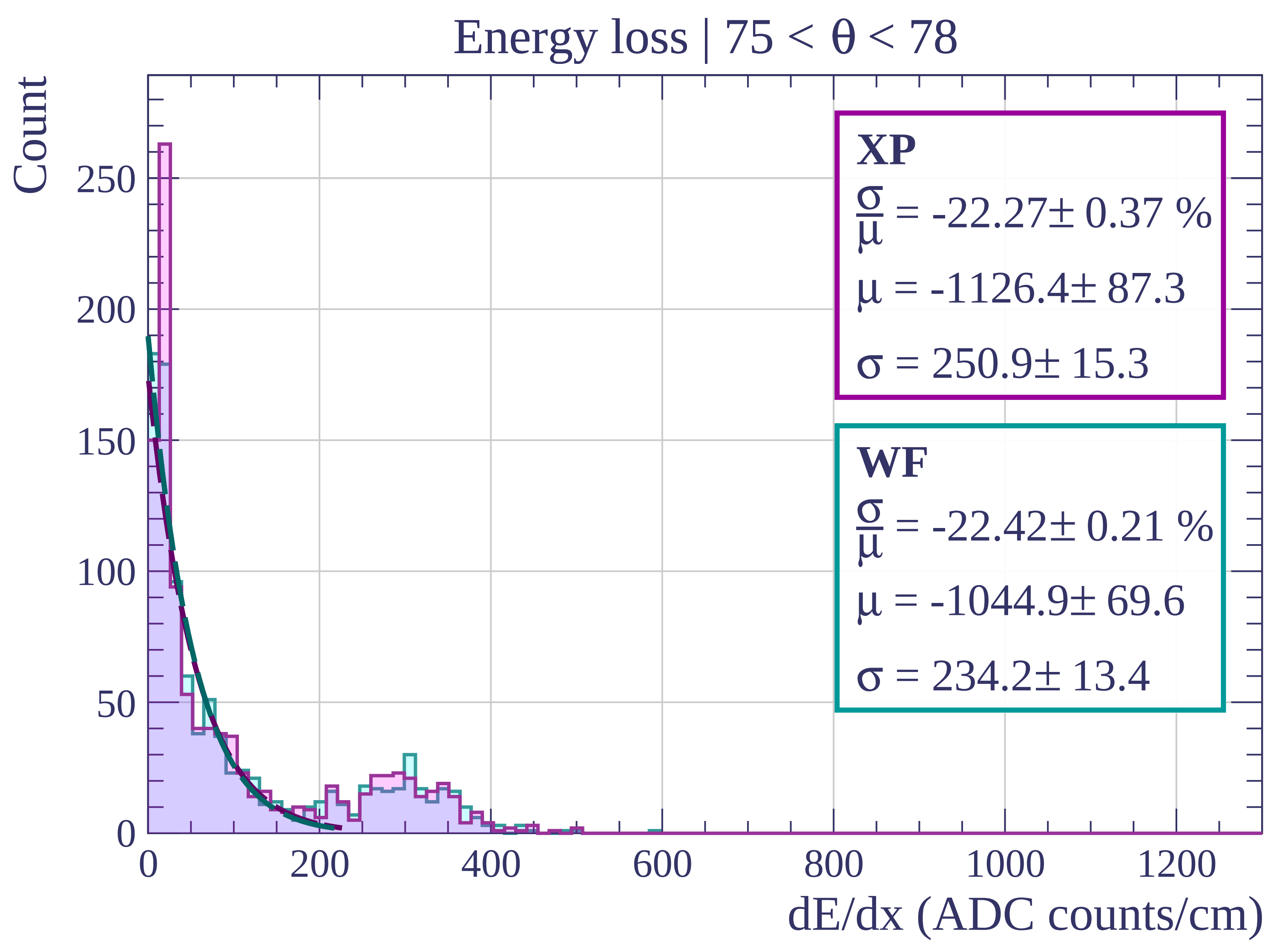
Energy loss | 66 < θ < 69
Count
XP
σ/μ = -150.13± 145229.95 %
μ = -200.7± 194331.7
σ = 301.3± 295.0
WF
σ/μ = -24.52± 0.38 %
μ = -4115.2± 433.6
σ = 1009.1± 90.8
dE/dx (ADC counts/cm)



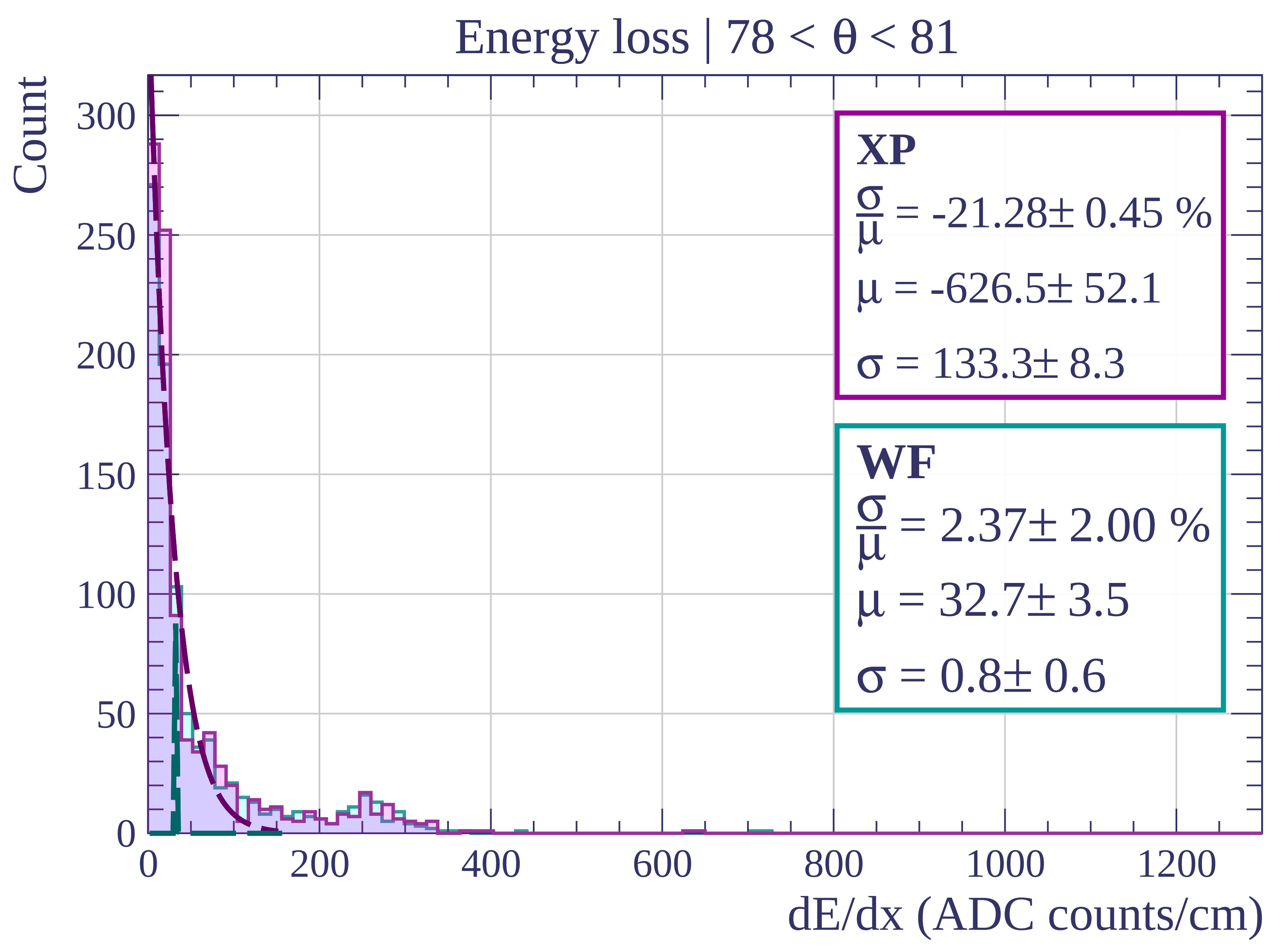
Energy loss | 69 < θ < 72
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
140
160
180
200
220
0
200
400
600
800
1000
1200
XP
σ/μ = -26.97± 1.97 %
μ = -2691.6± 541.0
σ = 726.0± 93.0
WF
σ/μ = -23.75± 0.43 %
μ = -2496.9± 276.7
σ = 593.1± 54.9



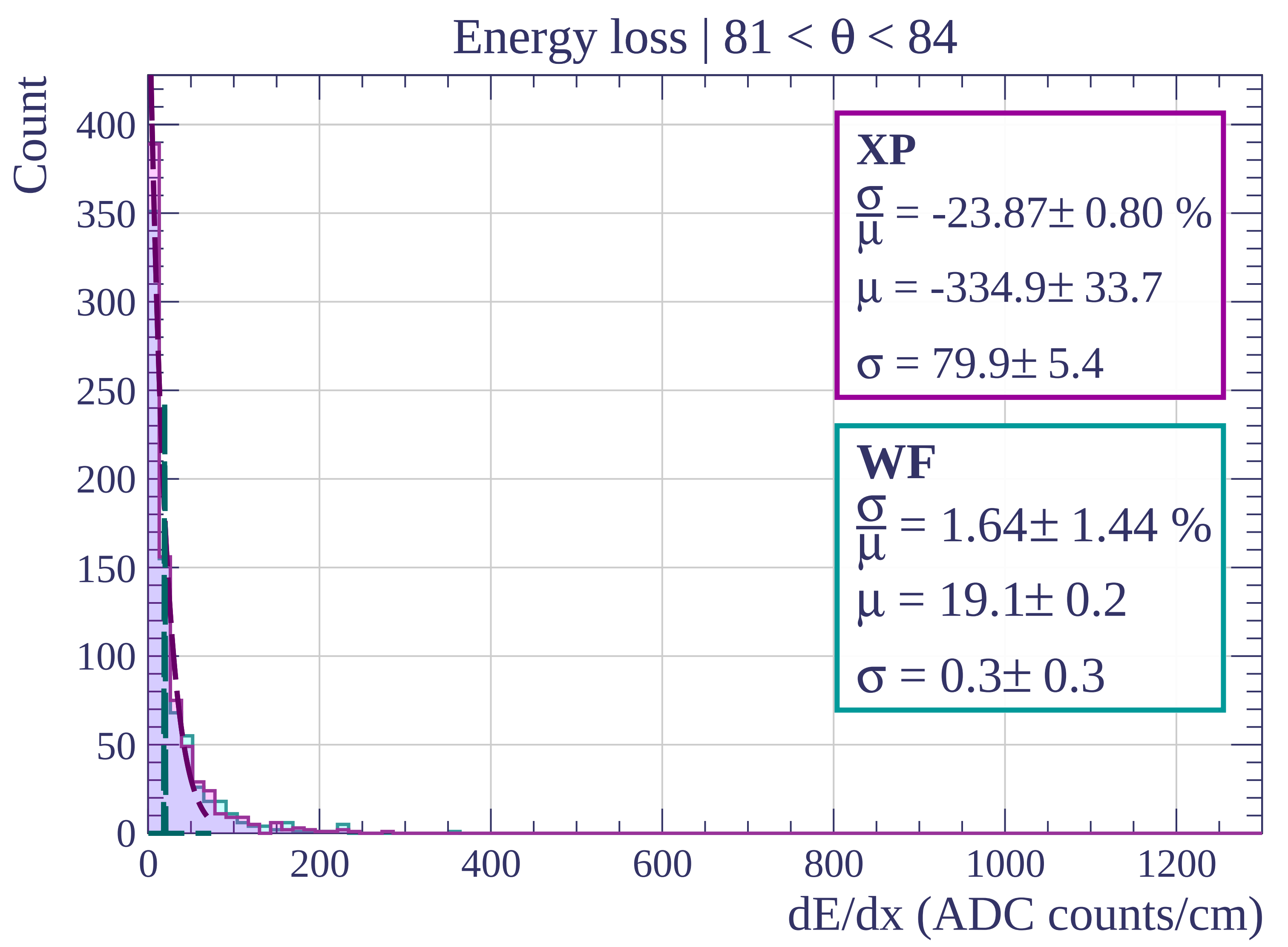
Energy loss | 72 < θ < 75
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = -23.82± 0.07 %
μ = -1760.5± 87.5
σ = 419.4± 19.6
WF
$\frac{\sigma}{\mu}$ = -22.06± 0.36 %
μ = -1987.3± 187.9
σ = 438.3± 34.3



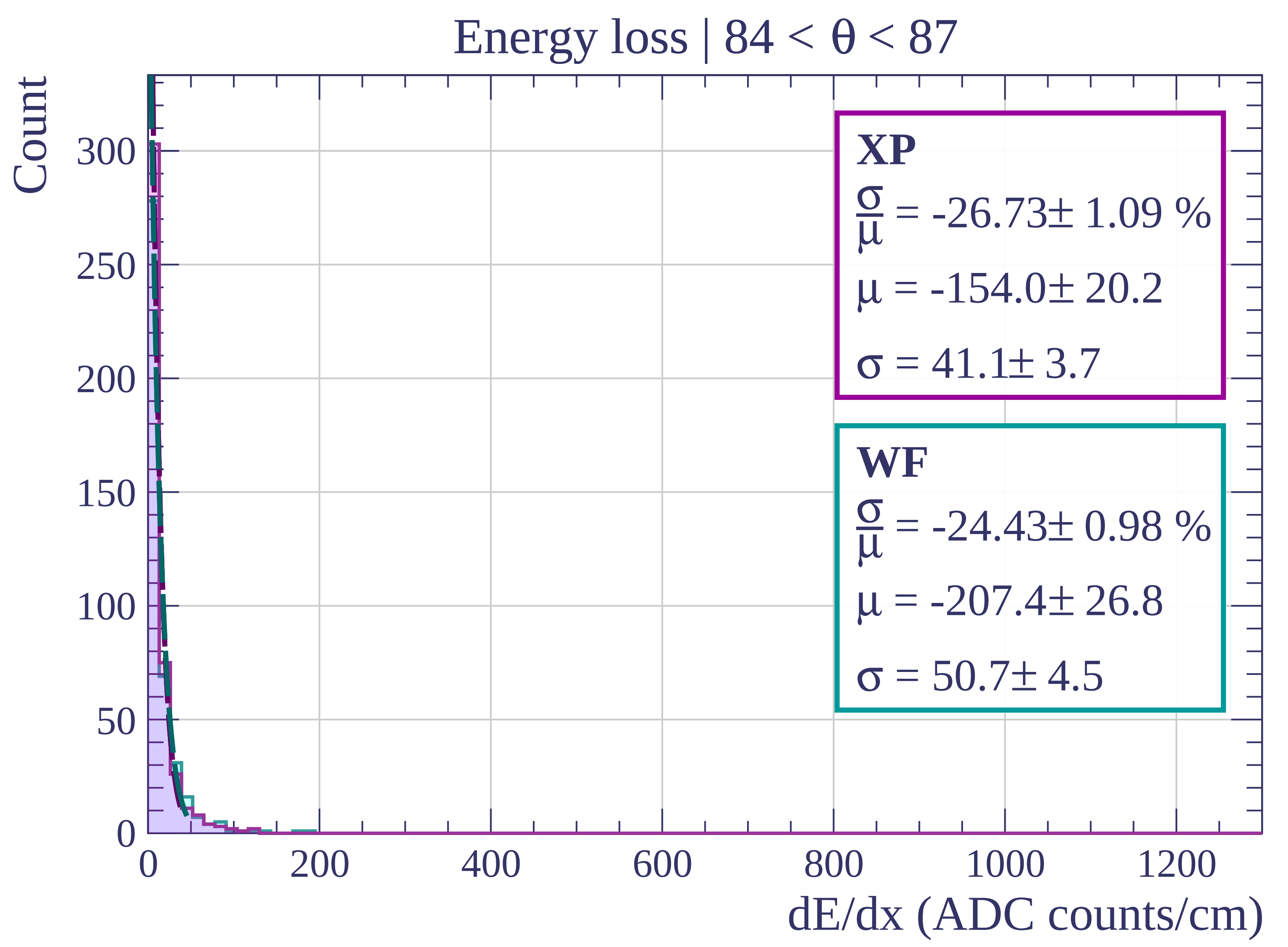
Energy loss | 75 < θ < 78
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = -22.27± 0.37 %
μ = -1126.4± 87.3
σ = 250.9± 15.3
WF
$\frac{\sigma}{\mu}$ = -22.42± 0.21 %
μ = -1044.9± 69.6
σ = 234.2± 13.4



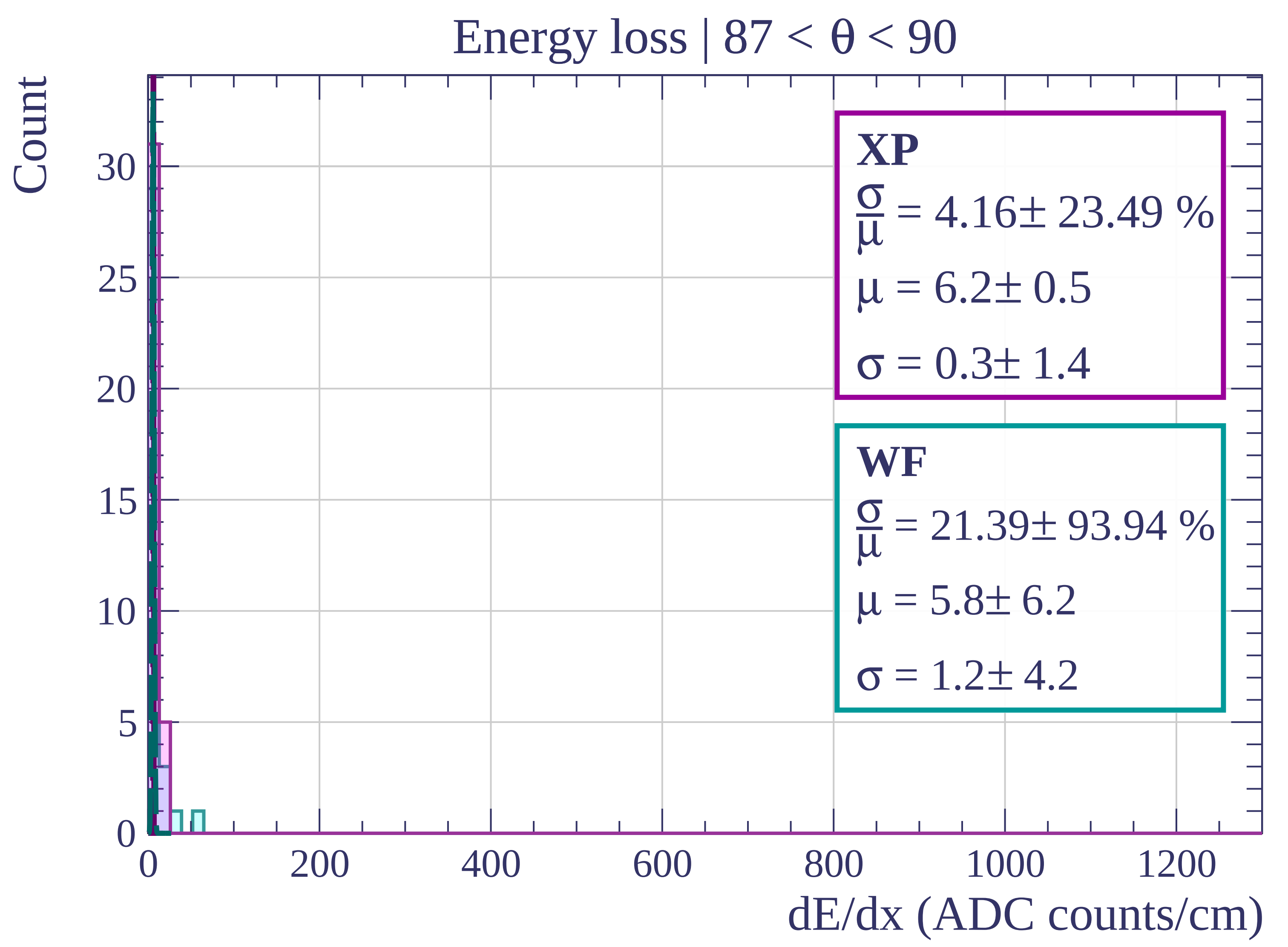
Energy loss | 78 < θ < 81
Count
300
250
200
150
100
50
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -21.28± 0.45 %
μ = -626.5± 52.1
σ = 133.3± 8.3
WF
σ/μ = 2.37± 2.00 %
μ = 32.7± 3.5
σ = 0.8± 0.6

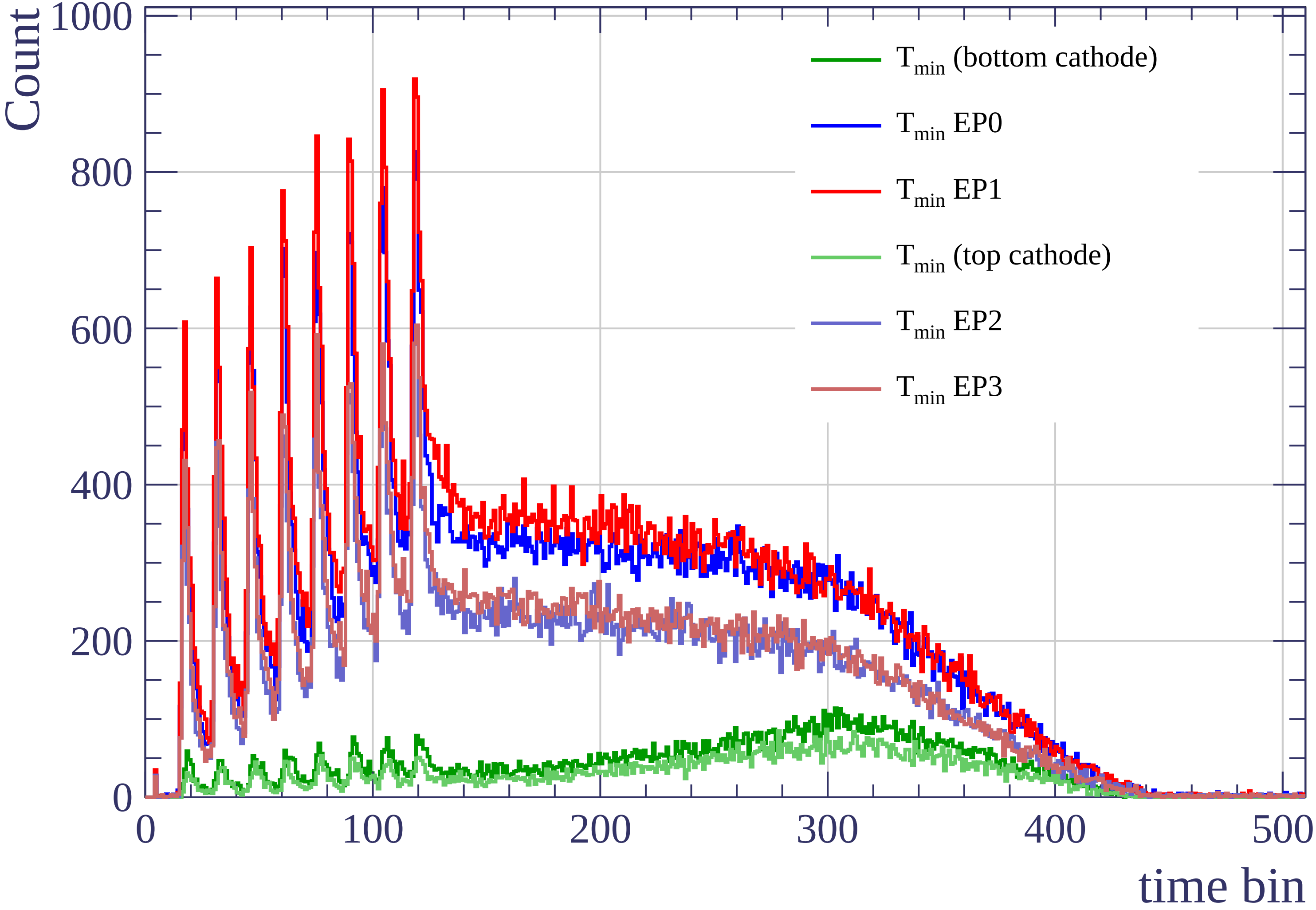


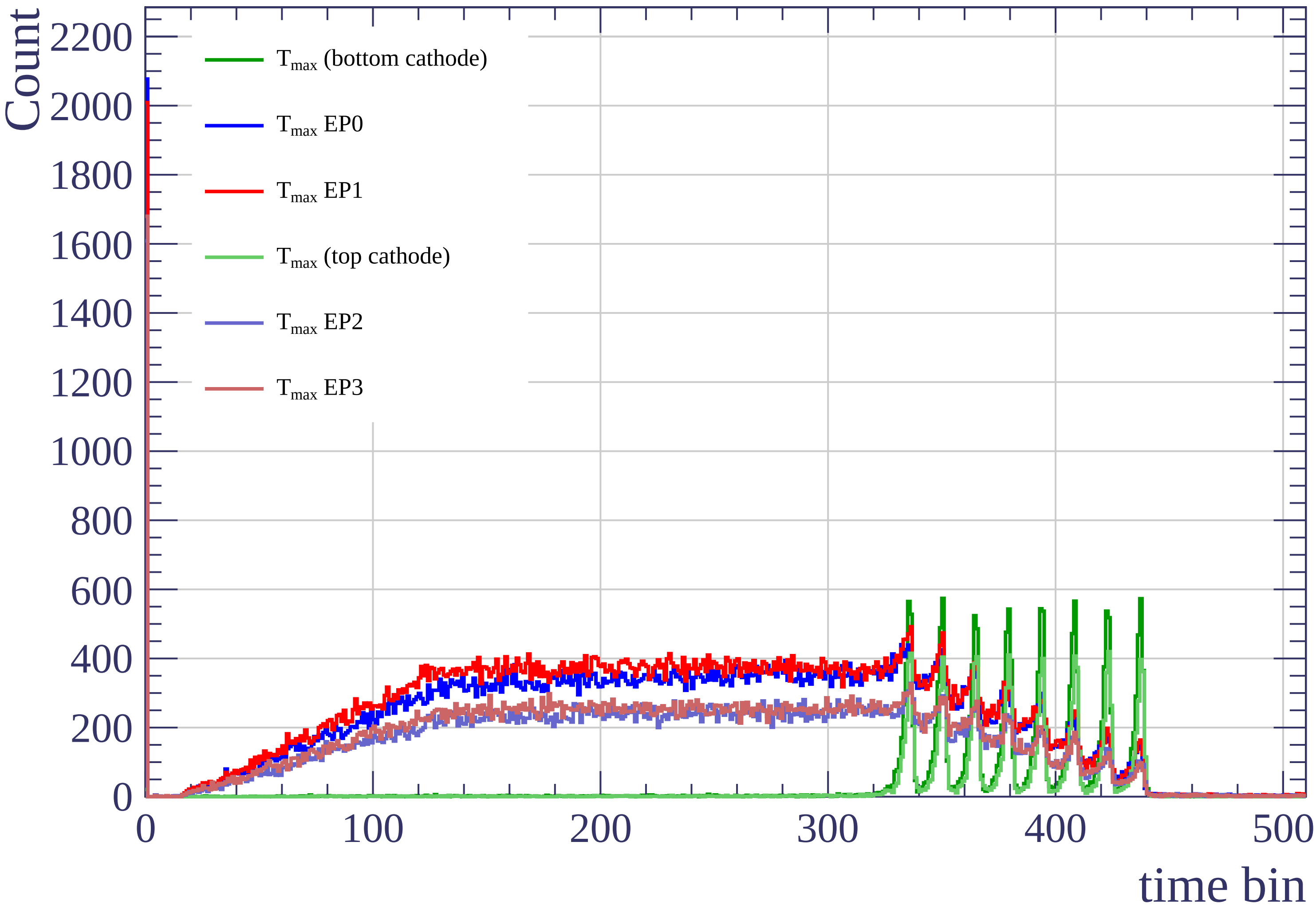
Energy loss | 81 < θ < 84
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
400
0
200
400
600
800
1000
1200
XP
σ/μ = -23.87± 0.80 %
μ = -334.9± 33.7
σ = 79.9± 5.4
WF
σ/μ = 1.64± 1.44 %
μ = 19.1± 0.2
σ = 0.3± 0.3



Energy loss | 84 < θ < 87
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
0
200
400
600
800
1000
1200
XP
σ/μ = -26.73± 1.09 %
μ = -154.0± 20.2
σ = 41.1± 3.7
WF
σ/μ = -24.43± 0.98 %
μ = -207.4± 26.8
σ = 50.7± 4.5


Energy loss | 87 < θ < 90
Count
dE/dx (ADC counts/cm)
0
5
10
15
20
25
30
0
200
400
600
800
1000
1200
XP
σ/μ = 4.16± 23.49 %
μ = 6.2± 0.5
σ = 0.3± 1.4
WF
σ/μ = 21.39± 93.94 %
μ = 5.8± 6.2
σ = 1.2± 4.2


Count
1000
800
600
400
200
0
0
100
200
300
400
500
time bin
T_min (bottom cathode)
T_min EP0
T_min EP1
T_min (top cathode)
T_min EP2
T_min EP3



Count
time bin
0
200
400
600
800
1000
1200
1400
1600
1800
2000
2200
0
100
200
300
400
500
T_{max} (bottom cathode)
T_{max} EP0
T_{max} EP1
T_{max} (top cathode)
T_{max} EP2
T_{max} EP3